AI & COMPUTING

A History of Mathematics

Florian Cajori

FAHAD.COM — Free Book Library

Public Domain Edition

The Project Gutenberg EBook of A History of Mathematics, by Florian Cajori This eBook is for the use of anyone anywhere at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org Title: A History of Mathematics Author: Florian Cajori Release Date: January 24, 2010 [EBook #31061] Language: English Character set encoding: ISO-8859-1 *** START OF THIS PROJECT GUTENBERG EBOOK A HISTORY OF MATHEMATICS ***

Produced by Andrew D. Hwang, Peter Vachuska, Carl Hudkins and the Online Distributed Proofreading Team at http://www.pgdp.net transcriber’s note Figures may have been moved with respect to the surrounding text. Minor typographical corrections and presentational changes have been made without comment. This PDF file is formatted for screen viewing, but may be easily formatted for printing. Please consult the preamble of the LATEX source file for instructions.

A HISTORY OF MATHEMATICS

A HISTORY OF M A T H E M A T I C S BY FLORIAN CAJORI, Ph.D. Formerly Professor of Applied Mathematics in the Tulane University of Louisiana; now Professor of Physics in Colorado College “I am sure that no subject loses more than mathematics by any attempt to dissociate it from its history.”—J. W. L. Glaisher New York THE MACMILLAN COMPANY LONDON: MACMILLAN & CO., Ltd. 1909 All rights reserved

Copyright, 1893, By MACMILLAN AND CO. Set up and electrotyped January, □□□□. Reprinted March, □□□□; October, □□□□; November, □□□□; January, □□□□; July, □□□□. Norwood Pre&: J. S. Cushing & Co.—Berwick & Smith. Norwood, Mass., U.S.A.

PREFACE. An increased interest in the history of the exact sciences manifested in recent years by teachers everywhere, and the attention given to historical inquiry in the mathematical class-rooms and seminaries of our leading universities, cause me to believe that a brief general History of Mathematics will be found acceptable to teachers and students. The pages treating—necessarily in a very condensed form— of the progress made during the present century, are put forth with great diffidence, although I have spent much time in the effort to render them accurate and reasonably complete. Many valuable suggestions and criticisms on the chapter on “Recent Times” have been made by Dr. E. W. Davis, of the University of Nebraska. The proof-sheets of this chapter have also been submitted to Dr. J. E. Davies and Professor C. A. Van Velzer, both of the University of Wisconsin; to Dr. G. B. Halsted, of the University of Texas; Professor L. M. Hoskins, of the Leland Stanford Jr. University; and Professor G. D. Olds, of Amherst College,—all of whom have afforded valuable assistance. I am specially indebted to Professor F. H. Loud, of Colorado College, who has read the proof-sheets throughout. To all the gentlemen above named, as well as to Dr. Carlo Veneziani of Salt Lake City, who read the first part of my work in manuscript, I desire to express my hearty thanks. But in acknowledging their kindness, I trust that I shall not seem to v

lay upon them any share in the responsibility for errors which I may have introduced in subsequent revision of the text. FLORIAN CAJORI. Colorado College, December, 1893.

TABLE OF CONTENTS Page INTRODUCTION . . . . . 1 ANTIQUITY . . . . . 5 The Babylonians . . . . . 5 The Egyptians . . . . . 10 The Greeks . . . . . 17 Greek Geometry . . . . . 17 The Ionic School . . . . . 19 The School of Pythagoras . . . . . 22 The Sophist School . . . . . 26 The Platonic School . . . . . 33 The First Alexandrian School . . . . . 39 The Second Alexandrian School . . . . . 62 Greek Arithmetic . . . . . 72 The Romans . . . . . 89 MIDDLE AGES . . . . . 97 The Hindoos . . . . . 97 The Arabs . . . . . 116 Europe During the Middle Ages . . . . . 135 Introduction of Roman Mathematics . . . . . 136 Translation of Arabic Manuscripts . . . . . 144 The First Awakening and its Sequel . . . . . 148 MODERN EUROPE . . . . . 160 The Renaissance . . . . . 161 Vieta to Descartes . . . . . 181 Descartes to Newton . . . . . 213 Newton to Euler . . . . . 231 vii

TABLE OF CONTENTS. viii Page Euler, Lagrange, and Laplace . . . . . 286 The Origin of Modern Geometry . . . . . 332 RECENT TIMES . . . . . 339 Synthetic Geometry . . . . . 341 Analytic Geometry . . . . . 358 Algebra . . . . . 367 Analysis . . . . . 386 Theory of Functions . . . . . 405 Theory of Numbers . . . . . 422 Applied Mathematics 435

BOOKS OF REFERENCE. The following books, pamphlets, and articles have been used in the preparation of this history. Reference to any of them is made in the text by giving the respective number. Histories marked with a star are the only ones of which extensive use has been made. 1. G¨unther, S. Ziele und Resultate der neueren Mathematisch- historischen Forschung. Erlangen, 1876. 2. Cajori, F. The Teaching and History of Mathematics in the U. S. Washington, 1890. 3. *Cantor, Moritz. Vorlesungen¨uber Geschichte der Mathematik. Leipzig. Bd. I., 1880; Bd. II., 1892. 4. Epping, J. Astronomisches aus Babylon. Unter Mitwirkung von P. J. R. Strassmaier. Freiburg, 1889. 5. Bretschneider, C. A. Die Geometrie und die Geometer vor Euklides. Leipzig, 1870. 6. *Gow, James. A Short History of Greek Mathematics.Cambridge, 1884. 7. *Hankel, Hermann. Zur Geschichte der Mathematik im Alterthum und Mittelalter. Leipzig, 1874. 8. *Allman, G. J. Greek Geometry from Thales to Euclid. Dublin, 1889. 9. De Morgan, A. “Euclides” in Smith’s Dictionary of Greek and Roman Biography and Mythology. 10. Hankel, Hermann. Theorie der Complexen Zahlensysteme. Leipzig, 1867. 11. Whewell, William. History of the Inductive Sciences. 12. Zeuthen, H. G. Die Lehre von den Kegelschnitten im Alterthum. Kopenhagen, 1886. ix

A HISTORY OF MATHEMATICS. x 13. *Chasles, M. Geschichte der Geometrie. Aus dem Franz¨osischen ¨ubertragen durch Dr. L. A. Sohncke. Halle, 1839. 14. Marie, Maximilien. Histoire des Sciences Math´ ematiques et Physiques. Tome I.–XII. Paris,

1883–1888. 15. Comte, A. Philosophy of Mathematics , translated by W. M. Gillespie. 16. Hankel, Hermann. Die Entwickelung der Mathematik in den letzten Jahrhunderten. T¨ubingen, 1884. 17. G¨unther, Siegmund und Windelband, W. Geschichte der antiken Naturwissenschaft und Philosophie. N¨ordlingen, 1888. 18. Arneth, A. Geschichte der reinen Mathematik. Stuttgart, 1852. 19. Cantor, Moritz. Mathematische Beitr¨age zum Kulturleben der V¨olker. Halle, 1863. 20. Matthiessen, Ludwig. Grundz¨uge der Antiken und Modernen Algebra der Litteralen Gleichungen. Leipzig, 1878. 21. Ohrtmann und M¨uller. Fortschritte der Mathematik. 22. Peacock, George. Article “Arithmetic,” in The Encyclopædia of Pure Mathematics. London, 1847. 23. Herschel, J. F. W. Article “Mathematics,” in Edinburgh Encyclopædia. 24. Suter, Heinrich. Geschichte der Mathematischen Wissenschaf- ten. Z¨urich, 1873–75. 25. Quetelet, A. Sciences Math´ ematiques et Physiques chez les Belges. Bruxelles, 1866. 26. Playfair, John. Article “Progress of the Mathematical and Physical Sciences,” in Encyclopædia Britannica , 7th edition, continued in the 8th edition by Sir John Leslie. 27. De Morgan, A. Arithmetical Books from the Invention of Printing to the Present Time. 28. Napier, Mark. Memoirs of John Napier of Merchiston. Edinburgh, 1834. 29. Halsted, G. B. “Note on the First English Euclid,” American Journal of Mathematics, Vol. II., 1879.

BOOKS OF REFERENCE. xi 30. Madame Perier. The Life of Mr. Paschal. Translated into English by W. A., London, 1744. 31. Montucla, J. F. Histoire des Math´ ematiques.Paris, 1802. 32. D¨uhring E. Kritische Geschichte der allgemeinen Principien der Mechanik. Leipzig, 1887. 33. Brewster, D. The Memoirs of Newton. Edinburgh, 1860. 34. Ball, W. W. R. A Short Account of the History of Mathematics. London, 1888, 2nd edition, 1893. 35. De Morgan, A. “On the Early History of Infinitesimals,” in the Philosophical Magazine, November, 1852. 36. Bibliotheca Mathematica, herausgegeben von Gustaf Enestr¨om, Stockholm. 37. G¨unther, Siegmund. Vermischte Untersuchungen zur Geschich- te der mathematischen Wissenschaften. Leipzig, 1876. 38. *Gerhardt, C. I. Geschichte der Mathematik in Deutschland. M¨unchen, 1877. 39. Gerhardt, C. I. Entdeckung der Differenzialrechnung durch Leibniz. Halle, 1848. 40. Gerhardt, K. I. “Leibniz in London,” in Sitzungsberichte der K¨oniglich Preussischen Academie der Wissenschaften zu Berlin, Februar, 1891. 41. De Morgan, A. Articles “Fluxions” and “Commercium Epis- tolicum,” in the Penny Cyclopædia. 42. *Todhunter, I. A History of the Mathematical Theory of Probability from the Time of Pascal to that of Laplace. Cambridge and London, 1865. 43. *Todhunter, I. A History of the Theory of Elasticity and of the Strength of Materials. Edited and completed by Karl Pearson. Cambridge, 1886. 44. Todhunter, I. “Note on the History of Certain Formulæ in Spherical Trigonometry,” Philosophical Magazine , February, 1873. 45. Die Basler Mathematiker, Daniel Bernoulli und Leonhard Euler. Basel, 1884.

A HISTORY OF MATHEMATICS. xii 46. Reiff, R. Geschichte der Unendlichen Reihen. T¨ubingen, 1889. 47. Waltershausen, W. Sartorius. Gauss, zum Ged ¨achtniss. Leipzig, 1856. 48. Baumgart, Oswald.Ueber das Quadratische Reciprocit¨atsgesetz. Leipzig, 1885. 49. Hathaway, A. S. “Early History of the Potential,” Bulletin of the N. Y. Mathematical Society, I. 3. 50. Wolf, Rudolf. Geschichte der Astronomie. M¨unchen, 1877. 51. Arago, D. F. J.“Eulogy on Laplace.”

Translated byB. Powell, Smithsonian Report, 1874. 52. Beaumont, M. ´Elie De. “Memoir of Legendre.” Translated by C. A. Alexander, Smithsonian Report, 1867. 53. Arago, D. F. J. “Joseph Fourier.” Smithsonian Report, 1871. 54. Wiener, Christian. Lehrbuch der Darstellenden Geometrie. Leipzig, 1884. 55. *Loria, Gino. Die Haupts ¨achlichsten Theorien der Geometrie in ihrer fr ¨uheren und heutigen Entwickelung , ins deutsche ¨ubertragen von Fritz Sch¨utte. Leipzig, 1888. 56. Cayley, Arthur. Inaugural Address before the British Associa- tion, 1883. 57. Spottiswoode, William. Inaugural Address before the British Association, 1878. 58. Gibbs, J. Willard. “Multiple Algebra,” Proceedings of the American Association for the Advancement of Science, 1886. 59. Fink, Karl. Geschichte der Elementar-Mathematik. T¨ubingen, 1890. 60. Wittstein, Armin. Zur Geschichte des Malfatti’schen Problems. N¨ordlingen, 1878. 61. Klein, Felix. Vergleichende Betrachtungen ¨uber neuere geome- trische Forschungen. Erlangen, 1872. 62. Forsyth, A. R. Theory of Functions of a Complex Variable. Cambridge, 1893. 63. Graham, R. H. Geometry of Position. London, 1891.

BOOKS OF REFERENCE. xiii 64. Schmidt, Franz. “Aus dem Leben zweier ungarischer Mathe- matiker Johann und Wolfgang Bolyai von Bolya.” Grunert’s Archiv, 48:2, 1868. 65. Favaro, Anton. “Justus Bellavitis,” Zeitschrift f¨ur Mathematik und Physik, 26:5, 1881. 66. Dronke, Ad. Julius Pl¨ ucker.Bonn, 1871. 67. Bauer, Gustav. Ged¨achtnissrede auf Otto Hesse. M¨unchen, 1882. 68. Alfred Clebsch. Versuch einer Darlegung und W ¨urdigung seiner wissenschaftlichen Leistungen von einigen seiner Freunde. Leipzig, 1873. 69. Haas, August. Versuch einer Darstellung der Geschichte des Kr¨ummungsmasses. T¨ubingen, 1881. 70. Fine, Henry B. The Number-System of Algebra. Boston and New York, 1890. 71. Schlegel, Victor. Hermann Grassmann, sein Leben und seine Werke. Leipzig, 1878. 72. Zahn, W. v. “Einige Worte zum Andenken an Hermann Hankel,” Mathematische Annalen, VII. 4, 1874. 73. Muir, Thomas. A Treatise on Determinants. 1882. 74. Salmon, George. “Arthur Cayley,” Nature, 28:21, September, 1883. 75. Cayley, A. “James Joseph Sylvester,” Nature, 39:10, January, 1889. 76. Burkhardt, Heinrich. “Die Anf ¨ange der Gruppentheorie und Paolo Ruffini,” Zeitschrift f ¨ur Mathematik und Physik , Supplement, 1892. 77. Sylvester, J. J.Inaugural Presidential Address to the Mathemat- ical and Physical Section of the British Association at Exeter. 1869. 78. Valson, C. A. La Vie et les travaux du Baron Cauchy. Tome I., II., Paris, 1868. 79. Sachse, Arnold. Versuch einer Geschichte der Darstellung willk¨urlicher Funktionen einer variablen durch trigonometrische Reihen. G¨ottingen, 1879.

A HISTORY OF MATHEMATICS. xiv 80. Bois-Reymond, Paul du.Zur Geschichte der Trigonometrischen Reihen, Eine Entgegnung. T¨ubingen. 81. Poincar´e, Henri. Notice sur les Travaux Scientifiques de Henri Poincar´ e.Paris, 1886. 82. Bjerknes, C. A. Niels-Henrik Abel, Tableau de sa vie et de son action scientifique. Paris, 1885. 83. Tucker, R. “Carl Friedrich Gauss,” Nature, April, 1877. 84. Dirichlet, Lejeune. Ged¨achtnissrede auf Carl Gustav Jacob Jacobi. 1852. 85. Enneper, Alfred. Elliptische Funktionen. Theorie und Ge- schichte. Halle a/S., 1876. 86. Henrici, O. “Theory of Functions,” Nature, 43:14 and 15, 1891. 87. Darboux, Gaston. Notice sur les Travaux Scientifiques de M. Gaston Darboux. Paris, 1884. 88. Kummer, E. E. Ged¨achtnissrede auf Gustav Peter Lejeune-Diri- chlet. Berlin, 1860. 89. Smith, H. J.

Stephen. “On the Present State and Prospects of Some Branches of Pure Mathematics,” Proceedings of the London Mathematical Society, Vol. VIII., Nos. 104, 105, 1876. 90. Glaisher, J. W. L. “Henry John Stephen Smith,” Monthly Notices of the Royal Astronomical Society, XLIV., 4, 1884. 91. Bessel als Bremer Handlungslehrling. Bremen, 1890. 92. Frantz, J. Festrede aus Veranlassung von Bessel’s hundertj ¨ahr-igem Geburtstag. K¨onigsberg, 1884. 93. Dziobek, O. Mathematical Theories of Planetary Motions. Translated into English by M. W. Harrington and W. J. Hussey. 94. Hermite, Ch. “Discours prononc´ e devant le pr´ esident de la R´ epublique,”Bulletin des Sciences Math´ ematiques, XIV., Janvier, 1890. 95. Schuster, Arthur. “The Influence of Mathematics on the Progress of Physics,” Nature, 25:17, 1882. 96. Kerbedz, E. de. “Sophie de Kowalevski,” Rendiconti del Circolo Matematico di Palermo, V., 1891. 97. Voigt, W. Zum Ged¨achtniss von G. Kirchhoff. G¨ottingen, 1888.

BOOKS OF REFERENCE. xv 98. Bˆocher, Maxime. “A Bit of Mathematical History,” Bulletin of the N. Y. Math. Soc., Vol. II., No. 5. 99. Cayley, Arthur. Report on the Recent Progress of Theoretical Dynamics. 1857. 100. Glazebrook, R. T. Report on Optical Theories. 1885. 101. Rosenberger, F. Geschichte der Physik. Braunschweig, 1887– 1890.

A HISTORY OF MATHEMATICS. Introduction. The contemplation of the various steps by which mankind has come into possession of the vast stock of mathematical knowledge can hardly fail to interest the mathematician. He takes pride in the fact that his science, more than any other, is an exact science, and that hardly anything ever done in mathematics has proved to be useless. The chemist smiles at the childish efforts of alchemists, but the mathematician finds the geometry of the Greeks and the arithmetic of the Hindoos as useful and admirable as any research of to-day. He is pleased to notice that though, in course of its development, mathematics has had periods of slow growth, yet in the main it has been pre-eminently a progressive science. The history of mathematics may be instructive as well as agreeable; it may not only remind us of what we have, but may also teach us how to increase our store. Says De Morgan, “The early history of the mind of men with regard to mathematics leads us to point out our own errors; and in this respect it is well to pay attention to the history of mathematics.” It warns us against hasty conclusions; it points out the importance of a good notation upon the progress of the science; it discourages excessive specialisation on the part of investigators, by showing how apparently distinct branches 1

A HISTORY OF MATHEMATICS. 2 have been found to possess unexpected connecting links; it saves the student from wasting time and energy upon problems which were, perhaps, solved long since; it discourages him from attacking an unsolved problem by the same method which has led other mathematicians to failure; it teaches that fortifications can be taken in other ways than by direct attack, that when repulsed from a direct assault it is well to reconnoitre and occupy the surrounding ground and to discover the secret paths by which the apparently unconquerable position can be taken. [1] The importance of this strategic rule may be emphasised by citing a case in which it has been violated. An untold amount of

intellectual energy has been expended on the quadrature of the circle, yet no conquest has been made by direct assault. The circle-squarers have existed in crowds ever since the period of Archimedes. After innumerable failures to solve the problem at a time, even, when investigators possessed that most powerful tool, the differential calculus, persons versed in mathematics dropped the subject, while those who still persisted were completely ignorant of its history and generally misunderstood the conditions of the problem. “Our problem,” says De Morgan, “is to square the circle with the old allowance of means : Euclid’s postulates and nothing more. We cannot remember an instance in which a question to be solved by a definite method was tried by the best heads, and answered at last, by that method , after thousands of complete failures.” But progress was made on this problem by approaching it from a different direction and by newly discovered paths. Lambert proved in 1761 that

INTRODUCTION. 3 the ratio of the circumference of a circle to its diameter is incommensurable. Some years ago, Lindemann demonstrated that this ratio is also transcendental and that the quadrature of the circle, by means of the ruler and compass only, is impossible. He thus showed by actual proof that which keen- minded mathematicians had long suspected; namely, that the great army of circle-squarers have, for two thousand years, been assaulting a fortification which is as indestructible as the firmament of heaven. Another reason for the desirability of historical study is the value of historical knowledge to the teacher of mathematics. The interest which pupils take in their studies may be greatly increased if the solution of problems and the cold logic of geometrical demonstrations are interspersed with historical remarks and anecdotes. A class in arithmetic will be pleased to hear about the Hindoos and their invention of the “Arabic notation”; they will marvel at the thousands of years which elapsed before people had even thought of introducing into the numeral notation that Columbus-egg—the zero; they will find it astounding that it should have taken so long to invent a notation which they themselves can now learn in a month. After the pupils have learned how to bisect a given angle, surprise them by telling of the many futile attempts which have been made to solve, by elementary geometry, the apparently very simple problem of the trisection of an angle. When they know how to construct a square whose area is double the area of a given square, tell them about the duplication of the cube—how the wrath of Apollo could be

A HISTORY OF MATHEMATICS. 4 appeased only by the construction of a cubical altar double the given altar, and how mathematicians long wrestled with this problem. After the class have exhausted their energies on the theorem of the right triangle, tell them the legend about its discoverer—how Pythagoras, jubilant over his great accomplishment, sacrificed a hecatomb to the Muses who inspired him. When the value of mathematical training is called in question, quote the inscription over the entrance into the academy of Plato, the philosopher: “Let no one who is unacquainted with geometry enter here.” Students in analytical geometry should know something of Descartes, and, after taking up the differential and integral calculus, they should become familiar with the parts that Newton, Leibniz, and Lagrange played in creating that science. In his historical talk it is possible for the teacher to make it plain to the student that mathematics is not a dead science, but a living one in which steady progress is made. [2] The history of mathematics is important also as a valuable

contribution to the history of civilisation. Human progress is closely identified with scientific thought. Mathematical and physical researches are a reliable record of intellectual progress. The history of mathematics is one of the large windows through which the philosophic eye looks into past ages and traces the line of intellectual development.

ANTIQUITY. THE BABYLONIANS. The fertile valley of the Euphrates and Tigris was one of the primeval seats of human society. Authentic history of the peoples inhabiting this region begins only with the foundation, in Chaldæa and Babylonia, of a united kingdom out of the previously disunited tribes. Much light has been thrown on their history by the discovery of the art of reading the cuneiform or wedge-shaped system of writing. In the study of Babylonian mathematics we begin with the notation of numbers. A vertical wedge stood for 1, while the characters and signified 10 and 100 respectively. Grotefend believes the character for 10 originally to have been the picture of two hands, as held in prayer, the palms being pressed together, the fingers close to each other, but the thumbs thrust out. In the Babylonian notation two principles were employed—the additive and multiplicative. Numbers below 100 were expressed by symbols whose respective values had to be added. Thus, stood for 2, for 3, for 4, for 23, for 30. Here the symbols of higher order appear always to the left of those of lower order. In writing the hundreds, on the other hand, asmaller symbol was placed to the left of the 100, and was, in that case, to be multiplied by 100. Thus, signified 10 times 100, or 1000. But 5

A HISTORY OF MATHEMATICS. 6 this symbol for 1000 was itself taken for a new unit, which could take smaller coefficients to its left. Thus, denoted, not 20 times 100, but 10 times 1000. Of the largest numbers written in cuneiform symbols, which have hitherto been found, none go as high as a million. [3] If, as is believed by most specialists, the early Sumerians were the inventors of the cuneiform writing, then they were, in all probability, also familiar with the notation of numbers. Most surprising, in this connection, is the fact that Sumerian inscriptions disclose the use, not only of the above decimal system, but also of a sexagesimal one. The latter was used chiefly in constructing tables for weights and measures. It is full of historical interest. Its consequential development, both for integers and fractions, reveals a high degree of mathematical insight. We possess two Babylonian tablets which exhibit its use. One of them, probably written between 2300 and 1600 b.c., contains a table of square numbers up to 602. The numbers 1, 4, 9, 16, 25, 36, 49, are given as the squares of the first seven integers respectively. We have next 1.4 = 8 2, 1.21 = 9 2, 1.40 = 10 2, 2.1 = 11 2, etc. This remains unintelligible, unless we assume the sexagesimal scale, which makes 1.4 = 60 + 4, 1.21 = 60 + 21, 2.1 = 2.60 + 1. The second tablet records the magnitude of the illuminated portion of the moon’s disc for every day from new to full moon, the whole disc being assumed to consist of 240 parts. The illuminated parts during the first five days are the series 5, 10, 20, 40, 1.20(= 80), which is a geometrical progression. From here on the series becomes an arithmetical progression, the numbers

THE BABYLONIANS. 7 from the fifth to the fifteenth day being respectively1.20, 1.36, 1.52, 1.8, 2.24, 2.40, 2.56, 3.12, 3.28, 3.44, 4. This table not only exhibits the

use of the sexagesimal system, but also indicates the acquaintance of the Babylonians with progressions. Not to be overlooked is the fact that in the sexagesimal notation of integers the “principle of position” was employed. Thus, in 1.4 (= 64), the 1 is made to stand for 60, the unit of the second order, by virtue of its position with respect to the 4. The introduction of this principle at so early a date is the more remarkable, because in the decimal notation it was not introduced till about the fifth or sixth century after Christ. The principle of position, in its general and systematic application, requires a symbol for zero. We ask, Did the Babylonians possess one? Had they already taken the gigantic step of representing by a symbol theabsence of units? Neither of the above tables answers this question, for they happen to contain no number in which there was occasion to use a zero. The sexagesimal system was used also in fractions. Thus, in the Babylonian inscriptions, 1 2 and 1 3 are designated by 30 and 20, the reader being expected, in his mind, to supply the word “sixtieths.” The Greek geometer Hypsicles and the Alexandrian astronomer Ptolemæus borrowed the sexagesimal notation of fractions from the Babylonians and introduced it into Greece. From that time sexagesimal frac- tions held almost full sway in astronomical and mathematical calculations until the sixteenth century, when they finally yielded their place to the decimal fractions. It may be asked, What led to the invention of the sexagesimal system? Why

A HISTORY OF MATHEMATICS. 8 was it that60 parts were selected? To this we have no positive answer. Ten was chosen, in the decimal system, because it represents the number of fingers. But nothing of the human body could have suggested 60. Cantor offers the following theory: At first the Babylonians reckoned the year at360 days. This led to the division of the circle into 360 degrees, each degree representing the daily amount of the supposed yearly revolution of the sun around the earth. Now they were, very probably, familiar with the fact that the radius can be applied to its circumference as a chord 6 times, and that each of these chords subtends an arc measuring exactly 60 degrees. Fixing their attention upon these degrees, the division into 60 parts may have suggested itself to them. Thus, when greater precision necessitated a subdivision of the degree, it was partitioned into 60 minutes. In this way the sexagesimal notation may have originated. The division of the day into 24 hours, and of the hour into minutes and seconds on the scale of 60, is due to the Babylonians. It appears that the people in the Tigro-Euphrates basin had made very creditable advance in arithmetic. Their knowledge of arithmetical and geometrical progressions has already been alluded to. Iamblichus attributes to them also a knowledge of proportion, and even the invention of the so-calledmusical proportion. Though we possess no conclusive proof, we have nevertheless reason to believe that in practical calculation they used the abacus. Among the races of middle Asia, even as far as China, the abacus is as old as fable. Now, Babylon was once a great commercial centre,—the metropolis

THE BABYLONIANS. 9 of many nations,—and it is, therefore, not unreasonable to suppose that her merchants employed this most improved aid to calculation. In geometry the Babylonians accomplished almost nothing. Besides the division of the circumference into 6 parts by its radius, and into 360 degrees, they had some knowledge of geometrical figures, such as the triangle and quadrangle, which they

used in their auguries. Like the Hebrews (1 Kin. 7:23), they took π = 3. Of geometrical demonstrations there is, of course, no trace. “As a rule, in the Oriental mind the intuitive powers eclipse the severely rational and logical.” The astronomy of the Babylonians has attracted much attention. They worshipped the heavenly bodies from the earliest historic times. When Alexander the Great, after the battle of Arbela (331 b.c.), took possession of Babylon, Callisthenes found there on burned brick astronomical records reaching back as far as 2234 b.c. Porphyrius says that these were sent to Aristotle. Ptolemy, the Alexandrian astronomer, possessed a Babylonian record of eclipses going back to 747 b.c. Recently Epping and Strassmaier [4] threw considerable light on Babylonian chronology and astronomy by explaining two calendars of the years 123b.c. and 111b.c., taken from cuneiform tablets coming, presumably, from an old observatory. These scholars have succeeded in giving an account of the Babylonian calculation of the new and full moon, and have identified by calculations the Babylonian names of the planets, and of the twelve zodiacal signs and twenty-eight normal stars which correspond to some extent

A HISTORY OF MATHEMATICS. 10 with the twenty-eightnakshatras of the Hindoos. We append part of an Assyrian astronomical report, as translated by Oppert:— “To the King, my lord, thy faithful servant, Mar-Istar.” “. . . On the first day, as the new moon’s day of the month Thammuz declined, the moon was again visible over the planet Mercury, as I had already predicted to my master the King. I erred not.” THE EGYPTIANS. Though there is great difference of opinion regarding the antiquity of Egyptian civilisation, yet all authorities agree in the statement that, however far back they go, they find no uncivilised state of society. “Menes, the first king, changes the course of the Nile, makes a great reservoir, and builds the temple of Phthah at Memphis.” The Egyptians built the pyramids at a very early period. Surely a people engaging in enterprises of such magnitude must have known something of mathematics—at least of practical mathematics. All Greek writers are unanimous in ascribing, without envy, to Egypt the priority of invention in the mathematical sciences. Plato in Phædrus says: “At the Egyptian city of Naucratis there was a famous old god whose name was Theuth; the bird which is called the Ibis was sacred to him, and he was the inventor of many arts, such as arithmetic and calculation and geometry and astronomy and draughts and dice, but his great discovery was the use of letters.” Aristotle says that mathematics had its birth in Egypt, because there the priestly class had the leisure needful for the

THE EGYPTIANS. 11 study of it. Geometry, in particular, is said by Herodotus, Diodorus, Diogenes Laertius, Iamblichus, and other ancient writers to have originated in Egypt. [5] In Herodotus we find this (II. c. 109): “They said also that this king [Sesostris] divided the land among all Egyptians so as to give each one a quadrangle of equal size and to draw from each his revenues, by imposing a tax to be levied yearly. But every one from whose part the river tore away anything, had to go to him and notify what had happened; he then sent the overseers, who had to measure out by how much the land had become smaller, in order that the owner might pay on what was left, in proportion to the entire tax imposed. In this way, it appears to me, geometry originated, which passed thence to Hellas.” We abstain from introducing additional Greek opinion regarding Egyptian mathematics, or from

indulging in wild conjectures. We rest our account on documentary evidence. A hieratic papyrus, included in the Rhind collection of the British Museum, was deciphered by Eisenlohr in 1877, and found to be a mathematical manual containing problems in arithmetic and geometry. It was written by Ahmes some time before 1700 b.c., and was founded on an older work believed by Birch to date back as far as 3400b.c.! This curious papyrus—the most ancient mathematical handbook known to us—puts us at once in contact with the mathematical thought in Egypt of three or five thousand years ago. It is entitled “Directions for obtaining the Knowledge of all Dark Things.” We see from it that the Egyptians cared but little for theoretical results. Theorems are not found in it at all. It

A HISTORY OF MATHEMATICS. 12 contains “hardly any general rules of procedure, but chiefly mere statements of results intended possibly to be explained by a teacher to his pupils.” [6] In geometry the forte of the Egyptians lay in making constructions and determining areas. The area of an isosceles triangle, of which the sides measure 10 ruths and the base 4 ruths, was erroneously given as 20 square ruths, or half the product of the base by one side. The area of an isosceles trapezoid is found, similarly, by multiplying half the sum of the parallel sides by one of the non-parallel sides. The area of a circle is found by deducting from the diameter 1 9 of its length and squaring the remainder. Here π is taken = (16 9)2 = 3.1604 . . ., a very fair approximation. [6] The papyrus explains also such problems as these,—To mark out in the field a right triangle whose sides are 10 and 4 units; or a trapezoid whose parallel sides are 6 and 4, and the non-parallel sides each20 units. Some problems in this papyrus seem to imply a rudimentary knowledge of proportion. The base-lines of the pyramids run north and south, and east and west, but probably only the lines running north and south were determined by astronomical observations. This, coupled with the fact that the wordharpedonaptæ, applied to Egyptian geometers, means “rope-stretchers,” would point to the conclusion that the Egyptian, like the Indian and Chinese geometers, constructed a right triangle upon a given line, by stretching around three pegs a rope consisting of three parts in the ratios 3 : 4 : 5, and thus forming a right triangle. [3] If this explanation is correct, then the Egyptians were familiar,

THE EGYPTIANS. 13 2000 years b.c., with the well-known property of the right triangle, for the special case at least when the sides are in the ratio 3 : 4 : 5. On the walls of the celebrated temple of Horus at Edfu have been found hieroglyphics, written about 100b.c., which enumerate the pieces of land owned by the priesthood, and give their areas. The area of any quadrilateral, however irregular, is there found by the formula a + b 2 · c + d 2 . Thus, for a quadrangle whose opposite sides are 5 and 8, 20 and 15, is given the area 113 1 2 1 4. [7] The incorrect formulæ of Ahmes of 3000 years b.c. yield generally closer approximations than those of the Edfu inscriptions, written 200 years after Euclid! The fact that the geometry of the Egyptians consists chiefly of constructions, goes far to explain certain of its great defects. The Egyptians failed in two essential points without which a science of geometry, in the true sense of the word, cannot exist. In the first place, they failed to construct a rigorously logical system of geometry, resting upon a few axioms and postulates. A great many of their rules, especially those in solid geometry, had

probably not been proved at all, but were known to be true merely from observation or as matters of fact. The second great defect was their inability to bring the numerous special cases under a more general view, and thereby to arrive at broader and more fundamental theorems. Some of the simplest geometrical truths were divided into numberless special cases of which each was supposed to require separate treatment. Some particulars about Egyptian geometry can be men-

A HISTORY OF MATHEMATICS. 14 tioned more advantageously in connection with the early Greek mathematicians who came to the Egyptian priests for instruction. An insight into Egyptian methods of numeration was ob- tained through the ingenious deciphering of the hieroglyphics by Champollion, Young, and their successors. The symbols used were the following: for 1, for 10, for 100, for 1000, for 10, 000, for 100, 000, for 1, 000, 000, for 10, 000, 000. [3] The symbol for 1 represents a vertical staff; that for 10, 000 a pointing finger; that for 100, 000 a burbot; that for 1, 000, 000, a man in astonishment. The significance of the remaining symbols is very doubtful. The writing of numbers with these hieroglyphics was very cumbrous. The unit symbol of each order was repeated as many times as there were units in that order. The principle employed was the additive. Thus, 23 was written . Besides the hieroglyphics, Egypt possesses thehieratic and demotic writings, but for want of space we pass them by. Herodotus makes an important statement concerning the mode of computing among the Egyptians. He says that they “calculate with pebbles by moving the hand from right to left, while the Hellenes move it from left to right.” Herein we recognise again that instrumental method of figuring so extensively used by peoples of antiquity. The Egyptians used the decimal scale. Since, in figuring, they moved their hands horizontally, it seems probable that they used ciphering-boards with vertical columns. In each column there must have been not more than nine pebbles, for ten pebbles would

THE EGYPTIANS. 15 be equal to one pebble in the column next to the left. The Ahmes papyrus contains interesting information on the way in which the Egyptians employed fractions. Their methods of operation were, of course, radically different from ours. Fractions were a subject of very great difficulty with the ancients. Simultaneous changes in both numerator and denominator were usually avoided. In manipulating fractions the Babylonians kept the denominators (60) constant. The Romans likewise kept them constant, but equal to 12. The Egyptians and Greeks, on the other hand, kept the numerators constant, and dealt with variable denominators. Ahmes used the term “fraction” in a restricted sense, for he applied it only to unit-fractions, or fractions having unity for the numerator. It was designated by writing the denominator and then placing over it a dot. Fractional values which could not be expressed by any one unit-fraction were expressed as the sum of two or more of them. Thus, he wrote 1 3 1 15 in place of 2 5. The first important problem naturally arising was, how to represent any fractional value as the sum of unit-fractions. This was solved by aid of a table, given in the papyrus, in which all fractions of the form 2 2n + 1 (where n designates successively all the numbers up to49) are reduced to the sum of unit-fractions. Thus, 2 7 = 1 4 1 28; 2 99 = 1 66 1 198. When, by whom, and how this table was calculated, we do not know. Probably it was compiled empirically at different times, by different persons. It will be seen that by repeated application of

this table, a fraction whose numerator exceeds two can be expressed in the desired form, provided that there is a fraction

A HISTORY OF MATHEMATICS. 16 in the table having the same denominator that it has. Take, for example, the problem, to divide5 by 21. In the first place, 5 = 1 + 2 + 2. From the table we get 2 21 = 1 14 1 42. Then 5 21 = 1 21 +(1 14 1 42)+(1 14 1 42) = 1 21 +(2 14 2 42) = 1 21 1 7 1 21 = 1 7 2 21 = 1 7 1 14 1 42. The papyrus contains problems in which it is required that fractions be raised by addition or multiplication to given whole numbers or to other fractions. For example, it is required to increase 1 4 1 8 1 10 1 30 1 45 to 1. The common denominator taken appears to be 45, for the numbers are stated as 11 1 4, 5 1 2 1 8, 4 1 2, 1 1 2, 1. The sum of these is 23 1 2 1 4 1 8 forty-fifths. Add to this 1 9 1 40, and the sum is 2 3. Add 1 3, and we have 1. Hence the quantity to be added to the given fraction is 1 3 1 9 1 40. Having finished the subject of fractions, Ahmes proceeds to the solution of equations of one unknown quantity. The unknown quantity is called ‘hau’ or heap. Thus the problem, “heap, its 1 7, its whole, it makes 19,” i.e. x 7 + x = 19. In this case, the solution is as follows: 8x 7 = 19; x 7 = 2 1 4 1 8; x = 16 1 2 1 8. But in other problems, the solutions are effected by various other methods. It thus appears that the beginnings of algebra are as ancient as those of geometry. The principal defect of Egyptian arithmetic was the lack of a simple, comprehensive symbolism—a defect which not even the Greeks were able to remove. The Ahmes papyrus doubtless represents the most advanced attainments of the Egyptians in arithmetic and geometry. It is remarkable that they should have reached so great proficiency in mathematics at so remote a period of antiquity. But

THE GREEKS. 17 strange, indeed, is the fact that, during the next two thousand years, they should have made no progress whatsoever in it. The conclusion forces itself upon us, that they resemble the Chinese in the stationary character , not only of their government, but also of their learning. All the knowledge of geometry which they possessed when Greek scholars visited them, six centuries b.c., was doubtless known to them two thousand years earlier, when they built those stupendous and gigantic structures—the pyramids. An explanation for this stagnation of learning has been sought in the fact that their early discoveries in mathematics and medicine had the misfortune of being entered upon their sacred books and that, in after ages, it was considered heretical to augment or modify anything therein. Thus the books themselves closed the gates to progress. THE GREEKS. GREEK GEOMETRY. About the seventh century b.c. an active commercial in- tercourse sprang up between Greece and Egypt. Naturally there arose an interchange of ideas as well as of merchan- dise. Greeks, thirsting for knowledge, sought the Egyptian priests for instruction. Thales, Pythagoras, Œnopides, Plato, Democritus, Eudoxus, all visited the land of the pyramids. Egyptian ideas were thus transplanted across the sea and there stimulated Greek thought, directed it into new lines, and gave to it a basis to work upon. Greek culture, therefore, is not

A HISTORY OF MATHEMATICS. 18 primitive. Not only in mathematics, but also in mythology and art, Hellas owes a debt to older countries. To Egypt Greece is indebted, among other things, for its elementary geometry. But this does not lessen

our admiration for the Greek mind. From the moment that Hellenic philosophers applied themselves to the study of Egyptian geometry, this science assumed a radically different aspect. “Whatever we Greeks receive, we improve and perfect,” says Plato. The Egyptians carried geometry no further than was absolutely necessary for their practical wants. The Greeks, on the other hand, had within them a strong speculative tendency. They felt a craving to discover the reasons for things. They found pleasure in the contemplation of ideal relations, and loved science as science. Our sources of information on the history of Greek geometry before Euclid consist merely of scattered notices in ancient writers. The early mathematicians, Thales and Pythagoras, left behind no written records of their discoveries. A full history of Greek geometry and astronomy during this period, written by Eudemus, a pupil of Aristotle, has been lost. It was well known to Proclus, who, in his commentaries on Euclid, gives a brief account of it. This abstract constitutes our most reliable information. We shall quote it frequently under the name of Eudemian Summary.

THE GREEKS. 19 The Ionic School. ToThales of Miletus (640–546b.c.), one of the “seven wise men,” and the founder of the Ionic school, falls the honour of having introduced the study of geometry into Greece. During middle life he engaged in commercial pursuits, which took him to Egypt. He is said to have resided there, and to have studied the physical sciences and mathematics with the Egyptian priests. Plutarch declares that Thales soon excelled his masters, and amazed King Amasis by measuring the heights of the pyramids from their shadows. According to Plutarch, this was done by considering that the shadow cast by a vertical staff of known length bears the same ratio to the shadow of the pyramid as the height of the staff bears to the height of the pyramid. This solution presupposes a knowledge of proportion, and the Ahmes papyrus actually shows that the rudiments of proportion were known to the Egyptians. According to Diogenes Laertius, the pyramids were measured by Thales in a different way; viz. by finding the length of the shadow of the pyramid at the moment when the shadow of a staff was equal to its own length. The Eudemian Summary ascribes to Thales the invention of the theorems on the equality of vertical angles, the equality of the angles at the base of an isosceles triangle, the bisection of a circle by any diameter, and the congruence of two triangles having a side and the two adjacent angles equal respectively. The last theorem he applied to the measurement of the distances of ships from the shore. Thus Thales was

A HISTORY OF MATHEMATICS. 20 the first to apply theoretical geometry to practical uses. The theorem that all angles inscribed in a semicircle are right angles is attributed by some ancient writers to Thales, by others to Pythagoras. Thales was doubtless familiar with other theorems, not recorded by the ancients. It has been inferred that he knew the sum of the three angles of a triangle to be equal to two right angles, and the sides of equiangular triangles to be proportional. [8] The Egyptians must have made use of the above theorems on the straight line, in some of their constructions found in the Ahmes papyrus, but it was left for the Greek philosopher to give these truths, which others saw, but did not formulate into words, an explicit, abstract expression, and to put into scientific language and subject to proof that which others merely felt to be true. Thales may be said to have created the geometry of lines, essentially abstract in its character, while the Egyptians studied

only the geometry of surfaces and the rudiments of solid geometry, empirical in their character. [8] With Thales begins also the study of scientific astronomy. He acquired great celebrity by the prediction of a solar eclipse in 585 b.c. Whether he predicted the day of the occurrence, or simply the year, is not known. It is told of him that while contemplating the stars during an evening walk, he fell into a ditch. The good old woman attending him exclaimed, “How canst thou know what is doing in the heavens, when thou seest not what is at thy feet?” The two most prominent pupils of Thales were Anaxi- mander (b. 611 b.c.) and Anaximenes (b. 570 b.c.). They

THE GREEKS. 21 studied chiefly astronomy and physical philosophy. OfAnax- agoras, a pupil of Anaximenes, and the last philosopher of the Ionic school, we know little, except that, while in prison, he passed his time attempting to square the circle. This is the first time, in the history of mathematics, that we find mention of the famous problem of the quadrature of the circle, that rock upon which so many reputations have been destroyed. It turns upon the determination of the exact value of π. Approximations to π had been made by the Chinese, Babylonians, Hebrews, and Egyptians. But the invention of a method to find its exact value, is the knotty problem which has engaged the attention of many minds from the time of Anaxagoras down to our own. Anaxagoras did not offer any solution of it, and seems to have luckily escaped paralogisms. About the time of Anaxagoras, but isolated from the Ionic school, flourished Œnopides of Chios. Proclus ascribes to him the solution of the following problems: From a point without, to draw a perpendicular to a given line, and to draw an angle on a line equal to a given angle. That a man could gain a reputation by solving problems so elementary as these, indicates that geometry was still in its infancy, and that the Greeks had not yet gotten far beyond the Egyptian constructions. The Ionic school lasted over one hundred years. The progress of mathematics during that period was slow, as compared with its growth in a later epoch of Greek history. A new impetus to its progress was given by Pythagoras.

A HISTORY OF MATHEMATICS. 22 The School of Pythagoras. Pythagoras (580?–500? b.c.) was one of those figures which impressed the imagination of succeeding times to such an extent that their real histories have become difficult to be discerned through the mythical haze that envelops them. The following account of Pythagoras excludes the most doubtful statements. He was a native of Samos, and was drawn by the fame of Pherecydes to the island of Syros. He then visited the ancient Thales, who incited him to study in Egypt. He sojourned in Egypt many years, and may have visited Babylon. On his return to Samos, he found it under the tyranny of Polycrates. Failing in an attempt to found a school there, he quitted home again and, following the current of civilisation, removed to Magna Græcia in South Italy. He settled at Croton, and founded the famous Pythagorean school. This was not merely an academy for the teaching of philosophy, mathematics, and natural science, but it was a brotherhood, the members of which were united for life. This brotherhood had observances approaching masonic peculiarity. They were forbidden to divulge the discoveries and doctrines of their school. Hence we are obliged to speak of the Pythagoreans as a body, and find it difficult to determine to whom each particular discovery is to be ascribed. The Pythagoreans themselves were in the habit of referring every discovery back to the great founder of the sect. This school grew

rapidly and gained considerable political ascendency. But the mystic and secret observances, intro-

THE GREEKS. 23 duced in imitation of Egyptian usages, and the aristocratic tendencies of the school, caused it to become an object of suspicion. The democratic party in Lower Italy revolted and destroyed the buildings of the Pythagorean school. Pythag- oras fled to Tarentum and thence to Metapontum, where he was murdered. Pythagoras has left behind no mathematical treatises, and our sources of information are rather scanty. Certain it is that, in the Pythagorean school, mathematics was the principal study. Pythagoras raised mathematics to the rank of a science. Arithmetic was courted by him as fervently as geometry. In fact, arithmetic is the foundation of his philosophic system. The Eudemian Summary says that “Pythagoras changed the study of geometry into the form of a liberal education, for he examined its principles to the bottom, and investigated its theorems in an immaterial and intellectual manner.” His geometry was connected closely with his arithmetic. He was especially fond of those geometrical relations which admitted of arithmetical expression. Like Egyptian geometry, the geometry of the Pythagoreans is much concerned with areas. To Pythagoras is ascribed the important theorem that the square on the hypotenuse of a right triangle is equal to the sum of the squares on the other two sides. He had probably learned from the Egyptians the truth of the theorem in the special case when the sides are 3, 4, 5, respectively. The story goes, that Pythagoras was so jubilant over this discovery that he sacrificed a hecatomb. Its authenticity is doubted, because the Pythagoreans believed

A HISTORY OF MATHEMATICS. 24 in the transmigration of the soul and opposed, therefore, the shedding of blood. In the later traditions of the Neo- Pythagoreans this objection is removed by replacing this bloody sacrifice by that of “an ox made of flour”! The proof of the law of three squares, given in Euclid’s Elements, I. 47, is due to Euclid himself, and not to the Pythagoreans. What the Pythagorean method of proof was has been a favourite topic for conjecture. The theorem on the sum of the three angles of a triangle, presumably known to Thales, was proved by the Pythagoreans after the manner of Euclid. They demonstrated also that the plane about a point is completely filled by six equilateral triangles, four squares, or three regular hexagons, so that it is possible to divide up a plane into figures of either kind. From the equilateral triangle and the square arise the solids, namely the tetraedron, octaedron, icosaedron, and the cube. These solids were, in all probability, known to the Egyptians, excepting, perhaps, the icosaedron. In Pythagorean philosophy, they represent respectively the four elements of the physical world; namely, fire, air, water, and earth. Later another regular solid was discovered, namely the dodecaedron, which, in absence of a fifth element, was made to represent the universe itself. Iamblichus states that Hippasus, a Pythagorean, perished in the sea, because he boasted that he first divulged “the sphere with the twelve pentagons.” The star-shaped pentagram was used as a symbol of recognition by the Pythagoreans, and was called by them Health. Pythagoras called the sphere the most beautiful of all

THE GREEKS. 25 solids, and the circle the most beautiful of all plane figures. The treatment of the subjects of proportion and of irrational quantities by him and his

school will be taken up under the head of arithmetic. According to Eudemus, the Pythagoreans invented the problems concerning the application of areas, including the cases of defect and excess, as in Euclid, VI. 28, 29. They were also familiar with the construction of a polygon equal in area to a given polygon and similar to another given polygon. This problem depends upon several important and somewhat advanced theorems, and testifies to the fact that the Pythagoreans made no mean progress in geometry. Of the theorems generally ascribed to the Italian school, some cannot be attributed to Pythagoras himself, nor to his earliest successors. The progress from empirical to reasoned solutions must, of necessity, have been slow. It is worth noticing that on the circle no theorem of any importance was discovered by this school. Though politics broke up the Pythagorean fraternity, yet the school continued to exist at least two centuries longer. Among the later Pythagoreans, Philolaus and Archytas are the most prominent. Philolaus wrote a book on the Pythagorean doctrines. By him were first given to the world the teachings of the Italian school, which had been kept secret for a whole century. The brilliantArchytasof Tarentum (428–347b.c.), known as a great statesman and general, and universally admired for his virtues, was the only great geometer among the Greeks when Plato opened his school. Archytas was

A HISTORY OF MATHEMATICS. 26 the first to apply geometry to mechanics and to treat the latter subject methodically. He also found a very ingenious mechanical solution to the problem of the duplication of the cube. His solution involves clear notions on the generation of cones and cylinders. This problem reduces itself to finding two mean proportionals between two given lines. These mean proportionals were obtained by Archytas from the section of a half-cylinder. The doctrine of proportion was advanced through him. There is every reason to believe that the later Pythagoreans exercised a strong influence on the study and development of mathematics at Athens. The Sophists acquired geometry from Pythagorean sources. Plato bought the works of Philolaus, and had a warm friend in Archytas. The Sophist School. After the defeat of the Persians under Xerxes at the battle of Salamis, 480 b.c., a league was formed among the Greeks to preserve the freedom of the now liberated Greek cities on the islands and coast of the Ægæan Sea. Of this league Athens soon became leader and dictator. She caused the separate treasury of the league to be merged into that of Athens, and then spent the money of her allies for her own aggrandisement. Athens was also a great commercial centre. Thus she became the richest and most beautiful city of antiquity. All menial work was performed by slaves. The citizen of Athens was well- to-do and enjoyed a large amount of leisure. The government

THE GREEKS. 27 being purely democratic, every citizen was a politician. To make his influence felt among his fellow-men he must, first of all, be educated. Thus there arose a demand for teachers. The supply came principally from Sicily, where Pythagorean doctrines had spread. These teachers were called Sophists, or “wise men.” Unlike the Pythagoreans, they accepted pay for their teaching. Although rhetoric was the principal feature of their instruction, they also taught geometry, astronomy, and philosophy. Athens soon became the headquarters of Grecian men of letters, and of mathematicians in particular. The home of mathematics among the

Greeks was first in the Ionian Islands, then in Lower Italy, and during the time now under consideration, at Athens. The geometry of the circle, which had been entirely ne- glected by the Pythagoreans, was taken up by the Sophists. Nearly all their discoveries were made in connection with their innumerable attempts to solve the following three famous problems:— (1) To trisect an arc or an angle. (2) To “double the cube,” i.e. to find a cube whose volume is double that of a given cube. (3) To “square the circle,”i.e. to find a square or some other rectilinear figure exactly equal in area to a given circle. These problems have probably been the subject of more discussion and research than any other problems in math- ematics. The bisection of an angle was one of the easiest problems in geometry. The trisection of an angle, on the other

A HISTORY OF MATHEMATICS. 28 hand, presented unexpected difficulties. A right angle had been divided into three equal parts by the Pythagoreans. But the general problem, though easy in appearance, transcended the power ofelementary geometry. Among the first to wrestle with it wasHippias of Elis, a contemporary of Socrates, and born about 460 b.c. Like all the later geometers, he failed in effecting the trisection by means of a ruler and compass only. Proclus mentions a man, Hippias, presumably Hippias of Elis, as the inventor of a transcendental curve which served to divide an angle not only into three, but into any number of equal parts. This same curve was used later by Deinostratus and others for the quadrature of the circle. On this account it is called the quadratrix. The Pythagoreans had shown that the diagonal of a square is the side of another square having double the area of the original one. This probably suggested the problem of the duplication of the cube, i.e. to find the edge of a cube having double the volume of a given cube. Eratosthenes ascribes to this problem a different origin. The Delians were once suffering from a pestilence and were ordered by the oracle to double a certain cubical altar. Thoughtless workmen simply constructed a cube with edges twice as long, but this did not pacify the gods. The error being discovered, Plato was consulted on the matter. He and his disciples searched eagerly for a solution to this “Delian Problem.”Hippocrates of Chios (about 430 b.c.), a talented mathematician, but otherwise slow and stupid, was the first to show that the problem could be reduced to finding two mean proportionals

THE GREEKS. 29 between a given line and another twice as long. For, in the proportion a : x = x : y = y : 2a, since x2 = ay and y2 = 2ax and x4 = a2y2, we have x4 = 2a3x and x3 = 2a3. But he failed to find the two mean proportionals. His attempt to square the circle was also a failure; for though he made himself celebrated by squaring a lune, he committed an error in attempting to apply this result to the squaring of the circle. In his study of the quadrature and duplication-problems, Hippocrates contributed much to the geometry of the circle. The subject of similar figures was studied and partly developed by Hippocrates. This involved the theory of proportion. Proportion had, thus far, been used by the Greeks only in numbers. They never succeeded in uniting the notions of numbers and magnitudes. The term “number” was used by them in a restricted sense. What we call irrational numbers was not included under this notion. Not even rational fractions were called numbers. They used the word in the same sense as we use “integers.” Hence numbers were

conceived as discontinuous, while magnitudes were continuous. The two notions appeared, therefore, entirely distinct. The chasm between them is exposed to full view in the statement of Euclid that “incommensurable magnitudes do not have the same ratio as numbers.” In Euclid’s Elements we find the theory of proportion of magnitudes developed and treated independent of that of numbers. The transfer of the theory of proportion from numbers to magnitudes (and to lengths in particular) was a difficult and important step. Hippocrates added to his fame by writing a geometrical

A HISTORY OF MATHEMATICS. 30 text-book, called the Elements. This publication shows that the Pythagorean habit of secrecy was being abandoned; secrecy was contrary to the spirit of Athenian life. The Sophist Antiphon, a contemporary of Hippocrates, introduced the process of exhaustion for the purpose of solving the problem of the quadrature. He did himself credit by remarking that by inscribing in a circle a square, and on its sides erecting isosceles triangles with their vertices in the circumference, and on the sides of these triangles erecting new triangles, etc., one could obtain a succession of regular polygons of 8, 16, 32, 64 sides, and so on, of which each approaches nearer to the circle than the previous one, until the circle is finally exhausted. Thus is obtained an inscribed polygon whose sides coincide with the circumference. Since there can be found squares equal in area to any polygon, there also can be found a square equal to the last polygon inscribed, and therefore equal to the circle itself. Bryson of Heraclea, a contemporary of Antiphon, advanced the problem of the quadrature considerably by circumscribing polygons at the same time that he inscribed polygons. He erred, however, in assuming that the area of a circle was the arithmetical mean between circumscribed and inscribed polygons. Unlike Bryson and the rest of Greek geometers, Antiphon seems to have believed it possible, by continually doubling the sides of an inscribed polygon, to obtain a polygon coinciding with the circle. This question gave rise to lively disputes in Athens. If a polygon can coincide with the circle, then, says Simplicius, we must put aside the notion that magnitudes are divisible

THE GREEKS. 31 ad infinitum. Aristotle always supported the theory of the infinite divisibility, while Zeno, the Stoic, attempted to show its absurdity by proving that if magnitudes are infinitely divisible, motion is impossible. Zeno argues that Achilles could not overtake a tortoise; for while he hastened to the place where the tortoise had been when he started, the tortoise crept some distance ahead, and while Achilles reached that second spot, the tortoise again moved forward a little, and so on. Thus the tortoise was always in advance of Achilles. Such arguments greatly confounded Greek geometers. No wonder they were deterred by such paradoxes from introducing the idea of infinity into their geometry. It did not suit the rigour of their proofs. The process of Antiphon and Bryson gave rise to the cum- brous but perfectly rigorous “method of exhaustion.” In determining the ratio of the areas between two curvilinear plane figures, say two circles, geometers first inscribed or circumscribed similar polygons, and then by increasing in- definitely the number of sides, nearly exhausted the spaces between the polygons and circumferences. From the theorem that similar polygons inscribed in circles are to each other as the squares on their diameters, geometers may have divined the theorem attributed to Hippocrates of Chios that the

circles, which differ but little from the last drawn polygons, must be to each other as the squares on their diameters. But in order to exclude all vagueness and possibility of doubt, later Greek geometers applied reasoning like that in Euclid, XII. 2, as follows: Let C and c, D and d be respectively the

A HISTORY OF MATHEMATICS. 32 circles and diameters in question. Then if the proportion D2 : d2 = C : c is not true, suppose that D2 : d2 = C : c′. If c′ < c , then a polygon p can be inscribed in the circle c which comes nearer to it in area than does c′. If P be the corresponding polygon in C, then P : p = D2 : d2 = C : c′, and P : C = p : c′. Since p > c′, we have P > C , which is absurd. Next they proved by this same method ofreductio ad absurdum the falsity of the supposition that c′ > c. Since c′ can be neither larger nor smaller than c, it must be equal to it, q.e.d. Hankel refers this Method of Exhaustion back to Hippocrates of Chios, but the reasons for assigning it to this early writer, rather than to Eudoxus, seem insufficient. Though progress in geometry at this period is traceable only at Athens, yet Ionia, Sicily, Abdera in Thrace, and Cyrene produced mathematicians who made creditable contributions to the science. We can mention here only Democritus of Abdera (about 460–370 b.c.), a pupil of Anaxagoras, a friend of Philolaus, and an admirer of the Pythagoreans. He visited Egypt and perhaps even Persia. He was a successful geometer and wrote on incommensurable lines, on geometry, on numbers, and on perspective. None of these works are extant. He used to boast that in the construction of plane figures with proof no one had yet surpassed him, not even the so-called harpedonaptæ (“rope-stretchers”) of Egypt. By this assertion he pays a flattering compliment to the skill and ability of the Egyptians.

THE GREEKS. 33 The Platonic School. During the Peloponnesian War (431–404b.c.) the progress of geometry was checked. After the war, Athens sank into the background as a minor political power, but advanced more and more to the front as the leader in philosophy, literature, and science. Plato was born at Athens in 429 b.c., the year of the great plague, and died in 348 b.c. He was a pupil and near friend of Socrates, but it was not from him that he acquired his taste for mathematics. After the death of Socrates, Plato travelled extensively. In Cyrene he studied mathematics under Theodorus. He went to Egypt, then to Lower Italy and Sicily, where he came in contact with the Pythagoreans. Archytas of Tarentum and Timæus of Locri became his intimate friends. On his return to Athens, about 389 b.c., he founded his school in the groves of theAcademia, and devoted the remainder of his life to teaching and writing. Plato’s physical philosophy is partly based on that of the Pythagoreans. Like them, he sought in arithmetic and geometry the key to the universe. When questioned about the occupation of the Deity, Plato answered that “He geometrises continually.” Accordingly, a knowledge of geometry is a necessary preparation for the study of philosophy. To show how great a value he put on mathematics and how necessary it is for higher speculation, Plato placed the inscription over his porch, “Let no one who is unacquainted with geometry enter here.” Xenocrates, a successor of Plato as teacher in the Academy, followed in his master’s footsteps, by declining

A HISTORY OF MATHEMATICS. 34 to admit a pupil who had no mathematical training, with the remark, “Depart, for thou hast not the grip of philosophy.” Plato

observed that geometry trained the mind for correct and vigorous thinking. Hence it was that theEudemian Summary says, “He filled his writings with mathematical discoveries, and exhibited on every occasion the remarkable connection between mathematics and philosophy.” With Plato as the head-master, we need not wonder that the Platonic school produced so large a number of mathematicians. Plato did little real original work, but he made valuable improvements in the logic and methods employed in geometry. It is true that the Sophist geometers of the previous century were rigorous in their proofs, but as a rule they did not reflect on the inward nature of their methods. They used the axioms without giving them explicit expression, and the geometrical concepts, such as the point, line, surface, etc., without assigning to them formal definitions. The Pythagoreans called a point “unity in position,” but this is a statement of a philosophical theory rather than a definition. Plato objected to calling a point a “geometrical fiction.” He defined a point as the “beginning of a line” or as “an indivisible line,” and a line as “length without breadth.” He called the point, line, surface, the ‘boundaries’ of the line, surface, solid, respectively. Many of the definitions in Euclid are to be ascribed to the Platonic school. The same is probably true of Euclid’s axioms. Aristotle refers to Plato the axiom that “equals subtracted from equals leave equals.” One of the greatest achievements of Plato and his school is

THE GREEKS. 35 the invention of analysis as a method of proof. To be sure, this method had been used unconsciously by Hippocrates and others; but Plato, like a true philosopher, turned the instinctive logic into a conscious, legitimate method. The termssynthesis and analysis are used in mathematics in a more special sense than in logic. In ancient mathematics they had a different meaning from what they now have. The oldest definition of mathematical analysis as opposed to synthesis is that given in Euclid, XIII. 5, which in all probability was framed by Eudoxus: “Analysis is the obtaining of the thing sought by assuming it and so reasoning up to an admitted truth; synthesis is the obtaining of the thing sought by reasoning up to the inference and proof of it.” The analytic method is not conclusive, unless all operations involved in it are known to be reversible. To remove all doubt, the Greeks, as a rule, added to the analytic process a synthetic one, consisting of a reversion of all operations occurring in the analysis. Thus the aim of analysis was to aid in the discovery of synthetic proofs or solutions. Plato is said to have solved the problem of the duplication of the cube. But the solution is open to the very same objection which he made to the solutions by Archytas, Eudoxus, and Menæchmus. He called their solutions not geometrical, but mechanical, for they required the use of other instruments than the ruler and compasses. He said that thereby “the good of geometry is set aside and destroyed, for we again reduce it to the world of sense, instead of elevating and imbuing it with the eternal and incorporeal images of thought, even as

A HISTORY OF MATHEMATICS. 36 it is employed by God, for which reason He always is God.” These objections indicate either that the solution is wrongly attributed to Plato or that he wished to show how easily non-geometric solutions of that character can be found. It is now generally admitted that the duplication problem, as well as the trisection and quadrature problems, cannot be solved by means of the ruler and compass only. Plato gave a healthful stimulus to the study of

stereometry, which until his time had been entirely neglected. The sphere and the regular solids had been studied to some extent, but the prism, pyramid, cylinder, and cone were hardly known to exist. All these solids became the subjects of investigation by the Platonic school. One result of these inquiries was epoch-making. Menæchmus, an associate of Plato and pupil of Eudoxus, invented the conic sections, which, in course of only a century, raised geometry to the loftiest height which it was destined to reach during antiquity. Menæchmus cut three kinds of cones, the 'right-angled,' 'acute-angled,' and 'obtuse-angled,' by planes at right angles to a side of the cones, and thus obtained the three sections which we now call the parabola, ellipse, and hyperbola. Judging from the two very elegant solutions of the "Delian Problem" by means of intersections of these curves, Menæchmus must have succeeded well in investigating their properties. Another great geometer was Dinostratus, the brother of Menæchmus and pupil of Plato. Celebrated is his mechanical solution of the quadrature of the circle, by means of the quadratrix of Hippias.

THE GREEKS. 37 Perhaps the most brilliant mathematician of this period was Eudoxus. He was born at Cnidus about 408 b.c., studied under Archytas, and later, for two months, under Plato. He was imbued with a true spirit of scientific inquiry, and has been called the father of scientific astronomical observation. From the fragmentary notices of his astronomical researches, found in later writers, Ideler and Schiaparelli succeeded in reconstructing the system of Eudoxus with its celebrated representation of planetary motions by "concentric spheres." Eudoxus had a school at Cyzicus, went with his pupils to Athens, visiting Plato, and then returned to Cyzicus, where he died 355 b.c. The fame of the academy of Plato is to a large extent due to Eudoxus's pupils of the school at Cyzicus, among whom are Menæchmus, Dinostratus, Athenæus, and Helicon. Diogenes Laertius describes Eudoxus as astronomer, physician, legislator, as well as geometer. The Eudemian Summary says that Eudoxus "first increased the number of general theorems, added to the three proportions three more, and raised to a considerable quantity the learning, begun by Plato, on the subject of the section, to which he applied the analytical method." By this 'section' is meant, no doubt, the "golden section" (sectio aurea), which cuts a line in extreme and mean ratio. The first five propositions in Euclid XIII. relate to lines cut by this section, and are generally attributed to Eudoxus. Eudoxus added much to the knowledge of solid geometry. He proved, says Archimedes, that a pyramid is exactly one-third of a prism, and a cone one-third of a cylinder, having equal base and altitude. The proof that spheres are to

A HISTORY OF MATHEMATICS. 38 each other as the cubes of their radii is probably due to him. He made frequent and skilful use of the method of exhaustion, of which he was in all probability the inventor. A scholiast on Euclid, thought to be Proclus, says further that Eudoxus practically invented the whole of Euclid's fifth book. Eudoxus also found two mean proportionals between two given lines, but the method of solution is not known. Plato has been called a maker of mathematicians. Besides the pupils already named, the Eudemian Summary mentions the following: Theætetus of Athens, a man of great natural gifts, to whom, no doubt, Euclid was greatly indebted in the composition of the 10th book, [8] treating of incommensurables; Leodamas of Thasos; Neocleides and his pupil Leon, who added much to the

work of their predecessors, for Leon wrote an Elements carefully designed, both in num- ber and utility of its proofs; Theudius of Magnesia, who composed a very good book of Elements and generalised propositions, which had been confined to particular cases; Hermotimus of Colophon, who discovered many proposi- tions of theElements and composed some onloci; and, finally, the names ofAmyclas of Heraclea, Cyzicenus of Athens, and Philippus of Mende. A skilful mathematician of whose life and works we have no details isAristæus, the elder, probably a senior contemporary of Euclid. The fact that he wrote a work on conic sections tends to show that much progress had been made in their study during the time of Menæchmus. Aristæus wrote also on regular solids and cultivated the analytic method. His

THE GREEKS. 39 works contained probably a summary of the researches of the Platonic school. [8] Aristotle (384–322 b.c.), the systematiser of deductive logic, though not a professed mathematician, promoted the science of geometry by improving some of the most difficult definitions. His Physics contains passages with suggestive hints of the principle of virtual velocities. About his time there appeared a work calledMechanica, of which he is regarded by some as the author. Mechanics was totally neglected by the Platonic school. The First Alexandrian School. In the previous pages we have seen the birth of geometry in Egypt, its transference to the Ionian Islands, thence to Lower Italy and to Athens. We have witnessed its growth in Greece from feeble childhood to vigorous manhood, and now we shall see it return to the land of its birth and there derive new vigour. During her declining years, immediately following the Peloponnesian War, Athens produced the greatest scientists and philosophers of antiquity. It was the time of Plato and Aristotle. In 338 b.c., at the battle of Chæronea, Athens was beaten by Philip of Macedon, and her power was broken forever. Soon after, Alexander the Great, the son of Philip, started out to conquer the world. In eleven years he built up a great empire which broke to pieces in a day. Egypt fell to the lot of Ptolemy Soter. Alexander had founded the

A HISTORY OF MATHEMATICS. 40 seaport of Alexandria, which soon became “the noblest of all cities.” Ptolemy made Alexandria the capital. The history of Egypt during the next three centuries is mainly the history of Alexandria. Literature, philosophy, and art were diligently cultivated. Ptolemy created the university of Alexandria. He founded the great Library and built laboratories, museums, a zo¨ ological garden, and promenades. Alexandria soon became the great centre of learning. Demetrius Phalereus was invited from Athens to take charge of the Library, and it is probable, says Gow, thatEuclid was invited with him to open the mathematical school. Euclid’s greatest activity was during the time of the first Ptolemy, who reigned from 306 to 283 b.c. Of the life of Euclid, little is known, except what is added by Proclus to the Eudemian Summary . Euclid, says Proclus, was younger than Plato and older than Eratosthenes and Archimedes, the latter of whom mentions him. He was of the Platonic sect, and well read in its doctrines. He collected the Elements, put in order much that Eudoxus had prepared, completed many things of Theætetus, and was the first who reduced to unobjectionable demonstration the imperfect attempts of his predecessors. When Ptolemy once asked him if geometry could not be mastered by an easier process than by studying the Elements, Euclid returned the answer, “There is no royal road to geometry.” Pappus states that Euclid

was distinguished by the fairness and kindness of his disposition, particularly toward those who could do anything to advance the mathematical sciences. Pappus is evidently making a

THE GREEKS. 41 contrast to Apollonius, of whom he more than insinuates the opposite character. [9] A pretty little story is related by Stobæus: [6] “A youth who had begun to read geometry with Euclid, when he had learnt the first proposition, inquired, ‘What do I get by learning these things?’ So Euclid called his slave and said, ‘Give him threepence, since he must make gain out of what he learns.’ ” These are about all the personal details preserved by Greek writers. Syrian and Arabian writers claim to know much more, but they are unreliable. At one time Euclid of Alexandria was universally confounded with Euclid of Megara, who lived a century earlier. The fame of Euclid has at all times rested mainly upon his book on geometry, called the Elements. This book was so far superior to the Elements written by Hippocrates, Leon, and Theudius, that the latter works soon perished in the struggle for existence. The Greeks gave Euclid the special title of “the author of the Elements.” It is a remarkable fact in the history of geometry, that theElements of Euclid, written two thousand years ago, are still regarded by many as the best introduction to the mathematical sciences. In England they are used at the present time extensively as a text-book in schools. Some editors of Euclid have, however, been inclined to credit him with more than is his due. They would have us believe that a finished and unassailable system of geometry sprang at once from the brain of Euclid, “an armed Minerva from the head of Jupiter.” They fail to mention the earlier eminent mathematicians from whom Euclid got his material. Comparatively few of the propositions and proofs in the

A HISTORY OF MATHEMATICS. 42 Elements are his own discoveries. In fact, the proof of the “Theorem of Pythagoras” is the only one directly ascribed to him. Allman conjectures that the substance of Books I., II., IV. comes from the Pythagoreans, that the substance of Book VI. is due to the Pythagoreans and Eudoxus, the latter contributing the doctrine of proportion as applicable to incommensurables and also the Method of Exhaustions (Book XII.), that Theætetus contributed much toward Books X. and XIII., that the principal part of the original work of Euclid himself is to be found in Book X. [8] Euclid was the greatest systematiser of his time. By careful selection from the material before him, and by logical arrangement of the propositions selected, he built up, from a few definitions and axioms, a proud and lofty structure. It would be erroneous to believe that he incorporated into his Elements all the elementary theorems known at his time. Archimedes, Apollonius, and even he himself refer to theorems not included in his Elements, as being well-known truths. The text of the Elements now commonly used is Theon’s edition. Theon of Alexandria, the father of Hypatia, brought out an edition, about 700 years after Euclid, with some alterations in the text. As a consequence, later commentators, especially Robert Simson, who laboured under the idea that Euclid must be absolutely perfect, made Theon the scapegoat for all the defects which they thought they could discover in the text as they knew it. But among the manuscripts sent by Napoleon I. from the Vatican to Paris was found a copy of the Elements believed to be anterior to Theon’s recension. Many

THE GREEKS. 43 variations from Theon's version were noticed therein, but they were not at all important, and showed that Theon generally made only verbal changes. The defects in the Elements for which Theon was blamed must, therefore, be due to Euclid himself. The Elements has been considered as offering models of scrupulously rigorous demonstrations. It is certainly true that in point of rigour it compares favourably with its modern rivals; but when examined in the light of strict mathematical logic, it has been pronounced by C. S. Peirce to be "riddled with fallacies." The results are correct only because the writer's experience keeps him on his guard. At the beginning of our editions of theElements, under the head of definitions, are given the assumptions of such notions as the point, line, etc., and some verbal explanations. Then follow three postulates or demands, and twelve axioms. The term 'axiom' was used by Proclus, but not by Euclid. He speaks, instead, of 'common notions'—common either to all men or to all sciences. There has been much controversy among ancient and modern critics on the postulates and axioms. An immense preponderance of manuscripts and the testimony of Proclus place the 'axioms' about right angles and parallels (Axioms 11 and 12) among the postulates. [9, 10] This is indeed their proper place, for they are reallyassumptions, and not common notions or axioms. The postulate aboutparallels plays an important rˆ ole in the history of non-Euclidean geometry. The only postulate which Euclid missed was the one of superposition, according to which figures can be moved about in space without any alteration in form or magnitude.

A HISTORY OF MATHEMATICS. 44 The Elements contains thirteen books by Euclid, and two, of which it is supposed that Hypsicles and Damascius are the authors. The first four books are on plane geometry. The fifth book treats of the theory of proportion as applied to magnitudes in general. The sixth book develops the geometry of similar figures. The seventh, eighth, ninth books are on the theory of numbers, or on arithmetic. In the ninth book is found the proof to the theorem that the number of primes is infinite. The tenth book treats of the theory of incommensurables. The next three books are on stereometry. The eleventh contains its more elementary theorems; the twelfth, the metrical relations of the pyramid, prism, cone, cylinder, and sphere. The thirteenth treats of the regular polygons, especially of the triangle and pentagon, and then uses them as faces of the five regular solids; namely, the tetraedron, octaedron, icosaedron, cube, and dodecaedron. The regular solids were studied so extensively by the Platonists that they received the name of "Platonic figures." The statement of Proclus that the whole aim of Euclid in writing the Elements was to arrive at the construction of the regular solids, is obviously wrong. The fourteenth and fifteenth books, treating of solid geometry, are apocryphal. A remarkable feature of Euclid's, and of all Greek geometry before Archimedes is that it eschews mensuration. Thus the theorem that the area of a triangle equals half the product of its base and its altitude is foreign to Euclid. Another extant book of Euclid is theData. It seems to have been written for those who, having completed the Elements,

THE GREEKS. 45 wish to acquire the power of solving new problems proposed to them. The Data is a course of practice in analysis. It contains little or nothing that an intelligent student could not pick up from theElements itself. Hence it contributes

little to the stock of scientific knowledge. The following are the other extant works generally attributed to Euclid: Phænomena, a work on spherical geometry and astronomy; Optics, which develops the hypothesis that light proceeds from the eye, and not from the object seen; Catoptrica, containing propositions on reflections from mirrors; De Divisionibus , a treatise on the division of plane figures into parts having to one another a given ratio; Sectio Canonis, a work on musical intervals. His treatise on Porisms is lost; but much learning has been expended by Robert Simson and M. Chasles in restoring it from numerous notes found in the writings of Pappus. The term 'porism' is vague in meaning. The aim of a porism is not to state some property or truth, like a theorem, nor to effect a construction, like a problem, but to find and bring to view a thing which necessarily exists with given numbers or a given construction, as, to find the centre of a given circle, or to find the G.C.D. of two given numbers. [6] His other lost works are Fallacies, containing exercises in detection of fallacies; Conic Sections, in four books, which are the foundation of a work on the same subject by Apollonius; and Loci on a Surface, the meaning of which title is not understood. Heiberg believes it to mean "loci which are surfaces." The immediate successors of Euclid in the mathematical school at Alexandria were probablyConon, Dositheus, and

A HISTORY OF MATHEMATICS. 46 Zeuxippus, but little is known of them. Archimedes (287?–212 b.c.), the greatest mathematician of antiquity, was born in Syracuse. Plutarch calls him a relation of King Hieron; but more reliable is the statement of Cicero, who tells us he was of low birth. Diodorus says he visited Egypt, and, since he was a great friend of Conon and Eratosthenes, it is highly probable that he studied in Alexandria. This belief is strengthened by the fact that he had the most thorough acquaintance with all the work previously done in mathematics. He returned, however, to Syracuse, where he made himself useful to his admiring friend and patron, King Hieron, by applying his extraordinary inventive genius to the construction of various war-engines, by which he inflicted much loss on the Romans during the siege of Marcellus. The story that, by the use of mirrors reflecting the sun's rays, he set on fire the Roman ships, when they came within bow-shot of the walls, is probably a fiction. The city was taken at length by the Romans, and Archimedes perished in the indiscriminate slaughter which followed. According to tradition, he was, at the time, studying the diagram to some problem drawn in the sand. As a Roman soldier approached him, he called out, "Don't spoil my circles." The soldier, feeling insulted, rushed upon him and killed him. No blame attaches to the Roman general Marcellus, who admired his genius, and raised in his honour a tomb bearing the figure of a sphere inscribed in a cylinder. When Cicero was in Syracuse, he found the tomb buried under rubbish. Archimedes was admired by his fellow-citizens chiefly for

THE GREEKS. 47 his mechanical inventions; he himself prized far more highly his discoveries in pure science. He declared that "every kind of art which was connected with daily needs was ignoble and vulgar." Some of his works have been lost. The following are the extant books, arranged approximately in chronological order: 1. Two books on Equiponderance of Planes or Centres of Plane Gravities , between which is inserted his treatise on the Quadrature of the Parabola ; 2. Two books on the

Sphere and Cylinder; 3. TheMeasurement of the Circle; 4. On Spirals; 5. Conoids and Spheroids; 6. The Sand-Counter; 7. Two books on Floating Bodies; 8. Fifteen Lemmas. In the book on the Measurement of the Circle, Archimedes proves first that the area of a circle is equal to that of a right triangle having the length of the circumference for its base, and the radius for its altitude. In this he assumes that there exists a straight line equal in length to the circumference—an assumption objected to by some ancient critics, on the ground that it is not evident that a straight line can equal a curved one. The finding of such a line was the next problem. He first finds an upper limit to the ratio of the circumference to the diameter, or π. To do this, he starts with an equilateral triangle of which the base is a tangent and the vertex is the centre of the circle. By successively bisecting the angle at the centre, by comparing ratios, and by taking the irrational square roots always a little too small, he finally arrived at the conclusion that π < 3 1 7. Next he finds a lower limit by inscribing in the circle regular polygons of 6, 12, 24, 48, 96 sides, finding for each successive polygon its perimeter,

A HISTORY OF MATHEMATICS. 48 which is, of course, always less than the circumference. Thus he finally concludes that “the circumference of a circle exceeds three times its diameter by a part which is less than 1 7 but more than 10 71 of the diameter.” This approximation is exact enough for most purposes. The Quadrature of the Parabola contains two solutions to the problem—one mechanical, the other geometrical. The method of exhaustion is used in both. Archimedes studied also the ellipse and accomplished its quadrature, but to the hyperbola he seems to have paid less attention. It is believed that he wrote a book on conic sections. Of all his discoveries Archimedes prized most highly those in hisSphere and Cylinder. In it are proved the new theorems, that the surface of a sphere is equal to four times a great circle; that the surface of a segment of a sphere is equal to a circle whose radius is the straight line drawn from the vertex of the segment to the circumference of its basal circle; that the volume and the surface of a sphere are 2 3 of the volume and surface, respectively, of the cylinder circumscribed about the sphere. Archimedes desired that the figure to the last proposition be inscribed on his tomb. This was ordered done by Marcellus. The spiral now called the “spiral of Archimedes,” and described in the bookOn Spirals, was discovered by Archimedes, and not, as some believe, by his friend Conon. [3] His treatise thereon is, perhaps, the most wonderful of all his works. Nowadays, subjects of this kind are made easy by the use of the infinitesimal calculus. In its stead the ancients used

THE GREEKS. 49 the method of exhaustion. Nowhere is the fertility of his genius more grandly displayed than in his masterly use of this method. With Euclid and his predecessors the method of exhaustion was only the means of proving propositions which must have been seen and believed before they were proved. But in the hands of Archimedes it became an instrument of discovery. [9] By the word ‘conoid,’ in his book onConoids and Spheroids, is meant the solid produced by the revolution of a parabola or a hyperbola about its axis. Spheroids are produced by the revolution of an ellipse, and are long or flat, according as the ellipse revolves around the major or minor axis. The book leads up to the cubature of these solids. We have now reviewed briefly all his extant works on geometry. His arithmetical treatise and problems will

be considered later. We shall now notice his works on mechanics. Archimedes is the author of the first sound knowledge on this subject. Archytas, Aristotle, and others attempted to form the known mechanical truths into a science, but failed. Aristotle knew the property of the lever, but could not establish its true mathematical theory. The radical and fatal defect in the speculations of the Greeks, says Whewell, was “that though they had in their possession facts and ideas, the ideas were not distinct and appropriate to the facts .” For instance, Aristotle asserted that when a body at the end of a lever is moving, it may be considered as having two motions; one in the direction of the tangent and one in the direction of the radius; the former motion is, he says, according to

A HISTORY OF MATHEMATICS. 50 nature, the latter contrary to nature . These inappropriate notions of ‘natural’ and ‘unnatural’ motions, together with the habits of thought which dictated these speculations, made the perception of the true grounds of mechanical properties impossible. [11] It seems strange that even after Archimedes had entered upon the right path, this science should have remained absolutely stationary till the time of Galileo—a period of nearly two thousand years. The proof of the property of the lever, given in his Equiponderance of Planes, holds its place in text-books to this day. His estimate of the efficiency of the lever is expressed in the saying attributed to him, “Give me a fulcrum on which to rest, and I will move the earth.” Whilethe Equiponderancetreatsofsolids, ortheequilibrium of solids, the book on Floating Bodies treats of hydrostatics. His attention was first drawn to the subject of specific gravity when King Hieron asked him to test whether a crown, professed by the maker to be pure gold, was not alloyed with silver. The story goes that our philosopher was in a bath when the true method of solution flashed on his mind. He immediately ran home, naked, shouting, “I have found it!” To solve the problem, he took a piece of gold and a piece of silver, each weighing the same as the crown. According to one author, he determined the volume of water displaced by the gold, silver, and crown respectively, and calculated from that the amount of gold and silver in the crown. According to another writer, he weighed separately the gold, silver, and crown, while immersed in water, thereby determining their loss of

THE GREEKS. 51 weight in water. From these data he easily found the solution. It is possible that Archimedes solved the problem by both methods. After examining the writings of Archimedes, one can well understand how, in ancient times, an ‘Archimedean problem’ came to mean a problem too deep for ordinary minds to solve, and how an ‘Archimedean proof’ came to be the synonym for unquestionable certainty. Archimedes wrote on a very wide range of subjects, and displayed great profundity in each. He is the Newton of antiquity. Eratosthenes, eleven years younger than Archimedes, was a native of Cyrene. He was educated in Alexandria under Callimachus the poet, whom he succeeded as custodian of the Alexandrian Library. His many-sided activity may be inferred from his works. He wrote on Good and Evil, Measurement of the Earth, Comedy, Geography, Chronology, Constellations, and theDuplication of the Cube. He was also a philologian and a poet. He measured the obliquity of the ecliptic and invented a device for finding prime numbers. Of his geometrical writings we possess only a letter to Ptolemy Euergetes, giving a history of the duplication problem and also the description of a very ingenious mechanical

contrivance of his own to solve it. In his old age he lost his eyesight, and on that account is said to have committed suicide by voluntary starvation. About forty years after Archimedes flourishedApollonius of Perga, whose genius nearly equalled that of his great predecessor. He incontestably occupies the second place in distinction among ancient mathematicians. Apollonius

A HISTORY OF MATHEMATICS. 52 was born in the reign of Ptolemy Euergetes and died under Ptolemy Philopator, who reigned 222–205b.c. He studied at Alexandria under the successors of Euclid, and for some time, also, at Pergamum, where he made the acquaintance of that Eudemus to whom he dedicated the first three books of his Conic Sections. The brilliancy of his great work brought him the title of the “Great Geometer.” This is all that is known of his life. His Conic Sections were in eight books, of which the first four only have come down to us in the original Greek. The next three books were unknown in Europe till the middle of the seventeenth century, when an Arabic translation, made about 1250, was discovered. The eighth book has never been found. In 1710 Halley of Oxford published the Greek text of the first four books and a Latin translation of the remaining three, together with his conjectural restoration of the eighth book, founded on the introductory lemmas of Pappus. The first four books contain little more than the substance of what earlier geometers had done. Eutocius tells us that Heraclides, in his life of Archimedes, accused Apollonius of having appropriated, in his Conic Sections , the unpublished discoveries of that great mathematician. It is difficult to believe that this charge rests upon good foundation. Eutocius quotes Geminus as replying that neither Archimedes nor Apollonius claimed to have invented the conic sections, but that Apollonius had introduced a real improvement. While the first three or four books were founded on the works of Menæchmus, Aristæus, Euclid, and

THE GREEKS. 53 Archimedes, the remaining ones consisted almost entirely of new matter. The first three books were sent to Eudemus at intervals, the other books (after Eudemus’s death) to one Attalus. The preface of the second book is interesting as showing the mode in which Greek books were ‘published’ at this time. It reads thus: “I have sent my son Apollonius to bring you (Eudemus) the second book of my Conics. Read it carefully and communicate it to such others as are worthy of it. If Philonides, the geometer, whom I introduced to you at Ephesus, comes into the neighbourhood of Pergamum, give it to him also.” [12] The first book, says Apollonius in his preface to it, “contains the mode of producing the three sections and the conjugate hyperbolas and their principal characteristics, more fully and generally worked out than in the writings of other authors.” We remember that Menæchmus, and all his successors down to Apollonius, considered only sections of right cones by a plane perpendicular to their sides, and that the three sections were obtained each from a different cone. Apollonius introduced an important generalisation. He produced all the sections from one and the same cone, whether right or scalene, and by sections which may or may not be perpendicular to its sides. The old names for the three curves were now no longer applicable. Instead of calling the three curves, sections of the ‘acute-angled,’ ‘right-angled,’ and ‘obtuse-angled’ cone, he called them ellipse, parabola, and hyperbola, respectively. To be sure, we find the words ‘parabola’ and ‘ellipse’ in the works of Archimedes, but they are probably only

interpolations.

A HISTORY OF MATHEMATICS. 54 The word ‘ellipse’ was applied because y2 < px , p being the parameter; the word ‘parabola’ was introduced because y2 = px, and the term ‘hyperbola’ because y2 > px. The treatise of Apollonius rests on a unique property of conic sections, which is derived directly from the nature of the cone in which these sections are found. How this property forms the key to the system of the ancients is told in a masterly way by M. Chasles. [13] “Conceive,” says he, “an oblique cone on a circular base; the straight line drawn from its summit to the centre of the circle forming its base is called the axis of the cone. The plane passing through the axis, perpendicular to its base, cuts the cone along two lines and determines in the circle a diameter; the triangle having this diameter for its base and the two lines for its sides, is called the triangle through the axis . In the formation of his conic sections, Apollonius supposed the cutting plane to be perpendicular to the plane of the triangle through the axis. The points in which this plane meets the two sides of this triangle are the vertices of the curve; and the straight line which joins these two points is a diameter of it. Apollonius called this diameter latus transversum. At one of the two vertices of the curve erect a perpendicular (latus rectum) to the plane of the triangle through the axis, of a certain length, to be determined as we shall specify later, and from the extremity of this perpendicular draw a straight line to the other vertex of the curve; now, through any point whatever of the diameter of the curve, draw at right angles anordinate: the square of this ordinate, comprehended between the diameter and the curve,

THE GREEKS. 55 will be equal to the rectangle constructed on the portion of the ordinate comprised between the diameter and the straight line, and the part of the diameter comprised between the first vertex and the foot of the ordinate. Such is the characteristic property which Apollonius recognises in his conic sections and which he uses for the purpose of inferring from it, by adroit transformations and deductions, nearly all the rest. It plays, as we shall see, in his hands, almost the same rˆ ole as the equation of the second degree with two variables (abscissa and ordinate) in the system of analytic geometry of Descartes. “It will be observed from this that the diameter of the curve and the perpendicular erected at one of its extremities suffice to construct the curve. These are the two elements which the ancients used, with which to establish their theory of conics. The perpendicular in question was called by them latus erectum; the moderns changed this name first to that of latus rectum, and afterwards to that ofparameter.” The first book of theConic Sections of Apollonius is almost wholly devoted to the generation of the three principal conic sections. The second book treats mainly of asymptotes, axes, and diameters. The third book treats of the equality or proportionality of triangles, rectangles, or squares, of which the component parts are determined by portions of transversals, chords, asymptotes, or tangents, which are frequently subject to a great number of conditions. It also touches the subject of foci of the ellipse and hyperbola.

A HISTORY OF MATHEMATICS. 56 In the fourth book, Apollonius discusses the harmonic division of straight lines. He also examines a system of two conics, and shows that they cannot cut each other in more than four points. He investigates the

various possible relative positions of two conics, as, for instance, when they have one or two points of contact with each other. The fifth book reveals better than any other the giant intellect of its author. Difficult questions of maxima and minima, of which few examples are found in earlier works, are here treated most exhaustively. The subject investigated is, to find the longest and shortest lines that can be drawn from a given point to a conic. Here are also found the germs of the subject of evolutes and centres of osculation. The sixth book is on the similarity of conics. The seventh book is on conjugate diameters. The eighth book, as restored by Halley, continues the subject of conjugate diameters. It is worthy of notice that Apollonius nowhere introduces the notion of directrix for a conic, and that, though he incidentally discovered the focus of an ellipse and hyperbola, he did not discover the focus of a parabola. [6] Conspicuous in his geometry is also the absence of technical terms and symbols, which renders the proofs long and cumbrous. The discoveries of Archimedes and Apollonius, says M. Chasles, [13] marked the most brilliant epoch of ancient geometry. Two questions which have occupied geometers of all periods may be regarded as having originated with them.

THE GREEKS. 57 The first of these is the quadrature of curvilinear figures, which gave birth to the infinitesimal calculus. The second is the theory of conic sections, which was the prelude to the theory of geometrical curves of all degrees, and to that portion of geometry which considers only the forms and situations of figures, and uses only the intersection of lines and surfaces and the ratios of rectilineal distances. These two great divisions of geometry may be designated by the names of Geometry of Measurements and Geometry of Forms and Situations, or, Geometry of Archimedes and of Apollonius. Besides the Conic Sections, Pappus ascribes to Apollonius the following works: On Contacts, Plane Loci, Inclinations, Section of an Area , Determinate Section, and gives lemmas from which attempts have been made to restore the lost originals. Two books on De Sectione Rationis have been found in the Arabic. The book on Contacts, as restored by Vieta, contains the so-called “Apollonian Problem”: Given three circles, to find a fourth which shall touch the three. Euclid, Archimedes, and Apollonius brought geometry to as high a state of perfection as it perhaps could be brought without first introducing some more general and more powerful method than the old method of exhaustion. A briefer symbolism, a Cartesian geometry, an infinitesimal calculus, were needed. The Greek mind was not adapted to the invention of general methods. Instead of a climb to still loftier heights we observe, therefore, on the part of later Greek geometers, a descent, during which they paused here and there to look around for details which had been passed by

A HISTORY OF MATHEMATICS. 58 in the hasty ascent. [3] Among the earliest successors of Apollonius was Nico- medes. Nothing definite is known of him, except that he invented the conchoid (“mussel-like”). He devised a little machine by which the curve could be easily described. With aid of the conchoid he duplicated the cube. The curve can also be used for trisecting angles in a way much resembling that in the eighth lemma of Archimedes. Proclus ascribes this mode of trisection to Nicomedes, but Pappus, on the other hand, claims it as his own. The conchoid was used by Newton in constructing curves of the third degree. About the time of Nicomedes, flourished also Diocles, the inventor of the cissoid (“ivy-like”). This curve he used for

finding two mean proportionals between two given straight lines. About the life ofPerseus we know as little as about that of Nicomedes and Diocles. He lived some time between 200 and 100 b.c. From Heron and Geminus we learn that he wrote a work on the spire, a sort of anchor-ring surface described by Heron as being produced by the revolution of a circle around one of its chords as an axis. The sections of this surface yield peculiar curves called spiral sections, which, according to Geminus, were thought out by Perseus. These curves appear to be the same as the Hippopede of Eudoxus. Probably somewhat later than Perseus lived Zenodorus. He wrote an interesting treatise on a new subject; namely, isoperimetrical figures. Fourteen propositions are preserved by Pappus and Theon. Here are a few of them: Of isoperimetrical,

THE GREEKS. 59 regular polygons, the one having the largest number of angles has the greatest area; the circle has a greater area than any regular polygon of equal periphery; of all isoperimetrical polygons of n sides, the regular is the greatest; of all solids having surfaces equal in area, the sphere has the greatest volume. Hypsicles (between 200 and 100 b.c.) was supposed to be the author of both the fourteenth and fifteenth books of Euclid, but recent critics are of opinion that the fifteenth book was written by an author who lived several centuries after Christ. The fourteenth book contains seven elegant theorems on regular solids. A treatise of Hypsicles on Risings is of interest because it is the first Greek work giving the division of the circumference into 360 degrees after the fashion of the Babylonians. Hipparchus of Nicæa in Bithynia was the greatest as- tronomer of antiquity. He established inductively the famous theory of epicycles and eccentrics. As might be expected, he was interested in mathematics, not per se, but only as an aid to astronomical inquiry. No mathematical writings of his are extant, but Theon of Alexandria informs us that Hipparchus originated the science of trigonometry, and that he calcu- lated a “table of chords” in twelve books. Such calculations must have required a ready knowledge of arithmetical and algebraical operations. About 100b.c. flourished Heron the Elderof Alexandria. He was the pupil of Ctesibius, who was celebrated for his ingenious mechanical inventions, such as the hydraulic organ,

A HISTORY OF MATHEMATICS. 60 the water-clock, and catapult. It is believed by some that Heron was a son of Ctesibius. He exhibited talent of the same order as did his master by the invention of the eolipile and a curious mechanism known as “Heron’s fountain.” Great uncertainty exists concerning his writings. Most authorities believe him to be the author of an important Treatise on the Dioptra , of which there exist three manuscript copies, quite dissimilar. But M. Marie [14] thinks that the Dioptra is the work of Heron the Younger, who lived in the seventh or eighth century after Christ, and that Geodesy, another book supposed to be by Heron, is only a corrupt and defective copy of the former work. Dioptra contains the important formula for finding the area of a triangle expressed in terms of its sides; its derivation is quite laborious and yet exceedingly ingenious. “It seems to me difficult to believe,” says Chasles, “that so beautiful a theorem should be found in a work so ancient as that of Heron the Elder, without that some Greek geometer should have thought to cite it.” Marie lays great stress on this silence of the ancient writers, and argues from it that the true author must be Heron the Younger or some writer much more recent

than Heron the Elder. But no reliable evidence has been found that there actually existed a second mathematician by the name of Heron. “Dioptra,” says Venturi, were instruments which had great resemblance to our modern theodolites. The book Dioptra is a treatise on geodesy containing solutions, with aid of these instruments, of a large number of questions in geometry, such as to find the distance between two points, of which one only

THE GREEKS. 61 is accessible, or between two points which are visible but both inaccessible; from a given point to draw a perpendicular to a line which cannot be approached; to find the difference of level between two points; to measure the area of a field without entering it. Heron was a practical surveyor. This may account for the fact that his writings bear so little resemblance to those of the Greek authors, who considered it degrading the science to apply geometry to surveying. The character of his geometry is not Grecian, but decidedly Egyptian. This fact is the more surprising when we consider that Heron demonstrated his familiarity with Euclid by writing a commentary on the Elements. [21] Some of Heron’s formulas point to an old Egyptian origin. Thus, besides the above exact formula for the area of a triangle in terms of its sides, Heron gives the formula a1 + a2 2 × b 2, which bears a striking likeness to the formula a1 + a2 2 × b1 + b2 2 for finding the area of a quadrangle, found in the Edfu inscriptions. There are, moreover, points of resemblance between Heron’s writings and the ancient Ahmes papyrus. Thus Ahmes used unit-fractions exclusively; Heron uses them oftener than other fractions. Like Ahmes and the priests at Edfu, Heron divides complicated figures into simpler ones by drawing auxiliary lines; like them, he shows, throughout, a special fondness for the isosceles trapezoid. The writings of Heron satisfied a practical want, and for that reason were borrowed extensively by other peoples. We find traces of them in Rome, in the Occident during the Middle

A HISTORY OF MATHEMATICS. 62 Ages, and even in India. Geminus of Rhodes (about 70 b.c.) published an astro- nomical work still extant. He wrote also a book, now lost, on the Arrangement of Mathematics, which contained many valuable notices of the early history of Greek mathematics. Proclus and Eutocius quote it frequently. Theodosius of Tripolis is the author of a book of little merit on the geometry of the sphere. Dionysodorus of Amisus in Pontus applied the intersection of a parabola and hyperbola to the solution of a problem which Archimedes, in his Sphere and Cylinder, had left incomplete. The problem is “to cut a sphere so that its segments shall be in a given ratio.” We have now sketched the progress of geometry down to the time of Christ. Unfortunately, very little is known of the history of geometry between the time of Apollonius and the beginning of the Christian era. The names of quite a number of geometers have been mentioned, but very few of their works are now extant. It is certain, however, that there were no mathematicians of real genius from Apollonius to Ptolemy, excepting Hipparchus and perhaps Heron. The Second Alexandrian School. The close of the dynasty of the Lagides which ruled Egypt from the time of Ptolemy Soter, the builder of Alexandria, for 300 years; the absorption of Egypt into the Roman Empire; the closer commercial relations between peoples of the East and of the West; the gradual decline of paganism and spread

THE GREEKS. 63 of Christianity,—these events were of far-reaching influence on the progress of the sciences, which then had their home in Alexandria. Alexandria became a commercial and intellectual emporium. Traders of all nations met in her busy streets, and in her magnificent Library, museums, lecture-halls, scholars from the East mingled with those of the West; Greeks began to study older literatures and to compare them with their own. In consequence of this interchange of ideas the Greek philosophy became fused with Oriental philosophy. Neo-Pythagoreanism and Neo-Platonism were the names of the modified systems. These stood, for a time, in opposition to Christianity. The study of Platonism and Pythagorean mysticism led to the revival of the theory of numbers. Perhaps the dispersion of the Jews and their introduction to Greek learning helped in bringing about this revival. The theory of numbers became a favourite study. This new line of mathematical inquiry ushered in what we may call a new school. There is no doubt that even now geometry continued to be one of the most important studies in the Alexandrian course. This Second Alexandrian School may be said to begin with the Christian era. It was made famous by the names of Claudius Ptolemæus, Diophantus, Pappus, Theon of Smyrna, Theon of Alexandria, Iamblichus, Porphyrius, and others. By the side of these we may place Serenus of Antissa, as having been connected more or less with this new school. He wrote on sections of the cone and cylinder, in two books, one of which treated only of the triangular section of the cone through the apex. He solved the problem, “given

A HISTORY OF MATHEMATICS. 64 a cone (cylinder), to find a cylinder (cone), so that the section of both by the same plane gives similar ellipses.” Of particular interest is the following theorem, which is the foundation of the modern theory of harmonics: If from D we A B C D EF G H J K draw DF , cutting the tri- angle ABC, and choose H on it, so that DE : DF = EH : HF , and if we draw the line AH, then every transversal through D, such as DG, will be di- vided by AH so that DK : DG = KJ : J G. Menelaus of Alexandria (about 98 a.d.) was the author of Sphærica, a work extant in Hebrew and Arabic, but not in Greek. In it he proves the theorems on the congruence of spherical triangles, and describes their properties in much the same way as Euclid treats plane triangles. In it are also found the theorems that the sum of the three sides of a spherical triangle is less than a great circle, and that the sum of the three angles exceeds two right angles. Celebrated are two theorems of his on plane and spherical triangles. The one on plane triangles is that, “if the three sides be cut by a straight line, the product of the three segments which have no common extremity is equal to the product of the other three.” The illustrious Carnot makes this proposition, known as the ‘lemma of Menelaus,’ the base of his theory of transversals. The corresponding theorem for spherical triangles, the so-called ‘regula sex quantitatum,’ is obtained from the above by reading “chords of three segments

THE GREEKS. 65 doubled,” in place of “three segments.” Claudius Ptolemæus, a celebrated astronomer, was a native of Egypt. Nothing is known of his personal history except that he flourished in Alexandria in 139 a.d. and that he made the earliest astronomical observations recorded in his works, in 125 a.d., the latest in 151 a.d. The chief of his works are the Syntaxis Mathematica (or the Almagest, as the Arabs call it) and the Geographica, both of which are extant. The former work is based partly on his own researches, but mainly on those of Hipparchus. Ptolemy

seems to have been not so much of an independent investigator, as a corrector and improver of the work of his great predecessors. The Almagest forms the foundation of all astronomical science down to Copernicus. The fundamental idea of his system, the “Ptolemaic System,” is that the earth is in the centre of the universe, and that the sun and planets revolve around the earth. Ptolemy did considerable for mathematics. He created, for astronomical use, a trigonometry remarkably perfect in form. The foundation of this science was laid by the illustrious Hipparchus. The Almagest is in 13 books. Chapter 9 of the first book shows how to calculate tables of chords. The circle is divided into 360 degrees, each of which is halved. The diameter is divided into120 divisions; each of these into60 parts, which are again subdivided into 60 smaller parts. In Latin, these parts were calledpartes minutæ primæand partes minutæ secundæ. Hence our names, ‘minutes’ and ‘seconds.’ [3] The sexagesimal method of dividing the circle is of Babylonian origin, and was

A HISTORY OF MATHEMATICS. 66 known to Geminus and Hipparchus. But Ptolemy’s method of calculating chords seems original with him. He first proved the proposition, now appended to Euclid VI. (D), that “the rectangle contained by the diagonals of a quadrilateral figure inscribed in a circle is equal to both the rectangles contained by its opposite sides.” He then shows how to find from the chords of two arcs the chords of their sum and difference, and from the chord of any arc that of its half. These theorems he applied to the calculation of his tables of chords. The proofs of these theorems are very pretty. Another chapter of the first book in the Almagest is devoted to trigonometry, and to spherical trigonometry in particular. Ptolemy proved the ‘lemma of Menelaus,’ and also the ‘regula sex quantitatum.’ Upon these propositions he built up his trigonometry. The fundamental theorem of plane trigonometry, that two sides of a triangle are to each other as the chords of double the arcs measuring the angles opposite the two sides, was not stated explicitly by him, but was contained implicitly in other theorems. More complete are the propositions in spherical trigonometry. The fact that trigonometry was cultivated not for its own sake, but to aid astronomical inquiry, explains the rather startling fact that spherical trigonometry came to exist in a developed state earlier than plane trigonometry. The remaining books of the Almagest are on astronomy. Ptolemy has written other works which have little or no bearing on mathematics, except one on geometry. Extracts from this book, made by Proclus, indicate that Ptolemy did

THE GREEKS. 67 not regard the parallel-axiom of Euclid as self-evident, and that Ptolemy was the first of the long line of geometers from ancient time down to our own who toiled in the vain attempt to prove it. Two prominent mathematicians of this time were Nico- machus and Theon of Smyrna. Their favourite study was theory of numbers. The investigations in this science culmi- nated later in the algebra of Diophantus. But no important geometer appeared after Ptolemy for 150 years. The only occupant of this long gap was Sextus Julius Africanus , who wrote an unimportant work on geometry applied to the art of war, entitledCestes. Pappus, probably born about 340a.d., in Alexandria, was the last great mathematician of the Alexandrian school. His genius was inferior to that of Archimedes, Apollonius, and Euclid, who flourished over 500 years earlier. But living, as he did, at a period when interest in geometry was declining, he towered above his contemporaries “like the

peak of Teneriffa above the Atlantic." He is the author of a Commentary on the Almagest , a Commentary on Euclid's Elements , a Commentary on the Analemma of Diodorus ,—a writer of whom nothing is known. All these works are lost. Proclus, probably quoting from the Commentary on Euclid, says that Pappus objected to the statement that an angle equal to a right angle is always itself a right angle. The only work of Pappus still extant is his Mathematical Collections. This was originally in eight books, but the first and portions of the second are now missing. TheMathematical

A HISTORY OF MATHEMATICS. 68 Collections seems to have been written by Pappus to supply the geometers of his time with a succinct analysis of the most difficult mathematical works and to facilitate the study of them by explanatory lemmas. But these lemmas are selected very freely, and frequently have little or no connection with the subject on hand. However, he gives very accurate summaries of the works of which he treats. TheMathematical Collections is invaluable to us on account of the rich information it gives on various treatises by the foremost Greek mathematicians, which are now lost. Mathematicians of the last century considered it possible to restore lost works from the r´ esum´ e by Pappus alone. We shall now cite the more important of those theorems in the Mathematical Collections which are supposed to be original with Pappus. First of all ranks the elegant theorem re-discovered byGuldin, over 1000 years later, that the volume generated by the revolution of a plane curve which lies wholly on one side of the axis, equals the area of the curve multiplied by the circumference described by its centre of gravity. Pappus proved also that the centre of gravity of a triangle is that of another triangle whose vertices lie upon the sides of the first and divide its three sides in the same ratio. In the fourth book are new and brilliant propositions on the quadratrix which indicate an intimate acquaintance with curved surfaces. He generates the quadratrix as follows: Let a spiral line be drawn upon a right circular cylinder; then the perpendiculars to the axis of the cylinder drawn from each point of the spiral line form the surface of a screw. A plane passed through one of

THE GREEKS. 69 these perpendiculars, making any convenient angle with the base of the cylinder, cuts the screw-surface in a curve, the orthogonal projection of which upon the base is thequadratrix. A second mode of generation is no less admirable: If we make the spiral of Archimedes the base of a right cylinder, and imagine a cone of revolution having for its axis the side of the cylinder passing through the initial point of the spiral, then this cone cuts the cylinder in a curve of double curvature. The perpendiculars to the axis drawn through every point in this curve form the surface of a screw which Pappus here calls the plectoidal surface. A plane passed through one of the perpendiculars at any convenient angle cuts that surface in a curve whose orthogonal projection upon the plane of the spiral is the required quadratrix. Pappus considers curves of double curvature still further. He produces a spherical spiral by a point moving uniformly along the circumference of a great circle of a sphere, while the great circle itself revolves uniformly around its diameter. He then finds the area of that portion of the surface of the sphere determined by the spherical spiral, "a complanation which claims the more lively admiration, if we consider that, although the entire surface of the sphere was known

since Archimedes' time, to measure portions thereof, such as spherical triangles, was then and for a long time afterwards an unsolved problem." [3] A question which was brought into prominence by Descartes and Newton is the "problem of Pappus." Given several straight lines in a plane, to find the locus of a point such that when perpendiculars (or, more generally, straight lines at given angles) are drawn

A HISTORY OF MATHEMATICS. 70 from it to the given lines, the product of certain ones of them shall be in a given ratio to the product of the remaining ones. It is worth noticing that it was Pappus who first found the focus of the parabola, suggested the use of the directrix, and propounded the theory of the involution of points. He solved the problem to draw through three points lying in the same straight line, three straight lines which shall form a triangle inscribed in a given circle. [3] From theMathematical Collections many more equally difficult theorems might be quoted which are original with Pappus as far as we know. It ought to be remarked, however, that he is known in three instances to have copied theorems without giving due credit, and that he may have done the same thing in other cases in which we have no data by which to ascertain the real discoverer. About the time of Pappus lived Theon of Alexandria. He brought out an edition of Euclid'sElements with notes, which he probably used as a text-book in his classes. His commentary on the Almagest is valuable for the many historical notices, and especially for the specimens of Greek arithmetic which it contains. Theon's daughter Hypatia, a woman celebrated for her beauty and modesty, was the last Alexandrian teacher of reputation, and is said to have been an abler philosopher and mathematician than her father. Her notes on the works of Diophantus and Apollonius have been lost. Her tragic death in 415 a.d. is vividly described in Kingsley's Hypatia. From now on, mathematics ceased to be cultivated in Alexandria. The leading subject of men's thoughts was

THE GREEKS. 71 Christian theology. Paganism disappeared, and with it pagan learning. The Neo-Platonic school at Athens struggled on a century longer. Proclus, Isidorus, and others kept up the "golden chain of Platonic succession."Proclus, the successor of Syrianus, at the Athenian school, wrote a commentary on Euclid's Elements. We possess only that on the first book, which is valuable for the information it contains on the history of geometry. Damascius of Damascus, the pupil of Isidorus, is now believed to be the author of the fifteenth book of Euclid. Another pupil of Isidorus was Eutocius of Ascalon, the commentator of Apollonius and Archimedes.Simplicius wrote a commentary on Aristotle'sDe Cœlo. In the year 529, Justinian, disapproving heathen learning, finally closed by imperial edict the schools at Athens. As a rule, the geometers of the last 500 years showed a lack of creative power. They were commentators rather than discoverers. The principal characteristics of ancient geometry are:— (1) A wonderful clearness and definiteness of its concepts and an almost perfect logical rigour of its conclusions. (2) A complete want of general principles and methods. Ancient geometry is decidedly special. Thus the Greeks possessed no general method of drawing tangents. "The determination of the tangents to the three conic sections did not furnish any rational assistance for drawing the tangent to any other new curve, such as the conchoid, the cissoid, etc." [15] In the demonstration of a theorem, there were, for the ancient geometers, as many different cases requiring

A HISTORY OF MATHEMATICS. 72 separate proof as there were different positions for the lines. The greatest geometers considered it necessary to treat all possible cases independently of each other, and to prove each with equal fulness. To devise methods by which the various cases could all be disposed of by one stroke, was beyond the power of the ancients. “If we compare a mathematical problem with a huge rock, into the interior of which we desire to penetrate, then the work of the Greek mathematicians appears to us like that of a vigorous stonecutter who, with chisel and hammer, begins with indefatigable perseverance, from without, to crumble the rock slowly into fragments; the modern mathematician appears like an excellent miner, who first bores through the rock some few passages, from which he then bursts it into pieces with one powerful blast, and brings to light the treasures within.” [16] GREEK ARITHMETIC. Greek mathematicians were in the habit of discriminating between the science of numbers and the art of calculation. The former they called arithmetica, the latter logistica. The drawing of this distinction between the two was very natural and proper. The difference between them is as marked as that between theory and practice. Among the Sophists the art of calculation was a favourite study. Plato, on the other hand, gave considerable attention to philosophical arithmetic, but pronounced calculation a vulgar and childish art. In sketching the history of Greek calculation, we shall first

THE GREEKS. 73 give a brief account of the Greek mode of counting and of writing numbers. Like the Egyptians and Eastern nations, the earliest Greeks counted on their fingers or with pebbles. In case of large numbers, the pebbles were probably arranged in parallel vertical lines. Pebbles on the first line represented units, those on the second tens, those on the third hundreds, and so on. Later, frames came into use, in which strings or wires took the place of lines. According to tradition, Pythagoras, who travelled in Egypt and, perhaps, in India, first introduced this valuable instrument into Greece. The abacus, as it is called, existed among different peoples and at different times, in various stages of perfection. An abacus is still employed by the Chinese under the name of Swan- pan. We possess no specific information as to how the Greek abacus looked or how it was used. Boethius says that the Pythagoreans used with the abacus certain nine signs called apices, which resembled in form the nine “Arabic numerals.” But the correctness of this assertion is subject to grave doubts. The oldest Grecian numerical symbols were the so-called Herodianic signs (after Herodianus, a Byzantine grammarian of about 200 a.d., who describes them). These signs occur frequently in Athenian inscriptions and are, on that account, now generally called Attic. For some unknown reason these symbols were afterwards replaced by thealphabetic numerals, in which the letters of the Greek alphabet were used, together with three strange and antique letters □ , , and , and the symbol M. This change was decidedly for the worse, for the old Attic numerals were less burdensome on the memory,

A HISTORY OF MATHEMATICS. 74 inasmuch as they contained fewer symbols and were better adapted to show forth analogies in numerical operations. The following table shows the Greek alphabetic numerals and their respective values:— α β γ δ ϵ ς ζ η θ ι κ λ μ ν ξ ο π 1 2 3 4 5 6 7 8 9 10 20 30 40 50 60 70 80 90 ρ σ τ υ φ χ ψ ω ,α ,β ,γ etc. 100 200 300 400 500 600 700 800 900 1000 2000 3000 M β M γ M etc. 10,000

20,000 30,000 It will be noticed that at 1000, the alphabet is begun over again, but, to prevent confusion, a stroke is now placed before the letter and generally somewhat below it. A horizontal line drawn over a number served to distinguish it more readily from words. The coefficient for M was sometimes placed before or behind instead of over the M. Thus 43, 678 was written δMÿ γχοη. It is to be observed that the Greeks had no zero. Fractions were denoted by first writing the numerator marked with an accent, then the denominator marked with two accents and written twice. Thus, ιγ′κθ′′κθ′′ = 13 29. In case of fractions having unity for the numerator, the α′ was omitted and the denominator was written only once. Thus μδ′′ = 1 44. Greek writers seldom refer to calculation with alphabetic nu- merals. Addition, subtraction, and even multiplication were probably performed on the abacus. Expert mathematicians

THE GREEKS. 75 may have used the symbols. Thus Eutocius, a commen- tator of the sixth century after Christ, gives a great many multiplications of which the following is a specimen: [6]— σ ξ ϵ 2 6 5 σ ξ ϵ 2 6 5 δ M α M ,β ,α 40000, 12000, 1000 α M ,β ,γχ τ 12000, 3600, 300 ,α τ κϵ 1000, 300, 25 ζ M σ κ ϵ 70225 The operation is explained sufficiently by the modern numerals appended. In case of mixed numbers, the pro- cess was still more clumsy. Divisions are found in Theon of Alexandria's commentary on the Almagest. As might be expected, the process is long and tedious. We have seen in geometry that the more advanced mathe- maticians frequently had occasion to extract the square root. Thus Archimedes in his Mensuration of the Circle gives a large number of square roots. He states, for instance, that$\sqrt{3} < 1351\ 780$ and $\sqrt{3} > 265\ 153$, but he gives no clue to the method by which he obtained these approximations. It is not improbable that the earlier Greek mathematicians found the square root by trial only. Eutocius says that the method of extracting it was given by Heron, Pappus, Theon, and other commentators on the Almagest. Theon's is the only ancient method known to us. It is the same as the one used nowadays, except that sexagesimal fractions are employed in place of our decimals. What the mode of procedure actually was when sexagesimal fractions were not used, has been the subject of conjecture on the part of numerous modern writers. [17] Of interest, in connection with arithmetical symbolism,

A HISTORY OF MATHEMATICS. 76 is the Sand-Counter (Arenarius), an essay addressed by Archimedes to Gelon, king of Syracuse. In it Archimedes shows that people are in error who think the sand cannot be counted, or that if it can be counted, the number cannot be expressed by arithmetical symbols. He shows that the number of grains in a heap of sand not only as large as the whole earth, but as large as the entire universe, can be arithmetically expressed. Assuming that 10, 000 grains of sand suffice to make a little solid of the magnitude of a poppy-seed, and that the diameter of a poppy-seed be not smaller than 1 40 part of a finger's breadth; assuming further, that the diameter of the universe (supposed to extend to the sun) be less than 10, 000 diameters of the earth, and that the latter be less than 1, 000, 000 stadia, Archimedes finds a number which would exceed the number of grains of sand in the sphere of the universe. He goes on even further. Supposing the universe to reach out to the fixed stars, he finds that the sphere, having the distance from the earth's centre to the fixed stars for its radius, would contain a number of grains of sand less than 1000 myriads of the eighth octad. In our notation, this number would

be 1063 or 1 with 63 ciphers after it. It can hardly be doubted that one object which Archimedes had in view in making this calculation was the improvement of the Greek symbolism. It is not known whether he invented some short notation by which to represent the above number or not. We judge from fragments in the second book of Pappus that Apollonius proposed an improvement in the Greek method of writing numbers, but its nature we do not know. Thus

THE GREEKS. 77 we see that the Greeks never possessed the boon of a clear, comprehensive symbolism. The honour of giving such to the world, once for all, was reserved by the irony of fate for a nameless Indian of an unknown time, and we know not whom to thank for an invention of such importance to the general progress of intelligence. [6] Passing from the subject of logistica to that of arithmetica, our attention is first drawn to the science of numbers of Pythagoras. Before founding his school, Pythagoras studied for many years under the Egyptian priests and familiarised himself with Egyptian mathematics and mysticism. If he ever was in Babylon, as some authorities claim, he may have learned the sexagesimal notation in use there; he may have picked up considerable knowledge on the theory of proportion, and may have found a large number of interesting astronomical observations. Saturated with that speculative spirit then pervading the Greek mind, he endeavoured to discover some principle of homogeneity in the universe. Before him, the philosophers of the Ionic school had sought it in the matter of things; Pythagoras looked for it in the structure of things. He observed various numerical relations or analogies between numbers and the phenomena of the universe. Being convinced that it was in numbers and their relations that he was to find the foundation to true philosophy, he proceeded to trace the origin of all things to numbers. Thus he observed that musical strings of equal length stretched by weights having the proportion of 1 2, 2 3, 3 4, produced intervals which were an octave, a fifth, and a fourth. Harmony, therefore, depends on

A HISTORY OF MATHEMATICS. 78 musical proportion; it is nothing but a mysterious numerical relation. Where harmony is, there are numbers. Hence the order and beauty of the universe have their origin in numbers. There are seven intervals in the musical scale, and also seven planets crossing the heavens. The same numerical relations which underlie the former must underlie the latter. But where numbers are, there is harmony. Hence his spiritual ear discerned in the planetary motions a wonderful ‘harmony of the spheres.’ The Pythagoreans invested particular numbers with extraordinary attributes. Thus one is the essence of things; it is an absolute number; hence the origin of all numbers and so of all things.Four is the most perfect number, and was in some mystic way conceived to correspond to the human soul. Philolaus believed that 5 is the cause of color, 6 of cold, 7 of mind and health and light, 8 of love and friendship. [6] In Plato’s works are evidences of a similar belief in religious relations of numbers. Even Aristotle referred the virtues to numbers. Enough has been said about these mystic speculations to show what lively interest in mathematics they must have created and maintained. Avenues of mathematical inquiry were opened up by them which otherwise would probably have remained closed at that time. The Pythagoreans classified numbers into odd and even. They observed that the sum of the series of odd numbers from 1 to 2n + 1was

always a complete square, and that by addition of the even numbers arises the series 2, 6, 12, 20, in which every number can be decomposed into two factors differing

THE GREEKS. 79 from each other by unity. Thus, 6 = 2·3, 12 = 3·4, etc. These latter numbers were considered of sufficient importance to receive the separate name ofheteromecic (not equilateral). [7] Numbers of the form n(n + 1) 2 were called triangular, because they could always be arranged thus, . Numbers which were equal to the sum of all their possible factors, such as6, 28, 496, were called perfect; those exceeding that sum, excessive; and those which were less, defective. Amicable numbers were those of which each was the sum of the factors in the other. Much attention was paid by the Pythagoreans to the subject of proportion. The quantities a, b, c, d were said to be in arithmetical proportion when a− b = c− d; in geometrical proportion, when a : b = c : d; in harmonic proportion, when a− b : b− c = a : c. It is probable that the Pythagoreans were also familiar with themusical proportion a : a + b 2 = 2ab a + b : b. Iamblichus says that Pythagoras introduced it from Babylon. In connection with arithmetic, Pythagoras made extensive investigations into geometry. He believed that an arithmetical fact had its analogue in geometry, andvice versa. In connection with his theorem on the right triangle he devised a rule by which integral numbers could be found, such that the sum of the squares of two of them equalled the square of the third. Thus, take for one side an odd number (2n + 1); then (2n + 1)2− 1 2 = 2n2 + 2n = the other side, and(2n2 + 2n+ 1) = hypotenuse. If 2n + 1 = 9, then the other two numbers are 40 and 41. But this rule only applies to cases in which the hypotenuse differs from one of the sides by 1. In the study of

A HISTORY OF MATHEMATICS. 80 the right triangle there doubtless arose questions of puzzling subtlety. Thus, given a number equal to the side of an isosceles right triangle, to find the number which the hypotenuse is equal to. The side may have been taken equal to 1, 2, 3 2, 6 5, or any other number, yet in every instance all efforts to find a number exactly equal to the hypotenuse must have remained fruitless. The problem may have been attacked again and again, until finally “some rare genius, to whom it is granted, during some happy moments, to soar with eagle’s flight above the level of human thinking,” grasped the happy thought that this problem cannot be solved. In some such manner probably arose the theory of irrational quantities , which is attributed by Eudemus to the Pythagoreans. It was indeed a thought of extraordinary boldness, to assume that straight lines could exist, differing from one another not only in length,—that is, in quantity,—but also in a quality, which, though real, was absolutely invisible. [7] Need we wonder that the Pythagoreans saw in irrationals a deep mystery, a symbol of the unspeakable? We are told that the one who first divulged the theory of irrationals, which the Pythagoreans kept secret, perished in consequence in a shipwreck. Its discovery is ascribed to Pythagoras, but we must remember that all important Pythagorean discoveries were, according to Pythagorean custom, referred back to him. The first incommensurable ratio known seems to have been that of the side of a square to its diagonal, as 1 : √ 2. Theodorus of Cyrene added to this the fact that the sides of squares represented in length by √ 3, √ 5, etc., up to √ 17,

THE GREEKS. 81 and Theætetus, that the sides of any square, represented by a surd, are incommensurable with the linear unit. Euclid (about 300 b.c.), in his Elements, X. 9, generalised still further: Two magnitudes whose squares are (or are not) to one another as a square number to a square number are commensurable (or incommensurable), and conversely. In the tenth book, he treats of incommensurable quantities at length. He investigates every possible variety of lines which can be represented by √√ a± √ b, a and b representing two commensurable lines, and obtains 25 species. Every individual of every species is incommensurable with all the individuals of every other species. “This book,” says De Morgan, “has a completeness which none of the others (not even the fifth) can boast of; and we could almost suspect that Euclid, having arranged his materials in his own mind, and having completely elaborated the tenth book, wrote the preceding books after it, and did not live to revise them thoroughly.” [9] The theory of incommensurables remained where Euclid left it, till the fifteenth century. Euclid devotes the seventh, eighth, and ninth books of his Elements to arithmetic. Exactly how much contained in these books is Euclid’s own invention, and how much is borrowed from his predecessors, we have no means of knowing. Without doubt, much is original with Euclid. The seventh book begins with twenty-one definitions. All except that for ‘prime’ numbers are known to have been given by the Pythagoreans. Next follows a process for finding the G.C.D. of two or more numbers. The eighth book deals with numbers in continued

A HISTORY OF MATHEMATICS. 82 proportion, and with the mutual relations of squares, cubes, and plane numbers. Thus, XXII., if three numbers are in continued proportion, and the first is a square, so is the third. In the ninth book, the same subject is continued. It contains the proposition that the number of primes is greater than any given number. After the death of Euclid, the theory of numbers remained almost stationary for 400 years. Geometry monopolised the attention of all Greek mathematicians. Only two are known to have done work in arithmetic worthy of mention. Eratosthenes (275–194 b.c.) invented a ‘sieve’ for finding prime numbers. All composite numbers are ‘sifted’ out in the following manner: Write down the odd numbers from 3 up, in succession. By striking out every third number after the 3, we remove all multiples of 3. By striking out every fifth number after the 5, we remove all multiples of 5. In this way, by rejecting multiples of 7, 11, 13, etc., we have left prime numbers only. Hypsicles (between 200 and 100 b.c.) worked at the subjects of polygonal numbers and arithmetical progressions, which Euclid entirely neglected. In his work on ‘risings of the stars,’ he showed (1) that in an arithmetical series of 2n terms, the sum of the last n terms exceeds the sum of the first n by a multiple of n2; (2) that in such a series of 2n + 1 terms, the sum of the series is the number of terms multiplied by the middle term; (3) that in such a series of 2n terms, the sum is half the number of terms multiplied by the two middle terms. [6] For two centuries after the time of Hypsicles, arithmetic

THE GREEKS. 83 disappears from history. It is brought to light again about 100 a.d. by Nicomachus, a Neo-Pythagorean, who inau- gurated the final era of Greek mathematics. From now on, arithmetic was a favourite study, while geometry was neglected. Nicomachus wrote a work entitled Introductio Arithmetica, which was very famous in its day. The great number of commentators it has received vouch for

its popu- larity. Boethius translated it into Latin. Lucian could pay no higher compliment to a calculator than this: “You reckon like Nicomachus of Gerasa.” The Introductio Arithmeticawas the first exhaustive work in which arithmetic was treated quite independently of geometry. Instead of drawing lines, like Euclid, he illustrates things by real numbers. To be sure, in his book the old geometrical nomenclature is retained, but the method is inductive instead of deductive. “Its sole business is classification, and all its classes are derived from, and exhibited by, actual numbers.” The work contains few results that are really original. We mention one important proposition which is probably the author’s own. He states that cubical numbers are always equal to the sum of successive odd numbers. Thus, 8 = 23 = 3 + 5, 27 = 33 = 7 + 9 + 11, 64 = 43 = 13 + 15 + 17 + 19, and so on. This theorem was used later for finding the sum of the cubical numbers themselves. Theon of Smyrna is the author of a treatise on “the mathematical rules necessary for the study of Plato.” The work is ill arranged and of little merit. Of interest is the theorem, that every square number, or that number minus 1, is divisible by 3 or 4 or both. A remarkable discovery is a proposition given by Iamblichus

A HISTORY OF MATHEMATICS. 84 in his treatise on Pythagorean philosophy. It is founded on the observation that the Pythagoreans called1, 10, 100, 1000, units of the first, second, third, fourth ‘course’ respectively. The theorem is this: If we add any three consecutive numbers, of which the highest is divisible by 3, then add the digits of that sum, then, again, the digits of that sum, and so on, the final sum will be 6. Thus, 61 + 62 + 63 = 186, 1 + 8 + 6 = 15, 1 + 5 = 6. This discovery was the more remarkable, because the ordinary Greek numerical symbolism was much less likely to suggest any such property of numbers than our “Arabic” notation would have been. The works of Nicomachus, Theon of Smyrna, Thymaridas, and others contain at times investigations of subjects which are really algebraic in their nature. Thymaridas in one place uses the Greek word meaning “unknown quantity” in a way which would lead one to believe that algebra was not far distant. Of interest in tracing the invention of algebra are the arithmetical epigrams in the Palatine Anthology, which contain about fifty problems leading to linear equations. Before the introduction of algebra these problems were propounded as puzzles. A riddle attributed to Euclid and contained in the Anthology is to this effect: A mule and a donkey were walking along, laden with corn. The mule says to the donkey, “If you gave me one measure, I should carry twice as much as you. If I gave you one, we should both carry equal burdens. Tell me their burdens, O most learned master of geometry.” [6] It will be allowed, says Gow, that this problem, if authentic, was not beyond Euclid, and the appeal to geometry smacks of

THE GREEKS. 85 antiquity. A far more difficult puzzle was the famous ‘cattle-problem,’ which Archimedes propounded to the Alexandrian mathematicians. The problem is indeterminate, for from only seven equations, eight unknown quantities in integral numbers are to be found. It may be stated thus: The sun had a herd of bulls and cows, of different colours. (1) Of Bulls, the white (W) were, in number, (1 2 + 1 3) of the blue (B) and yellow (Y): the B were (1 4 + 1 5) of the Y and piebald (P): the P were (1 6 + 1 7) of the W and Y . (2) Of Cows, which had the same colours (w, b, y, p), w = (1 3 + 1 4)(B + b) : b = (1 4 + 1 5)(P + p) : p = (1 5 + 1 6)(Y + y) : y = (1 6

+ 1 7)(W + w). Find the number of bulls and cows. [6] Another problem in the Anthology is quite familiar to school-boys: “Of four pipes, one fills the cistern in one day, the next in two days, the third in three days, the fourth in four days: if all run together, how soon will they fill the cistern?” A great many of these problems, puzzling to an arithmetician, would have been solved easily by an algebraist. They became very popular about the time of Diophantus, and doubtless acted as a powerful stimulus on his mind. Diophantus was one of the last and most fertile mathematicians of the second Alexandrian school. He died about 330 a.d. His age was eighty-four, as is known from an epitaph to this effect: Diophantus passed 1 6 of his life in childhood, 1 12 in youth, and 1 7 more as a bachelor; five years after his marriage was born a son who died four years before his father,

A HISTORY OF MATHEMATICS. 86 at half his father’s age. The place of nativity and parentage of Diophantus are unknown. If his works were not written in Greek, no one would think for a moment that they were the product of Greek mind. There is nothing in his works that reminds us of the classic period of Greek mathematics. His were almost entirely new ideas on a new subject. In the circle of Greek mathematicians he stands alone in his specialty. Except for him, we should be constrained to say that among the Greeks algebra was always an unknown science. Of his works we have lost the Porisms, but possess a fragment of Polygonal Numbers, and seven books of his great work on Arithmetica, said to have been written in 13 books. If we except the Ahmes papyrus, which contains the first suggestions of algebraic notation, and of the solution of equations, then his Arithmetica is the earliest treatise on algebra now extant. In this work is introduced the idea of an algebraic equation expressed in algebraic symbols. His treatment is purely analytical and completely divorced from geometrical methods. He is, as far as we know, the first to state that “a negative number multiplied by a negative number gives a positive number.” This is applied to the multiplication of differences, such as(x− 1)(x− 2). It must be remarked, however, that Diophantus had no notion whatever of negative numbers standing by themselves. All he knew were differences, such as (2x− 10), in which 2x could not be smaller than 10 without leading to an absurdity. He appears to be the first who could perform such operations as (x−1)×(x−2) without reference to geometry. Such identities

THE GREEKS. 87 as (a + b)2 = a2 + 2ab + b2, which with Euclid appear in the elevated rank of geometric theorems, are with Diophantus the simplest consequences of the algebraic laws of operation. His sign for subtraction was , for equality ι. For unknown quantities he had only one symbol, ς. He had no sign for addition except juxtaposition. Diophantus used but few symbols, and sometimes ignored even these by describing an operation in words when the symbol would have answered just as well. In the solution of simultaneous equations Diophantus adroitly managed with only one symbol for the unknown quantities and arrived at answers, most commonly, by the method of tentative assumption, which consists in assigning to some of the unknown quantities preliminary values, that satisfy only one or two of the conditions. These values lead to expressions palpably wrong, but which generally suggest some stratagem by which values can be secured satisfying all the conditions of the problem. Diophantus also solved determinate equations of the second degree. We are

ignorant of his method, for he nowhere goes through with the whole process of solution, but merely states the result. Thus, “ 84x2 + 7x = 7, whence x is found = 1 4.” Notice he gives only one root. His failure to observe that a quadratic equation has two roots, even when both roots are positive, rather surprises us. It must be remembered, however, that this same inability to perceive more than one out of the several solutions to which a problem may point is common to all Greek mathematicians. Another point to

A HISTORY OF MATHEMATICS. 88 be observed is that he never accepts as an answer a quantity which is negative or irrational. Diophantus devotes only the first book of hisArithmetica to the solution of determinate equations. The remaining books extant treat mainly of indeterminate quadratic equations of the form Ax2 + Bx + C = y2, or of two simultaneous equations of the same form. He considers several but not all the possible cases which may arise in these equations. The opinion of Nesselmann on the method of Diophantus, as stated by Gow, is as follows: “(1) Indeterminate equations of the second degree are treated completely only when the quadratic or the absolute term is wanting: his solution of the equations Ax2 + C = y2 and Ax2 + Bx + C = y2 is in many respects cramped. (2) For the ‘double equation’ of the second degree he has a definite rule only when the quadratic term is wanting in both expressions: even then his solution is not general. More complicated expressions occur only under specially favourable circumstances.” Thus, he solvesBx + C2 = y2, B1x+ C2 1 = y2 1. The extraordinary ability of Diophantus lies rather in another direction, namely, in his wonderful ingenuity to reduce all sorts of equations to particular forms which he knows how to solve. Very great is the variety of problems considered. The 130 problems found in the great work of Diophantus contain over 50 different classes of problems, which are strung together without any attempt at classification. But still more multifarious than the problems are the solutions. General methods are unknown to Diophantus. Each problem has its own distinct method, which is often useless for the

THE ROMANS. 89 most closely related problems. “It is, therefore, difficult for a modern, after studying 100 Diophantine solutions, to solve the 101st.” [7] That which robs his work of much of its scientific value is the fact that he always feels satisfied with one solution, though his equation may admit of an indefinite number of values. Another great defect is the absence of general methods. Modern mathematicians, such as Euler, Lagrange, Gauss, had to begin the study of indeterminate analysis anew and received no direct aid from Diophantus in the formulation of methods. In spite of these defects we cannot fail to admire the work for the wonderful ingenuity exhibited therein in the solution of particular equations. It is still an open question and one of great difficulty whether Diophantus derived portions of his algebra from Hindoo sources or not. THE ROMANS. Nowhere is the contrast between the Greek and Roman mind shown forth more distinctly than in their attitude toward the mathematical science. The sway of the Greek was a flowering time for mathematics, but that of the Roman a period of sterility. In philosophy, poetry, and art the Roman was an imitator. But in mathematics he did not even rise to the desire for imitation. The mathematical fruits of Greek genius lay before him untasted. In him a science which had no direct bearing on practical life could awake no interest. As a

A HISTORY OF MATHEMATICS. 90 consequence, not only the higher geometry of Archimedes and Apollonius, but even the Elements of Euclid, were entirely neglected. What little mathematics the Romans possessed did not come from the Greeks, but from more ancient sources. Exactly where and how it originated is a matter of doubt. It seems most probable that the “Roman notation,” as well as the practical geometry of the Romans, came from the old Etruscans, who, at the earliest period to which our knowledge of them extends, inhabited the district between the Arno and Tiber. Livy tells us that the Etruscans were in the habit of representing the number of years elapsed, by driving yearly a nail into the sanctuary of Minerva, and that the Romans continued this practice. A less primitive mode of designating numbers, presumably of Etruscan origin, was a notation resembling the present “Roman notation.” This system is noteworthy from the fact that a principle is involved in it which is not met with in any other; namely, the principle of subtraction. If a letter be placed before another of greater value, its value is not to be added to, but subtracted from, that of the greater. In the designation of large numbers a horizontal bar placed over a letter was made to increase its value one thousand fold. In fractions the Romans used the duodecimal system. Of arithmetical calculations, the Romans employed three different kinds: Reckoning on the fingers, upon the abacus, and by tables prepared for the purpose. [3] Finger-symbolism was known as early as the time of King Numa, for he had

THE ROMANS. 91 erected, says Pliny, a statue of the double-faced Janus, of which the fingers indicated 365 (355?), the number of days in a year. Many other passages from Roman authors point out the use of the fingers as aids to calculation. In fact, a finger-symbolism of practically the same form was in use not only in Rome, but also in Greece and throughout the East, certainly as early as the beginning of the Christian era, and continued to be used in Europe during the Middle Ages. We possess no knowledge as to where or when it was invented. The second mode of calculation, by the abacus, was a subject of elementary instruction in Rome. Passages in Roman writers indicate that the kind of abacus most commonly used was covered with dust and then divided into columns by drawing straight lines. Each column was supplied with pebbles (calculi, whence ‘calculare’ and ‘calculate’) which served for calculation. Additions and subtractions could be performed on the abacus quite easily, but in multiplication the abacus could be used only for adding the particular products, and in division for performing the subtractions occurring in the process. Doubtless at this point recourse was made to mental operations and to the multiplication table. Possibly finger-multiplication may also have been used. But the multiplication of large numbers must, by either method, have been beyond the power of the ordinary arithmetician. To obviate this difficulty, the arithmetical tables mentioned above were used, from which the desired products could be copied at once. Tables of this kind were prepared by Victorius of Aquitania. His tables contain a peculiar notation for fractions, which continued in

A HISTORY OF MATHEMATICS. 92 use throughout the Middle Ages. Victorius is best known for his canon paschalis, a rule for finding the correct date for Easter, which he published in 457a.d. Payments of interest and problems in interest were very old among the Romans. The Roman laws of inheritance gave rise to numerous

arithmetical examples. Especially unique is the following: A dying man wills that, if his wife, being with child, gives birth to a son, the son shall receive 2 3 and she 1 3 of his estates; but if a daughter is born, she shall receive 1 3 and his wife 2 3. It happens that twins are born, a boy and a girl. How shall the estates be divided so as to satisfy the will? The celebrated Roman jurist, Salvianus Julianus, decided that the estates shall be divided into seven equal parts, of which the son receives four, the wife two, the daughter one. We next consider Roman geometry. He who expects to find in Rome a science of geometry, with definitions, axioms, theorems, and proofs arranged in logical order, will be disappointed. The only geometry known was a practical geometry, which, like the old Egyptian, consisted only of empirical rules. This practical geometry was employed in surveying. Treatises thereon have come down to us, compiled by the Roman surveyors, called agrimensores or gromatici. One would naturally expect rules to be clearly formulated. But no; they are left to be abstracted by the reader from a mass of numerical examples. “The total impression is as though the Roman gromatic were thousands of years older than Greek geometry, and as though a deluge were lying between the two.” Some of their rules were probably inherited from

THE ROMANS. 93 the Etruscans, but others are identical with those of Heron. Among the latter is that for finding the area of a triangle from its sides and the approximate formula, 13 30 a2, for the area of equilateral triangles (a being one of the sides). But the latter area was also calculated by the formulas1 2(a2 + a) and 1 2 a2, the first of which was unknown to Heron. Probably the expression 1 2 a2 was derived from the Egyptian formulaa + b 2 · c + d 2 for the determination of the surface of a quadrilateral. This Egyptian formula was used by the Romans for finding the area, not only of rectangles, but of any quadrilaterals whatever. Indeed, the gromatici considered it even sufficiently accurate to determine the areas of cities, laid out irregularly, simply by measuring their circumferences. [7] Whatever Egyptian geometry the Romans possessed was transplanted across the Mediterranean at the time ofJulius Cæsar, who ordered a survey of the whole empire to secure an equitable mode of taxation. Cæsar also reformed the calendar, and, for that purpose, drew from Egyptian learning. He secured the services of the Alexandrian astronomer, Sosigenes. In the fifth century, the Western Roman Empire was fast falling to pieces. Three great branches—Spain, Gaul, and the province of Africa—broke off from the decaying trunk. In 476, the Western Empire passed away, and the Visigothic chief, Odoacer, became king. Soon after, Italy was conquered by the Ostrogoths under Theodoric. It is remarkable that this very period of political humiliation should be the one during which Greek science was studied in Italy most zealously. School-books began to be compiled from the elements of

A HISTORY OF MATHEMATICS. 94 Greek authors. These compilations are very deficient, but are of absorbing interest, from the fact that, down to the twelfth century, they were the only sources of mathematical knowledge in the Occident. Foremost among these writers is Boethius (died 524). At first he was a great favourite of King Theodoric, but later, being charged by envious courtiers with treason, he was imprisoned, and at last decapitated. While in prison he wrote On the Consolations of Philosophy. As a mathematician, Boethius was a Brobdingnagian among Roman

scholars, but a Liliputian by the side of Greek masters. He wrote an Institutis Arithmetica , which is essentially a translation of the arithmetic of Nicomachus, and a Geometry in several books. Some of the most beautiful results of Nicomachus are omitted in Boethius' arithmetic. The first book on geometry is an extract from Euclid'sElements, which contains, in addition to definitions, postulates, and axioms, the theorems in the first three books, without proofs. How can this omission of proofs be accounted for? It has been argued by some that Boethius possessed an incomplete Greek copy of the Elements; by others, that he had Theon's edition before him, and believed that only the theorems came from Euclid, while the proofs were supplied by Theon. The second book, as also other books on geometry attributed to Boethius, teaches, from numerical examples, the mensuration of plane figures after the fashion of the agrimensores. A celebrated portion in the geometry of Boethius is that pertaining to an abacus, which he attributes to the Pythago- reans. A considerable improvement on the old abacus is there

THE ROMANS. 95 introduced. Pebbles are discarded, and apices (probably small cones) are used. Upon each of these apices is drawn a numeral giving it some value below 10. The names of these numerals are pure Arabic, or nearly so, but are added, apparently, by a later hand. These figures are obviously the parents of our modern "Arabic" numerals. The 0 is not men- tioned by Boethius in the text. These numerals bear striking resemblance to the Gubar-numerals of the West-Arabs, which are admittedly of Indian origin. These facts have given rise to an endless controversy. Some contended that Pythagoras was in India, and from there brought the nine numerals to Greece, where the Pythagoreans used them secretly. This hypothesis has been generally abandoned, for it is not certain that Pythagoras or any disciple of his ever was in India, nor is there any evidence in any Greek author, that the apices were known to the Greeks, or that numeral signs of any sort were used by them with the abacus. It is improbable, moreover, that the Indian signs, from which the apices are derived, are so old as the time of Pythagoras. A second theory is that the Geometry attributed to Boethius is a forgery; that it is not older than the tenth, or possibly the ninth, century, and that the apices are derived from the Arabs. This theory is based on contradictions between passages in the Arithmetica and others in theGeometry. But there is an Encyclopædia written by Cassiodorius (died about 570) in which both the arithmetic and geometry of Boethius are mentioned. There appears to be no good reason for doubting the trustworthiness of this passage in the Encyclopædia. A third theory (Woepcke's) is

A HISTORY OF MATHEMATICS. 96 that the Alexandrians either directly or indirectly obtained the nine numerals from the Hindoos, about the second cen- tury a.d., and gave them to the Romans on the one hand, and to the Western Arabs on the other. This explanation is the most plausible.

MIDDLE AGES. THE HINDOOS. The first people who distinguished themselves in math- ematical research, after the time of the ancient Greeks, belonged, like them, to the Aryan race. It was, however, not a European, but an Asiatic nation, and had its seat in far-off India. Unlike the Greek, Indian society was fixed into castes. The only castes enjoying the privilege and leisure for advanced study and thinking were

theBrahmins, whose prime business was religion and philosophy, and theKshatriyas, who attended to war and government. Of the development of Hindoo mathematics we know but little. A few manuscripts bear testimony that the Indians had climbed to a lofty height, but their path of ascent is no longer traceable. It would seem that Greek mathematics grew up under more favourable conditions than the Hindoo, for in Greece it attained an independent existence, and was studied for its own sake, while Hindoo mathematics always remained merely a servant to astronomy. Furthermore, in Greece mathematics was a science of the people, free to be cultivated by all who had a liking for it; in India, as in Egypt, it was in the hands chiefly of the priests. Again, the Indians were in the habit of putting into verse all mathematical results they 97

A HISTORY OF MATHEMATICS. 98 obtained, and of clothing them in obscure and mystic language, which, though well adapted to aid the memory of him who already understood the subject, was often unintelligible to the uninitiated. Although the great Hindoo mathematicians doubtless reasoned out most or all of their discoveries, yet they were not in the habit of preserving the proofs, so that the naked theorems and processes of operation are all that have come down to our time. Very different in these respects were the Greeks. Obscurity of language was generally avoided, and proofs belonged to the stock of knowledge quite as much as the theorems themselves. Very striking was the difference in the bent of mind of the Hindoo and Greek; for, while the Greek mind was pre-eminently geometrical, the Indian was first of all arithmetical. The Hindoo dealt with number, the Greek with form. Numerical symbolism, the science of numbers, and algebra attained in India far greater perfection than they had previously reached in Greece. On the other hand, we believe that there was little or no geometry in India of which the source may not be traced back to Greece. Hindoo trigonometry might possibly be mentioned as an exception, but it rested on arithmetic more than on geometry. An interesting but difficult task is the tracing of the relation between Hindoo and Greek mathematics. It is well known that more or less trade was carried on between Greece and India from early times. After Egypt had become a Roman province, a more lively commercial intercourse sprang up between Rome and India, by way of Alexandria. A priori, it does not seem improbable, that with the traffic

THE HINDOOS. 99 of merchandise there should also be an interchange of ideas. That communications of thought from the Hindoos to the Alexandrians actually did take place, is evident from the fact that certain philosophic and theologic teachings of the Manicheans, Neo-Platonists, Gnostics, show unmistakable likeness to Indian tenets. Scientific facts passed also from Alexandria to India. This is shown plainly by the Greek origin of some of the technical terms used by the Hindoos. Hindoo astronomy was influenced by Greek astronomy. Most of the geometrical knowledge which they possessed is traceable to Alexandria, and to the writings of Heron in particular. In algebra there was, probably, a mutual giving and receiving. We suspect that Diophantus got the first glimpses of algebraic knowledge from India. On the other hand, evidences have been found of Greek algebra among the Brahmins. The earliest knowledge of algebra in India may possibly have been of Babylonian origin. When we consider that Hindoo scientists looked upon arithmetic and algebra merely

as tools useful in astronomical research, there appears deep irony in the fact that these secondary branches were after all the only ones in which they won real distinction, while in their pet science of astronomy they displayed an inaptitude to observe, to collect facts, and to make inductive investigations. We shall now proceed to enumerate the names of the leading Hindoo mathematicians, and then to review briefly Indian mathematics. We shall consider the science only in its complete state, for our data are not sufficient to trace the history of the development of methods. Of the great Indian

A HISTORY OF MATHEMATICS. 100 mathematicians, or rather, astronomers,—for India had no mathematicians proper,— Aryabhatta is the earliest. He was born 476 a.d., at Pataliputra, on the upper Ganges. His celebrity rests on a work entitled Aryabhattiyam, of which the third chapter is devoted to mathematics. About one hundred years later, mathematics in India reached the highest mark. At that time flourished Brahmagupta (born 598). In 628 he wrote his Brahma-sphuta-siddhanta ("The Revised System of Brahma"), of which the twelfth and eighteenth chapters belong to mathematics. To the fourth or fifth century belongs an anonymous astronomical work, called Surya-siddhanta ("Knowledge from the Sun"), which by native authorities was ranked second only to the Brahma-siddhanta, but is of interest to us merely as furnishing evidence that Greek science influenced Indian science even before the time of Aryabhatta. The following centuries produced only two names of importance; namely,Cridhara, who wrote a Ganita-sara ("Quintessence of Calculation"), and Padmanabha, the author of an algebra. The science seems to have made but little progress at this time; for a work entitled Siddhantaciromani ("Diadem of an Astronomical System"), written by Bhaskara Acarya in 1150, stands little higher than that of Brahmagupta, written over 500 years earlier. The two most important mathematical chapters in this work are the Lilavati (= "the beautiful," i.e. the noble science) and Viga-ganita (= "root-extraction"), devoted to arithmetic and algebra. From now on, the Hindoos in the Brahmin schools seemed to content themselves with studying

THE HINDOOS. 101 the masterpieces of their predecessors. Scientific intelligence decreases continually, and in modern times a very deficient Arabic work of the sixteenth century has been held in great authority. [7] The mathematical chapters of the Brahma-siddhanta and Siddhantaciromani were translated into English by H. T. Colebrooke, London, 1817. The Surya-siddhanta was trans- lated by E. Burgess, and annotated by W. D. Whitney, New Haven, Conn., 1860. The grandest achievement of the Hindoos and the one which, of all mathematical inventions, has contributed most to the general progress of intelligence, is the invention of the principle of position in writing numbers. Generally we speak of our notation as the "Arabic" notation, but it should be called the "Hindoo" notation, for the Arabs borrowed it from the Hindoos. That the invention of this notation was not so easy as we might suppose at first thought, may be inferred from the fact that, of other nations, not even the keen-minded Greeks possessed one like it. We inquire, who invented this ideal symbolism, and when? But we know neither the inventor nor the time of invention. That our system of notation is of Indian origin is the only point of which we are certain. From the evolution of ideas in general we may safely infer that our notation

did not spring into existence a completely armed Minerva from the head of Jupiter. The nine figures for writing the units are supposed to have been introduced earliest, and the sign of zero and the principle of position to be of later origin. This view receives support from the fact that

A HISTORY OF MATHEMATICS. 102 on the island of Ceylon a notation resembling the Hindoo, but without the zero has been preserved. We know that Buddhism and Indian culture were transplanted to Ceylon about the third century after Christ, and that this culture remained stationary there, while it made progress on the continent. It seems highly probable, then, that the numerals of Ceylon are the old, imperfect numerals of India. In Ceylon, nine figures were used for the units, nine others for the tens, one for 100, and also one for 1000. These 20 characters enabled them to write all the numbers up to9999. Thus, 8725 would have been written with six signs, representing the following numbers: 8, 1000, 7, 100, 20, 5. These Singhalesian signs, like the old Hindoo numerals, are supposed originally to have been the initial letters of the corresponding numeral adjectives. There is a marked resemblance between the notation of Ceylon and the one used by Aryabhatta in the first chapter of his work, and there only. Although the zero and the principle of position were unknown to the scholars of Ceylon, they were probably known to Aryabhatta; for, in the second chapter, he gives directions for extracting the square and cube roots, which seem to indicate a knowledge of them. It would appear that the zero and the accompanying principle of position were introduced about the time of Aryabhatta. These are the inventions which give the Hindoo system its great superiority, its admirable perfection. There appear to have been several notations in use in different parts of India, which differed, not in principle, but merely in the forms of the signs employed. Of interest is also

THE HINDOOS. 103 a symbolical system of position, in which the figures generally were not expressed by numerical adjectives, but by objects suggesting the particular numbers in question. Thus, for 1 were used the words moon, Brahma, Creator, or form; for 4, the words Veda, (because it is divided into four parts) or ocean, etc. The following example, taken from the Surya- siddhanta, illustrates the idea. The number 1, 577, 917, 828 is expressed from right to left as follows: Vasu (a class of 8 gods) + two + eight + mountains (the 7 mountain-chains) + form + digits (the 9 digits) + seven + mountains + lunar days (half of which equal 15). The use of such notations made it possible to represent a number in several different ways. This greatly facilitated the framing of verses containing arithmetical rules or scientific constants, which could thus be more easily remembered. At an early period the Hindoos exhibited great skill in calculating, even with large numbers. Thus, they tell us of an examination to which Buddha, the reformer of the Indian religion, had to submit, when a youth, in order to win the maiden he loved. In arithmetic, after having astonished his examiners by naming all the periods of numbers up to the 53d, he was asked whether he could determine the number of primary atoms which, when placed one against the other, would form a line one mile in length. Buddha found the required answer in this way: 7 primary atoms make a very minute grain of dust, 7 of these make a minute grain of dust, 7 of these a grain of dust whirled up by the wind, and so on. Thus he proceeded, step by step, until he finally reached the

A HISTORY OF MATHEMATICS. 104 length of a mile. The multiplication of all the factors gave for the multitude of primary atoms in a mile a number consisting of 15 digits. This problem reminds one of the ‘Sand-Counter’ of Archimedes. After the numerical symbolism had been perfected, figuring was made much easier. Many of the Indian modes of operation differ from ours. The Hindoos were generally inclined to follow the motion from left to right, as in writing. Thus, they added the left-hand columns first, and made the necessary corrections as they proceeded. For instance, they would have added 254 and 663 thus: 2 + 6 = 8, 5 + 6 = 11, which changes 8 into 9, 4 + 3 = 7. Hence the sum 917. In subtraction they had two methods. Thus in 821−348 they would say, 8 from 11 = 3 , 4 from 11 = 7, 3 from 7 = 4. Or they would say,8 from 11 = 3, 5 from 12 = 7 , 4 from 8 = 4 . In multiplication of a number by another of only one digit, say 569 by 5, they generally said, 5·5 = 25 , 5·6 = 30 , which changes 25 into 28, 5·9 = 45 , hence the 0 must be increased by 4. The product is 2845. In the multiplication with each other of many-figured numbers, they first multiplied, in the manner just indicated, with the left-hand digit of the multiplier, which was written above the multiplicand, and placed the product above the multiplier. On multiplying with the next digit of the multiplier, the product was not placed in a new row, as with us, but the first product obtained was corrected, as the process continued, by erasing, whenever necessary, the old digits, and replacing them by new ones, until finally the whole product was obtained. We who possess the modern luxuries of pencil and paper, would

THE HINDOOS. 105 not be likely to fall in love with this Hindoo method. But the Indians wrote “with a cane-pen upon a small blackboard with a white, thinly liquid paint which made marks that could be easily erased, or upon a white tablet, less than a foot square, strewn with red flour, on which they wrote the figures with a small stick, so that the figures appeared white on a red ground.” [7] Since the digits had to be quite large to be distinctly legible, and since the boards were small, it was desirable to have a method which would not require much space. Such a one was the above method of multiplication. Figures could be easily erased and replaced by others without sacrificing neatness. But the Hindoos had also other ways of multiplying, of which we mention the following: The 2 1 7 3 5 4 6 0 7 3 5 1 1 8 8 2 0 tablet was divided into squares like a chess-board. Diagonals were also drawn, as seen in the figure. The multiplication of 12× 735 = 8820 is exhibited in the adjoining dia- gram. [3] The manuscripts extant give no information of howdivisions were executed. The correctness of their additions, sub- tractions, and multiplications was tested “by excess of 9’s.” In writing fractions, the numerator was placed above the denominator, but no line was drawn between them. We shall now proceed to the consideration of some arith- metical problems and the Indian modes of solution. A favourite method was that ofinversion. With laconic brevity, Aryabhatta describes it thus: “Multiplication becomes divi-

A HISTORY OF MATHEMATICS. 106 sion, division becomes multiplication; what was gain becomes loss, what loss, gain; inversion.” Quite different from this quotation in style is the following problem from Aryabhatta, which illustrates the method: [3] “Beautiful maiden with beaming eyes, tell me, as thou understandst the right method of inversion, which is the number which multiplied by3, then increased by 3 4 of the

product, divided by 7, diminished by 1 3 of the quotient, multiplied by itself, diminished by 52, the square root extracted, addition of 8, and division by 10, gives the number 2?" The process consists in beginning with 2 and working backwards. Thus, (2·10− 8)2 + 52 = 196, √ 196 = 14, and 14· 3 2·7· 4 7÷ 3 = 28, the answer. Here is another example taken from Lilavati, a chapter in Bhaskara's great work: "The square root of half the number of bees in a swarm has flown out upon a jessamine-bush, 8 9 of the whole swarm has remained behind; one female bee flies about a male that is buzzing within a lotus-flower into which he was allured in the night by its sweet odour, but is now imprisoned in it. Tell me the number of bees." Answer, 72. The pleasing poetic garb in which all arithmetical problems are clothed is due to the Indian practice of writing all school- books in verse, and especially to the fact that these problems, propounded as puzzles, were a favourite social amusement. Says Brahmagupta: "These problems are proposed simply for pleasure; the wise man can invent a thousand others, or he can solve the problems of others by the rules given here. As the sun eclipses the stars by his brilliancy, so the man of knowledge will eclipse the fame of others in assemblies of the

THE HINDOOS. 107 people if he proposes algebraic problems, and still more if he solves them." The Hindoos solved problems in interest, discount, part- nership, alligation, summation of arithmetical and geometric series, devised rules for determining the numbers of combina- tions and permutations, and invented magic squares. It may here be added that chess, the profoundest of all games, had its origin in India. The Hindoos made frequent use of the "rule of three," and also of the method of "falsa positio," which is almost identical with that of the "tentative assumption" of Diophantus. These and other rules were applied to a large number of problems. Passing now to algebra, we shall first take up the symbols of operation. Addition was indicated simply by juxtaposition as in Diophantine algebra; subtraction, by placing a dot over the subtrahend; multiplication, by putting after the factors, bha, the abbreviation of the word bhavita, "the product"; division, by placing the divisor beneath the dividend; square- root, by writing ka, from the word karana (irrational), before the quantity. The unknown quantity was called by Brahmagupta yˆ avattˆ avat(quantum tantum). When several unknown quantities occurred, he gave, unlike Diophantus, to each a distinct name and symbol. The first unknown was designated by the general term "unknown quantity." The rest were distinguished by names of colours, as the black, blue, yellow, red, or green unknown. The initial syllable of each word constituted the symbol for the respective unknown quantity. Thusyˆ ameant x; kˆ a(from kˆ alaka= black) meanty;

A HISTORY OF MATHEMATICS. 108 yˆ a kˆ a bha, "x times y"; ka 15 ka 10, " √ 15− √ 10." The Indians were the first to recognise the existence of ab- solutely negative quantities. They brought out the difference between positive and negative quantities by attaching to the one the idea of 'possession,' to the other that of 'debts.' The conception also of opposite directions on a line, as an interpre- tation of + and− quantities, was not foreign to them. They advanced beyond Diophantus in observing that a quadratic has always two roots. Thus Bhaskara givesx = 50 and x =−5 for the roots of x2− 45x = 250. "But," says he, "the second value is in this case not to be taken, for it is inadequate; people do not approve of negative roots." Commentators

speak of this as if negative roots were seen, but not admitted. Another important generalisation, says Hankel, was this, that the Hindoos never confined their arithmetical operations to rational numbers. For instance, Bhaskara showed how, by the formula √ a + √ b = √ a + √ a2− b 2 + √ a− √ a2− b 2 the square root of the sum of rational and irrational numbers could be found. The Hindoos never discerned the dividing line between numbers and magnitudes, set up by the Greeks, which, though the product of a scientific spirit, greatly re- tarded the progress of mathematics. They passed from magnitudes to numbers and from numbers to magnitudes without anticipating that gap which to a sharply discriminat- ing mind exists between the continuous and discontinuous.

THE HINDOOS. 109 Yet by doing so the Indians greatly aided the general progress of mathematics. “Indeed, if one understands by algebra the application of arithmetical operations to complex magnitudes of all sorts, whether rational or irrational numbers or space- magnitudes, then the learned Brahmins of Hindostan are the real inventors of algebra.” [7] Let us now examine more closely the Indian algebra. In extracting the square and cube roots they used the formulas (a + b)2 = a2 + 2ab + b2 and (a + b)3 = a3 + 3a2b + 3ab2 + b3. In this connection Aryabhatta speaks of dividing a number into periods of two and three digits. From this we infer that the principle of position and the zero in the numeral notation were already known to him. In figuring with zeros, a statement of Bhaskara is interesting. A fraction whose denominator is zero, says he, admits of no alteration, though much be added or subtracted. Indeed, in the same way, no change takes place in the infinite and immutable Deity when worlds are destroyed or created, even though numerous orders of beings be taken up or brought forth. Though in this he apparently evinces clear mathematical notions, yet in other places he makes a complete failure in figuring with fractions of zero denominator. In the Hindoo solutions of determinate equations, Cantor thinks he can see traces of Diophantine methods. Some technical terms betray their Greek origin. Even if it be true that the Indians borrowed from the Greeks, they deserve great credit for improving and generalising the solutions of linear and quadratic equations. Bhaskara advances far beyond the

A HISTORY OF MATHEMATICS. 110 Greeks and even beyond Brahmagupta when he says that “the square of a positive, as also of a negative number, is positive; that the square root of a positive number is twofold, positive and negative. There is no square root of a negative number, for it is not a square.” Of equations of higher degrees, the Indians succeeded in solving only some special cases in which both sides of the equation could be made perfect powers by the addition of certain terms to each. Incomparably greater progress than in the solution of determinate equations was made by the Hindoos in the treatment ofindeterminate equations. Indeterminate analysis was a subject to which the Hindoo mind showed a happy adaptation. We have seen that this very subject was a favourite with Diophantus, and that his ingenuity was almost inexhaustible in devising solutions for particular cases. But the glory of having invented general methods in this most subtle branch of mathematics belongs to the Indians. The Hindoo indeterminate analysis differs from the Greek not only in method, but also in aim. The object of the former was to find all possible integral solutions. Greek analysis, on the other hand, demanded not necessarily

integral, but simply rational answers. Diophantus was content with a single solution; the Hindoos endeavoured to find all solutions possible. Aryabhatta gives solutions in integers to linear equations of the form ax± by = c, where a, b, c are integers. The rule employed is called the pulveriser. For this, as for most other rules, the Indians give no proof. Their solution is essentially the same as the one of Euler. Euler's process of

THE HINDOOS. 111 reducing a b to a continued fraction amounts to the same as the Hindoo process of finding the greatest common divisor of a and b by division. This is frequently called the Diophantine method. Hankel protests against this name, on the ground that Diophantus not only never knew the method, but did not even aim at solutions purely integral. [7] These equations probably grew out of problems in astronomy. They were applied, for instance, to determine the time when a certain constellation of the planets would occur in the heavens. Passing by the subject of linear equations with more than two unknown quantities, we come to indeterminate quadratic equations. In the solution ofxy = ax + by + c, they applied the method re-invented later by Euler, of decomposing (ab + c) into the product of two integers m·n and of placing x = m + b and y = n + a. Remarkable is the Hindoo solution of the quadratic equation cy2 = ax2 + b. With great keenness of intellect they recognised in the special case y2 = ax2 + 1 a fundamental problem in indeterminate quadratics. They solved it by the cyclic method. "It consists," says De Morgan, "in a rule for finding an indefinite number of solutions of y2 = ax2 + 1 (a being an integer which is not a square), by means of one solution given or found, and of feeling for one solution by making a solution of y2 = ax2 + b give a solution of y2 = ax2 + b2. It amounts to the following theorem: If p and q be one set of values ofx and y in y2 = ax2 + b and p′ and q′ the same or another set, then qp + pq′ and app′ + qq′ are values of x and y in y2 = ax2 + b2. From this it is obvious that one solution of y2 = ax2 + 1 may

A HISTORY OF MATHEMATICS. 112 be made to give any number, and that if, takingb at pleasure, y2 = ax2 + b2 can be solved so that x and y are divisible by b, then one preliminary solution of y2 = ax2 + 1 can be found. Another mode of trying for solutions is a combination of the preceding with the cuttaca (pulveriser)." These calculations were used in astronomy. Doubtless this "cyclic method" constitutes the greatest invention in the theory of numbers before the time of Lagrange. The perversity of fate has willed it, that the equationy2 = ax2+ 1 should now be called Pell's problem, while in recognition of Brahmin scholarship it ought to be called the "Hindoo problem." It is a problem that has exercised the highest faculties of some of our greatest modern analysts. By them the work of the Hindoos was done over again; for, unfortunately, the Arabs transmitted to Europe only a small part of Indian algebra and the original Hindoo manuscripts, which we now possess, were unknown in the Occident. Hindoo geometry is far inferior to the Greek. In it are found no definitions, no postulates, no axioms, no logical chain of reasoning or rigid form of demonstration, as with Euclid. Each theorem stands by itself as an independent truth. Like the early Egyptian, it is empirical. Thus, in the proof of the theorem of the right triangle, Bhaskara draws the right triangle four times in the square of the hypotenuse, so that in the middle there remains a square whose side equals

THE HINDOOS. 113 the difference between the two sides of the right triangle. Arranging this square and the four triangles in a different way, they are seen, together, to make up the sum of the square of the two sides. “Behold!” says Bhaskara, without adding another word of explanation. Bretschneider conjectures that the Pythagorean proof was substantially the same as this. In another place, Bhaskara gives a second demonstration of this theorem by drawing from the vertex of the right angle a perpendicular to the hypotenuse, and comparing the two triangles thus obtained with the given triangle to which they are similar. This proof was unknown in Europe till Wallis re- discovered it. The Brahmins never inquired into the properties of figures. They considered only metrical relations applicable in practical life. In the Greek sense, the Brahmins never had a science of geometry. Of interest is the formula given by Brahmagupta for the area of a triangle in terms of its sides. In the great work attributed to Heron the Elder this formula is first found. Whether the Indians themselves invented it, or whether they borrowed it from Heron, is a disputed question. Several theorems are given by Brahmagupta on quadrilaterals which are true only of those which can be inscribed on a circle—a limitation which he omits to state. Among these is the proposition of Ptolemæus, that the product of the diagonals is equal to the sum of the products of the opposite sides. The Hindoos were familiar with the calculation of the areas of circles and their segments, of the length of chords and perimeters of regular inscribed polygons. An old Indian tradition makes π = 3, also = √ 10; but Aryabhatta gives the

A HISTORY OF MATHEMATICS. 114 value 31416 10000. Bhaskara gives two values,—the ‘accurate,’ 3927 1250, and the ‘inaccurate,’ Archimedean value, 22 7 . A commentator on Lilavati says that these values were calculated by beginning with a regular inscribed hexagon, and applying repeatedly the formula AD = √ 2− √ 4− AB2, wherein AB is the side of the given polygon, and AD that of one with double the number of sides. In this way were obtained the perimeters of the inscribed polygons of 12, 24, 48, 96, 192, 384 sides. Taking the radius = 100, the perimeter of the last one gives the value which Aryabhatta used for π. Greater taste than for geometry was shown by the Hindoos for trigonometry. Like the Babylonians and Greeks, they divided the circle into quadrants, each quadrant into90 degrees and 5400 minutes. The whole circle was therefore made up of 21, 600 equal parts. From Bhaskara’s ‘accurate’ value for π it was found that the radius contained 3438 of these circular parts. This last step was not Grecian. The Greeks might have had scruples about taking a part of a curve as the measure of a straight line. Each quadrant was divided into 24 equal parts, so that each part embraced 225 units of the whole circumference, and corresponds to 3 3 4 degrees. Notable is the fact that the Indians never reckoned, like the Greeks, with the whole chord of double the arc, but always with the sine (joa) and versed sine. Their mode of calculating tables was theoretically very simple. The sine of 90◦ was equal to the radius, or 3438; the sine of 30◦ was evidently half that, or 1719. Applying the formula sin2 a + cos2 a = r2,

THE HINDOOS. 115 they obtained sin 45◦ = √ r2 2 = 2431. Substituting for cos a its equal sin(90− a), and making a = 60◦, they obtained sin 60◦ = √ 3r2 2 = 2978. With the sines of 90, 60, 45, and 30 as starting-points, they reckoned the sines of half the angles by the formula ver sin 2a = 2 sin2 a, thus obtaining the sines of 22◦ 30′,

11◦ 15′, 7◦ 30′, 3◦ 45′. They now figured out the sines of the complements of these angles, namely, the sines of 86◦ 15′, 82◦ 30′, 78◦ 45′, 75◦, 67◦ 30′; then they calculated the sines of half these angles; then of their complements; then, again, of half their complements; and so on. By this very simple process they got the sines of angles at intervals of 3◦ 45′. In this table they discovered the unique law that if a, b, c be three successive arcs such thata− b = b− c = 3◦ 45′, then sin a− sin b = (sin b− sin c)− sin b 225 . This formula was afterwards used whenever a re-calculation of tables had to be made. No Indian trigonometrical treatise on the triangle is extant. In astronomy they solved plane and spherical right triangles. [18] It is remarkable to what extent Indian mathematics enters into the science of our time. Both the form and the spirit of the arithmetic and algebra of modern times are essentially Indian and not Grecian. Think of that most perfect of mathematical symbolisms—the Hindoo notation, think of the Indian arithmetical operations nearly as perfect as our own, think of their elegant algebraical methods, and then judge whether the Brahmins on the banks of the Ganges are not entitled to some credit. Unfortunately, some of the

A HISTORY OF MATHEMATICS. 116 most brilliant of Hindoo discoveries in indeterminate analysis reached Europe too late to exert the influence they would have exerted, had they come two or three centuries earlier. THE ARABS. After the flight of Mohammed from Mecca to Medina in 622 a.d., an obscure people of Semitic race began to play an important part in the drama of history. Before the lapse of ten years, the scattered tribes of the Arabian peninsula were fused by the furnace blast of religious enthusiasm into a powerful nation. With sword in hand the united Arabs subdued Syria and Mesopotamia. Distant Persia and the lands beyond, even unto India, were added to the dominions of the Saracens. They conquered Northern Africa, and nearly the whole Spanish peninsula, but were finally checked from further progress in Western Europe by the firm hand of Charles Martel (732 a.d.). The Moslem dominion extended now from India to Spain; but a war of succession to the caliphate ensued, and in 755 the Mohammedan empire was divided,—one caliph reigning at Bagdad, the other at Cordova in Spain. Astounding as was the grand march of conquest by the Arabs, still more so was the ease with which they put aside their former nomadic life, adopted a higher civilisation, and assumed the sovereignty over cultivated peoples. Arabic was made the written language throughout the conquered lands. With the rule of the Abbasides in the East began a new period in the history of learning. The capital, Bagdad, situated

THE ARABS. 117 on the Euphrates, lay half-way between two old centres of scientific thought,—India in the East, and Greece in the West. The Arabs were destined to be the custodians of the torch of Greek and Indian science, to keep it ablaze during the period of confusion and chaos in the Occident, and afterwards to pass it over to the Europeans. Thus science passed from Aryan to Semitic races, and then back again to the Aryan. The Mohammedans have added but little to the knowledge in mathematics which they received. They now and then explored a small region to which the path had been previously pointed out, but they were quite incapable of discovering new fields. Even the more elevated regions in which the Hellenes and Hindoos delighted to wander—namely, the Greek conic sections and the Indian indeterminate analysis—were seldom entered upon by the Arabs. They were

less of a speculative, and more of a practical turn of mind. The Abbasides at Bagdad encouraged the introduction of the sciences by inviting able specialists to their court, irrespective of nationality or religious belief. Medicine and astronomy were their favourite sciences. Thus Haroun-al- Raschid, the most distinguished Saracen ruler, drew Indian physicians to Bagdad. In the year 772 there came to the court of Caliph Almansur a Hindoo astronomer with astronomical tables which were ordered to be translated into Arabic. These tables, known by the Arabs as the Sindhind, and probably taken from the Brahma-sphuta-siddhanta of Brahmagupta, stood in great authority. They contained the important Hindoo table of sines.

A HISTORY OF MATHEMATICS. 118 Doubtless at this time, and along with these astronomical tables, the Hindoo numerals, with the zero and the principle of position, were introduced among the Saracens. Before the time of Mohammed the Arabs had no numerals. Numbers were written out in words. Later, the numerous computations connected with the financial administration over the conquered lands made a short symbolism indispensable. In some localities, the numerals of the more civilised conquered nations were used for a time. Thus in Syria, the Greek notation was retained; in Egypt, the Coptic. In some cases, the numeral adjectives may have been abbreviated in writing. The Diwani-numerals, found in an Arabic-Persian dictionary, are supposed to be such abbreviations. Gradually it became the practice to employ the 28 Arabic letters of the alphabet for numerals, in analogy to the Greek system. This notation was in turn superseded by the Hindoo notation, which quite early was adopted by merchants, and also by writers on arithmetic. Its superiority was so universally recognised, that it had no rival, except in astronomy, where the alphabetic notation continued to be used. Here the alphabetic notation offered no great disadvantage, since in the sexagesimal arithmetic, taken from the Almagest, numbers of generally only one or two places had to be written. [7] As regards the form of the so-called Arabic numerals, the statement of the Arabic writer Albiruni (died 1039), who spent many years in India, is of interest. He says that the shape of the numerals, as also of the letters in India, differed in different localities, and that the Arabs selected from the

THE ARABS. 119 various forms the most suitable. An Arabian astronomer says there was among people much difference in the use of symbols, especially of those for 5, 6, 7, and 8. The symbols used by the Arabs can be traced back to the tenth century. We find material differences between those used by the Saracens in the East and those used in the West. But most surprising is the fact that the symbols of both the East and of the West Arabs deviate so extraordinarily from the Hindoo Devanagari numerals (= divine numerals) of to-day, and that they resemble much more closely the apices of the Roman writer Boethius. This strange similarity on the one hand, and dissimilarity on the other, is difficult to explain. The most plausible theory is the one of Woepcke: (1) that about the second century after Christ, before the zero had been invented, the Indian numerals were brought to Alexandria, whence they spread to Rome and also to West Africa; (2) that in the eighth century, after the notation in India had been already much modified and perfected by the invention of the zero, the Arabs at Bagdad got it from the Hindoos; (3) that the Arabs of the West borrowed the Columbus-egg, the zero, from those in the East, but retained the old forms of the nine

numerals, if for no other reason, simply to be contrary to their political enemies of the East; (4) that the old forms were remembered by the West-Arabs to be of Indian origin, and were hence called Gubar-numerals (= dust-numerals, in memory of the Brahmin practice of reckoning on tablets strewn with dust or sand; (5) that, since the eighth century, the numerals in India underwent further changes, and assumed the greatly modified

A HISTORY OF MATHEMATICS. 120 forms of the modern Devanagari-numerals. [3] This is rather a bold theory, but, whether true or not, it explains better than any other yet propounded, the relations between the apices, the Gubar, the East-Arabic, and Devanagari numerals. It has been mentioned that in 772 the IndianSiddhanta was brought to Bagdad and there translated into Arabic. There is no evidence that any intercourse existed between Arabic and Indian astronomers either before or after this time, excepting the travels of Albiruni. But we should be very slow to deny the probability that more extended communications actually did take place. Better informed are we regarding the way in which Greek science, in successive waves, dashed upon and penetrated Arabic soil. In Syria the sciences, especially philosophy and medicine, were cultivated by Greek Christians. Celebrated were the schools at Antioch and Emesa, and, first of all, the flourishing Nestorian school at Edessa. From Syria, Greek physicians and scholars were called to Bagdad. Translations of works from the Greek began to be made. A large number of Greek manuscripts were secured by Caliph Al Mamun (813–833) from the emperor in Constantinople and were turned over to Syria. The successors of Al Mamun continued the work so auspiciously begun, until, at the beginning of the tenth century, the more important philosophic, medical, mathematical, and astronomical works of the Greeks could all be read in the Arabic tongue. The translations of mathematical works must have been very deficient at first, as it was evidently difficult to secure translators who were

THE ARABS. 121 masters of both the Greek and Arabic and at the same time proficient in mathematics. The translations had to be revised again and again before they were satisfactory. The first Greek authors made to speak in Arabic were Euclid and Ptolemæus. This was accomplished during the reign of the famous Haroun-al-Raschid. A revised translation of Euclid’s Elements was ordered by Al Mamun. As this revision still contained numerous errors, a new translation was made, either by the learned Honein ben Ishak, or by his son, Ishak ben Honein. To the thirteen books of theElements were added the fourteenth, written by Hypsicles, and the fifteenth by Damascius. But it remained for Tabit ben Korra to bring forth an Arabic Euclid satisfying every need. Still greater difficulty was experienced in securing an intelligible translation of theAlmagest. Among other important translations into Arabic were the works of Apollonius, Archimedes, Heron, and Diophantus. Thus we see that in the course of one century the Arabs gained access to the vast treasures of Greek science. Having been little accustomed to abstract thought, we need not marvel if, during the ninth century, all their energy was exhausted merely in appropriating the foreign material. No attempts were made at original work in mathematics until the next century. In astronomy, on the other hand, great activity in original research existed as early as the ninth century. The religious observances demanded by Mohammedanism presented to as- tronomers several practical

problems. The Moslem dominions being of such enormous extent, it remained in some localities for the astronomer to determine which way the “Believer”

A HISTORY OF MATHEMATICS. 122 must turn during prayer that he may be facing Mecca. The prayers and ablutions had to take place at definite hours during the day and night. This led to more accurate determinations of time. To fix the exact date for the Mohammedan feasts it became necessary to observe more closely the motions of the moon. In addition to all this, the old Oriental superstition that extraordinary occurrences in the heavens in some myste- rious way affect the progress of human affairs added increased interest to the prediction of eclipses. [7] For these reasons considerable progress was made. Astro- nomical tables and instruments were perfected, observatories erected, and a connected series of observations instituted. This intense love for astronomy and astrology continued during the whole Arabic scientific period. As in India, so here, we hardly ever find a man exclusively devoted to pure mathematics. Most of the so-called mathematicians were first of all astronomers. The first notable author of mathematical books was Mohammed ben Musa Al Hovarezmi, who lived during the reign of Caliph Al Mamun (813–833). He was engaged by the caliph in making extracts from the Sindhind, in revising the tablets of Ptolemæus, in taking observations at Bagdad and Damascus, and in measuring a degree of the earth’s meridian. Important to us is his work on algebra and arithmetic. The portion on arithmetic is not extant in the original, and it was not till 1857 that a Latin translation of it was found. It begins thus: “Spoken has Algoritmi. Let us give deserved praise to God, our leader and defender.” Here the name of the author,

THE ARABS. 123 Al Hovarezmi, has passed into Algoritmi, from which comes our modern word, algorithm, signifying the art of computing in any particular way. The arithmetic of Hovarezmi, being based on the principle of position and the Hindoo method of calculation, “excels,” says an Arabic writer, “all others in brevity and easiness, and exhibits the Hindoo intellect and sagacity in the grandest inventions.” This book was followed by a large number of arithmetics by later authors, which differed from the earlier ones chiefly in the greater variety of methods. Arabian arithmetics generally contained the four operations with integers and fractions, modelled after the Indian processes. They explained the operation of casting out the 9’s, which was sometimes called the “Hindoo proof.” They contained also the regula falsa and the regula duorum falsorum , by which algebraical examples could be solved without algebra. Both these methods were known to the Indians. The regula falsaor falsa positio was the assigning of an assumed value to the unknown quantity, which value, if wrong, was corrected by some process like the “rule of three.” Diophantus used a method almost identical with this. The regula duorum falsorum was as follows: [7] To solve an equation f(x) = V , assume, for the moment, two values forx; namely, x = a and x = b. Then form f(a) = A and f(b) = B, and determine the errorsV− A = Ea and V− B = Eb; then the required x = bEa− aEb Ea− Eb is generally a close approximation, but is absolutely accurate whenever f(x) is a linear function of x. We now return to Hovarezmi, and consider the other part

A HISTORY OF MATHEMATICS. 124 of his work,—the algebra. This is the first book known to contain this word itself as title. Really the title consists of two words, aldshebr walmukabala, the nearest English translation of which is “restoration” and “reduction.” By “restoration” was meant the transposing of negative terms to the other side of the equation; by “reduction,” the uniting of similar terms. Thus, x2−2x = 5x+6 passes by aldshebr intox2 = 5x+2 x+6; and this, by walmukabala, into x2 = 7 x + 6. The work on algebra, like the arithmetic, by the same author, contains nothing original. It explains the elementary operations and the solutions of linear and quadratic equations. From whom did the author borrow his knowledge of algebra? That it came entirely from Indian sources is impossible, for the Hindoos had no such rules like the “restoration” and “reduction.” They were, for instance, never in the habit of making all terms in an equation positive, as is done by the process of “restoration.” Diophantus gives two rules which resemble somewhat those of our Arabic author, but the probability that the Arab got all his algebra from Diophantus is lessened by the considerations that he recognised both roots of a quadratic, while Diophantus noticed only one; and that the Greek algebraist, unlike the Arab, habitually rejected irrational solutions. It would seem, therefore, that the algebra of Hovarezmi was neither purely Indian nor purely Greek, but was a hybrid of the two, with the Greek element predominating. The algebra of Hovarezmi contains also a few meagre fragments on geometry. He gives the theorem of the right triangle, but proves it after Hindoo fashion and only for the

THE ARABS. 125 simplest case, when the right triangle is isosceles. He then calculates the areas of the triangle, parallelogram, and circle. For π he uses the value 3 1 7, and also the two Indian, π = √ 10 and π = 62832 20000. Strange to say, the last value was afterwards forgotten by the Arabs, and replaced by others less accurate. This bit of geometry doubtless came from India. Later Arabic writers got their geometry almost entirely from Greece. Next to be noticed are the three sons of Musa ben Sakir, who lived in Bagdad at the court of the Caliph Al Mamun. They wrote several works, of which we mention a geometry in which is also contained the well-known formula for the area of a triangle expressed in terms of its sides. We are told that one of the sons travelled to Greece, probably to collect astronomical and mathematical manuscripts, and that on his way back he made acquaintance with Tabit ben Korra. Recognising in him a talented and learned astronomer, Mohammed procured for him a place among the astronomers at the court in Bagdad. Tabit ben Korra (836–901) was born at Harran in Mesopotamia. He was proficient not only in astronomy and mathematics, but also in the Greek, Arabic, and Syrian languages. His translations of Apollonius, Archimedes, Euclid, Ptolemy, Theodosius, rank among the best. His dissertation on amicable numbers (of which each is the sum of the factors of the other) is the first known specimen of original work in mathematics on Arabic soil. It shows that he was familiar with the Pythagorean theory of numbers. Tabit invented the following rule for finding amicable numbers: If p = 3·2n− 1, q = 3·2n−1− 1, r = 9·22n−1− 1 (n being a whole

A HISTORY OF MATHEMATICS. 126 number) are three primes, thena = 2npq, b = 2nr are a pair of amicable numbers. Thus, if n = 2, then p = 11, q = 5, r = 71, and a = 220, b = 284. Tabit also trisected an angle. Foremost among the astronomers of the

ninth century ranked Al Battani, called Albategnius by the Latins. Battan in Syria was his birthplace. His observations were celebrated for great precision. His work, De scientia stellarum , was translated into Latin by Plato Tiburtinus, in the twelfth century. Out of this translation sprang the word ‘sinus,’ as the name of a trigonometric function. The Arabic word for “sine,” dschiba, was derived from the Sanscrit jiva, and resembled the Arabic word dschaib, meaning an indentation or gulf. Hence the Latin “sinus.” [3] Al Battani was a close student of Ptolemy, but did not follow him altogether. He took an important step for the better, when he introduced the Indian “sine” orhalf the chord, in place of thewhole chord of Ptolemy. Another improvement on Greek trigonometry made by the Arabs points likewise to Indian influences. Propositions and operations which were treated by the Greeks geometrically are expressed by the Arabs algebraically. Thus, Al Battani at once gets from an equation sin θ cos θ = D, the value of θ by means of sin θ = D√ 1 + D2 ,—a process unknown to the ancients. He knows, of course, all the formulas for spherical triangles given in theAlmagest, but goes further, and adds an important one of his own for oblique-angled triangles; namely, cos a = cos b cos c + sin b sin c cos A. At the beginning of the tenth century political troubles

THE ARABS. 127 arose in the East, and as a result the house of the Abbasides lost power. One province after another was taken, till, in 945, all possessions were wrested from them. Fortunately, the new rulers at Bagdad, the Persian Buyides, were as much interested in astronomy as their predecessors. The progress of the sciences was not only unchecked, but the conditions for it became even more favourable. The EmirAdud-ed-daula (978–983) gloried in having studied astronomy himself. His son Saraf-ed-daula erected an observatory in the garden of his palace, and called thither a whole group of scholars. [7] Among them were Abul Wefa, Al Kuhi, Al Sagani. Abul Wefa(940–998) was born at Buzshan in Chorassan, a region among the Persian mountains, which has brought forth many Arabic astronomers. He forms an important exception to the unprogressive spirit of Arabian scientists by his brilliant discovery of the variation of the moon, an inequality usually supposed to have been first discovered by Tycho Brahe. [11] Abul Wefa translated Diophantus. He is one of the last Arabic translators and commentators of Greek authors. The fact that he esteemed the algebra of Mohammed ben Musa Hovarezmi worthy of his commentary indicates that thus far algebra had made little or no progress on Arabic soil. Abul Wefa invented a method for computing tables of sines which gives the sine of half a degree correct to nine decimal places. He did himself credit by introducing the tangent into trigonometry and by calculating a table of tangents. The first step toward this had been taken by Al Battani. Unfortunately, this innovation and the discovery of

A HISTORY OF MATHEMATICS. 128 the moon’s variation excited apparently no notice among his contemporaries and followers. “We can hardly help looking upon this circumstance as an evidence of a servility of intellect belonging to the Arabian period.” A treatise by Abul Wefa on “geometric constructions” indicates that efforts were being made at that time to improve draughting. It contains a neat construction of the corners of the regular polyhedrons on the circumscribed sphere. Here, for the first time, appears the condition which afterwards became very famous in the

Occident, that the construction be effected with a single opening of the compass. Al Kuhi , the second astronomer at the observatory of the emir at Bagdad, was a close student of Archimedes and Apollonius. He solved the problem, to construct a segment of a sphere equal in volume to a given segment and having a curved surface equal in area to that of another given segment. He, Al Sagani, and Al Biruni made a study of the trisection of angles. Abul Gud, an able geometer, solved the problem by the intersection of a parabola with an equilateral hyperbola. The Arabs had already discovered the theorem that the sum of two cubes can never be a cube. Abu Mohammed Al Hogendi of Chorassan thought he had proved this, but we are told that the demonstration was defective. Creditable work in theory of numbers and algebra was done byAl Karhi of Bagdad, who lived at the beginning of the eleventh century. His treatise on algebra is the greatest algebraic work of the Arabs. In it he appears as a disciple of Diophantus. He was the first to operate with higher roots and to solve equations

THE ARABS. 129 of the form x2n + axn = b. For the solution of quadratic equations he gives both arithmetical and geometric proofs. He was the first Arabic author to give and prove the theorems on the summation of the series:— 12 + 22 + 32 +⋯ + n2 = (1 + 2 +⋯ + n) 2n + 1 3 , 13 + 23 + 33 +⋯ + n3 = (1 + 2 +⋯ + n)2. Al Karhi also busied himself with indeterminate analysis. He showed skill in handling the methods of Diophantus, but added nothing whatever to the stock of knowledge already on hand. As a subject for original research, indeterminate analysis was too subtle for even the most gifted of Arabian minds. Rather surprising is the fact that Al Karhi’s algebra shows no traces whatever of Hindoo indeterminate analysis. But most astonishing it is, that an arithmetic by the same author completely excludes the Hindoo numerals. It is constructed wholly after Greek pattern. Abul Wefa also, in the second half of the tenth century, wrote an arithmetic in which Hindoo numerals find no place. This practice is the very opposite to that of other Arabian authors. The question, why the Hindoo numerals were ignored by so eminent authors, is certainly a puzzle. Cantor suggests that at one time there may have been rival schools, of which one followed almost exclusively Greek mathematics, the other Indian. The Arabs were familiar with geometric solutions of quad- ratic equations. Attempts were now made to solve cubic equations geometrically. They were led to such solutions

A HISTORY OF MATHEMATICS. 130 by the study of questions like the Archimedean problem, demanding the section of a sphere by a plane so that the two segments shall be in a prescribed ratio. The first to state this problem in form of a cubic equation was Al Mahani of Bagdad, while Abu Gafar Al Hazin was the first Arab to solve the equation by conic sections. Solutions were given also by Al Kuhi, Al Hasan ben Al Haitam, and others. [20] Another difficult problem, to determine the side of a regular heptagon, required the construction of the side from the equation x3− x2− 2x + 1 = 0. It was attempted by many and at last solved by Abul Gud. The one who did most to elevate to a method the solution of algebraic equations by intersecting conics, was Omar al Hayyami of Chorassan, about 1079 a.d. He divides cubics intotwoclasses, thetrinomialandquadrinomial, andeachclass into families and species. Each species is treated separately but according to a general plan. He believed that cubics could not be solved by calculation, nor biquadratics by geometry.

He rejected negative roots and often failed to discover all the positive ones. Attempts at biquadratic equations were made by Abul Wefa, [20] who solved geometricallyx4 = a and x4 + ax3 = b. The solution of cubic equations by intersecting conics was the greatest achievement of the Arabs in algebra. The foundation to this work had been laid by the Greeks, for it was Menæchmus who first constructed the roots ofx3− a = 0 or x3− 2a3 = 0 . It was not his aim to find the number corresponding to x, but simply to determine the side x of

THE ARABS. 131 a cube double another cube of side a. The Arabs, on the other hand, had another object in view: to find the roots of given numerical equations. In the Occident, the Arabic solutions of cubics remained unknown until quite recently. Descartes and Thomas Baker invented these constructions anew. The works of Al Hayyami, Al Karhi, Abul Gud, show how the Arabs departed further and further from the Indian methods, and placed themselves more immediately under Greek influences. In this way they barred the road of progress against themselves. The Greeks had advanced to a point where material progress became difficult with their methods; but the Hindoos furnished new ideas, many of which the Arabs now rejected. With Al Karhi and Omar Al Hayyami, mathematics among the Arabs of the East reached flood-mark, and now it begins to ebb. Between 1100 and 1300 a.d. come the crusades with war and bloodshed, during which European Christians profited much by their contact with Arabian culture, then far superior to their own; but the Arabs got no science from the Christians in return. The crusaders were not the only adversaries of the Arabs. During the first half of the thirteenth century, they had to encounter the wild Mongolian hordes, and, in 1256, were conquered by them under the leadership of Hulagu. The caliphate at Bagdad now ceased to exist. At the close of the fourteenth century still another empire was formed by Timur or Tamerlane, the Tartar. During such sweeping turmoil, it is not surprising that science declined. Indeed, it is a marvel that it existed at all. During the supremacy

A HISTORY OF MATHEMATICS. 132 of Hulagu, lived Nasir Eddin (1201–1274), a man of broad culture and an able astronomer. He persuaded Hulagu to build him and his associates a large observatory at Maraga. Treatises on algebra, geometry, arithmetic, and a translation of Euclid’s Elements, were prepared by him. Even at the court of Tamerlane in Samarkand, the sciences were by no means neglected. A group of astronomers was drawn to this court. Ulug Beg (1393–1449), a grandson of Tamerlane, was himself an astronomer. Most prominent at this time was Al Kaschi, the author of an arithmetic. Thus, during intervals of peace, science continued to be cultivated in the East for several centuries. The last Oriental writer was Beha Eddin (1547–1622). His Essence of Arithmetic stands on about the same level as the work of Mohammed ben Musa Hovarezmi, written nearly 800 years before. “Wonderful is the expansive power of Oriental peoples, with which upon the wings of the wind they conquer half the world, but more wonderful the energy with which, in less than two generations, they raise themselves from the lowest stages of cultivation to scientific efforts.” During all these centuries, astronomy and mathematics in the Orient greatly excel these sciences in the Occident. Thus far we have spoken only of the Arabs in the East. Between the Arabs of the East and of the West, which were under separate governments, there generally existed consid- erable political animosity. In

consequence of this, and of the enormous distance between the two great centres of learning, Bagdad and Cordova, there was less scientific intercourse

THE ARABS. 133 among them than might be expected to exist between peoples having the same religion and written language. Thus the course of science in Spain was quite independent of that in Persia. While wending our way westward to Cordova, we must stop in Egypt long enough to observe that there, too, scientific activity was rekindled. Not Alexandria, but Cairo with its library and observatory, was now the home of learning. Foremost among her scientists ranked Ben Junus (died 1008), a contemporary of Abul Wefa. He solved some difficult problems in spherical trigonometry. Another Egyptian astronomer was Ibn Al Haitam (died 1038), who wrote on geometric loci. Travelling westward, we meet in Morocco Abul Hasan Ali, whose treatise 'on astronomical instruments' discloses a thorough knowledge of theConics of Apollonius. Arriving finally in Spain at the capital, Cordova, we are struck by the magnificent splendour of her architecture. At this renowned seat of learning, schools and libraries were founded during the tenth century. Little is known of the progress of mathematics in Spain. The earliest name that has come down to us isAl Madshriti(died 1007), the author of a mystic paper on 'amicable numbers.' His pupils founded schools at Cordova, Dania, and Granada. But the only great astronomer among the Saracens in Spain is Gabir ben Aflahof Sevilla, frequently calledGeber. He lived in the second half of the eleventh century. It was formerly believed that he was the inventor of algebra, and that the word algebra came from 'Gabir' or 'Geber.' He ranks among the most eminent astronomers of this time, but, like so many

A HISTORY OF MATHEMATICS. 134 of his contemporaries, his writings contain a great deal of mysticism. His chief work is an astronomy in nine books, of which the first is devoted to trigonometry. In his treatment of spherical trigonometry, he exercises great independence of thought. He makes war against the time-honoured procedure adopted by Ptolemy of applying "the rule of six quantities," and gives a new way of his own, based on the 'rule of four quantities.' This is: If P P1 and QQ1 be two arcs of great circles intersecting in A, and if P Q and P1Q1 be arcs of great circles drawn perpendicular to QQ1, then we have the proportion sin AP : sin P Q = sin AP1 : sin P1Q1. From this he derives the formulas for spherical right triangles. To the four fundamental formulas already given by Ptolemy, he added a fifth, discovered by himself. If a, b, c, be the sides, and A, B, C, the angles of a spherical triangle, right-angled at A, then cos B = cos b sin C. This is frequently called "Geber's Theorem." Radical and bold as were his innovations in spherical trigonometry, in plane trigonometry he followed slavishly the old beaten path of the Greeks. Not even did he adopt the Indian 'sine' and 'cosine,' but still used the Greek 'chord of double the angle.' So painful was the departure from old ideas, even to an independent Arab! After the time of Gabir ben Aflah there was no mathematician among the Spanish Saracens of any reputation. In the year in which Columbus discovered America, the Moors lost their last foothold on Spanish soil. We have witnessed a laudable intellectual activity among

EUROPE DURING THE MIDDLE AGES. 135 the Arabs. They had the good fortune to possess rulers who, by their munificence, furthered scientific research. At the courts of the caliphs, scientists were supplied with libraries and observatories. A large number of astronomical and mathematical works were written by Arabic authors. Yet we fail to find a single important principle in mathematics brought forth by the Arabic mind. Whatever discoveries they made, were in fields previously traversed by the Greeks or the Indians, and consisted of objects which the latter had overlooked in their rapid march. The Arabic mind did not possess that penetrative insight and invention by which mathematicians in Europe afterwards revolutionised the science. The Arabs were learned, but not original. Their chief service to science consists in this, that they adopted the learning of Greece and India, and kept what they received with scrupulous care. When the love for science began to grow in the Occident, they transmitted to the Europeans the valuable treasures of antiquity. Thus a Semitic race was, during the Dark Ages, the custodian of the Aryan intellectual possessions. EUROPE DURING THE MIDDLE AGES. With the third century after Christ begins an era of migration of nations in Europe. The powerful Goths quit their swamps and forests in the North and sweep onward in steady southwestern current, dislodging the Vandals, Sueves, and Burgundians, crossing the Roman territory, and stopping and

A HISTORY OF MATHEMATICS. 136 recoiling only when reaching the shores of the Mediterranean. From the Ural Mountains wild hordes sweep down on the Danube. The Roman Empire falls to pieces, and the Dark Ages begin. But dark though they seem, they are the germinating season of the institutions and nations of modern Europe. The Teutonic element, partly pure, partly intermixed with the Celtic and Latin, produces that strong and luxuriant growth, the modern civilisation of Europe. Almost all the various nations of Europe belong to the Aryan stock. As the Greeks and the Hindoos—both Aryan races—were the great thinkers of antiquity, so the nations north of the Alps became the great intellectual leaders of modern times. Introduction of Roman Mathematics. We shall now consider how these as yet barbaric nations of the North gradually came in possession of the intellectual treasures of antiquity. With the spread of Christianity the Latin language was introduced not only in ecclesiastical but also in scientific and all important worldly transactions. Naturally the science of the Middle Ages was drawn largely from Latin sources. In fact, during the earlier of these ages Roman authors were the only ones read in the Occident. Though Greek was not wholly unknown, yet before the thirteenth century not a single Greek scientific work had been read or translated into Latin. Meagre indeed was the science which could be gotten from Roman writers, and we must wait several centuries before any substantial progress is made in

EUROPE DURING THE MIDDLE AGES. 137 mathematics. After the time of Boethius and Cassiodorius mathematical activity in Italy died out. The first slender blossom of science among tribes that came from the North was an encyclopædia entitled Origines, written by Isidorus (died 636 as bishop of Seville). This work is modelled after the Roman encyclopædias of Martianus Capella of Carthage and of Cassiodorius. Part of it is devoted to the quadrivium, arithmetic, music, geometry, and astronomy. He gives definitions and grammatical explications of technical terms,

but does not describe the modes of computation then in vogue. After Isidorus there follows a century of darkness which is at last dissipated by the appearance of Bede the Venerable (672–735), the most learned man of his time. He was a native of Ireland, then the home of learning in the Occident. His works contain treatises on the Computus, or the computation of Easter-time, and on finger-reckoning. It appears that a finger- symbolism was then widely used for calculation. The correct determination of the time of Easter was a problem which in those days greatly agitated the Church. It became desirable to have at least one monk at each monastery who could determine the day of religious festivals and could compute the calendar. Such determinations required some knowledge of arithmetic. Hence we find that the art of calculating always found some little corner in the curriculum for the education of monks. The year in which Bede died is also the year in whichAlcuin (735–804) was born. Alcuin was educated in Ireland, and was

A HISTORY OF MATHEMATICS. 138 called to the court of Charlemagne to direct the progress of education in the great Frankish Empire. Charlemagne was a great patron of learning and of learned men. In the great sees and monasteries he founded schools in which were taught the psalms, writing, singing, computation (computus), and grammar. By computus was here meant, probably, not merely the determination of Easter-time, but the art of computation in general. Exactly what modes of reckoning were then employed we have no means of knowing. It is not likely that Alcuin was familiar with the apices of Boethius or with the Roman method of reckoning on the abacus. He belongs to that long list of scholars who dragged the theory of numbers into theology. Thus the number of beings created by God, who created all things well, is6, because 6 is a perfect number (the sum of its divisors being 1 + 2 + 3 = 6); 8, on the other hand, is an imperfect number (1 + 2 + 4< 8); hence the second origin of mankind emanated from the number 8, which is the number of souls said to have been in Noah’s ark. There is a collection of “Problems for Quickening the Mind” (propositiones ad acuendos iuvenes), which are certainly as old as 1000 a.d. and possibly older. Cantor is of the opinion that they were written much earlier and by Alcuin. The following is a specimen of these “Problems”: A dog chasing a rabbit, which has a start of150 feet, jumps9 feet every time the rabbit jumps 7. In order to determine in how many leaps the dog overtakes the rabbit,150 is to be divided by2. In this collection of problems, the areas of triangular and quadrangular pieces of land are found by the same formulas of approximation as

EUROPE DURING THE MIDDLE AGES. 139 those used by the Egyptians and given by Boethius in his geometry. An old problem is the “cistern-problem” (given the time in which several pipes can fill a cistern singly, to find the time in which they fill it jointly), which has been found previously in Heron, in the Greek Anthology, and in Hindoo works. Many of the problems show that the collection was compiled chiefly from Roman sources. The problem which, on account of its uniqueness, gives the most positive testimony regarding the Roman origin is that on the interpretation of a will in a case where twins are born. The problem is identical with the Roman, except that different ratios are chosen. Of the exercises for recreation, we mention the one of the wolf, goat, and cabbage, to be rowed across a river in a boat holding only one besides the ferry-man. Query: How must he carry them across so that the goat shall

not eat the cabbage, nor the wolf the goat? The solutions of the “problems for quickening the mind” require no further knowledge than the recollection of some few formulas used in surveying, the ability to solve linear equations and to perform the four fundamental operations with integers. Extraction of roots was nowhere demanded; fractions hardly ever occur. [3] The great empire of Charlemagne tottered and fell almost immediately after his death. War and confusion ensued. Scientific pursuits were abandoned, not to be resumed until the close of the tenth century, when under Saxon rule in Germany and Capetian in France, more peaceful times began. The thick gloom of ignorance commenced to disappear. The zeal with which the study of mathematics was now taken up

A HISTORY OF MATHEMATICS. 140 by the monks is due principally to the energy and influence of one man,—Gerbert. He was born in Aurillac in Auvergne. After receiving a monastic education, he engaged in study, chiefly of mathematics, in Spain. On his return he taught school at Rheims for ten years and became distinguished for his profound scholarship. By King Otto I. and his successors Gerbert was held in highest esteem. He was elected bishop of Rheims, then of Ravenna, and finally was made Pope under the name of Sylvester II. by his former pupil Emperor Otho III. He died in 1003, after a life intricately involved in many political and ecclesiastical quarrels. Such was the career of the greatest mathematician of the tenth century in Europe. By his contemporaries his mathematical knowledge was considered wonderful. Many even accused him of criminal intercourse with evil spirits. Gerbert enlarged the stock of his knowledge by procuring copies of rare books. Thus in Mantua he found the geometry of Boethius. Though this is of small scientific value, yet it is of great importance in history. It was at that time the only book from which European scholars could learn the elements of geometry. Gerbert studied it with zeal, and is generally believed himself to be the author of a geometry. H. Weissenborn denies his authorship, and claims that the book in question consists of three parts which cannot come from one and the same author. [21] This geometry contains nothing more than the one of Boethius, but the fact that occasional errors in the latter are herein corrected shows that the author had mastered the subject. “The first mathematical

EUROPE DURING THE MIDDLE AGES. 141 paper of the Middle Ages which deserves this name,” says Hankel, “is a letter of Gerbert to Adalbold, bishop of Utrecht,” in which is explained the reason why the area of a triangle, obtained “geometrically” by taking the product of the base by half its altitude, differs from the area calculated “arithmetically,” according to the formula 1 2 a(a + 1), used by surveyors, where a stands for a side of an equilateral triangle. He gives the correct explanation that in the latter formula all the small squares, in which the triangle is supposed to be divided, are counted in wholly, even though parts of them project beyond it. Gerbert made a careful study of the arithmetical works of Boethius. He himself published two works,— Rule of Computation on the Abacus , and A Small Book on the Division of Numbers. They give an insight into the methods of calculation practised in Europe before the introduction of the Hindoo numerals. Gerbert used the abacus, which was probably unknown to Alcuin. Bernelinus, a pupil of Gerbert, describes it as consisting of a smooth board upon which geometricians were

accustomed to strew blue sand, and then to draw their diagrams. For arithmetical purposes the board was divided into 30 columns, of which 3 were reserved for fractions, while the remaining 27 were divided into groups with 3 columns in each. In every group the columns were marked respectively by the letters C (centum), D (decem), and S (singularis) or M (monas). Bernelinus gives the nine numerals used, which are the apices of Boethius, and then remarks that the Greek letters may be used in their place. [3]

A HISTORY OF MATHEMATICS. 142 By the use of these columns any number can be written without introducing a zero, and all operations in arithmetic can be performed in the same way as we execute ours without the columns, but with the symbol for zero. Indeed, the methods of adding, subtracting, and multiplying in vogue among the abacists agree substantially with those of to-day. But in a division there is very great difference. The early rules for division appear to have been framed to satisfy the following three conditions: (1) The use of the multiplication table shall be restricted as far as possible; at least, it shall never be required to multiply mentally a figure of two digits by another of one digit. (2) Subtractions shall be avoided as much as possible and replaced by additions. (3) The operation shall proceed in a purely mechanical way, without requiring trials. [7] That it should be necessary to make such conditions seems strange to us; but it must be remembered that the monks of the Middle Ages did not attend school during childhood and learn the multiplication table while the memory was fresh. Gerbert’s rules for division are the oldest extant. They are so brief as to be very obscure to the uninitiated. They were probably intended simply to aid the memory by calling to mind the successive steps in the work. In later manuscripts they are stated more fully. In dividing any number by another of one digit, say 668 by 6, the divisor was first increased to 10 by adding 4. The process is exhibited in the adjoining figure. [3] As it continues, we must imagine the digits which are crossed out, to be erased and then replaced by the ones beneath. It is as follows: 600÷ 10 = 60, but, to

EUROPE DURING THE MIDDLE AGES. 143 rectify the error, 4× 60, or 240, must be added; 200÷ 10 = 20, but 4×20, or80, must be added. We now write for60+40+80 ,))) C D S 6 4 6 6 8 6/ 6/ 8/ 2/ 4/ 4/ 1/ 8/ 8/ 1/ 4/ 8/ 2/ 2 4/ 6/ 2/ 2/ 6/ 6/ 2/ 2/ 1/ 2/ 1 1 its sum 180, and continue thus: 100÷ 10 = 10; the correction necessary is 4× 10, or 40, which, added to 80, gives 120. Now 100÷ 10 = 10 , and the correction 4× 10, together with the20, gives 60. Proceeding as before, 60÷ 10 = 6 ; the correction is 4× 6 = 24. Now 20÷ 10 = 2, the correction being 4× 2 = 8. In the column of units we have now 8 + 4 + 8, or 20. As before, 20÷ 10 = 2; the correction is 2× 4 = 8, which is not divisible by 10, but only by 6, giving the quotient 1 and the remainder 2. All the partial quotients taken together give 60 + 20 + 10 + 10 + 6 + 2 + 2 + 1 = 111, and the remainder 2. Similar but more complicated, is the process when the divisor contains two or more digits. Were the divisor 27, then the next higher multiple of 10, or 30, would be taken for the divisor, but corrections would be required for the 3. He who has the patience to carry such a division through to the end, will understand why it has been said of Gerbert that “Regulas dedit, quæ a sudantibus abacistis vix intelliguntur.” He will also perceive why the Arabic method of division, when first introduced, was called the divisio aurea, but the one on the abacus, the divisio ferrea.

A HISTORY OF MATHEMATICS. 144 In his book on the abacus, Bernelinus devotes a chapter to fractions. These are, of course, the duodecimals, first used by the Romans. For want of a suitable notation, calculation with them was exceedingly difficult. It would be so even to us, were we accustomed, like the early abacists, to express them, not by a numerator or denominator, but by the application of names, such as uncia for 1 12, quincunx for 5 12, dodrans for 9 12. In the tenth century, Gerbert was the central figure among the learned. In his time the Occident came into secure possession of all mathematical knowledge of the Romans. During the eleventh century it was studied assiduously. Though numerous works were written on arithmetic and geometry, mathematical knowledge in the Occident was still very insignificant. Scanty indeed were the mathematical treasures obtained from Roman sources. Translation of Arabic Manuscripts. By his great erudition and phenomenal activity, Gerbert infused new life into the study not only of mathematics, but also of philosophy. Pupils from France, Germany, and Italy gathered at Rheims to enjoy his instruction. When they themselves became teachers, they taught of course not only the use of the abacus and geometry, but also what they had learned of the philosophy of Aristotle. His philosophy was known, at first, only through the writings of Boethius. But the growing enthusiasm for it created a demand for his complete works. Greek texts were wanting. But the Latins heard

EUROPE DURING THE MIDDLE AGES. 145 that the Arabs, too, were great admirers of Peripatetism, and that they possessed translations of Aristotle’s works and commentaries thereon. This led them finally to search for and translate Arabic manuscripts. During this search, mathematical works also came to their notice, and were translated into Latin. Though some few unimportant works may have been translated earlier, yet the period of greatest activity began about 1100. The zeal displayed in acquiring the Mohammedan treasures of knowledge excelled even that of the Arabs themselves, when, in the eighth century, they plundered the rich coffers of Greek and Hindoo science. Among the earliest scholars engaged in translating man- uscripts into Latin was Athelard of Bath. The period of his activity is the first quarter of the twelfth century. He travelled extensively in Asia Minor, Egypt, and Spain, and braved a thousand perils, that he might acquire the language and science of the Mohammedans. He made the earliest translations, from the Arabic, of Euclid’s Elements and of the astronomical tables of Mohammed ben Musa Hovarezmi. In 1857, a manuscript was found in the library at Cambridge, which proved to be the arithmetic by Mohammed ben Musa in Latin. This translation also is very probably due to Athelard. At about the same time flourished Plato of Tivoli or Plato Tiburtinus. He effected a translation of the astronomy of Al Battani and of the Sphærica of Theodosius. Through the former, the term sinus was introduced into trigonometry. About the middle of the twelfth century there was a group of Christian scholars busily at work at Toledo, under the

A HISTORY OF MATHEMATICS. 146 leadership of Raymond, then archbishop of Toledo. Among those who worked under his direction, John of Seville was most prominent. He translated works chiefly on Aristotelian philosophy. Of importance to us is aliber algorismi, compiled by him from Arabic authors. On comparing works like this with those of the abacists, we notice at once the most striking difference, which

shows that the two parties drew from independent sources. It is argued by some that Gerbert got his apices and his arithmetical knowledge, not from Boethius, but from the Arabs in Spain, and that part or the whole of the geometry of Boethius is a forgery, dating from the time of Gerbert. If this were the case, then the writings of Gerbert would betray Arabic sources, as do those of John of Seville. But no points of resemblance are found. Gerbert could not have learned from the Arabs the use of the abacus, because all evidence we have goes to show that they did not employ it. Nor is it probable that he borrowed from the Arabs the apices, because they were never used in Europe except on the abacus. In illustrating an example in division, mathematicians of the tenth and eleventh centuries state an example in Roman numerals, then draw an abacus and insert in it the necessary numbers with the apices. Hence it seems probable that the abacus and apices were borrowed from the same source. The contrast between authors like John of Seville, drawing from Arabic works, and the abacists, consists in this, that, unlike the latter, the former mention the Hindoos, use the term algorism, calculate with the zero, and do not employ the abacus. The former teach the extraction of roots, the

EUROPE DURING THE MIDDLE AGES. 147 abacists do not; they teach the sexagesimal fractions used by the Arabs, while the abacists employ the duodecimals of the Romans. [3] A little later than John of Seville flourished Gerard of Cremona in Lombardy. Being desirous to gain possession of the Almagest, he went to Toledo, and there, in 1175, translated this great work of Ptolemy. Inspired by the richness of Mohammedan literature, he gave himself up to its study. He translated into Latin over 70 Arabic works. Of mathematical treatises, there were among these, besides the Almagest, the 15 books of Euclid, theSphærica of Theodosius, a work of Menelaus, the algebra of Mohammed ben Musa Hovarezmi, the astronomy of Dshabir ben Aflah, and others less important. In the thirteenth century, the zeal for the acquisition of Arabic learning continued. Foremost among the patrons of science at this time ranked Emperor Frederick II. of Hohenstaufen (died 1250). Through frequent contact with Mohammedan scholars, he became familiar with Arabic science. He employed a number of scholars in translating Arabic manuscripts, and it was through him that we came in possession of a new translation of the Almagest. Another royal head deserving mention as a zealous promoter of Arabic science was Alfonso X. of Castile (died 1284). He gathered around him a number of Jewish and Christian scholars, who translated and compiled astronomical works from Arabic sources. Rabbi Zag and Iehuda ben Mose Cohen were the most prominent among them. Astronomical tables

A HISTORY OF MATHEMATICS. 148 prepared by these two Jews spread rapidly in the Occident, and constituted the basis of all astronomical calculation till the sixteenth century. [7] The number of scholars who aided in transplanting Arabic science upon Christian soil was large. But we mention only one more. Giovanni Campano of Novara (about 1260) brought out a new translation of Euclid, which drove the earlier ones from the field, and which formed the basis of the printed editions. [7] At the close of the twelfth century, the Occident was in possession of the so-called Arabic notation. The Hindoo methods of calculation began to supersede the cumbrous methods inherited from Rome. Algebra, with its rules for solving linear and

quadratic equations, had been made accessible to the Latins. The geometry of Euclid, theSphærica of Theodosius, the astronomy of Ptolemy, and other works were now accessible in the Latin tongue. Thus a great amount of new scientific material had come into the hands of the Christians. The talent necessary to digest this heterogeneous mass of knowledge was not wanting. The figure of Leonardo of Pisa adorns the vestibule of the thirteenth century. It is important to notice that no work either on mathematics or astronomy was translated directly from the Greek previous to the fifteenth century. The First Awakening and its Sequel. Thus far, France and the British Isles have been the headquarters of mathematics in Christian Europe. But at the

EUROPE DURING THE MIDDLE AGES. 149 beginning of the thirteenth century the talent and activity of one man was sufficient to assign the mathematical science a new home in Italy. This man was not a monk, like Bede, Alcuin, or Gerbert, but a merchant, who in the midst of business pursuits found time for scientific study. Leonardo of Pisa is the man to whom we owe the first renaissance of mathematics on Christian soil. He is also called Fibonacci, i.e. son of Bonaccio. His father was secretary at one of the numerous factories erected on the south and east coast of the Mediterranean by the enterprising merchants of Pisa. He made Leonardo, when a boy, learn the use of the abacus. The boy acquired a strong taste for mathematics, and, in later years, during his extensive business travels in Egypt, Syria, Greece, and Sicily, collected from the various peoples all the knowledge he could get on this subject. Of all the methods of calculation, he found the Hindoo to be unquestionably the best. Returning to Pisa, he published, in 1202, his great work, the Liber Abaci. A revised edition of this appeared in 1228. This work contains about all the knowledge the Arabs possessed in arithmetic and algebra, and treats the subject in a free and independent way. This, together with the other books of Leonardo, shows that he was not merely a compiler, or, like other writers of the Middle Ages, a slavish imitator of the form in which the subject had been previously presented, but that he was an original worker of exceptional power. He was the first great mathematician to advocate the adoption of the “Arabic notation.” The calculation with the zero was the portion of Arabic mathematics earliest adopted

A HISTORY OF MATHEMATICS. 150 by the Christians. The minds of men had been prepared for the reception of this by the use of the abacus and the apices. The reckoning with columns was gradually abandoned, and the very word abacus changed its meaning and became a synonym for algorism. For the zero, the Latins adopted the name zephirum, from the Arabic sifr (sifra=empty); hence our English word cipher. The new notation was accepted readily by the enlightened masses, but, at first, rejected by the learned circles. The merchants of Italy used it as early as the thirteenth century, while the monks in the monasteries adhered to the old forms. In 1299, nearly 100 years after the publication of Leonardo’s Liber Abaci, the Florentine merchants were forbidden the use of the Arabic numerals in book-keeping, and ordered either to employ the Roman numerals or to write the numeral adjectives out in full. In the fifteenth century the abacus with its counters ceased to be used in Spain and Italy. In France it was used later, and it did not disappear in England and Germany before the middle of the seventeenth century. [22] Thus, in theWinter’s Tale

(iv. 3), Shakespeare lets the clown be embarrassed by a problem which he could not do without counters. Iago (in Othello, i. 1) expresses his contempt for Michael Cassio, “forsooth a great mathematician,” by calling him a “counter- caster.” So general, indeed, says Peacock, appears to have been the practice of this species of arithmetic, that its rules and principles form an essential part of the arithmetical treatises of that day. The real fact seems to be that the old methods were used long after the Hindoo numerals were in common

EUROPE DURING THE MIDDLE AGES. 151 and general use. With such dogged persistency does man cling to the old! The Liber Abaci was, for centuries, the storehouse from which authors got material for works on arithmetic and algebra. In it are set forth the most perfect methods of calculation with integers and fractions, known at that time; the square and cube root are explained; equations of the first and second degree leading to problems, either determinate or indeterminate, are solved by the methods of ‘single’ or ‘double position,’ and also by real algebra. The book contains a large number of problems. The following was proposed to Leonardo of Pisa by a magister in Constantinople, as a difficult problem: If A gets from B 7 denare, then A’s sum is five-fold B’s; if B gets from A 5 denare, then B’s sum is seven-fold A’s. How much has each? The Liber Abaci contains another problem, which is of historical interest, because it was given with some variations by Ahmes, 3000 years earlier: 7 old women go to Rome; each woman has 7 mules, each mule carries 7 sacks, each sack contains 7 loaves, with each loaf are 7 knives, each knife is put up in 7 sheaths. What is the sum total of all named? Ans. 137, 256. [3] In 1220, Leonardo of Pisa published his Practica Ge- ometriæ, which contains all the knowledge of geometry and trigonometry transmitted to him. The writings of Euclid and of some other Greek masters were known to him, either from Arabic manuscripts directly or from the translations made by his countrymen, Gerard of Cremona and Plato of Tivoli. Leonardo’s Geometry contains an elegant geometrical

A HISTORY OF MATHEMATICS. 152 demonstration of Heron’s formula for the area of a triangle, as a function of its three sides. Leonardo treats the rich material before him with skill and Euclidean rigour. Of still greater interest than the preceding works are those containing Fibonacci’s original investigations. We must here preface that after the publication of theLiber Abaci, Leonardo was presented by the astronomer Dominicus to Emperor Frederick II. of Hohenstaufen. On that occasion, John of Palermo, an imperial notary, proposed several problems, which Leonardo solved promptly. The first problem was to find a number x, such that x2 + 5 and x2− 5 are each square numbers. The answer is x = 3 5 12; for (3 5 12)2 + 5 = (4 1 12)2, (3 5 12)2− 5 = (2 7 12)2. His masterly solution of this is given in his liber quadratorum, a copy of which work was sent by him to Frederick II. The problem was not original with John of Palermo, since the Arabs had already solved similar ones. Some parts of Leonardo’s solution may have been borrowed from the Arabs, but the method which he employed of building squares by the summation of odd numbers is original with him. The second problem proposed to Leonardo at the famous scientific tournament which accompanied the presentation of this celebrated algebraist to that great patron of learning, Emperor Frederick II., was the solving of the equation x3 + 2x2 + 10x = 20. As yet cubic equations had not been solved algebraically. Instead of brooding

stubbornly over this knotty problem, and after many failures still entertaining new hopes of success, he changed his method of inquiry and

EUROPE DURING THE MIDDLE AGES. 153 showed by clear and rigorous demonstration that the roots of this equation could not be represented by the Euclidean irrational quantities, or, in other words, that they could not be constructed with the ruler and compass only. He contented himself with finding a very close approximation to the required root. His work on this cubic is found in theFlos, together with the solution of the following third problem given him by John of Palermo: Three men possess in common an unknown sum of money t; the share of the first is t 2; that of the second, t 3; that of the third, t 6. Desirous of depositing the sum at a safer place, each takes at hazard a certain amount; the first takesx, but deposits only x 2; the second carriesy, but deposits only y 3; the third takes z, and deposits z 6. Of the amount deposited each one must receive exactly 1 3, in order to possess his share of the whole sum. Find x, y, z. Leonardo shows the problem to be indeterminate. Assuming 7 for the sum drawn by each from the deposit, he finds t = 47, x = 33, y = 13, z = 1. One would have thought that after so brilliant a beginning, the sciences transplanted from Mohammedan to Christian soil would have enjoyed a steady and vigorous development. But this was not the case. During the fourteenth and fifteenth centuries, the mathematical science was almost stationary. Long wars absorbed the energies of the people and thereby kept back the growth of the sciences. The death of Frederick II. in 1254 was followed by a period of confusion in Germany. The German emperors and the popes were continually quarrelling, and Italy was inevitably drawn into the struggles between the Guelphs and the Ghibellines. France and England were

A HISTORY OF MATHEMATICS. 154 engaged in the Hundred Years' War (1338–1453). Then followed in England the Wars of the Roses. The growth of science was retarded not only by war, but also by the injurious influence of scholastic philosophy. The intellectual leaders of those times quarrelled over subtle subjects in metaphysics and theology. Frivolous questions, such as "How many angels can stand on the point of a needle?" were discussed with great interest. Indistinctness and confusion of ideas characterised the reasoning during this period. Among the mathematical productions of the Middle Ages, the works of Leonardo of Pisa appear to us like jewels among quarry-rubbish. The writers on mathematics during this period were not few in number, but their scientific efforts were vitiated by the method of scholastic thinking. Though they possessed the Elements of Euclid, yet the true nature of a mathematical proof was so little understood, that Hankel believes it no exaggeration to say that "since Fibonacci, not a single proof, not borrowed from Euclid, can be found in the whole literature of these ages, which fulfils all necessary conditions." The only noticeable advance is a simplification of numerical operations and a more extended application of them. Among the Italians are evidences of an early maturity of arithmetic. Peacock [22] says: The Tuscans generally, and the Florentines in particular, whose city was the cradle of the literature and arts of the thirteenth and fourteenth centuries, were celebrated for their knowledge of arithmetic and book-keeping, which were so necessary for their extensive commerce; the Italians were in familiar possession of commercial arithmetic long

EUROPE DURING THE MIDDLE AGES. 155 before the other nations of Europe; to them we are indebted for the formal introduction into books of arithmetic, under distinct heads, of questions in the single and double rule of three, loss and gain, fellowship, exchange, simple and compound interest, discount, and so on. There was also a slow improvement in the algebraic no- tation. The Hindoo algebra possessed a tolerable symbolic notation, which was, however, completely ignored by the Mohammedans. In this respect, Arabic algebra approached much more closely to that of Diophantus, which can scarcely be said to employ symbols in a systematic way. Leonardo of Pisa possessed no algebraic symbolism. Like the Arabs, he expressed the relations of magnitudes to each other by lines or in words. But in the mathematical writings of the monkLuca Pacioli (also called Lucas de Burgo sepulchri) symbols began to appear. They consisted merely in abbreviations of Italian words, such asp for piu (more), m for meno (less), co for cosa (the thing or unknown quantity). “Our present notation has arisen by almost insensible degrees as convenience suggested different marks of abbreviation to different authors; and that perfect symbolic language which addresses itself solely to the eye, and enables us to take in at a glance the most complicated relations of quantity, is the result of a large series of small improvements.” [23] We shall now mention a few authors who lived during the thirteenth and fourteenth and the first half of the fifteenth centuries. About the time of Leonardo of Pisa (1200 a.d.), lived the German monk Jordanus Nemorarius, who wrote

A HISTORY OF MATHEMATICS. 156 a once famous work on the properties of numbers (1496), modelled after the arithmetic of Boethius. The most trifling numeral properties are treated with nauseating pedantry and prolixity. A practical arithmetic based on the Hindoo notation was also written by him. John Halifax (Sacro Bosco, died 1256) taught in Paris and made an extract from the Almagest containing only the most elementary parts of that work. This extract was for nearly 400 years a work of great popularity and standard authority. Other prominent writers are Albertus Magnus and George Purbach in Germany, and Roger Bacon in England. It appears that here and there some of our modern ideas were anticipated by writers of the Middle Ages. Thus, Nicole Oresme, a bishop in Normandy (died 1382), first conceived a notation of fractional powers, afterwards re-discovered by Stevinus, and gave rules for operating with them. His notation was totally different from ours. Thomas Bradwardine, archbishop of Canterbury, studied star-polygons,—a subject which has recently received renewed attention. The first appearance of such polygons was with Pythagoras and his school. We next meet with such polygons in the geometry of Boethius and also in the translation of Euclid from the Arabic by Athelard of Bath. Bradwardine’s philosophic writings contain discussions on the infinite and the infinitesimal— subjects never since lost sight of. To England falls the honour of having produced the earliest European writers on trigonometry. The writings of Bradwardine, of Richard of Wallingford, and John Maudith, both professors at Oxford,

EUROPE DURING THE MIDDLE AGES. 157 and of Simon Bredon of Winchecombe, contain trigonometry drawn from Arabic sources. The works of the Greek monk Maximus Planudes, who lived in the first half of the fourteenth century, are of interest only as showing that the Hindoo numerals were then known in Greece.

A writer belonging, like Planudes, to the Byzantine school, was Moschopulus, who lived in Constantinople in the early part of the fifteenth century. To him appears to be due the introduction into Europe of magic squares. He wrote a treatise on this subject. Magic squares were known to the Arabs, and perhaps to the Hindoos. Mediæval astrologers and physicians believed them to possess mystical properties and to be a charm against plague, when engraved on silver plate. In 1494 wasprinted the Summa de Arithmetica, Geometria, Proportione et Proportionalita, written by the Tuscan monk Lucas Pacioli, who, as we remarked, first introduced symbols in algebra. This contains all the knowledge of his day on arithmetic, algebra, and trigonometry, and is the first comprehensive work which appeared after the Liber Abaci of Fibonacci. It contains little of importance which cannot be found in Fibonacci's great work, published three centuries earlier. [1] Perhaps the greatest result of the influx of Arabic learning was the establishment of universities. What was their attitude toward mathematics? The University of Paris, so famous at the beginning of the twelfth century under the teachings of Abelard, paid but little attention to this science during the

A HISTORY OF MATHEMATICS. 158 Middle Ages. Geometry was neglected, and Aristotle's logic was the favourite study. In 1336, a rule was introduced that no student should take a degree without attending lectures on mathematics, and from a commentary on the first six books of Euclid, dated 1536, it appears that candidates for the degree of A.M. had to give an oath that they had attended lectures on these books. [7] Examinations, when held at all, probably did not extend beyond the first book, as is shown by the nickname "magister matheseos," applied to the Theorem of Pythagoras, the last in the first book. More attention was paid to mathematics at the University of Prague , founded 1384. For the Baccalaureate degree, students were required to take lectures on Sacro Bosco's famous work on astronomy. Of candidates for the A.M. were required not only the six books of Euclid, but an additional knowledge of applied mathematics. Lectures were given on the Almagest. At the University of Leipzig, the daughter of Prague, and atCologne, less work was required, and, as late as the sixteenth century, the same requirements were made at these as at Prague in the fourteenth. The universities of Bologna, Padua, Pisa, occupied similar positions to the ones in Germany, only that purely astrological lectures were given in place of lectures on the Almagest. At Oxford, in the middle of the fifteenth century, the first two books of Euclid were read. [6] Thus it will be seen that the study of mathematics was maintained at the universities only in a half-hearted manner. No great mathematician and teacher appeared, to inspire the students. The best energies of the schoolmen were expended

EUROPE DURING THE MIDDLE AGES. 159 upon the stupid subtleties of their philosophy. The genius of Leonardo of Pisa left no permanent impress upon the age, and another Renaissance of mathematics was wanted.

MODERN EUROPE. We find it convenient to choose the time of the capture of Constantinople by the Turks as the date at which the Middle Ages ended and Modern Times began. In 1453, the Turks battered the walls of this celebrated metropolis with cannon, and finally captured the city; the Byzantine Empire fell, to rise no more.

Calamitous as was this event to the East, it acted favourably upon the progress of learning in the West. A great number of learned Greeks fled into Italy, bringing with them precious manuscripts of Greek literature. This contributed vastly to the reviving of classic learning. Up to this time, Greek masters were known only through the often very corrupt Arabic manuscripts, but now they began to be studied from original sources and in their own language. The first English translation of Euclid was made in 1570 from the Greek by Sir Henry Billingsley , assisted by John Dee . [29] About the middle of the fifteenth century, printing was invented; books became cheap and plentiful; the printing-press transformed Europe into an audience- room. Near the close of the fifteenth century, America was discovered, and, soon after, the earth was circumnavigated. The pulse and pace of the world began to quicken. Men’s minds became less servile; they became clearer and stronger. The indistinctness of thought, which was the characteristic feature of mediæval learning, began to be remedied chiefly by the steady cultivation of Pure Mathematics and Astronomy. 160

THE RENAISSANCE. 161 Dogmatism was attacked; there arose a long struggle with the authority of the Church and the established schools of philosophy. The Copernican System was set up in opposition to the time-honoured Ptolemaic System. The long and eager contest between the two culminated in a crisis at the time of Galileo, and resulted in the victory of the new system. Thus, by slow degrees, the minds of men were cut adrift from their old scholastic moorings and sent forth on the wide sea of scientific inquiry, to discover new islands and continents of truth. THE RENAISSANCE. With the sixteenth century began a period of increased intellectual activity. The human mind made a vast effort to achieve its freedom. Attempts at its emancipation from Church authority had been made before, but they were stifled and rendered abortive. The first great and successful revolt against ecclesiastical authority was made in Germany. The new desire for judging freely and independently in matters of religion was preceded and accompanied by a growing spirit of scientific inquiry. Thus it was that, for a time, Germany led the van in science. She produced Regiomontanus, Copernicus, Rhæticus , Kepler, and Tycho Brahe, at a period when France and England had, as yet, brought forth hardly any great scientific thinkers. This remarkable scientific productiveness was no doubt due, to a great extent, to the commercial prosperity of Germany.

A HISTORY OF MATHEMATICS. 162 Material prosperity is an essential condition for the progress of knowledge. As long as every individual is obliged to collect the necessaries for his subsistence, there can be no leisure for higher pursuits. At this time, Germany had accumulated considerable wealth. The Hanseatic League commanded the trade of the North. Close commercial relations existed between Germany and Italy. Italy, too, excelled in commercial activity and enterprise. We need only mention Venice, whose glory began with the crusades, and Florence, with her bankers and her manufacturers of silk and wool. These two cities became great intellectual centres. Thus, Italy, too, produced men in art, literature, and science, who shone forth in fullest splendour. In fact, Italy was the fatherland of what is termed the Renaissance. For the first great contributions to the mathematical sciences we must, therefore, look to Italy and Germany. In Italy brilliant accessions were made to algebra, in Germany to astronomy and trigonometry. On the threshold

of this new era we meet in Germany with the figure of John Mueller, more generally calledRegiomon- tanus (1436–1476). Chiefly to him we owe the revival of trigonometry. He studied astronomy and trigonometry at Vienna under the celebrated George Purbach. The latter perceived that the existing Latin translations of theAlmagest were full of errors, and that Arabic authors had not remained true to the Greek original. Purbach therefore began to make a translation directly from the Greek. But he did not live to finish it. His work was continued by Regiomontanus, who

THE RENAISSANCE. 163 went beyond his master. Regiomontanus learned the Greek language from Cardinal Bessarion, whom he followed to Italy, where he remained eight years collecting manuscripts from Greeks who had fled thither from the Turks. In addition to the translation of and the commentary on the Almagest, he prepared translations of the Conics of Apollonius, of Archi- medes, and of the mechanical works of Heron. Regiomontanus and Purbach adopted the Hindoo sine in place of the Greek chord of double the arc. The Greeks and afterwards the Arabs divided the radius into60 equal parts, and each of these again into 60 smaller ones. The Hindoos expressed the length of the radius by parts of the circumference, saying that of the21, 600 equal divisions of the latter, it took 3438 to measure the ra- dius. Regiomontanus, to secure greater precision, constructed one table of sines on a radius divided into 600, 000 parts, and another on a radius divided decimally into 10, 000, 000 divisions. He emphasised the use of the tangent in trigono- metry. Following out some ideas of his master, he calculated a table of tangents. German mathematicians were not the first Europeans to use this function. In England it was known a century earlier to Bradwardine, who speaks of tangent (umbra recta) and cotangent (umbra versa), and to John Maudith. Regiomontanus was the author of an arithmetic and also of a complete treatise on trigonometry, containing solutions of both plane and spherical triangles. The form which he gave to trigonometry has been retained, in its main features, to the present day. Regiomontanus ranks among the greatest men that Ger-

A HISTORY OF MATHEMATICS. 164 many has ever produced. His complete mastery of astronomy and mathematics, and his enthusiasm for them, were of far-reaching influence throughout Germany. So great was his reputation, that Pope Sixtus IV. called him to Italy to improve the calendar. Regiomontanus left his beloved city of N¨ urnberg for Rome, where he died in the following year. After the time of Purbach and Regiomontanus, trigono- metry and especially the calculation of tables continued to occupy German scholars. More refined astronomical in- struments were made, which gave observations of greater precision; but these would have been useless without trigono- metrical tables of corresponding accuracy. Of the several tables calculated, that by Georg Joachim of Feldkirch in Tyrol, generally called Rhæticus, deserves special mention. He calculated a table of sines with the radius= 10, 000, 000, 000 and from 10″ to 10″; and, later on, another with the radius = 1 , 000, 000, 000, 000, 000, and proceeding from 10″ to 10″. He began also the construction of tables of tangents and secants, to be carried to the same degree of accuracy; but he died before finishing them. For twelve years he had had in continual employment several calculators. The work was completed by his pupil, Valentine Otho, in 1596. This was indeed a gigantic work,—a monument of German diligence

and indefatigable perseverance. The tables were republished in 1613 by Pitiscus, who spared no pains to free them of errors. Astronomical tables of so great a degree of accu- racy had never been dreamed of by the Greeks, Hindoos, or Arabs. That Rhæticus was not a ready calculator only, is

THE RENAISSANCE. 165 indicated by his views on trigonometrical lines. Up to his time, the trigonometric functions had been considered always with relation to the arc; he was the first to construct the right triangle and to make them depend directly upon its angles. It was from the right triangle that Rhæticus got his idea of calculating the hypotenuse;i.e. he was the first to plan a table of secants. Good work in trigonometry was done also by Vieta and Romanus. We shall now leave the subject of trigonometry to witness the progress in the solution of algebraical equations. To do so, we must quit Germany for Italy. The first comprehensive algebra printed was that of Lucas Pacioli. He closes his book by saying that the solution of the equations x3 + mx = n, x3 + n = mx is as impossible at the present state of science as the quadrature of the circle. This remark doubtless stimulated thought. The first step in the algebraic solution of cubics was taken byScipio Ferro(died 1526), a professor of mathematics at Bologna, who solved the equation x3 + mx = n. Nothing more is known of his discovery than that he imparted it to his pupil, Floridas, in 1505. It was the practice in those days and for two centuries afterwards to keep discoveries secret, in order to secure by that means an advantage over rivals by proposing problems beyond their reach. This practice gave rise to numberless disputes regarding the priority of inventions. A second solution of cubics was given by Nicolo of Brescia (1506(?)–1557). When a boy of six, Nicolo was so badly cut by a French soldier that he never again gained the free use of his tongue. Hence he was called Tartaglia,

A HISTORY OF MATHEMATICS. 166 i.e. the stammerer. His widowed mother being too poor to pay his tuition in school, he learned to read and picked up a knowledge of Latin, Greek, and mathematics by himself. Possessing a mind of extraordinary power, he was able to appear as teacher of mathematics at an early age. In 1530, one Colla proposed him several problems, one leading to the equation x3 + px2 = q. Tartaglia found an imperfect method for solving this, but kept it secret. He spoke about his secret in public and thus caused Ferro's pupil, Floridas, to proclaim his own knowledge of the form x3 + mx = n. Tartaglia, believing him to be a mediocrist and braggart, challenged him to a public discussion, to take place on the 22d of February, 1535. Hearing, meanwhile, that his rival had gotten the method from a deceased master, and fearing that he would be beaten in the contest, Tartaglia put in all the zeal, industry, and skill to find the rule for the equations, and he succeeded in it ten days before the appointed date, as he himself modestly says. [7] The most difficult step was, no doubt, the passing from quadratic irrationals, used in operating from time of old, to cubic irrationals. Placing x = 3√ t− 3√u, Tartaglia perceived that the irrationals disappeared from the equation x3 + mx = n, making n = t− u. But this last equality, together with (1 3 m)3 = tu, gives at once t = √(n 2)3 + (m 3)3 + n 2 , u = √(n 2)2 + (m 2)3 − n 2 . This is Tartaglia's solution of x3 + mx = n. On the 13th of February, he found a similar solution for x3 = mx + n. The contest began on the 22d. Each contestant proposed thirty

THE RENAISSANCE. 167 problems. The one who could solve the greatest number within fifty days should be the victor. Tartaglia solved the thirty problems proposed by Floridas in two hours; Floridas could not solve any of Tartaglia's. From now on, Tartaglia studied cubic equations with a will. In 1541 he discovered a general solution for the cubic x3± px2 =±q, by transforming it into the form x3± mx =±n. The news of Tartaglia's victory spread all over Italy. Tartaglia was entreated to make known his method, but he declined to do so, saying that after his completion of the translation from the Greek of Euclid and Archimedes, he would publish a large algebra containing his method. But a scholar from Milan, named Hieronimo Cardano (1501–1576), after many solicitations, and after giving the most solemn and sacred promises of secrecy, succeeded in obtaining from Tartaglia a knowledge of his rules. At this time Cardan was writing his Ars Magna, and he knew no better way to crown his work than by inserting the much sought for rules for solving cubics. Thus Cardan broke his most solemn vows, and published in 1545 in hisArs Magna Tartaglia's solution of cubics. Tartaglia became desperate. His most cherished hope, of giving to the world an immortal work which should be the monument of his deep learning and power for original research, was suddenly destroyed; for the crown intended for his work had been snatched away. His first step was to write a history of his invention; but, to completely annihilate his enemies, he challenged Cardan and his pupil Lodovico Ferrari to a contest: each party should

A HISTORY OF MATHEMATICS. 168 propose thirty-one questions to be solved by the other within fifteen days. Tartaglia solved most questions in seven days, but the other party did not send in their solution before the expiration of the fifth month; moreover, all their solutions except one were wrong. A replication and a rejoinder followed. Endless were the problems proposed and solved on both sides. The dispute produced much chagrin and heart-burnings to the parties, and to Tartaglia especially, who met with many other disappointments. After having recovered himself again, Tartaglia began, in 1556, the publication of the work which he had had in his mind for so long; but he died before he reached the consideration of cubic equations. Thus the fondest wish of his life remained unfulfilled; the man to whom we owe the greatest contribution to algebra made in the sixteenth century was forgotten, and his method came to be regarded as the discovery of Cardan and to be called Cardan's solution. Remarkable is the great interest that the solution of cubics excited throughout Italy. It is but natural that after this great conquest mathematicians should attack biquadratic equations. As in the case of cubics, so here, the first impulse was given by Colla, who, in 1540, proposed for solution the equation x4 + 6x2 + 36 = 60 x. To be sure, Cardan had studied particular cases as early as 1539. Thus he solved the equation 13x2 = x4 + 2x3 + 2x + 1 by a process similar to that employed by Diophantus and the Hindoos; namely, by adding to both sides 3x2 and thereby rendering both numbers complete squares. But Cardan failed to find a general solution; it remained for his pupil Ferrari to prop

THE RENAISSANCE. 169 the reputation of his master by the brilliant discovery of the general solution of biquadratic equations. Ferrari reduced Colla's equation to the form (x2 + 6)2 = 60x + 6x2. In order to give also the right member the form of a complete square he added to both members the expression 2(x2 + 6)y + y2,

containing a new unknown quantity y. This gave him (x2 + 6 + y)2 = (6 + 2y)x2 + 60x + (12y + y2). The condition that the right member be a complete square is expressed by the cubic equation (2y + 6)(12y + y2) = 900. Extracting the square root of the biquadratic, he gotx2+6+ y = x√2y + 6+ 900√2y + 6. Solving the cubic for y and substituting, it remained only to determine x from the resulting quadratic. Ferrari pursued a similar method with other numerical biquadratic equations. [7] Cardan had the pleasure of publishing this discovery in his Ars Magna in 1545. Ferrari's solution is sometimes ascribed to Bombelli, but he is no more the discoverer of it than Cardan is of the solution called by his name. To Cardan algebra is much indebted. In his Ars Magna he takes notice of negative roots of an equation, calling them fictitious, while the positive roots are called real. Imaginary roots he does not consider; cases where they appear he calls impossible. Cardan also observed the difficulty in the irreducible case in the cubics, which, like the quadrature of the circle, has since "so much tormented the perverse ingenuity of mathematicians." But he did not understand its nature. It remained for Raphael Bombelli of Bologna, who published in 1572 an algebra of great merit, to point out the reality of the apparently imaginary expression which the root assumes,

A HISTORY OF MATHEMATICS. 170 and thus to lay the foundation of a more intimate knowledge of imaginary quantities. After this brilliant success in solving equations of the third and fourth degrees, there was probably no one who doubted, that with aid of irrationals of higher degrees, the solution of equations of any degree whatever could be found. But all attempts at the algebraic solution of the quintic were fruitless, and, finally, Abel demonstrated that all hopes of finding algebraic solutions to equations of higher than the fourth degree were purely Utopian. Since no solution by radicals of equations of higher degrees could be found, there remained nothing else to be done than the devising of rules by which at least the numerical values of the roots could be ascertained. Cardan applied the Hindoo rule of "false position" (called by him regula aurea) to the cubic, but this mode of approximating was exceedingly rough. An incomparably better method was invented byFranciscus Vieta, a French mathematician, whose transcendent genius enriched mathematics with several important innovations. Taking the equation f(x) = Q, wherein f(x) is a polynomial containing different powers of x, with numerical coefficients, and Q is a given number, Vieta first substitutes in f(x) a known approximate value of the root, and then shows that another figure of the root can be obtained by division. A repetition of the same process gives the next figure of the root, and so on. Thus, in x2 + 14x = 7929, taking 80 for the

THE RENAISSANCE. 171 approximate root, and placing x = 80 + b, we get (80 + b)2 + 14(80 + b) = 7929, or 174b + b2 = 409. Since 174b is much greater than b2, we place 174b = 409 , and obtain thereby b = 2. Hence the second approximation is 82. Put x = 82 + c, then (82 + c)2 + 14(82 + c) = 7929 , or 178c + c2 = 57. As before, place 178c = 57, then c = .3, and the third approximation gives 82.3. Assuming x = 82.3 + d, and substituting, gives 178.6d + d2 = 3.51, and 178.6d = 3.51, ∴ d = .01; giving for the fourth approximation 82.31. In the same way, e = .009, and the value for the root of the given equation is 82.319 . . .. For this process, Vieta was greatly admired by his contemporaries. It was employed by Harriot, Oughtred,

Pell, and others. Its principle is identical with the main principle involved in the methods of approximation of Newton and Horner. The only change lies in the arrangement of the work. This alteration was made to afford facility and security in the process of evolution of the root. We pause a moment to sketch the life of Vieta, the most eminent French mathematician of the sixteenth century. He was born in Poitou in 1540, and died in 1603 at Paris. He was employed throughout life in the service of the state, under Henry III. and Henry IV. He was, therefore, not a mathematician by profession, but his love for the science was so great that he remained in his chamber studying, sometimes several days in succession, without eating and sleeping more than was necessary to sustain himself. So great devotion to

A HISTORY OF MATHEMATICS. 172 abstract science is the more remarkable, because he lived at a time of incessant political and religious turmoil. During the war against Spain, Vieta rendered service to Henry IV. by deciphering intercepted letters written in a species of cipher, and addressed by the Spanish Court to their governor of Netherlands. The Spaniards attributed the discovery of the key to magic. An ambassador from Netherlands once told Henry IV. that France did not possess a single geometer capable of solving a problem propounded to geometers by a Belgian mathematician, Adrianus Romanus. It was the solution of the equation of the forty-fifth degree:— 45y− 3795y3 + 95634y5−⋯ + 945y41− 45y43 + y45 = C. Henry IV. called Vieta, who, having already pursued similar investigations, saw at once that this awe-inspiring problem was simply the equation by whichC = 2 sinφ was expressed in terms of y = 2 sin 1 45 φ; that, since 45 = 3·3·5, it was necessary only to divide an angle once into 5 equal parts, and then twice into 3,—a division which could be effected by corresponding equations of the fifth and third degrees. Brilliant was the discovery by Vieta of 23 roots to this equation, instead of only one. The reason why he did not find 45 solutions, is that the remaining ones involve negative sines, which were unintelligible to him. Detailed investigations on the famous old problem of the section of an angle into an odd number of equal parts, led Vieta to the discovery of a trigonometrical solution of Cardan’s irreducible case in cubics. He applied the

THE RENAISSANCE. 173 equation (2 cos 1 3 φ)3− 3 (2 cos 1 3 φ) = 2 cos φ to the solution of x3− 3a2x = a2b, when a > 1 2 b, by placing x = 2a cos 1 3 φ, and determining φ from b = 2a cos φ. The main principle employed by him in the solution of equations is that of reduction. He solves the quadratic by making a suitable substitution which will remove the term containing x to the first degree. Like Cardan, he reduces the general expression of the cubic to the form x3 + mx + n = 0; then, assuming x = (1 3 a− z2) ÷ z and substituting, he gets z6− bz3− 1 27 a3 = 0 . Putting z3 = y, he has a quadratic. In the solution of biquadratics, Vieta still remains true to his principle of reduction. This gives him the well-known cubic resolvent. He thus adheres throughout to his favourite principle, and thereby introduces into algebra a uniformity of method which claims our lively admiration. In Vieta’s algebra we discover a partial knowledge of the relations existing between the coefficients and the roots of an equation. He shows that if the coefficient of the second term in an equation of the second degree is minus the sum of two numbers whose product is the third term, then the two numbers are roots of the equation. Vieta rejected all except

positive roots; hence it was impossible for him to fully perceive the relations in question. The most epoch-making innovation in algebra due to Vieta is the denoting of general or indefinite quantities by letters of the alphabet. To be sure, Regiomontanus and Stifel in Germany, and Cardan in Italy, used letters before him, but Vieta extended the idea and first made it an essential part of

A HISTORY OF MATHEMATICS. 174 algebra. The new algebra was called by himlogistica speciosa in distinction to the oldlogistica numerosa. Vieta's formalism differed considerably from that of to-day. The equation a3 + 3a2b + 3ab2 + b3 = (a + b)3 was written by him "a cubus + b in a quadr. 3 + a in b quadr. 3 + b cubo æqualia a + b cubo." In numerical equations the unknown quantity was denoted by N, its square by Q, and its cube by C. Thus the equation x3− 8x2 + 16x = 40 was written 1C− 8Q + 16N æqual. 40. Observe that exponents and our symbol (=) for equality were not yet in use; but that Vieta employed the Maltese cross (+) as the short-hand symbol for addition, and the (−) for subtraction. These two characters had not been in general use before the time of Vieta. "It is very singular," says Hallam, "that discoveries of the greatest convenience, and, apparently, not above the ingenuity of a village schoolmaster, should have been overlooked by men of extraordinary acuteness like Tartaglia, Cardan, and Ferrari; and hardly less so that, by dint of that acuteness, they dispensed with the aid of these contrivances in which we suppose that so much of the utility of algebraic expression consists." Even after improvements in notation were once proposed, it was with extreme slowness that they were admitted into general use. They were made oftener by accident than design, and their authors had little notion of the effect of the change which they were making. The introduction of the + and− symbols seems to be due to the Germans, who, although they did not enrich algebra during the Renaissance with great inventions, as did the Italians, still cultivated it with great zeal. The arithmetic of

THE RENAISSANCE. 175 John Widmann, printeda.d. 1489 in Leipzig, is the earliest book in which the + and− symbols have been found. There are indications leading us to surmise that they were in use first among merchants. They occur again in the arithmetic of Grammateus, a teacher at the University of Vienna. His pupil, Christoff Rudolff, the writer of the first text-book on algebra in the German language (printed in 1525), employs these symbols also. So did Stifel, who brought out a second edition of Rudolff's Coss in 1553. Thus, by slow degrees, their adoption became universal. There is another short-hand symbol of which we owe the origin to the Germans. In a manuscript published sometime in the fifteenth century, a dot placed before a number is made to signify the extraction of a root of that number. This dot is the embryo of our present symbol for the square root. Christoff Rudolff, in his algebra, remarks that "the radix quadrata is, for brevity, designated in his algorithm with the character √ , as √ 4." Here the dot has grown into a symbol much like our own. This same symbol was used by Michael Stifel . Our sign of equality is due to Robert Recorde (1510–1558), the author ofThe Whetstone of Witte (1557), which is the first English treatise on algebra. He selected this symbol because no two things could be more equal than two parallel lines =. The sign ÷ for division was first used byJohann Heinrich Rahn, a Swiss, in 1659, and was introduced in England by John Pell in 1668. Michael Stifel (1486?–1567), the greatest German alge- braist of

the sixteenth century, was born in Esslingen, and died in Jena. He was educated in the monastery of his native

A HISTORY OF MATHEMATICS. 176 place, and afterwards became Protestant minister. The study of the significance of mystic numbers in Revelation and in Daniel drew him to mathematics. He studied German and Italian works, and published in 1544, in Latin, a book entitled Arithmetica integra. Melanchthon wrote a preface to it. Its three parts treat respectively of rational numbers, irrational numbers, and algebra. Stifel gives a table containing the numerical values of the binomial coefficients for powers below the 18th. He observes an advantage in letting a geometric progression correspond to an arithmetical progression, and arrives at the designation of integral powers by numbers. Here are the germs of the theory of exponents. In 1545 Stifel published an arithmetic in German. His edition of Rudolff's Coss contains rules for solving cubic equations, derived from the writings of Cardan. We remarked above that Vieta discarded negative roots of equations. Indeed, we find few algebraists before and during the Renaissance who understood the significance even of negative quantities. Fibonacci seldom uses them. Pacioli states the rule that "minus times minus gives plus," but applies it really only to the development of the product of (a− b)(c− d); purely negative quantities do not appear in his work. The great German "Cossist" (algebraist), Michael Stifel , speaks as early as 1544 of numbers which are "absurd" or "fictitious below zero," and which arise when "real numbers above zero" are subtracted from zero. Cardan, at last, speaks of a "pure minus"; "but these ideas," says Hankel, "remained sparsely, and until the beginning of

THE RENAISSANCE. 177 the seventeenth century, mathematicians dealt exclusively with absolute positive quantities." The first algebraist who occasionally places a purely negative quantity by itself on one side of an equation, is Harriot in England. As regards the recognition of negative roots, Cardan and Bombelli were far in advance of all writers of the Renaissance, including Vieta. Yet even they mentioned these so-called false or fictitious roots only in passing, and without grasping their real significance and importance. On this subject Cardan and Bombelli had advanced to about the same point as had the Hindoo Bhaskara, who saw negative roots, but did not approve of them. The generalisation of the conception of quantity so as to include the negative, was an exceedingly slow and difficult process in the development of algebra. We shall now consider the history of geometry during the Renaissance. Unlike algebra, it made hardly any progress. The greatest gain was a more intimate knowledge of Greek geometry. No essential progress was made before the time of Descartes. Regiomontanus, Xylander of Augsburg, Tartaglia, Commandinus of Urbino in Italy, Maurolycus, and others, made translations of geometrical works from the Greek.John Wernerof N¨ urnberg published in 1522 the first work on conics which appeared in Christian Europe. Unlike the geometers of old, he studied the sections in relation with the cone, and derived their properties directly from it. This mode of studying the conics was followed byMaurolycus of Messina (1494–1575). The latter is, doubtless, the greatest geometer of the sixteenth century. From the notes of Pappus, he

A HISTORY OF MATHEMATICS. 178 attempted to restore the missing fifth book of Apollonius on maxima and minima. His chief work is his masterly and original treatment of the conic sections, wherein he discusses tangents and asymptotes more fully than Apollonius had done, and applies them to various physical and astronomical problems. The foremost geometrician of Portugal was Nonius; of France, before Vieta, was Peter Ramus, who perished in the massacre of St. Bartholomew. Vieta possessed great familiarity with ancient geometry. The new form which he gave to algebra, by representing general quantities by letters, enabled him to point out more easily how the construction of the roots of cubics depended upon the celebrated ancient problems of the duplication of the cube and the trisection of an angle. He reached the interesting conclusion that the former problem includes the solutions of all cubics in which the radical in Tartaglia's formula is real, but that the latter problem includes only those leading to the irreducible case. The problem of the quadrature of the circle was revived in this age, and was zealously studied even by men of eminence and mathematical ability. The army of circle-squarers became most formidable during the seventeenth century. Among the first to revive this problem was the German Cardinal Nicolaus Cusanus (died 1464), who had the reputation of being a great logician. His fallacies were exposed to full view by Regiomontanus. As in this case, so in others, every quadrator of note raised up an opposing mathematician: Orontius was met by Buteo and Nonius; Joseph Scaliger by

THE RENAISSANCE. 179 Vieta, Adrianus Romanus, and Clavius; A. Quercu by Peter Metius. Two mathematicians of Netherlands, Adrianus Romanus and Ludolph van Ceulen, occupied themselves with approximating to the ratio between the circumference and the diameter. The former carried the value π to 15, the latter to 35, places. The value of π is therefore often named "Ludolph's number." His performance was considered so extraordinary, that the numbers were cut on his tomb-stone in St. Peter's church-yard, at Leyden. Romanus was the one who propounded for solution that equation of the forty-fifth degree solved by Vieta. On receiving Vieta's solution, he at once departed for Paris, to make his acquaintance with so great a master. Vieta proposed to him the Apollonian problem, to draw a circle touching three given circles. "Adrianus Romanus solved the problem by the intersection of two hyperbolas; but this solution did not possess the rigour of the ancient geometry. Vieta caused him to see this, and then, in his turn, presented a solution which had all the rigour desirable." [25] Romanus did much toward simplifying spherical trigonometry by reducing, by means of certain projections, the28 cases in triangles then considered to only six. Mention must here be made of the improvements of the Julian calendar. The yearly determination of the movable feasts had for a long time been connected with an untold amount of confusion. The rapid progress of astronomy led to the consideration of this subject, and many new calendars were proposed. Pope Gregory XIII. convoked a large number of mathematicians, astronomers, and prelates, who decided

A HISTORY OF MATHEMATICS. 180 upon the adoption of the calendar proposed by the Jesuit Lilius Clavius. To rectify the errors of the Julian calendar it was agreed to write in the new calendar the 15th of October immediately after the 4th of October of the year 1582. The Gregorian calendar met with a great deal of opposition both

among scientists and among Protestants. Clavius, who ranked high as a geometer, met the objections of the former most ably and effectively; the prejudices of the latter passed away with time. The passion for the study of mystical properties of num- bers descended from the ancients to the moderns. Much was written on numerical mysticism even by such eminent men as Pacioli and Stifel. The Numerorum Mysteria of Peter Bungus covered 700 quarto pages. He worked with great industry and satisfaction on 666, which is the num- ber of the beast in Revelation (xiii. 18), the symbol of Antichrist. He reduced the name of the 'impious' Martin Luther to a form which may express this formidable number. Placing a = 1 , b = 2 , etc, k = 10 , l = 20 , etc., he finds, after misspelling the name, that M(30)A(1)R(80)T(100)I(9)N(40) L(20)V(200)T(100)E(5)R(80)A(1) constitutes the number re- quired. These attacks on the great reformer were not unprovoked, for his friend, Michael Stifel, the most acute and original of the early mathematicians of Germany, exercised an equal ingenuity in showing that the above number referred to Pope Leo X.,—a demonstration which gave Stifel unspeakable comfort. [22] Astrology also was still a favourite study. It is well known

VIETA TO DESCARTES. 181 that Cardan, Maurolycus, Regiomontanus, and many other eminent scientists who lived at a period even later than this, engaged in deep astrological study; but it is not so generally known that besides the occult sciences already named, men engaged in the mystic study of star-polygons and magic squares. "The pentagramma gives you pain," says Faust to Mephistopheles. It is of deep psychological interest to see scientists, like the great Kepler, demonstrate on one page a theorem on star-polygons, with strict geometric rigour, while on the next page, perhaps, he explains their use as amulets or in conjurations. [1] Playfair, speaking of Cardan as an astrologer, calls him "a melancholy proof that there is no folly or weakness too great to be united to high intellectual attainments." [26] Let our judgment not be too harsh. The period under consideration is too near the Middle Ages to admit of complete emancipation from mysticism even among scientists. Scholars like Kepler, Napier, Albrecht D¨ urer, while in the van of progress and planting one foot upon the firm ground of truly scientific inquiry, were still resting with the other foot upon the scholastic ideas of preceding ages. VIETA TO DESCARTES. The ecclesiastical power, which in the ignorant ages was an unmixed benefit, in more enlightened ages became a serious evil. Thus, in France, during the reigns preced- ing that of Henry IV., the theological spirit predominated. This is painfully shown by the massacres of Vassy and of

A HISTORY OF MATHEMATICS. 182 St. Bartholomew. Being engaged in religious disputes, people had no leisure for science and for secular literature. Hence, down to the time of Henry IV., the French "had not put forth a single work, the destruction of which would now be a loss to Europe." In England, on the other hand, no religious wars were waged. The people were comparatively indifferent about religious strifes; they concentrated their ability upon secular matters, and acquired, in the sixteenth century, a literature which is immortalised by the genius of Shakespeare and Spenser. This great literary age in England was followed by a great scientific age. At the close of the sixteenth cen- tury, the shackles of ecclesiastical authority were thrown off by France. The ascension of Henry IV. to the throne was followed in 1598

by the Edict of Nantes, granting freedom of worship to the Huguenots, and thereby terminating religious wars. The genius of the French nation now began to blossom. Cardinal Richelieu, during the reign of Louis XIII., pursued the broad policy of not favouring the opinions of any sect, but of promoting the interests of the nation. His age was remarkable for the progress of knowledge. It produced that great secular literature, the counterpart of which was found in England in the sixteenth century. The seventeenth century was made illustrious also by the great French mathematicians, Roberval, Descartes, Desargues, Fermat, and Pascal. More gloomy is the picture in Germany. The great changes which revolutionised the world in the sixteenth century, and which led England to national greatness, led Germany to degradation. The first effects of the Reformation there

VIETA TO DESCARTES. 183 were salutary. At the close of the fifteenth and during the sixteenth century, Germany had been conspicuous for her scientific pursuits. She had been the leader in astronomy and trigonometry. Algebra also, excepting for the discoveries in cubic equations, was, before the time of Vieta, in a more advanced state there than elsewhere. But at the beginning of the seventeenth century, when the sun of science began to rise in France, it set in Germany. Theologic disputes and religious strife ensued. The Thirty Years' War (1618–1648) proved ruinous. The German empire was shattered, and became a mere lax confederation of petty despotisms. Commerce was destroyed; national feeling died out. Art disappeared, and in literature there was only a slavish imitation of French artificiality. Nor did Germany recover from this low state for 200 years; for in 1756 began another struggle, the Seven Years' War, which turned Prussia into a wasted land. Thus it followed that at the beginning of the seventeenth century, the great Kepler was the only German mathematician of eminence, and that in the interval of200 years between Kepler and Gauss, there arose no great mathematician in Germany excepting Leibniz. Up to the seventeenth century, mathematics was cultivated but little in Great Britain. During the sixteenth century, she brought forth no mathematician comparable with Vieta, Stifel, or Tartaglia. But with the time of Recorde, the English became conspicuous for numerical skill. The first important arithmetical work of English authorship was published in Latin in 1522 by Cuthbert Tonstall (1474–1559). He had

A HISTORY OF MATHEMATICS. 184 studied at Oxford, Cambridge, and Padua, and drew freely from the works of Pacioli and Regiomontanus. Reprints of his arithmetic appeared in England and France. After Recorde the higher branches of mathematics began to be studied. Later, Scotland brought forth Napier, the inventor of logarithms. The instantaneous appreciation of their value is doubtless the result of superiority in calculation. In Italy, and especially in France, geometry, which for a long time had been an almost stationary science, began to be studied with success. Galileo, Torricelli, Roberval, Fermat, Desargues, Pascal, Descartes, and the English Wallis are the great revolutioners of this science. Theoretical mechanics began to be studied. The foundations were laid by Fermat and Pascal for the theory of numbers and the theory of probability. We shall first consider the improvements made in the art of calculating. The nations of antiquity experimented thousands of years upon numeral notations before they happened to strike upon the so-called "Arabic

notation.” In the simple expedient of the cipher, which was introduced by the Hindoos about the fifth or sixth century after Christ, mathematics received one of the most powerful impulses. It would seem that after the “Arabic notation” was once thoroughly understood, decimal fractions would occur at once as an obvious extension of it. But “it is curious to think how much science had attempted in physical research and how deeply numbers had been pondered, before it was perceived that the all-powerful simplicity of the ‘Arabic notation’ was as valuable and as manageable in an infinitely descending as

VIETA TO DESCARTES. 185 in an infinitely ascending progression.” [28] Simple as decimal fractions appear to us, the invention of them is not the result of one mind or even of one age. They came into use by almost imperceptible degrees. The first mathematicians identified with their history did not perceive their true nature and importance, and failed to invent a suitable notation. The idea of decimal fractions makes its first appearance in methods for approximating to the square roots of numbers. Thus John of Seville, presumably in imitation of Hindoo rules, adds 2 n ciphers to the number, then finds the square root, and takes this as the numerator of a fraction whose denominator is 1 followed by n ciphers. The same method was followed by Cardan, but it failed to be generally adopted even by his Italian contemporaries; for otherwise it would certainly have been at least mentioned by Cataldi (died 1626) in a work devoted exclusively to the extraction of roots. Cataldi finds the square root by means of continued fractions—a method ingenious and novel, but for practical purposes inferior to Cardan’s. Orontius Finaeus (died 1555) in France, and William Buckley (died about 1550) in England extracted the square root in the same way as Cardan and John of Seville. The invention of decimals is frequently attributed to Regiomontanus, on the ground that instead of placing the sinus totus, in trigonometry, equal to a multiple of60, like the Greeks, he put it= 100, 000. But here the trigonometrical lines were expressed in integers, and not in fractions. Though he adopted a decimal division of the radius, he and his successors did not apply the idea outside of trigonometry and, indeed,

A HISTORY OF MATHEMATICS. 186 had no notion whatever of decimal fractions. To Simon Stevin of Bruges in Belgium (1548–1620), a man who did a great deal of work in most diverse fields of science, we owe the first systematic treatment of decimal fractions. In his La Disme (1585) he describes in very express terms the advantages, not only of decimal fractions, but also of the decimal division in systems of weights and measures. Stevin applied the new fractions “to all the operations of ordinary arithmetic.” [25] What he lacked was a suitable notation. In place of our decimal point, he used a cipher; to each place in the fraction was attached the corresponding index. Thus, in his notation, the number 5.912 would be 0 5 1 9 2 1 3 2 or 5①0 9①1 1①2 2①3 . These indices, though cumbrous in practice, are of interest, because they are the germ of an important innovation. To Stevin belongs the honour of inventing our present mode of designating powers and also of introducing fractional exponents into algebra. Strictly speaking, this had been done much earlier by Oresme, but it remained wholly unnoticed. Not even Stevin’s innovations were immediately appreciated or at once accepted, but, unlike Oresme’s, they remained a secure possession. No improvement was made in the notation of decimals till the

beginning of the seventeenth century. After Stevin, decimals were used by Joost B¨ urgi, a Swiss by birth, who prepared a manuscript on arithmetic soon after 1592, and by Johann Hartmann Beyer , who assumes the invention as his own. In 1603, he published at Frankfurt on the Main a Logistica Decimalis. With B¨ urgi, a zero placed underneath the digit in unit's place answers as

VIETA TO DESCARTES. 187 sign of separation. Beyer's notation resembles Stevin's. The decimal point, says Peacock, is due to Napier, who in 1617 published his Rabdologia, containing a treatise on decimals, wherein the decimal point is used in one or two instances. In the English translation of Napier's Mirifici logarithmorum canonis descriptio, executed by Edward Wright in 1616, and corrected by the author, the decimal point occurs in the tables. There is no mention of decimals in English arithmetics between 1619 and 1631. Oughtred in 1631 designates the fraction .56 thus, 0 56. Albert Girard, a pupil of Stevin, in 1629 uses the point on one occasion. John Wallis in 1657 writes 12 345 , but afterwards in his algebra adopts the usual point. De Morgan says that "to the first quarter of the eighteenth century we must refer not only the complete and final victory of the decimal point, but also that of the now universal method of performing the operations of division and extraction of the square root. [27] We have dwelt at some length on the progress of the decimal notation, because "the history of language . . . is of the highest order of interest, as well as utility: its suggestions are the best lesson for the future which a reflecting mind can have." [27] The miraculous powers of modern calculation are due to three inventions: the Arabic Notation, Decimal Fractions, and Logarithms. The invention of logarithms in the first quarter of the seventeenth century was admirably timed, for Kepler was then examining planetary orbits, and Galileo had just turned the telescope to the stars. During the Renaissance German mathematicians had constructed trigonometrical

A HISTORY OF MATHEMATICS. 188 tables of great accuracy, but this greater precision enormously increased the work of the calculator. It is no exaggeration to say that the invention of logarithms "by shortening the labours doubled the life of the astronomer." Logarithms were invented byJohn Napier, Baron of Merchiston, in Scotland (1550–1617). It is one of the greatest curiosities of the history of science that Napier constructed logarithms before exponents were used. To be sure, Stifel and Stevin made some attempts to denote powers by indices, but this notation was not generally known,—not even to Harriot, whose algebra appeared long after Napier's death. That logarithms flow naturally from the exponential symbol was not observed until much later. It was Euler who first considered logarithms as being indices of powers. What, then, was Napier's line of thought? Let AB be a definite line, DE a line extending from D indefinitely. Imagine two points starting at the same A BC D EF moment; the one mov- ing from A toward B, the other from D to- ward E. Let the ve- locity during the first moment be the same for both: let that of the point on lineDE be uniform; but the velocity of the point on AB decreasing in such a way that when it arrives at any pointC, its velocity is proportional to the remaining distance BC. While the first point moves over a distance AC, the second one moves over a distance DF . Napier calls DF the logarithm of BC.

VIETA TO DESCARTES. 189 Napier’s process is so unique and so different from all other modes of presenting the subject that there cannot be the shadow of a doubt that this invention is entirely his own; it is the result of unaided, isolated speculation. He first sought the logarithms only of sines; the line AB was the sine of 90◦ and was taken = 10 7; BC was the sine of the arc, and DF its logarithm. We notice that as the motion proceeds, BC decreases in geometrical progression, while DF increases in arithmetical progression. Let AB = a = 10 7, let x = DF , y = BC, then AC = a− y. The velocity of the point C is d(a− y) dt = y; this gives− nat. log y = t + c. When t = 0, then y = a and c =− nat. log a. Again, let dx dt = a be the velocity of the point F, then x = at. Substituting for t and c their values and remembering that a = 107 and that by definition x = Nap. log y, we get Nap. log y = 107 nat. log 107 y . It is evident from this formula that Napier’s logarithms are not the same as the natural logarithms. Napier’s logarithms increase as the number itself decreases. He took the logarithm of sin 90 = 0; i.e. the logarithm of 107 = 0. The logarithm of sin α increased from zero as α decreased from 90◦. Napier’s genesis of logarithms from the conception of two flowing points reminds us of Newton’s doctrine of fluxions. The relation between geometric and arithmetical progressions, so skilfully utilised by Napier, had been observed by Archimedes, Stifel, and others. Napier did not determine the base to his system of logarithms. The notion of a “base” in fact never suggested

A HISTORY OF MATHEMATICS. 190 itself to him. The one demanded by his reasoning is the reciprocal of that of the natural system, but such a base would not reproduce accurately all of Napier’s figures, owing to slight inaccuracies in the calculation of the tables. Napier’s great invention was given to the world in 1614 in a work entitled Mirifici logarithmorum canonis descriptio. In it he explained the nature of his logarithms, and gave a logarithmic table of the natural sines of a quadrant from minute to minute. Henry Briggs (1556–1631), in Napier’s time professor of geometry at Gresham College, London, and afterwards professor at Oxford, was so struck with admiration of Napier’s book, that he left his studies in London to do homage to the Scottish philosopher. Briggs was delayed in his journey, and Napier complained to a common friend, “Ah, John, Mr. Briggs will not come.” At that very moment knocks were heard at the gate, and Briggs was brought into the lord’s chamber. Almost one-quarter of an hour was spent, each beholding the other without speaking a word. At last Briggs began: “My lord, I have undertaken this long journey purposely to see your person, and to know by what engine of wit or ingenuity you came first to think of this most excellent help in astronomy, viz. the logarithms; but, my lord, being by you found out, I wonder nobody found it out before, when now known it is so easy.” [28] Briggs suggested to Napier the advantage that would result from retaining zero for the logarithm of the whole sine, but choosing 10, 000, 000, 000 for the logarithm of the10th part of that same sine, i.e. of 5◦ 44′22′′. Napier said that he had already thought

VIETA TO DESCARTES. 191 of the change, and he pointed out a slight improvement on Briggs’ idea; viz. that zero should be the logarithm of 1, and 10, 000, 000, 000 that of the whole sine, thereby making the characteristic of numbers greater than unity positive and not negative, as suggested by Briggs. Briggs admitted this to be more convenient. The invention of “Briggian logarithms” occurred, therefore, to

Briggs and Napier independently. The great practical advantage of the new system was that its fundamental progression was accommodated to the base, 10, of our numerical scale. Briggs devoted all his energies to the construction of tables upon the new plan. Napier died in 1617, with the satisfaction of having found in Briggs an able friend to bring to completion his unfinished plans. In 1624 Briggs published hisArithmetica logarithmica, containing the logarithms to 14 places of numbers, from1 to 20, 000 and from 90, 000 to 100, 000. The gap from 20, 000 to 90, 000 was filled up by that illustrious successor of Napier and Briggs,Adrian Vlacq of Gouda in Holland. He published in 1628 a table of logarithms from 1 to 100, 000, of which 70, 000 were calculated by himself. The first publication of Briggian logarithms of trigonometric functions was made in 1620 by Gunter, a colleague of Briggs, who found the logarithmic sines and tangents for every minute to seven places. Gunter was the inventor of the words cosine and cotangent. Briggs devoted the last years of his life to calculating more extensive Briggian logarithms of trigonometric functions, but he died in 1631, leaving his work unfinished. It was carried on by the English Henry Gellibrand, and then published by Vlacq at his own

A HISTORY OF MATHEMATICS. 192 expense. Briggs divided a degree into 100 parts, but owing to the publication by Vlacq of trigonometrical tables constructed on the old sexagesimal division, Briggs' innovation remained unrecognised. Briggs and Vlacq published four fundamental works, the results of which "have never been superseded by any subsequent calculations." The first logarithms upon thenatural base e were published by John Speidell in his New Logarithmes (London, 1619), which contains the natural logarithms of sines, tangents, and secants. The only possible rival of John Napier in the invention of logarithms was the Swiss Justus Byrgius (Joost B¨ urgi). He published a rude table of logarithms six years after the appearance of the Canon Mirificus , but it appears that he conceived the idea and constructed that table as early, if not earlier, than Napier did his. But he neglected to have the results published until Napier's logarithms were known and admired throughout Europe. Among the various inventions of Napier to assist the memory of the student or calculator, is "Napier's rule of circular parts" for the solution of spherical right triangles. It is, perhaps, "the happiest example of artificial memory that is known." The most brilliant conquest in algebra during the sixteenth century had been the solution of cubic and biquadratic equations. All attempts at solving algebraically equations of higher degrees remaining fruitless, a new line of inquiry—the properties of equations and their roots—was gradually opened up. We have seen that Vieta had attained a partial knowledge

VIETA TO DESCARTES. 193 of the relations between roots and coefficients. Peletarius, a Frenchman, had observed as early as 1558, that the root of an equation is a divisor of the last term. One who extended the theory of equations somewhat further than Vieta, wasAlbert Girard (1590–1634), a Flemish mathematician. Like Vieta, this ingenious author applied algebra to geometry, and was the first who understood the use of negative roots in the solution of geometric problems. He spoke of imaginary quantities; inferred by induction that every equation has as many roots as there are units in the number expressing its degree; and first showed how to express the sums of their powers in terms of the coefficients. Another algebraist of

considerable power was the English Thomas Harriot(1560–1621). He accompanied the first colony sent out by Sir Walter Raleigh to Virginia. After having surveyed that country he returned to England. As a mathematician, he was the boast of his country. He brought the theory of equations under one comprehensive point of view by grasping that truth in its full extent to which Vieta and Girard only approximated; viz. that in an equation in its simplest form, the coefficient of the second term with its sign changed is equal to the sum of the roots; the coefficient of the third is equal to the sum of the products of every two of the roots; etc. He was the first to decompose equations into their simple factors; but, since he failed to recognise imaginary and even negative roots, he failed also to prove that every equation could be thus decomposed. Harriot made some changes in algebraic notation, adopting small letters of the alphabet in place of the capitals used by Vieta. The symbols of inequality

A HISTORY OF MATHEMATICS. 194 > and < were introduced by him. Harriot's work, Artis Analyticæ praxis, was published in 1631, ten years after his death. William Oughtred (1574–1660) contributed vastly to the propagation of mathematical knowledge in England by his treatises, which were long used in the universities. He introduced× as symbol of multiplication, and :: as that of proportion. By him ratio was expressed by only one dot. In the eighteenth century Christian Wolf secured the general adoption of the dot as a symbol of multiplication, and the sign for ratio was thereupon changed to two dots. Oughtred's ministerial duties left him but little time for the pursuit of mathematics during daytime, and evenings his economical wife denied him the use of a light. Algebra was now in a state of sufficient perfection to enable Descartes to take that important step which forms one of the grand epochs in the history of mathematics,—the application of algebraic analysis to define the nature and investigate the properties of algebraic curves. In geometry, the determination of the areas of curvilinear figures was diligently studied at this period. Paul Guldin (1577–1643), a Swiss mathematician of considerable note, re-discovered the following theorem, published in hisCentro- baryca, which has been named after him, though first found in the Mathematical Collections of Pappus: The volume of a solid of revolution is equal to the area of the generating figure, multiplied by the circumference described by the centre of gravity. We shall see that this method excels that of Kepler and Cavalieri in following a more exact and natural course;

VIETA TO DESCARTES. 195 but it has the disadvantage of necessitating the determination of the centre of gravity, which in itself may be a more difficult problem than the original one of finding the volume. Guldin made some attempts to prove his theorem, but Cavalieri pointed out the weakness of his demonstration. Johannes Kepler (1571–1630) was a native of W¨ urtem- berg and imbibed Copernican principles while at the Uni- versity of T¨ ubingen. His pursuit of science was repeatedly interrupted by war, religious persecution, pecuniary embar- rassments, frequent changes of residence, and family troubles. In 1600 he became for one year assistant to the Danish as- tronomer, Tycho Brahe, in the observatory near Prague. The relation between the two great astronomers was not always of an agreeable character. Kepler's publications are voluminous. His first attempt to explain the solar system was made in 1596, when he thought he had discovered a curious relation between the five regular solids and the number and distance of the planets. The

publication of this pseudo-discovery brought him much fame. Maturer reflection and intercourse with Tycho Brahe and Galileo led him to investigations and results more worthy of his genius—“Kepler’s laws.” He enriched pure mathematics as well as astronomy. It is not strange that he was interested in the mathematical science which had done him so much service; for “if the Greeks had not cultivated conic sections, Kepler could not have superseded Ptolemy.” [11] The Greeks never dreamed that these curves would ever be of practical use; Aristæus and Apollonius studied them merely to satisfy their intellectual cravings after

A HISTORY OF MATHEMATICS. 196 the ideal; yet the conic sections assisted Kepler in tracing the march of the planets in their elliptic orbits. Kepler made also extended use of logarithms and decimal fractions, and was enthusiastic in diffusing a knowledge of them. At one time, while purchasing wine, he was struck by the inaccuracy of the ordinary modes of determining the contents of kegs. This led him to the study of the volumes of solids of revolution and to the publication of the Stereometria Doliorum in 1615. In it he deals first with the solids known to Archimedes and then takes up others. Kepler introduced a new idea into geometry; namely, that of infinitely great and infinitely small quantities. Greek mathematicians always shunned this notion, but with it modern mathematicians have completely revolutionised the science. In comparing rectilinear figures, the method of superposition was employed by the ancients, but in comparing rectilinear and curvilinear figures with each other, this method failed because no addition or subtraction of rectilinear figures could ever produce curvilinear ones. To meet this case, they devised the Method of Exhaustion, which was long and difficult; it was purely synthetical, and in general required that the conclusion should be known at the outset. The new notion of infinity led gradually to the invention of methods immeasurably more powerful. Kepler conceived the circle to be composed of an infinite number of triangles having their common vertices at the centre, and their bases in the circumference; and the sphere to consist of an infinite number of pyramids. He applied conceptions of this kind to the determination of the areas and volumes of figures generated

VIETA TO DESCARTES. 197 by curves revolving about any line as axis, but succeeded in solving only a few of the simplest out of the 84 problems which he proposed for investigation in hisStereometria. Other points of mathematical interest in Kepler’s works are (1) the statement of the earliest problem of inverse tangents; (2) an investigation which amounts to the evaluation of the definite integral $\int \varphi\ 0 \sin \varphi\ d\varphi = 1 - \cos \varphi$; (3) the assertion that the circumference of an ellipse, whose axes are 2a and 2b, is nearly $\pi(a + b)$; (4) a passage from which it has been inferred that Kepler knew the variation of a function near its maximum value to disappear; (5) the assumption of the principle of continuity (which differentiates modern from ancient geometry), when he shows that a parabola has a focus at infinity, that lines radiating from this “cæcus focus” are parallel and have no other point at infinity. The Stereometria led Cavalieri, an Italian Jesuit, to the consideration of infinitely small quantities. Bonaventura Cavalieri (1598–1647), a pupil of Galileo and professor at Bologna, is celebrated for his Geometria indivisibilibus continuorum nova quadam ratione promota , 1635. This work expounds his method of Indivisibles, which

occupies an intermediate place between the method of exhaustion of the Greeks and the methods of Newton and Leibniz. He considers lines as composed of an infinite number of points, surfaces as composed of an infinite number of lines, and solids of an infinite number of planes. The relative magnitude of two solids or surfaces could then be found simply by the summation of

A HISTORY OF MATHEMATICS. 198 series of planes or lines. For example, he finds the sum of the squares of all lines making up a triangle equal to one-third the sum of the squares of all lines of a parallelogram of equal base and altitude; for if in a triangle, the first line at the apex be 1, then the second is 2, the third is 3, and so on; and the sum of their squares is 12 + 22 + 32 +⋯ + n2 = n(n + 1)(2n + 1)÷ 6. In the parallelogram, each of the lines is n and their number is n; hence the total sum of their squares is n3. The ratio between the two sums is therefore n(n + 1)(2n + 1)÷ 6n3 = 1 3 , since n is infinite. From this he concludes that the pyramid or cone is respectively 1 3 of a prism or cylinder of equal base and altitude, since the polygons or circles composing the former decrease from the base to the apex in the same way as the squares of the lines parallel to the base in a triangle decrease from base to apex. By the Method of Indivisibles, Cavalieri solved the majority of the problems proposed by Kepler. Though expeditious and yielding correct results, Cavalieri's method lacks a scientific foundation. If a line has absolutely no width, then no number, however great, of lines can ever make up an area; if a plane has no thickness whatever, then even an infinite number of planes cannot form a solid. The reason why this method led to correct conclusions is that one area is to another area in the same ratio as the sum of the series of lines in the one is to the sum of the series of lines in the other.

VIETA TO DESCARTES. 199 Though unscientific, Cavalieri's method was used for fifty years as a sort of integral calculus. It yielded solutions to some difficult problems. Guldin made a severe attack on Cavalieri and his method. The latter published in 1647, after the death of Guldin, a treatise entitled Exercitationes geometricæ sex, in which he replied to the objections of his opponent and attempted to give a clearer explanation of his method. Guldin had never been able to demonstrate the theorem named after him, except by metaphysical reasoning, but Cavalieri proved it by the method of indivisibles. A revised edition of the Geometry of Indivisibles appeared in 1653. There is an important curve, not known to the ancients, which now began to be studied with great zeal. Roberval gave it the name of "trochoid," Pascal the name of "roulette," Galileo the name of "cycloid." The invention of this curve seems to be due to Galileo, who valued it for the graceful form it would give to arches in architecture. He ascertained its area by weighing paper figures of the cycloid against that of the generating circle, and found thereby the first area to be nearly but not exactly thrice the latter. A mathematical determination was made by his pupil,Evangelista Torricelli (1608–1647), who is more widely known as a physicist than as a mathematician. By the Method of Indivisibles he demonstrated its area to be triple that of the revolving circle, and published his solution. This same quadrature had been effected a few years earlier by Roberval in France, but his solution was not known to the Italians. Roberval, being a man of irritable and violent

A HISTORY OF MATHEMATICS. 200 disposition, unjustly accused the mild and amiable Torricelli of stealing the proof. This accusation of plagiarism created so much chagrin with Torricelli that it is considered to have been the cause of his early death. Vincenzo Viviani, another prominent pupil of Galileo, determined the tangent to the cycloid. This was accomplished in France by Descartes and Fermat. In France, where geometry began to be cultivated with greatest success, Roberval, Fermat, Pascal, employed the Method of Indivisibles and made new improvements in it. Giles Persone de Roberval (1602–1675), for forty years professor of mathematics at the College of France in Paris, claimed for himself the invention of the Method of Indivisibles. Since his complete works were not published until after his death, it is difficult to settle questions of priority. Montucla and Chasles are of the opinion that he invented the method independent of and earlier than the Italian geometer, though the work of the latter was published much earlier than Roberval's. Marie finds it difficult to believe that the Frenchman borrowed nothing whatever from the Italian, for both could not have hit independently upon the word Indivisibles, which is applicable to infinitely small quantities, as conceived by Cavalieri, but not as conceived by Roberval. Roberval and Pascal improved the rational basis of the Method of Indivisibles, by considering an area as made up of an indefinite number of rectangles instead of lines, and a solid as composed of indefinitely small solids instead of surfaces. Roberval applied the method to the finding of areas,

VIETA TO DESCARTES. 201 volumes, and centres of gravity. He effected the quadrature of a parabola of any degree $ym = am-1x$, and also of a parabola $ym = am-nxn$. We have already mentioned his quadrature of the cycloid. Roberval is best known for his method of drawing tangents. He was the first to apply motion to the resolution of this important problem. His method is allied to Newton's principle of fluxions. Archimedes conceived his spiral to be generated by a double motion. This idea Roberval extended to all curves. Plane curves, as for instance the conic sections, may be generated by a point acted upon by two forces, and are the resultant of two motions. If at any point of the curve the resultant be resolved into its components, then the diagonal of the parallelogram determined by them is the tangent to the curve at that point. The greatest difficulty connected with this ingenious method consisted in resolving the resultant into components having the proper lengths and directions. Roberval did not always succeed in doing this, yet his new idea was a great step in advance. He broke off from the ancient definition of a tangent as a straight line having only one point in common with a curve,—a definition not valid for curves of higher degrees, nor apt even in curves of the second degree to bring out the properties of tangents and the parts they may be made to play in the generation of the curves. The subject of tangents received special attention also from Fermat, Descartes, and Barrow, and reached its highest development after the invention of the differential calculus. Fermat and Descartes defined tangents as secants whose two points of intersection with the curve coincide;

A HISTORY OF MATHEMATICS. 202 Barrow considered a curve a polygon, and called one of its sides produced a tangent. A profound scholar in all branches of learning and a mathematician of exceptional powers wasPierre de Fermat (1601–1665). He studied law at Toulouse, and in 1631 was made councillor for the

parliament of Toulouse. His leisure time was mostly devoted to mathematics, which he studied with irresistible passion. Unlike Descartes and Pascal, he led a quiet and unaggressive life. Fermat has left the impress of his genius upon all branches of mathematics then known. A great contribution to geometry was his De maximis et minimis . About twenty years earlier, Kepler had first observed that the increment of a variable, as, for instance, the ordinate of a curve, is evanescent for values very near a maximum or a minimum value of the variable. Developing this idea, Fermat obtained his rule for maxima and minima. He substituted x + e for x in the given function ofx and then equated to each other the two consecutive values of the function and divided the equation by e. If e be taken 0, then the roots of this equation are the values ofx, making the function a maximum or a minimum. Fermat was in possession of this rule in 1629. The main difference between it and the rule of the differential calculus is that it introduces the indefinite quantity e instead of the infinitely small dx. Fermat made it the basis for his method of drawing tangents. Owing to a want of explicitness in statement, Fermat's method of maxima and minima, and of tangents, was severely attacked by his great contemporary, Descartes, who could

VIETA TO DESCARTES. 203 never be brought to render due justice to his merit. In the ensuing dispute, Fermat found two zealous defenders in Roberval and Pascal, the father; while Mydorge, Desargues, and Hardy supported Descartes. Since Fermat introduced the conception of infinitely small differences between consecutive values of a function and arrived at the principle for finding the maxima and minima, it was maintained by Lagrange, Laplace, and Fourier, that Fermat may be regarded as the first inventor of the differential calculus. This point is not well taken, as will be seen from the words of Poisson, himself a Frenchman, who rightly says that the differential calculus "consists in a system of rules proper for finding the differentials of all functions, rather than in the use which may be made of these infinitely small variations in the solution of one or two isolated problems." A contemporary mathematician, whose genius excelled even that of the great Fermat, was Blaise Pascal (1623–1662). He was born at Clermont in Auvergne. In 1626 his father retired to Paris, where he devoted himself to teaching his son, for he would not trust his education to others. Blaise Pascal's genius for geometry showed itself when he was but twelve years old. His father was well skilled in mathematics, but did not wish his son to study it until he was perfectly acquainted with Latin and Greek. All mathematical books were hidden out of his sight. The boy once asked his father what mathematics treated of, and was answered, in general, "that it was the method of making figures with exactness, and of finding out what proportions they relatively had to one

A HISTORY OF MATHEMATICS. 204 another." He was at the same time forbidden to talk any more about it, or ever to think of it. But his genius could not submit to be confined within these bounds. Starting with the bare fact that mathematics taught the means of making figures infallibly exact, he employed his thoughts about it and with a piece of charcoal drew figures upon the tiles of the pavement, trying the methods of drawing, for example, an exact circle or equilateral triangle. He gave names of his own to these figures and then formed axioms, and, in short, came to

make perfect demonstrations. In this way he arrived unaided at the theorem that the sum of the three angles of a triangle is equal to two right angles. His father caught him in the act of studying this theorem, and was so astonished at the sublimity and force of his genius as to weep for joy. The father now gave him Euclid’s Elements, which he, without assistance, mastered easily. His regular studies being languages, the boy employed only his hours of amusement on the study of geometry, yet he had so ready and lively a penetration that, at the age of sixteen, he wrote a treatise upon conics, which passed for such a surprising effort of genius, that it was said nothing equal to it in strength had been produced since the time of Archimedes. Descartes refused to believe that it was written by one so young as Pascal. This treatise was never published, and is now lost. Leibniz saw it in Paris and reported on a portion of its contents. The precocious youth made vast progress in all the sciences, but the constant application at so tender an age greatly impaired his health. Yet he continued working, and at nineteen invented his famous machine for performing

VIETA TO DESCARTES. 205 arithmetical operations mechanically. This continued strain from overwork resulted in a permanent indisposition, and he would sometimes say that from the time he was eighteen, he never passed a day free from pain. At the age of twenty-four he resolved to lay aside the study of the human sciences and to consecrate his talents to religion. His Provincial Letters against the Jesuits are celebrated. But at times he returned to the favourite study of his youth. Being kept awake one night by a toothache, some thoughts undesignedly came into his head concerning the roulette or cycloid; one idea followed another; and he thus discovered properties of this curve even to demonstration. A correspondence between him and Fermat on certain problems was the beginning of the theory of probability. Pascal’s illness increased, and he died at Paris at the early age of thirty-nine years. [30] By him the answer to the objection to Cavalieri’s Method of Indivisibles was put in the clearest form. Like Roberval, he explained “the sum of right lines” to mean “the sum of infinitely small rectangles.” Pascal greatly advanced the knowledge of the cycloid. He determined the area of a section produced by any line parallel to the base; the volume generated by it revolving around its base or around the axis; and, finally, the centres of gravity of these volumes, and also of half these volumes cut by planes of symmetry. Before publishing his results, he sent, in 1658, to all mathematicians that famous challenge offering prizes for the first two solutions of these problems. Only Wallis and A. La Lou` ere competed for them. The latter was quite unequal to the task; the former, being pressed for time, made numerous

A HISTORY OF MATHEMATICS. 206 mistakes: neither got a prize. Pascal then published his own solutions, which produced a great sensation among scientific men. Wallis, too, published his, with the errors corrected. Though not competing for the prizes, Huygens, Wren, and Fermat solved some of the questions. The chief discoveries of Christopher Wren(1632–1723), the celebrated architect of St. Paul’s Cathedral in London, were the rectification of a cycloidal arc and the determination of its centre of gravity. Fermat found the area generated by an arc of the cycloid. Huygens invented the cycloidal pendulum. The beginning of the seventeenth century witnessed also a revival of synthetic geometry. One who treated conics still by ancient methods, but who succeeded in greatly simplifying many prolix proofs of

Apollonius, was Claude Mydorge in Paris (1585–1647), a friend of Descartes. But it remained for Girard Desargues (1593–1662) of Lyons, and for Pascal, to leave the beaten track and cut out fresh paths. They introduced the important method of Perspective. All conics on a cone with circular base appear circular to an eye at the apex. Hence Desargues and Pascal conceived the treatment of the conic sections as projections of circles. Two important and beautiful theorems were given by Desargues: The one is on the "involution of the six points," in which a transversal meets a conic and an inscribed quadrangle; the other is that, if the vertices of two triangles, situated either in space or in a plane, lie on three lines meeting in a point, then their sides meet in three points lying on a line; and conversely. This last theorem has been employed in recent times by Brianchon, Sturm,

VIETA TO DESCARTES. 207 Gergonne, and Poncelet. Poncelet made it the basis of his beautiful theory of homoligical figures. We owe to Desargues the theory of involution and of transversals; also the beautiful conception that the two extremities of a straight line may be considered as meeting at infinity, and that parallels differ from other pairs of lines only in having their points of intersection at infinity. Pascal greatly admired Desargues' results, saying (in his Essais pour les Coniques), "I wish to acknowledge that I owe the little that I have discovered on this subject, to his writings." Pascal's and Desargues' writings contained the fundamental ideas of modern synthetic geometry. In Pascal's wonderful work on conics, written at the age of sixteen and now lost, were given the theorem on the anharmonic ratio, first found in Pappus, and also that celebrated proposition on the mystic hexagon, known as "Pascal's theorem," viz. that the opposite sides of a hexagon inscribed in a conic intersect in three points which are collinear. This theorem formed the keystone to his theory. He himself said that from this alone he deduced over 400 corollaries, embracing the conics of Apollonius and many other results. Thus the genius of Desargues and Pascal uncovered several of the rich treasures of modern synthetic geometry; but owing to the absorbing interest taken in the analytical geometry of Descartes and later in the differential calculus, the subject was almost entirely neglected until the present century. In the theory of numbers no new results of scientific value had been reached for over 1000 years, extending from the times of Diophantus and the Hindoos until the beginning of

A HISTORY OF MATHEMATICS. 208 the seventeenth century. But the illustrious period we are now considering produced men who rescued this science from the realm of mysticism and superstition, in which it had been so long imprisoned; the properties of numbers began again to be studied scientifically. Not being in possession of the Hindoo indeterminate analysis, many beautiful results of the Brahmins had to be re-discovered by the Europeans. Thus a solution in integers of linear indeterminate equations was re-discovered by the FrenchmanBachet de M´ eziriac(1581– 1638), who was the earliest noteworthy European Diophantist. In 1612 he published Probl` emes plaisants et d´ electables qui se font par les nombres, and in 1621 a Greek edition ofDiophantus with notes. The father of the modern theory of numbers is Fermat. He was so uncommunicative in disposition, that he generally concealed his methods and made known his results only. In some cases later analysts have been greatly puzzled in the attempt of supplying the proofs. Fermat owned a

copy of Bachet’s Diophantus, in which he entered numerous marginal notes. In 1670 these notes were incorporated in a new edition of Diophantus, brought out by his son. Other theorems on numbers, due to Fermat, were published in his Opera varia (edited by his son) and in Wallis’s Commercium epistolicum of 1658. Of the following theorems, the first seven are found in the marginal notes:— (1) xn + yn = zn is impossible for integral values of x, y, and z, when n > 2. Remark: “I have found for this a truly wonderful proof, but the margin is too small to hold it.” Repeatedly was this theorem made the prize question

VIETA TO DESCARTES. 209 of learned societies. It has given rise to investigations of great interest and difficulty on the part of Euler, Lagrange, Dirichlet, and Kummer. (2) A prime of the form4n + 1 is only once the hypothenuse of a right triangle; its square is twice; its cube is three times, etc. Example: 52 = 3 2 + 42; 252 = 15 2 + 202 = 7 2 + 242; 1252 = 752 + 1002 = 352 + 1202 = 442 + 1172. (3) A prime of the form 4n + 1 can be expressed once, and only once, as the sum of two squares. Proved by Euler. (4) A number composed of two cubes can be resolved into two other cubes in an infinite multiplicity of ways. (5) Every number is either a triangular number or the sum of two or three triangular numbers; either a square or the sum of two, three, or four squares; either a pentagonal number or the sum of two, three, four, or five pentagonal numbers; similarly for polygonal numbers in general. The proof of this and other theorems is promised by Fermat in a future work which never appeared. This theorem is also given, with others, in a letter of 1637(?) addressed to Pater Mersenne. (6) As many numbers as you please may be found, such that the square of each remains a square on the addition to or subtraction from it of the sum of all the numbers. (7) x4 + y4 = z2 is impossible. (8) In a letter of 1640 he gives the celebrated theorem generally known as “Fermat’s theorem,” which we state in Gauss’s notation: If p is prime, and a is prime to p, then $a^{p-1} \equiv 1 \pmod{p}$. It was proved by Euler.

A HISTORY OF MATHEMATICS. 210 (9) Fermat died with the belief that he had found a long- sought-for law of prime numbers in the formula 22n + 1 = a prime, but he admitted that he was unable to prove it rigorously. The law is not true, as was pointed out by Euler in the example225 + 1 = 4, 294, 967, 297 = 6, 700, 417 times 641. The American lightning calculator Zerah Colburn, when a boy, readily found the factors, but was unable to explain the method by which he made his marvellous mental computation. (10) An odd prime number can be expressed as the difference of two squares in one, and only one, way. This theorem, given in the Relation, was used by Fermat for the decomposition of large numbers into prime factors. (11) If the integers a, b, c represent the sides of a right triangle, then its area cannot be a square number. This was proved by Lagrange. (12) Fermat’s solution of ax2 + 1 = y2, where a is integral but not a square, has come down in only the broadest outline, as given in the Relation. He proposed the problem to the Frenchman, Bernhard Frenicle de Bessy , and in 1657 to all living mathematicians. In England, Wallis and Lord Brounker conjointly found a laborious solution, which was published in 1658, and also in 1668, in an algebraical work brought out by John Pell. Though Pell had no other connection with the problem, it went by the name of “Pell’s problem.” The first solution was given by the Hindoos. We are not sure that Fermat subjected all his

theorems to rigorous proof. His methods of proof were entirely lost until 1879, when a document was found buried among the

VIETA TO DESCARTES. 211 manuscripts of Huygens in the library of Leyden, entitled Relation des d´ ecouvertes en la science des nombres. It appears from it that he used an inductive method, called by him la descente infinie ou indefinie. He says that this was particularly applicable in proving the impossibility of certain relations, as, for instance, Theorem 11, given above, but that he succeeded in using the method also in proving affirmative statements. Thus he proved Theorem 3 by showing that if we suppose there be a prime 4n + 1 which does not possess this property, then there will be a smaller prime of the form 4n + 1 not possessing it; and a third one smaller than the second, not possessing it; and so on. Thus descending indefinitely, he arrives at the number 5, which is the smallest prime factor of the form 4n + 1. From the above supposition it would follow that 5 is not the sum of two squares—a conclusion contrary to fact. Hence the supposition is false, and the theorem is established. Fermat applied this method of descent with success in a large number of theorems. By this method Euler, Legendre, Dirichlet, proved several of his enunciations and many other numerical propositions. A correspondence between Pascal and Fermat relating to a certain game of chance was the germ of the theory of prob- abilities, which has since attained a vast growth. Chevalier de M´ er´ e proposed to Pascal the fundamental problem, to determine the probability which each player has, at any given stage of the game, of winning the game. Pascal and Fermat supposed that the players have equal chances of winning a single point.

A HISTORY OF MATHEMATICS. 212 The former communicated this problem to Fermat, who studied it with lively interest and solved it by the theory of combinations, a theory which was diligently studied both by him and Pascal. The calculus of probabilities engaged the attention also of Huygens. The most important theorem reached by him was that, if A has p chances of winning a sum a, and q chances of winning a sum b, then he may expect to win the sum ap + bq p + q . The next great work on the theory of probability was the Ars conjectandi of Jakob Bernoulli. Among the ancients, Archimedes was the only one who attained clear and correct notions on theoretical statics. He had acquired firm possession of the idea of pressure, which lies at the root of mechanical science. But his ideas slept nearly twenty centuries, until the time ofStevin and Galileo. Stevin determined accurately the force necessary to sustain a body on a plane inclined at any angle to the horizon. He was in possession of a complete doctrine of equilibrium. While Stevin investigated statics, Galileo pursued principally dynamics. Galileo was the first to abandon the Aristotelian idea that bodies descend more quickly in proportion as they are heavier; he established the first law of motion; determined the laws of falling bodies; and, having obtained a clear notion of acceleration and of the independence of different motions, was able to prove that projectiles move in parabolic curves. Up to his time it was believed that a cannon-ball moved forward at first in a straight line and then suddenly fell vertically to the ground. Galileo had an understanding of centrifugal forces, and gave a correct definition of momentum. Though

DESCARTES TO NEWTON. 213 he formulated the fundamental principle of statics, known as the parallelogram of forces, yet he did not fully recognise its scope. The principle of virtual velocities was partly conceived by Guido Ubaldo (died 1607), and afterwards more fully by Galileo. Galileo is the founder of the science of dynamics. Among his contemporaries it was chiefly the novelties he detected in the sky that made him celebrated, but Lagrange claims that his astronomical discoveries required only a telescope and perseverance, while it took an extraordinary genius to discover laws from phenomena, which we see constantly and of which the true explanation escaped all earlier philosophers. The first contributor to the science of mechanics after Galileo was Descartes. DESCARTES TO NEWTON. Among the earliest thinkers of the seventeenth and eigh- teenth centuries, who employed their mental powers toward the destruction of old ideas and the up-building of new ones, ranks Ren´ e Descartes(1596–1650). Though he professed orthodoxy in faith all his life, yet in science he was a profound sceptic. He found that the world's brightest thinkers had been long exercised in metaphysics, yet they had discovered nothing certain; nay, had even flatly contradicted each other. This led him to the gigantic resolution of taking nothing whatever on authority, but of subjecting everything to scrutinous exami- nation, according to new methods of inquiry. The certainty of

A HISTORY OF MATHEMATICS. 214 the conclusions in geometry and arithmetic brought out in his mind the contrast between the true and false ways of seeking the truth. He thereupon attempted to apply mathematical reasoning to all sciences. "Comparing the mysteries of nature with the laws of mathematics, he dared to hope that the secrets of both could be unlocked with the same key." Thus he built up a system of philosophy called Cartesianism. Great as was Descartes' celebrity as a metaphysician, it may be fairly questioned whether his claim to be remembered by posterity as a mathematician is not greater. His philosophy has long since been superseded by other systems, but the analytical geometry of Descartes will remain a valuable possession forever. At the age of twenty-one, Descartes enlisted in the army of Prince Maurice of Orange. His years of soldiering were years of leisure, in which he had time to pursue his studies. At that time mathematics was his favourite science. But in 1625 he ceased to devote himself to pure mathematics. Sir William Hamilton is in error when he states that Descartes considered mathematical studies absolutely pernicious as a means of internal culture. In a letter to Mersenne, Descartes says: "M. Desargues puts me under obligations on account of the pains that it has pleased him to have in me, in that he shows that he is sorry that I do not wish to study more in geometry, but I have resolved to quit only abstract geometry, that is to say, the consideration of questions which serve only to exercise the mind , and this, in order to study another kind of geometry, which has for its object the explanation of the phenomena of nature. . . . You

DESCARTES TO NEWTON. 215 know that all my physics is nothing else than geometry." The years between 1629 and 1649 were passed by him in Holland in the study, principally, of physics and metaphysics. His residence in Holland was during the most brilliant days of the Dutch state. In 1637 he published his Discours de la M´ ethode, containing among others an essay of 106 pages on geometry. His Geometry is not easy reading. An edition appeared subsequently with notes by his friend De

Beaune, which were intended to remove the difficulties. It is frequently stated that Descartes was the first to apply algebra to geometry. This statement is inaccurate, for Vieta and others had done this before him. Even the Arabs sometimes used algebra in connection with geometry. The new step that Descartes did take was the introduction into geometry of an analytical method based on the notion of variables and constants, which enabled him to represent curves by algebraic equations. In the Greek geometry, the idea of motion was wanting, but with Descartes it became a very fruitful conception. By him a point on a plane was determined in position by its distances from two fixed right lines or axes. These distances varied with every change of position in the point. This geometric idea of co-ordinate representation, together with the algebraic idea of two variables in one equation having an indefinite number of simultaneous values, furnished a method for the study of loci, which is admirable for the generality of its solutions. Thus the entire conic sections of Apollonius is wrapped up and contained in a single equation of the second degree.

A HISTORY OF MATHEMATICS. 216 The Latin term for “ordinate” used by Descartes comes from the expression lineæ ordinatæ , employed by Roman surveyors for parallel lines. The term abscissa occurs for the first time in a Latin work of 1659, written by Stefano degli Angeli (1623–1697), a professor of mathematics in Rome. [3] Descartes’ geometry was called “analytical geometry,” partly because, unlike the synthetic geometry of the ancients, it is actually analytical, in the sense that the word is used in logic; and partly because the practice had then already arisen, of designating by the term analysis the calculus with general quantities. The first important example solved by Descartes in his geometry is the “problem of Pappus”; viz. “Given several straight lines in a plane, to find the locus of a point such that the perpendiculars, or more generally, straight lines at given angles, drawn from the point to the given lines, shall satisfy the condition that the product of certain of them shall be in a given ratio to the product of the rest.” Of this celebrated problem, the Greeks solved only the special case when the number of given lines is four, in which case the locus of the point turns out to be a conic section. By Descartes it was solved completely, and it afforded an excellent example of the use which can be made of his analytical method in the study of loci. Another solution was given later by Newton in the Principia. The methods of drawing tangents invented by Roberval and Fermat were noticed earlier. Descartes gave a third method. Of all the problems which he solved by his geometry,

DESCARTES TO NEWTON. 217 none gave him as great pleasure as his mode of constructing tangents. It is profound but operose, and, on that account, inferior to Fermat’s. His solution rests on the method of Indeterminate Coefficients, of which he bears the honour of invention. Indeterminate coefficients were employed by him also in solving biquadratic equations. The essays of Descartes on dioptrics and geometry were sharply criticised by Fermat, who wrote objections to the former, and sent his own treatise on “maxima and minima” to show that there were omissions in the geometry. Descartes thereupon made an attack on Fermat’s method of tangents. Descartes was in the wrong in this attack, yet he continued the controversy with obstinacy. He had a controversy also with Roberval on the cycloid. This curve has been called the “Helen of geometers,” on account of its beautiful properties and the

controversies which their discovery occasioned. Its quadrature by Roberval was generally considered a brilliant achievement, but Descartes commented on it by saying that any one moderately well versed in geometry might have done this. He then sent a short demonstration of his own. On Roberval's intimating that he had been assisted by a knowledge of the solution, Descartes constructed the tangent to the curve, and challenged Roberval and Fermat to do the same. Fermat accomplished it, but Roberval never succeeded in solving this problem, which had cost the genius of Descartes but a moderate degree of attention. He studied some new curves, now called "ovals of Descartes," which were intended by him to serve in the construction of

A HISTORY OF MATHEMATICS. 218 converging lenses, but which yielded no results of practical value. The application of algebra to the doctrine of curved lines reacted favourably upon algebra. As an abstract science, Descartes improved it by the systematic use of exponents and by the full interpretation and construction of negative quantities. Descartes also established some theorems on the theory of equations. Celebrated is his "rule of signs" for determining the number of positive and negative roots; viz. an equation may have as many + roots as there are variations of signs, and as many − roots as there are permanencies of signs. Descartes was charged by Wallis with availing himself, without acknowledgment, of Harriot's theory of equations, particularly his mode of generating equations; but there seems to be no good ground for the charge. Wallis also claimed that Descartes failed to observe that the above rule of signs is not true whenever the equation has imaginary roots; but Descartes does not say that the equationalways has, but that it may have so many roots. It is true that Descartes does not consider the case of imaginaries directly, but further on in his Geometry he gives incontestable evidence of being able to handle this case also. In mechanics, Descartes can hardly be said to have advanced beyond Galileo. The latter had overthrown the ideas of Aristotle on this subject, and Descartes simply "threw himself upon the enemy" that had already been "put to the rout." His statement of the first and second laws of motion was an improvement in form, but his third law is false in substance.

DESCARTES TO NEWTON. 219 The motions of bodies in their direct impact was imperfectly understood by Galileo, erroneously given by Descartes, and first correctly stated by Wren, Wallis, and Huygens. One of the most devoted pupils of Descartes was the learned Princess Elizabeth, daughter of Frederick V. She applied the new analytical geometry to the solution of the "Apollonian problem." His second royal follower wasQueen Christina, the daughter of Gustavus Adolphus. She urged upon Descartes to come to the Swedish court. After much hesitation he accepted the invitation in 1649. He died at Stockholm one year later. His life had been one long warfare against the prejudices of men. It is most remarkable that the mathematics and philoso- phy of Descartes should at first have been appreciated less by his countrymen than by foreigners. The indiscreet tem- per of Descartes alienated the great contemporary French mathematicians, Roberval, Fermat, Pascal. They continued in investigations of their own, and on some points strongly opposed Descartes. The universities of France were under strict ecclesiastical control and did nothing to introduce his mathematics and philosophy. It was in the youthful universi-

ties of Holland that the effect of Cartesian teachings was most immediate and strongest. The only prominent Frenchman who immediately followed in the footsteps of the great master was De Beaune (1601– 1652). He was one of the first to point out that the properties of a curve can be deduced from the properties of its tangent. This mode of inquiry has been called the inverse method

A HISTORY OF MATHEMATICS. 220 of tangents . He contributed to the theory of equations by considering for the first time the upper and lower limits of the roots of numerical equations. In the Netherlands a large number of distinguished mathematicians were at once struck with admiration for the Cartesian geometry. Foremost among these arevan Schooten, John de Witt, van Heuraet, Sluze, andHudde. Van Schooten (died 1660), professor of mathematics at Leyden, brought out an edition of Descartes' geometry, together with the notes thereon by De Beaune. His chief work is his Exercitationes Mathematicæ, in which he applies the analytical geometry to the solution of many interesting and difficult problems. The noble-hearted Johann de Witt, grand-pensioner of Holland, celebrated as a statesman and for his tragical end, was an ardent geometrician. He conceived a new and ingenious way of generating conics, which is essentially the same as that by projective pencils of rays in modern synthetic geometry. He treated the subject not synthetically, but with aid of the Cartesian analysis. Ren´ e Fran¸ cois de Sluze(1622–1685) and Johann Hudde (1633–1704) made some improvements on Descartes' and Fermat's methods of drawing tangents, and on the theory of maxima and minima. With Hudde, we find the first use of three variables in analytical geometry. He is the author of an ingenious rule for finding equal roots. We illustrate it by the equation $x3- x2- 8x + 12 = 0$. Taking an arithmetical progression 3, 2, 1, 0, of which the highest term is equal to the degree of the equation, and multiplying each term of the equation respectively by the corresponding term of the

DESCARTES TO NEWTON. 221 progression, we get $3x3- 2x2- 8x = 0$, or $3x2- 2x- 8 = 0$. This last equation is by one degree lower than the original one. Find the G.C.D. of the two equations. This is $x- 2$; hence 2 is one of the two equal roots. Had there been no common divisor, then the original equation would not have possessed equal roots. Hudde gave a demonstration for this rule. [24] Heinrich van Heuraet must be mentioned as one of the earliest geometers who occupied themselves with success in the rectification of curves. He observed in a general way that the two problems of quadrature and of rectification are really identical, and that the one can be reduced to the other. Thus he carried the rectification of the hyperbola back to the quadrature of the hyperbola. The semi-cubical parabola $y3 = ax2$ was the first curve that was ever rectified absolutely. This appears to have been accomplished independently by Van Heuraet in Holland and by William Neil (1637–1670) in England. According to Wallis the priority belongs to Neil. Soon after, the cycloid was rectified by Wren and Fermat. The prince of philosophers in Holland, and one of the greatest scientists of the seventeenth century, wasChristian Huygens (1629–1695), a native of the Hague. Eminent as a physicist and astronomer, as well as mathematician, he was a worthy predecessor of Sir Isaac Newton. He studied at Leyden under the youngerVan Schooten. The perusal of some of his earliest theorems led Descartes to predict his future greatness. In 1651 Huygens wrote a treatise in

which he pointed out the fallacies of Gregory St. Vincent (1584–1667) on the subject of quadratures. He himself gave a remarkably

A HISTORY OF MATHEMATICS. 222 close and convenient approximation to the length of a circular arc. In 1660 and 1663 he went to Paris and to London. In 1666 he was appointed by Louis XIV. member of the French Academy of Sciences. He was induced to remain in Paris from that time until 1681, when he returned to his native city, partly for consideration of his health and partly on account of the revocation of the Edict of Nantes. The majority of his profound discoveries were made with aid of the ancient geometry, though at times he used the geometry of Descartes or of Cavalieri and Fermat. Thus, like his illustrious friend, Sir Isaac Newton, he always showed partiality for the Greek geometry. Newton and Huygens were kindred minds, and had the greatest admiration for each other. Newton always speaks of him as the "Summus Hugenius." To the two curves (cubical parabola and cycloid) previously rectified he added a third,—the cissoid. He solved the problem of the catenary, determined the surface of the parabolic and hyperbolic conoid, and discovered the properties of the logarithmic curve and the solids generated by it. Huygens'De horologio oscillatorio(Paris, 1673) is a work that ranks second only to the Principia of Newton and constitutes historically a necessary introduction to it. [13] The book opens with a description of pendulum clocks, of which Huygens is the inventor. Then follows a treatment of accelerated motion of bodies falling free, or sliding on inclined planes, or on given curves,—culminating in the brilliant discovery that the cycloid is the tautochronous curve. To the theory of curves he

DESCARTES TO NEWTON. 223 added the important theory of "evolutes." After explaining that the tangent of the evolute is normal to the involute, he applied the theory to the cycloid, and showed by simple reasoning that the evolute of this curve is an equal cycloid. Then comes the complete general discussion of the centre of oscillation. This subject had been proposed for investigation by Mersenne and discussed by Descartes and Roberval. In Huygens' assumption that the common centre of gravity of a group of bodies, oscillating about a horizontal axis, rises to its original height, but no higher, is expressed for the first time one of the most beautiful principles of dynamics, afterwards called the principle of the conservation of vis viva. [32] The thirteen theorems at the close of the work relate to the theory of centrifugal force in circular motion. This theory aided Newton in discovering the law of gravitation. Huygens wrote the first formal treatise on probability. He proposed the wave-theory of light and with great skill applied geometry to its development. This theory was long neglected, but was revived and successfully worked out by Young and Fresnel a century later. Huygens and his brother improved the telescope by devising a better way of grinding and polishing lenses. With more efficient instruments he determined the nature of Saturn's appendage and solved other astronomical questions. Huygens' Opuscula posthuma appeared in 1703. Passing now from Holland to England, we meet there one of the most original mathematicians of his day— John Wallis (1616–1703). He was educated for the Church at Cambridge and entered Holy Orders. But his genius was employed chiefly

A HISTORY OF MATHEMATICS. 224 in the study of mathematics. In 1649 he was appointed Savilian professor of geometry at Oxford. He was one of the original members of the Royal Society, which was founded in 1663. Wallis thoroughly grasped the mathematical methods both of Cavalieri and Descartes. His Conic Sections is the earliest work in which these curves are no longer considered as sections of a cone, but as curves of the second degree, and are treated analytically by the Cartesian method of co-ordinates. In this work Wallis speaks of Descartes in the highest terms, but in his Algebra he, without good reason, accuses Descartes of plagiarising from Harriot. We have already mentioned elsewhere Wallis's solution of the prize questions on the cycloid, which were proposed by Pascal. The Arithmetic of Infinites , published in 1655, is his greatest work. By the application of analysis to the Method of Indivisibles, he greatly increased the power of this instrument for effecting quadratures. He advanced beyond Kepler by making more extended use of the "law of continuity" and placing full reliance in it. By this law he was led to regard the denominators of fractions as powers with negative exponents. Thus, the descending geometrical progressionx3, x2, x1, x0, if continued, gives x−1, x−2, x−3, etc.; which is the same thing as 1 x, 1 x2 , 1 x3 . The exponents of this geometric series are in continued arithmetical progression, 3, 2, 1, 0, −1, −2,−3. He also used fractional exponents, which, like the negative, had been invented long before, but had failed to be generally introduced. The symbol∞ for infinity is due to him. Cavalieri and the French geometers had ascertained the

DESCARTES TO NEWTON. 225 formula for squaring the parabola of any degree, y = xm, m being a positive integer. By the summation of the powers of the terms of infinite arithmetical series, it was found that the curve y = xm is to the area of the parallelogram having the same base and altitude as 1 is to m + 1. Aided by the law of continuity, Wallis arrived at the result that this formula holds true not only when m is positive and integral, but also when it is fractional or negative. Thus, in the parabola y = √ px, m = 1 2; hence the area of the parabolic segment is to that of the circumscribed rectangle as 1 : 1 1 2, or as 2 : 3 . Again, suppose that in y = xm, m =− 1 2; then the curve is a kind of hyperbola referred to its asymptotes, and the hyperbolic space between the curve and its asymptotes is to the corresponding parallelogram as 1 : 1 2. If m =−1, as in the common equilateral hyperbola y = x−1 or xy = 1, then this ratio is 1 :−1 + 1, or 1 : 0, showing that its asymptotic space is infinite. But in the case when m is greater than unity and negative, Wallis was unable to interpret correctly his results. For example, if m =−3, then the ratio becomes 1 :−2, or as unity to a negative number. What is the meaning of this? Wallis reasoned thus: If the denominator is only zero, then the area is already infinite; but if it is less than zero, then the area must be more than infinite. It was pointed out later by Varignon, that this space, supposed to exceed infinity, is really finite, but taken negatively; that is, measured in a contrary direction. [31] The method of Wallis was easily extended to cases such asy = ax m n + bx p q by performing the quadrature for each term separately, and then adding the results.

A HISTORY OF MATHEMATICS. 226 The manner in which Wallis studied the quadrature of the circle and arrived at his expression for the value of π is extraordinary. He found that the areas comprised between the axes, the ordinate

corresponding to x, and the curves represented by the equations y = (1− x2)0, y = (1− x2)1, y = (1− x2)2, y = (1− x2)3, etc., are expressed in functions of the circumscribed rectangles havingx and y for their sides, by the quantities forming the series x, x− 1 3 x3, x− 2 3 x3 + 1 5 x5, x− 3 3 x3 + 3 5 x5− 1 7 x7, etc. When x = 1, these values become respectively 1, 2 3, 8 15, 48 105, etc. Now since the ordinate of the circle is y = (1− x2) 1 2 , the exponent of which is 1 2 or the mean value between 0 and 1, the question of this quadrature reduced itself to this: If 0, 1, 2, 3, etc., operated upon by a certain law, give 1, 2 3, 8 15, 48 105, what will 1 2 give, when operated upon by the same law? He attempted to solve this by interpolation, a method first brought into prominence by him, and arrived by a highly complicated and difficult analysis at the following very remarkable expression: π 2 = 2·2·4·4·6·6·8·8··· 1·3·3·5·5·7·7·9··· He did not succeed in making the interpolation itself, because he did not employ literal or general exponents, and

DESCARTES TO NEWTON. 227 could not conceive a series with more than one term and less than two, which it seemed to him the interpolated series must have. The consideration of this difficulty led Newton to the discovery of the Binomial Theorem. This is the best place to speak of that discovery. Newton virtually assumed that the same conditions which underlie the general expressions for the areas given above must also hold for the expression to be interpolated. In the first place, he observed that in each expression the first term is x, that x increases in odd powers, that the signs alternate + and−, and that the second terms 0 3 x3, 1 3 x3, 2 3 x3, 3 3 x3, are in arithmetical progression. Hence the first two terms of the interpolated series must bex− 1 2 x3 3 . He next considered that the denominators 1, 3, 5, 7, etc., are in arithmetical progression, and that the coefficients in the numerators in each expression are the digits of some power of the number 11; namely, for the first expression, 110 or 1; for the second, 111 or 1, 1; for the third, 112 or 1, 2, 1; for the fourth, 113 or 1, 3, 3, 1; etc. He then discovered that, having given the second digit (call itm), the remaining digits can be found by continual multiplication of the terms of the series m− 0 1 · m− 1 2 · m− 2 3 · m− 3 4 · etc. Thus, if m = 4, then 4· m− 1 2 gives 6; 6· m− 2 3 gives 4; 4· m− 3 4 gives 1. Applying this rule to the required series, since the second term is 1 2 x3 3 , we have m = 1 2, and then get for the succeeding coefficients in the numerators respectively− 1 8, + 1 16,− 5 128, etc.; hence the required area for the circular segment isx− 1 2 x3 3 − 1 8 x5 5 − 1 16 x7 7 −

A HISTORY OF MATHEMATICS. 228 etc. Thus he found the interpolated expression to be an infinite series, instead of one having more than one term and less than two, as Wallis believed it must be. This interpolation suggested to Newton a mode of expanding(1− x2) 1 2 , or, more generally, (1− x2)m, into a series. He observed that he had only to omit from the expression just found the denominators 1, 3, 5, 7, etc., and to lower each power of x by unity, and he had the desired expression. In a letter to Oldenburg (June 13, 1676), Newton states the theorem as follows: The extraction of roots is much shortened by the theorem (P + P Q) m n = P m n + m n AQ + m− n 2n BQ + m− 2n 3n CQ + etc., where A means the first term, P m n , B the second term, C the third term, etc. He verified it by actual multiplication, but gave no regular proof of it. He gave it for any exponent whatever, but made no distinction between the case when the exponent is positive and integral, and the others. It

should here be mentioned that very rude beginnings of the binomial theorem are found very early. The Hindoos and Arabs used the expansions of(a+b)2 and (a+b)3 for extracting roots; Vieta knew the expansion of(a + b)4; but these were the results of simple multiplication without the discovery of any law. The binomial coefficients for positive whole exponents were known to some Arabic and European mathematicians. Pascal derived the coefficients from the method of what is called the "arithmetical triangle." Lucas de Burgo, Stifel, Stevinus, Briggs, and others, all possessed something from which one would think the binomial theorem could have been

DESCARTES TO NEWTON. 229 gotten with a little attention, "if we did not know that such simple relations were difficult to discover." Though Wallis had obtained an entirely new expression for π, he was not satisfied with it; for instead of a finite number of terms yielding an absolute value, it contained merely an infinite number, approaching nearer and nearer to that value. He therefore induced his friend, Lord Brouncker (1620?- 1684), the first president of the Royal Society, to investigate this subject. Of course Lord Brouncker did not find what they were after, but he obtained the following beautiful equality:— π = 4 1 + 1 2 + 9 2 + 25 2 + 49 2 + etc. Continued fractions, both ascending and descending, appear to have been known already to the Greeks and Hindoos, though not in our present notation. Brouncker's expression gave birth to the theory of continued fractions. Wallis' method of quadratures was diligently studied by his disciples. Lord Brouncker obtained the first infinite series for the area of an equilateral hyperbola between its asymptotes. Nicolaus Mercatorof Holstein, who had settled in England, gave, in his Logarithmotechnia (London, 1668), a similar series. He started with the grand property of the equilateral hyperbola, discovered in 1647 by Gregory St. Vincent, which connected the hyperbolic space between the asymptotes with

A HISTORY OF MATHEMATICS. 230 the natural logarithms and led to these logarithms being called hyperbolic. By it Mercator arrived at the logarithmic series, which Wallis had attempted but failed to obtain. He showed how the construction of logarithmic tables could be reduced to the quadrature of hyperbolic spaces. Following up some suggestions of Wallis, William Neil succeeded in rectifying the cubical parabola, and Wren in rectifying any cycloidal arc. A prominent English mathematician and contemporary of Wallis was Isaac Barrow (1630-1677). He was professor of mathematics in London, and then in Cambridge, but in 1669 he resigned his chair to his illustrious pupil, Isaac Newton, and renounced the study of mathematics for that of divinity. As a mathematician, he is most celebrated for his method of tangents. He simplified the method of Fermat by introducing two infinitesimals instead of one, and approximated to the course of reasoning afterwards followed by Newton in his doctrine on Ultimate Ratios. He considered the infinitesimal right triangle ABB′ having T P P ′ B B′ A for its sides the difference between two successive ordi- nates, the distance between them, and the portion of the curve intercepted by them. This triangle is similar to BP T, formed by the ordi- nate, the tangent, and the sub-tangent. Hence, if we know the ratio of B′A to BA, then

NEWTON TO EULER. 231 we know the ratio of the ordinate and the sub-tangent, and the tangent can be constructed at once. For any curve, say y2 = px, the ratio ofB′A to BA is determined from its equation as follows: If x receives an infinitesimal increment P P′ = e, then y receives an increment B′A = a, and the equation for the ordinate B′P′ becomes y2 + 2ay + a2 = px + pe. Since y2 = px, we get 2ay + a2 = pe; neglecting higher powers of the infinitesimals, we have2ay = pe, which gives a : e = p : 2y = p : 2√px. But a : e = the ordinate : the sub-tangent; hence p : 2√px =√px : sub-tangent, giving 2x for the value of the sub-tangent. This method differs from that of the differential calculus only in notation. [31] NEWTON TO EULER. It has been seen that in France prodigious scientific progress was made during the beginning and middle of the seventeenth century. The toleration which marked the reign of Henry IV. and Louis XIII. was accompanied by intense intellectual activity. Extraordinary confidence came to be placed in the power of the human mind. The bold intellectual conquests of Descartes, Fermat, and Pascal enriched mathematics with imperishable treasures. During the early part of the reign of Louis XIV. we behold the sunset splendour of this glorious period. Then followed a night of mental effeminacy. This

A HISTORY OF MATHEMATICS. 232 lack of great scientific thinkers during the reign of Louis XIV. may be due to the simple fact that no great minds were born; but, according to Buckle, it was due to the paternalism, to the spirit of dependence and subordination, and to the lack of toleration, which marked the policy of Louis XIV. In the absence of great French thinkers, Louis XIV. sur- rounded himself by eminent foreigners. R¨ omer from Denmark, Huygens from Holland, Dominic Cassini from Italy, were the mathematicians and astronomers adorning his court. They were in possession of a brilliant reputation before going to Paris. Simply because they performed scientific work in Paris, that work belongs no more to France than the discoveries of Descartes belong to Holland, or those of Lagrange to Germany, or those of Euler and Poncelet to Russia. We must look to other countries than France for the great scientific men of the latter part of the seventeenth century. About the time when Louis XIV. assumed the direction of the French government Charles II. became king of England. At this time England was extending her commerce and navi- gation, and advancing considerably in material prosperity. A strong intellectual movement took place, which was unwit- tingly supported by the king. The age of poetry was soon followed by an age of science and philosophy. In two successive centuries England produced Shakespeare and Newton! Germany still continued in a state of national degradation. The Thirty Years’ War had dismembered the empire and brutalised the people. Yet this darkest period of Germany’s history produced Leibniz, one of the greatest geniuses of

NEWTON TO EULER. 233 modern times. There are certain focal points in history toward which the lines of past progress converge, and from which radiate the advances of the future. Such was the age of Newton and Leibniz in the history of mathematics. During fifty years preceding this era several of the brightest and acutest mathematicians bent the force of their genius in a direction which finally led to the discovery of the infinitesimal calculus by Newton and Leibniz. Cavalieri, Roberval, Fermat, Descartes, Wallis, and others had each contributed to the new

geometry. So great was the advance made, and so near was their approach toward the invention of the infinitesimal analysis, that both Lagrange and Laplace pronounced their countryman, Fermat, to be the true inventor of it. The differential calculus, therefore, was not so much an individual discovery as the grand result of a succession of discoveries by different minds. Indeed, no great discovery ever flashed upon the mind at once, and though those of Newton will influence mankind to the end of the world, yet it must be admitted that Pope's lines are only a "poetic fancy":— " Nature and Nature's laws lay hid in night; God said, 'Let Newton be,' and all was light." Isaac Newton (1642-1727) was born at Woolsthorpe, in Lincolnshire, the same year in which Galileo died. At his birth he was so small and weak that his life was despaired of. His mother sent him at an early age to a village school, and in his twelfth year to the public school at Grantham. At first he seems to have been very inattentive to his studies and

A HISTORY OF MATHEMATICS. 234 very low in the school; but when, one day, the little Isaac received a severe kick upon his stomach from a boy who was above him, he laboured hard till he ranked higher in school than his antagonist. From that time he continued to rise until he was the head boy. [33] At Grantham, Isaac showed a decided taste for mechanical inventions. He constructed a water-clock, a wind-mill, a carriage moved by the person who sat in it, and other toys. When he had attained his fifteenth year his mother took him home to assist her in the management of the farm, but his great dislike for farm-work and his irresistible passion for study, induced her to send him back to Grantham, where he remained till his eighteenth year, when he entered Trinity College, Cambridge (1660). Cambridge was the real birthplace of Newton's genius. Some idea of his strong intuitive powers may be drawn from the fact that he regarded the theorems of ancient geometry as self-evident truths, and that, without any preliminary study, he made himself master of Descartes' Geometry. He afterwards regarded this neglect of elementary geometry a mistake in his mathematical studies, and he expressed to Dr. Pemberton his regret that "he had applied himself to the works of Descartes and other algebraic writers before he had considered the Elements of Euclid with that attention which so excellent a writer deserves." Besides Descartes' Geometry, he studied Oughtred's Clavis, Kepler's Optics, the works of Vieta, Schooten's Miscellanies, Barrow's Lectures, and the works of Wallis. He was particularly delighted with Wallis' Arithmetic of Infinites, a treatise fraught with rich and varied

NEWTON TO EULER. 235 suggestions. Newton had the good fortune of having for a teacher and fast friend the celebrated Dr. Barrow, who had been elected professor of Greek in 1660, and was made Lucasian professor of mathematics in 1663. The mathematics of Barrow and of Wallis were the starting-points from which Newton, with a higher power than his masters', moved onward into wider fields. Wallis had effected the quadrature of curves whose ordinates are expressed by any integral and positive power of $(1- x2)$. We have seen how Wallis attempted but failed to interpolate between the areas thus calculated, the areas of other curves, such as that of the circle; how Newton attacked the problem, effected the interpolation, and discovered the Binomial Theorem, which afforded a much easier and direct access to the quadrature of curves than did the method of interpolation; for even though the

binomial expression for the ordinate be raised to a fractional or negative power, the binomial could at once be expanded into a series, and the quadrature of each separate term of that series could be effected by the method of Wallis. Newton introduced the system of literal indices. Newton's study of quadratures soon led him to another and most profound invention. He himself says that in 1665 and 1666 he conceived the method of fluxions and applied them to the quadrature of curves. Newton did not communicate the invention to any of his friends till 1669, when he placed in the hands of Barrow a tract, entitled De Analysi per Æquationes Numero Terminorum Infinitas, which was sent by Barrow to Collins, who greatly admired it. In this treatise the principle

A HISTORY OF MATHEMATICS. 236 of fluxions, though distinctly pointed out, is only partially developed and explained. Supposing the abscissa to increase uniformly in proportion to the time, he looked upon the area of a curve as a nascent quantity increasing by continued fluxion in the proportion of the length of the ordinate. The expression which was obtained for the fluxion he expanded into a finite or infinite series of monomial terms, to which Wallis' rule was applicable. Barrow urged Newton to publish this treatise; "but the modesty of the author, of which the excess, if not culpable, was certainly in the present instance very unfortunate, prevented his compliance." [26] Had this tract been published then, instead of forty-two years later, there would probably have been no occasion for that long and deplorable controversy between Newton and Leibniz. For a long time Newton's method remained unknown, except to his friends and their correspondents. In a letter to Collins, dated December 10th, 1672, Newton states the fact of his invention with one example, and then says: "This is one particular, or rather corollary, of a general method, which extends itself, without any troublesome calculation, not only to the drawing of tangents to any curve lines, whether geometrical or mechanical, or anyhow respecting right lines or other curves, but also to the resolving other abstruser kinds of problems about the crookedness, areas, lengths, centres of gravity of curves, etc.; nor is it (as Hudden's method of Maximis and Minimis) limited to equations which are free from surd quantities. This method I have interwoven with that other of working in equations, by reducing them to

NEWTON TO EULER. 237 infinite series." These last words relate to a treatise he composed in the year 1671, entitled Method of Fluxions , in which he aimed to represent his method as an independent calculus and as a completesystem. Thistractwasintendedasanintroductionto an edition of KinckhuysensAlgebra, which he had undertaken to publish. "But the fear of being involved in disputes about this new discovery, or perhaps the wish to render it more complete, or to have the sole advantage of employing it in his physical researches, induced him to abandon this design." [33] Excepting two papers on optics, all of his works appear to have been published only after the most pressing solicitations of his friends and against his own wishes. [34] His researches on light were severely criticised, and he wrote in 1675: "I was so persecuted with discussions arising out of my theory of light that I blamed my own imprudence for parting with so substantial a blessing as my quiet to run after a shadow." The Method of Fluxions , translated by J. Colson from Newton's Latin, was first published in 1736, or sixty-five years after it was written. In it he explains first

the expansion into series of fractional and irrational quantities,—a subject which, in his first years of study, received the most careful attention. He then proceeds to the solution of the two following mechanical problems, which constitute the pillars, so to speak, of the abstract calculus:— “I. The length of the space described being continually (i.e. at all times) given; to find the velocity of the motion at any time proposed.

A HISTORY OF MATHEMATICS. 238 “II. The velocity of the motion being continually given; to find the length of the space described at any time proposed.” Preparatory to the solution, Newton says: “Thus, in the equation y = x2, if y represents the length of the space at any time described, which (time) another space x, by increasing with an uniform celerity ẋ, measures and exhibits as described: then 2x ẋ will represent the celerity by which the space y, at the same moment of time, proceeds to be described; and contrarywise.” “But whereas we need not consider the time here, any farther than it is expounded and measured by an equable local motion; and besides, whereas only quantities of the same kind can be compared together, and also their velocities of increase and decrease; therefore, in what follows I shall have no regard to time formally considered, but I shall suppose some one of the quantities proposed, being of the same kind, to be increased by an equable fluxion, to which the rest may be referred, as it were to time; and, therefore, by way of analogy, it may not improperly receive the name of time.” In this statement of Newton there is contained a satisfactory answer to the objection which has been raised against his method, that it introduces into analysis the foreign idea of motion. A quantity thus increasing by uniform fluxion, is what we now call an independent variable. Newton continues: “Now those quantities which I consider as gradually and indefinitely increasing, I shall hereafter call fluents, or flowing quantities , and shall represent them by the final letters of the alphabet, v, x, y, and z; . . . and the

NEWTON TO EULER. 239 velocities by which every fluent is increased by its generating motion (which I may call fluxions, or simply velocities, or celerities), I shall represent by the same letters pointed, thus, v̇, ẋ, ẏ, ż. That is, for the celerity of the quantityv I shall put v̇, and so for the celerities of the other quantities x, y, and z, I shall put ẋ, ẏ, and ż, respectively.” It must here be observed that Newton does not take the fluxions themselves infinitely small. The “moments of fluxions,” a term introduced further on, are infinitely small quantities. These “moments,” as defined and used in the Method of Fluxions, are substantially the differentials of Leibniz. De Morgan points out that no small amount of confusion has arisen from the use of the word fluxion and the notation ẋ by all the English writers previous to 1704, excepting Newton and Cheyne, in the sense of an infinitely small increment. [35] Strange to say, even in the Commercium Epistolicum the words moment and fluxion appear to be used as synonymous. After showing by examples how to solve the first problem, Newton proceeds to the demonstration of his solution:— “The moments of flowing quantities (that is, their indef- initely small parts, by the accession of which, in infinitely small portions of time, they are continually increased) are as the velocities of their flowing or increasing. “Wherefore, if the moment of any one (asx) be represented by the product of its celerity ẋ into an infinitely small quantity 0 (i.e.by ẋ0), the moments of the others, v, y, z, will be represented by v̇0, ẏ0, ż0;

because ˙v0, ˙x0, ˙y0, and ˙z0 are to each other as ˙v, ˙x, ˙y, and ˙z.

A HISTORY OF MATHEMATICS. 240 “Now since the moments, as ˙x0 and ˙y0, are the indefinitely little accessions of the flowing quantities x and y, by which those quantities are increased through the several indefinitely little intervals of time, it follows that those quantities,x and y, after any indefinitely small interval of time, become x + ˙x0 and y + ˙y0, and therefore the equation, which at all times indifferently expresses the relation of the flowing quantities, will as well express the relation betweenx + ˙x0 and y + ˙y0, as between x and y; so that x + ˙x0 and y + ˙y0 may be substituted in the same equation for those quantities, instead of x and y. Thus let any equation x3− ax2 + axy− y3 = 0 be given, and substitute x + ˙x0 for x, and y + ˙y0 for y, and there will arise x3 + 3x2 ˙x0 + 3x ˙x0 ˙x0 + ˙x303 −ax2− 2ax ˙x0− a ˙x0 ˙x0 +axy + ay ˙x0 + a ˙x0 ˙y0 + ax ˙y0 −y3 − 3y2 ˙y0− 3y ˙y0 ˙y0− ˙y303 □ □□□□□□□□□□ □□□□□□□□□□ = 0. “Now, by supposition, x3− ax2 + axy− y3 = 0 , which therefore, being expunged and the remaining terms being divided by 0, there will remain 3x2 ˙x− 2ax ˙x + ay ˙x + ax ˙y− 3y2 ˙y + 3x ˙x ˙x0− a ˙x ˙x0 + a ˙x ˙y0 − 3y ˙y ˙y0 + ˙x300− ˙y300 = 0. But whereas zero is supposed to be infinitely little, that it may represent the moments of quantities, the terms that are multiplied by it will be nothing in respect of the rest (termini

NEWTON TO EULER. 241 in eam ducti pro nihilo possunt haberi cum aliis collati); therefore I reject them, and there remains 3x2 ˙x− 2ax ˙x + ay ˙x + ax ˙y− 3y2 ˙y = 0, as above in Example I.” Newton here uses infinitesimals. Much greater than in the first problem were the difficulties encountered in the solution of the second problem, involving, as it does, inverse operations which have been taxing the skill of the best analysts since his time. Newton gives first a special solution to the second problem in which he resorts to a rule for which he has given no proof. In the general solution of his second problem, Newton assumed homogeneity with respect to the fluxions and then considered three cases: (1) when the equation contains two fluxions of quantities and but one of the fluents; (2) when the equation involves both the fluents as well as both the fluxions; (3) when the equation contains the fluents and the fluxions of three or more quantities. The first case is the easiest since it requires simply the integration of dy dx = f(x), to which his “special solution” is applicable. The second case demanded nothing less than the general solution of a differential equation of the first order. Those who know what efforts were afterwards needed for the complete exploration of this field in analysis, will not depreciate Newton’s work even though he resorted to solutions in form of infinite series. Newton’s third case comes now under the solution of partial differential equations. He took the equation 2 ˙x− ˙z + x ˙y = 0 and succeeded in finding a particular integral of it.

A HISTORY OF MATHEMATICS. 242 The rest of the treatise is devoted to the determination of maxima and minima, the radius of curvature of curves, and other geometrical applications of his fluxionary calculus. All this was done previous to the year 1672. It must be observed that in the Method of Fluxions (as well as in his De Analysi and all earlier papers) the method employed by Newton is strictly infinitesimal, and in substance like that of Leibniz. Thus, the original conception of the calculus in England, as well as on the Continent, was based on infinitesimals. The

fundamental principles of the fluxionary calculus were first given to the world in thePrincipia; but its peculiar notation did not appear until published in the second volume of Wallis' Algebra in 1693. The exposition given in the Algebra was substantially a contribution of Newton; it rests on infinitesimals. In the first edition of the Principia (1687) the description of fluxions is likewise founded on infinitesimals, but in the second (1713) the foundation is somewhat altered. In Book II. Lemma II. of the first edition we read: "Cave tamen intellexeris particulas finitas. Momenta quam primum finitæ sunt magnitudinis, desinunt esse momenta. Finiri enim repugnat aliquatenus perpetuo eorum incremento vel decremento.Intelligenda sunt principia jamjam nascentia finitorum magnitudinum." In the second edition the two sentences which we print in italics are replaced by the following: "Particulæ finitæ non sunt momenta sed quantitates ipsæ ex momentis genitæ." Through the difficulty of the phrases in both extracts, this much distinctly appears, that in the first, moments are infinitely small quantities. What

NEWTON TO EULER. 243 else they are in the second is not clear. [35] In theQuadrature of Curves of 1704, the infinitely small quantity is completely abandoned. It has been shown that in theMethod of Fluxions Newton rejected terms involving the quantity 0, because they are infinitely small compared with other terms. This reasoning is evidently erroneous; for as long as 0 is a quantity, though ever so small, this rejection cannot be made without affecting the result. Newton seems to have felt this, for in the Quadrature of Curves he remarked that "in mathematics the minutest errors are not to be neglected" (errores quam minimi in rebus mathematicis non sunt contemnendi). The early distinction between the system of Newton and Leibniz lies in this, that Newton, holding to the conception of velocity or fluxion, used the infinitely small increment as a means of determining it, while with Leibniz the relation of the infinitely small increments is itself the object of determination. The difference between the two rests mainly upon a difference in the mode of generating quantities. [35] We give Newton's statement of the method of fluxions or rates, as given in the introduction to his Quadrature of Curves. "I consider mathematical quantities in this place not as consisting of very small parts, but as described by a continued motion. Lines are described, and thereby generated, not by the apposition of parts, but by the continued motion of points; superficies by the motion of lines; solids by the motion of superficies; angles by the rotation of the sides; portions of time by continual flux: and so on in other quantities. These geneses really take place in the nature of things, and are daily

A HISTORY OF MATHEMATICS. 244 seen in the motion of bodies. . . . "Fluxions are, as near as we please (quam proxime), as the increments of fluents generated in times, equal and as small as possible, and to speak accurately, they are in the prime ratio of nascent increments; yet they can be expressed by any lines whatever, which are proportional to them." Newton exemplifies this last assertion by the problem of tangency: Let AB be the abscissa, BC the ordinate, V CH the tangent, Ec the increment of the ordinate, which produced meets V H at T, and Cc the increment of the curve. The right line Cc being produced to K, there are formed three small triangles, the rectilinear CEc, the mixtilinear CEc, and the rectilinear CET . Of these, the first is evidently the smallest, and the last the greatest. Now suppose the

ordinatebc to move into the place BC, so that the point c exactly coincides with V B b EC c H K A T the point C; CK, and therefore the curve Cc, is coin- cident with the tangent CH , Ec is abso- lutely equal to ET , and the mixtilinear evanescent triangle CEc is, in the last form, similar to the triangle CET ; and its evanescent sides CE, Ec, Cc, will be proportional to CE, ET , and CT , the sides of the triangle CET . Hence it follows that the fluxions of the lines AB, BC, AC, being in the last ratio of their evanescent increments, are proportional to the sides

NEWTON TO EULER. 245 of the triangle CET , or, which is all one, of the triangle V BC similar thereunto. As long as the points C and c are distant from each other by an interval, however small, the line CK will stand apart by a small angle from the tangent CH . But when CK coincides with CH , and the lines CE, Ec, cC reach their ultimate ratios, then the points C and c accurately coincide and are one and the same. Newton then adds that “in mathematics the minutest errors are not to be neglected.” This is plainly a rejection of the postulates of Leibniz. The doctrine of infinitely small quantities is here renounced in a manner which would lead one to suppose that Newton had never held it himself. Thus it appears that Newton’s doctrine was different in different periods. Though, in the above reasoning, the Charybdis of infinitesimals is safely avoided, the dangers of a Scylla stare us in the face. We are required to believe that a point may be considered a triangle, or that a triangle can be inscribed in a point; nay, that three dissimilar triangles become similar and equal when they have reached their ultimate form in one and the same point. In the introduction to theQuadrature of Curves the fluxion of xn is determined as follows:— “In the same time that x, by flowing, becomes x + 0, the power xn becomes (x+ 0)n, i.e. by the method of infinite series xn + n0xn−1 + n2− n 2 02xn−2 + etc., and the increments 0 and n0xn−1 + n2− n 2 02xn−2 + etc.,

A HISTORY OF MATHEMATICS. 246 are to one another as 1 to nxn−1 + n2− n 2 0xn−2 + etc. “Let now the increments vanish, and their last proportion will be 1 to nxn−1: hence the fluxion of the quantity x is to the fluxion of the quantityxn as 1 : nxn−1. “The fluxion of lines, straight or curved, in all cases whatever, as also the fluxions of superficies, angles, and other quantities, can be obtained in the same manner by the method of prime and ultimate ratios. But to establish in this way the analysis of infinite quantities, and to investigate prime and ultimate ratios of finite quantities, nascent or evanescent, is in harmony with the geometry of the ancients; and I have endeavoured to show that, in the method of fluxions, it is not necessary to introduce into geometry infinitely small quantities.” This mode of differentiating does not remove all the difficulties connected with the subject. When 0 becomes nothing, then we get the ratio0 0 = nxn−1, which needs further elucidation. Indeed, the method of Newton, as delivered by himself, is encumbered with difficulties and objections. Among the ablest admirers of Newton, there have been obstinate disputes respecting his explanation of his method of “prime and ultimate ratios.” The so-called “method of limits” is frequently attributed to Newton, but the pure method of limits was never adopted by him as his method of constructing the calculus. All he did was to establish in his Principia certain principles which are applicable to that method, but which he used for a different

NEWTON TO EULER. 247 purpose. The first lemma of the first book has been made the foundation of the method of limits:— “Quantities and the ratios of quantities, which in any finite time converge continually to equality, and before the end of that time approach nearer the one to the other than by any given difference, become ultimately equal.” In this, as well as in the lemmas following this, there are obscurities and difficulties. Newton appears to teach that a variable quantity and its limit will ultimately coincide and be equal. But it is now generally agreed that in the clearest statements which have been made of the theory of limits, the variable does not actually reach its limit, though the variable may approach it as near as we please. The full title of Newton’sPrincipia is Philosophiæ Naturalis Principia Mathematica. It was printed in 1687 under the direction, and at the expense, of Dr. Edmund Halley. A second edition was brought out in 1713 with many alterations and improvements, and accompanied by a preface from Mr. Cotes. It was sold out in a few months, but a pirated edition published in Amsterdam supplied the demand. [34] The third and last edition which appeared in England during Newton’s lifetime was published in 1726 by Henry Pemberton. The Principia consists of three books, of which the first two, constituting the great bulk of the work, treat of the mathematical principles of natural philosophy, namely, the laws and conditions of motions and forces. In the third book is drawn up the constitution of the universe as deduced from the foregoing principles. The great principle underlying this

A HISTORY OF MATHEMATICS. 248 memorable work is that of universal gravitation. The first book was completed on April 28, 1686. After the remarkably short period of three months, the second book was finished. The third book is the result of the next nine or ten months’ labours. It is only a sketch of a much more extended elaboration of the subject which he had planned, but which was never brought to completion. The law of gravitation is enunciated in the first book. Its discovery envelops the name of Newton in a halo of perpetual glory. The current version of the discovery is as follows: it was conjectured by Hooke, Huygens, Halley, Wren, Newton, and others, that, if Kepler’s third law was true (its absolute accuracy was doubted at that time), then the attraction between the earth and other members of the solar system varied inversely as the square of the distance. But the proof of the truth or falsity of the guess was wanting. In 1666 Newton reasoned, in substance, that if g represent the acceleration of gravity on the surface of the earth, r be the earth’s radius, R the distance of the moon from the earth,T the time of lunar revolution, and a a degree at the equator, then, if the law is true, g r2 R2 = 4π2 R T 2 , or g = 4π T 2 (R r)3 ·180a. The data at Newton’s command gaveR = 60.4r, T = 2, 360, 628 seconds, but a only 60 instead of 69 1 2 English miles. This wrong value of a rendered the calculated value of g smaller than its true value, as known from actual measurement. It looked as though the law of inverse squares were not the true

NEWTON TO EULER. 249 law, and Newton laid the calculation aside. In 1684 he casually ascertained at a meeting of the Royal Society that Jean Picard had measured an arc of the meridian, and obtained a more accurate value for the earth’s radius. Taking the corrected value for a, he found a figure for g which corresponded to the known value. Thus the law of inverse squares was verified. In a scholium in the Principia, Newton acknowledged his indebtedness to Huygens for the laws on

centrifugal force employed in his calculation. The perusal by the astronomer Adams of a great mass of unpublished letters and manuscripts of Newton forming the Portsmouth collection (which remained private property until 1872, when its owner placed it in the hands of the University of Cambridge) seems to indicate that the difficulties encountered by Newton in the above calculation were of a different nature. According to Adams, Newton's numerical verification was fairly complete in 1666, but Newton had not been able to determine what the attraction of a spherical shell upon an external point would be. His letters to Halley show that he did not suppose the earth to attract as though all its mass were concentrated into a point at the centre. He could not have asserted, therefore, that the assumed law of gravity was verified by the figures, though for long distances he might have claimed that it yielded close approximations. When Halley visited Newton in 1684, he requested Newton to determine what the orbit of a planet would be if the law of attraction were that of inverse squares. Newton had solved a similar problem for Hooke in 1679, and replied at once that it

A HISTORY OF MATHEMATICS. 250 was an ellipse. After Halley's visit, Newton, with Picard's new value for the earth's radius, reviewed his early calculation, and was able to show that if the distances between the bodies in the solar system were so great that the bodies might be considered as points, then their motions were in accordance with the assumed law of gravitation. In 1685 he completed his discovery by showing that a sphere whose density at any point depends only on the distance from the centre attracts an external point as though its whole mass were concentrated at the centre. [34] Newton's unpublished manuscripts in the Portsmouth col- lection show that he had worked out, by means of fluxions and fluents, his lunar calculations to a higher degree of approxima- tion than that given in the Principia, but that he was unable to interpret his results geometrically. The papers in that col- lection throw light upon the mode by which Newton arrived at some of the results in the Principia, as, for instance, the famous construction in Book II., Prop. 25, which is unproved in the Principia, but is demonstrated by him twice in a draft of a letter to David Gregory, of Oxford. [34] It is chiefly upon the Principia that the fame of Newton rests. Brewster calls it "the brightest page in the records of human reason." Let us listen, for a moment, to the comments of Laplace, the foremost among those followers of Newton who grappled with the subtle problems of the motions of planets under the influence of gravitation: "Newton has well established the existence of the principle which he had the merit of discovering, but the development of its consequences

NEWTON TO EULER. 251 and advantages has been the work of the successors of this great mathematician. The imperfection of the infinitesimal calculus, when first discovered, did not allow him completely to resolve the difficult problems which the theory of the universe offers; and he was oftentimes forced to give mere hints, which were always uncertain till confirmed by rigorous analysis. Notwithstanding these unavoidable defects, the importance and the generality of his discoveries respecting the system of the universe, and the most interesting points of natural philosophy, the great number of profound and original views, which have been the origin of the most brilliant discoveries of the mathematicians of the last century, which were all

presented with much elegance, will insure to the Principia a lasting pre-eminence over all other productions of the human mind." Newton's Arithmetica Universalis, consisting of algebraical lectures delivered by him during the first nine years he was professor at Cambridge, were published in 1707, or more than thirty years after they were written. This work was published by Mr. Whiston. We are not accurately informed how Mr. Whiston came in possession of it, but according to some authorities its publication was a breach of confidence on his part. The Arithmetica Universalis contains new and important results on the theory of equations. His theorem on the sums of powers of roots is well known. Newton showed that in equations with real coefficients, imaginary roots always occur in pairs. His inventive genius is grandly displayed in his rule

A HISTORY OF MATHEMATICS. 252 for determining the inferior limit of the number of imaginary roots, and the superior limits for the number of positive and negative roots. Though less expeditious than Descartes', Newton's rule always gives as close, and generally closer, limits to the number of positive and negative roots. Newton did not prove his rule. It awaited demonstration for a century and a half, until, at last, Sylvester established a remarkable general theorem which includes Newton's rule as a special case. The treatise on Method of Fluxions contains Newton's method of approximating to the roots of numerical equations. This is simply the method of Vieta improved. The same treatise contains "Newton's parallelogram," which enabled him, in an equation, f(x, y) = 0, to find a series in powers ofx equal to the variable y. The great utility of this rule lay in its determining the form of the series; for, as soon as the law was known by which the exponents in the series vary, then the expansion could be effected by the method of indeterminate coefficients. The rule is still used in determining the infinite branches to curves, or their figure at multiple points. Newton gave no proof for it, nor any clue as to how he discovered it. The proof was supplied half a century later, by Kaestner and Cramer, independently. [37] In 1704 was published, as an appendix to the Opticks, the Enumeratio linearum tertii ordinis , which contains the- orems on the theory of curves. Newton divides cubics into seventy-two species, arranged in larger groups, for which his commentators have supplied the names "genera" and

NEWTON TO EULER. 253 "classes," recognising fourteen of the former and seven (or four) of the latter. He overlooked six species demanded by his principles of classification, and afterwards added by Stirling, Murdoch, and Cramer. He enunciates the remarkable theorem that the five species which he names "divergent parabolas" give by their projection every cubic curve whatever. As a rule, the tract contains no proofs. It has been the subject of frequent conjecture how Newton deduced his results. Recently we have gotten at the facts, since much of the analysis used by Newton and a few additional theorems have been discovered among the Portsmouth papers. An account of the four holograph manuscripts on this subject has been published by W. W. Rouse Ball, in the Transactions of the London Mathematical Society (vol. xx., pp. 104–143). It is interesting to observe how Newton begins his research on the classification of cubic curves by the algebraic method, but, finding it laborious, attacks the problem geometrically, and afterwards returns again to analysis. [36] Space does not permit us to do more than merely men- tion Newton's

prolonged researches in other departments of science. He conducted a long series of experiments in optics and is the author of the corpuscular theory of light. The last of a number of papers on optics, which he contributed to the Royal Society, 1687, elaborates the theory of “fits.” He explained the decomposition of light and the theory of the rainbow. By him were invented the reflecting telescope and the sextant (afterwards re-discovered by Thomas Godfrey of Philadelphia [2] and by John Hadley). He deduced a theo-

A HISTORY OF MATHEMATICS. 254 retical expression for the velocity of sound in air, engaged in experiments on chemistry, elasticity, magnetism, and the law of cooling, and entered upon geological speculations. During the two years following the close of 1692, Newton suffered from insomnia and nervous irritability. Some thought that he laboured under temporary mental aberration. Though he recovered his tranquillity and strength of mind, the time of great discoveries was over; he would study out questions propounded to him, but no longer did he by his own accord enter upon new fields of research. The most noted investigation after his sickness was the testing of his lunar theory by the observations of Flamsteed, the astronomer royal. In 1695 he was appointed warden, and in 1699 master, of the mint, which office he held until his death. His body was interred in Westminster Abbey, where in 1731 a magnificent monument was erected, bearing an inscription ending with, “Sibi gratulentur mortales tale tantumque exstitisse humani generis decus.” It is not true that the Binomial Theorem is also engraved on it. We pass to Leibniz, the second and independent inventor of the calculus. Gottfried Wilhelm Leibniz (1646–1716) was born in Leipzig. No period in the history of any civilised nation could have been less favourable for literary and scientific pursuits than the middle of the seventeenth century in Germany. Yet circumstances seem to have happily combined to bestow on the youthful genius an education hardly otherwise obtainable during this darkest period of German history. He was brought early in contact with the best of

NEWTON TO EULER. 255 the culture then existing. In his fifteenth year he entered the University of Leipzig. Though law was his principal study, he applied himself with great diligence to every branch of knowledge. Instruction in German universities was then very low. The higher mathematics was not taught at all. We are told that a certain John Kuhn lectured on Euclid’sElements, but that his lectures were so obscure that none except Leibniz could understand them. Later on, Leibniz attended, for a half- year, at Jena, the lectures of Erhard Weigel, a philosopher and mathematician of local reputation. In 1666 Leibniz published a treatise, De Arte Combinatoria, in which he does not pass beyond the rudiments of mathematics. Other theses written by him at this time were metaphysical and juristical in character. A fortunate circumstance led Leibniz abroad. In 1672 he was sent by Baron Boineburg on a political mission to Paris. He there formed the acquaintance of the most distinguished men of the age. Among these was Huygens, who presented a copy of his work on the oscillation of the pendulum to Leibniz, and first led the gifted young German to the study of higher mathematics. In 1673 Leibniz went to London, and remained there from January till March. He there became incidentally acquainted with the mathematician Pell, to whom he explained a method he had found on the summation of series of numbers by their differences. Pell told him that a similar

formula had been published by Mouton as early as 1670, and then called his attention to Mercator's work on the rectification of the parabola. While in London, Leibniz exhibited to the Royal Society his arithmetical machine, which

A HISTORY OF MATHEMATICS. 256 was similar to Pascal's, but more efficient and perfect. After his return to Paris, he had the leisure to study mathematics more systematically. With indomitable energy he set about removing his ignorance of higher mathematics. Huygens was his principal master. He studied the geometric works of Descartes, Honorarius Fabri, Gregory St. Vincent, and Pascal. A careful study of infinite series led him to the discovery of the following expression for the ratio of the circumference to the diameter of the circle, previously discovered by James Gregory:— π 4 = 1− 1 3 + 1 5− 1 7 + 1 9− etc. This elegant series was found in the same way as Mercator's on the hyperbola. Huygens was highly pleased with it and urged him on to new investigations. Leibniz entered into a detailed study of the quadrature of curves and thereby became intimately acquainted with the higher mathematics. Among the papers of Leibniz is still found a manuscript on quadratures, written before he left Paris in 1676, but which was never printed by him. The more important parts of it were embodied in articles published later in theActa Eruditorum. In the study of Cartesian geometry the attention of Leibniz was drawn early to the direct and inverse problems of tangents. The direct problem had been solved by Descartes for the simplest curves only; while the inverse had completely transcended the power of his analysis. Leibniz investigated both problems for any curve; he constructed what he called the triangulum characteristicum—an infinitely small triangle between the infinitely small part of the curve coinciding

NEWTON TO EULER. 257 with the tangent, and the differences of the ordinates and abscissas. A curve is here considered to be a polygon. The triangulum characteristicum is similar to the triangle formed by the tangent, the ordinate of the point of contact, and the sub-tangent, as well as to that between the ordinate, normal, and sub-normal. It was first employed by Barrow in England, but appears to have been re-invented by Leibniz. From it Leibniz observed the connection existing between the direct and inverse problems of tangents. He saw also that the latter could be carried back to the quadrature of curves. All these results are contained in a manuscript of Leibniz, written in 1673. One mode used by him in effecting quadratures was as follows: The rectangle formed by a sub-tangent p and an element a (i.e. infinitely small part of the abscissa) is equal to the rectangle formed by the ordinate y and the element l of that ordinate; or in symbols, pa = yl. But the summation of these rectangles from zero on gives a right triangle equal to half the square of the ordinate. Thus, using Cavalieri's notation, he gets omn. pa = omn. yl = y2 2 (omn. meaning omnia, all). But y = omn. l; hence omn. omn. l l a = omn. l2 2a . This equation is especially interesting, since it is here that Leibniz first introduces a new notation. He says: "It will be useful to write ∫ for omn., as ∫ l for omn. l, that is, the sum of

A HISTORY OF MATHEMATICS. 258 the l's"; he then writes the equation thus:— ∫ l2 2a = ∫ ∫ ¯l l a . From this he deduced the simplest integrals, such as ∫ x = x2 2 , ∫ (x + y) = ∫ x + ∫ y. Since the symbol of summation ∫ raises the dimensions, he

concluded that the opposite calculus, or that of differences d, would lower them. Thus, if ∫ l = ya, then l = ya d . The symbol d was at first placed by Leibniz in the denominator, because the lowering of the power of a term was brought about in ordinary calculation by division. The manuscript giving the above is dated October 29th, 1675. [39] This, then, was the memorable day on which the notation of the new calculus came to be,—a notation which contributed enormously to the rapid growth and perfect development of the calculus. Leibniz proceeded to apply his new calculus to the solution of certain problems then grouped together under the name of the Inverse Problems of Tangents. He found the cubical parabola to be the solution to the following: To find the curve in which the sub-normal is reciprocally proportional to the ordinate. The correctness of his solution was tested by him by applyingtotheresultSluze'smethodoftangentsandreasoning backwards to the original supposition. In the solution of the third problem he changes his notation fromx d to the now usual notation dx. It is worthy of remark that in these investigations, Leibniz nowhere explains the significance ofdx and dy, except

NEWTON TO EULER. 259 at one place in a marginal note: "Idem est dx et x d, id est, differentia inter duas x proximas." Nor does he use the term differential, but always difference. Not till ten years later, in the Acta Eruditorum, did he give further explanations of these symbols. What he aimed at principally was to determine the change an expression undergoes when the symbol ∫ or d is placed before it. It may be a consolation to students wrestling with the elements of the differential calculus to know that it required Leibniz considerable thought and attention [39] to determine whether dx dy is the same asd(xy), and dx dy the same as d x y . After considering these questions at the close of one of his manuscripts, he concluded that the expressions were not the same, though he could not give the true value for each. Ten days later, in a manuscript dated November 21, 1675, he found the equation y dx = d xy− x dy, giving an expression for d(xy), which he observed to be true for all curves. He succeeded also in eliminating dx from a differential equation, so that it contained only dy, and thereby led to the solution of the problem under consideration. "Behold, a most elegant way by which the problems of the inverse methods of tangents are solved, or at least are reduced to quadratures!" Thus he saw clearly that the inverse problems of tangents could be solved by quadratures, or, in other words, by the integral calculus. In course of a half-year he discovered that the direct problem of tangents, too, yielded to the power of his new calculus, and that thereby a more general solution than that of Descartes could be obtained. He succeeded in solving all the special problems of this kind, which had been left unsolved by

A HISTORY OF MATHEMATICS. 260 Descartes. Of these we mention only the celebrated problem proposed to Descartes by De Beaune, viz. to find the curve whose ordinate is to its sub-tangent as a given line is to that part of the ordinate which lies between the curve and a line drawn from the vertex of the curve at a given inclination to the axis. Such was, in brief, the progress in the evolution of the new calculus made by Leibniz during his stay in Paris. Before his departure, in October, 1676, he found himself in possession of the most elementary rules and formulæ of the infinitesimal calculus. From Paris, Leibniz returned to Hanover by way of London and Amsterdam. In London he met Collins, who showed him a part of his scientific correspondence. Of

this we shall speak later. In Amsterdam he discussed mathematics with Sluze, and became satisfied that his own method of constructing tangents not only accomplished all that Sluze's did, but even more, since it could be extended to three variables, by which tangent planes to surfaces could be found; and especially, since neither irrationals nor fractions prevented the immediate application of his method. In a paper of July 11, 1677, Leibniz gave correct rules for the differentiation of sums, products, quotients, powers, and roots. He had given the differentials of a few negative and fractional powers, as early as November, 1676, but had made some mistakes. For d√x he had given the erroneous value 1√x, and in another place the value− 1 2 x− 1 2 ; for d 1 x3 occurs in one place the wrong value, − 2 x2 , while a few lines lower is given

NEWTON TO EULER. 261 − 3 x4 , its correct value. In 1682 was founded in Berlin the Acta Eruditorum , a journal usually known by the name of Leipzig Acts. It was a partial imitation of the French Journal des Savans (founded in 1665), and the literary and scientific review published in Germany. Leibniz was a frequent contributor. Tschirnhaus, who had studied mathematics in Paris with Leibniz, and who was familiar with the new analysis of Leibniz, published in the Acta Eruditorum a paper on quadratures, which consists principally of subject-matter communicated by Leibniz to Tschirnhaus during a controversy which they had had on this subject. Fearing that Tschirnhaus might claim as his own and publish the notation and rules of the differential calculus, Leibniz decided, at last, to make public the fruits of his inventions. In 1684, or nine years after the new calculus first dawned upon the mind of Leibniz, and nineteen years after Newton first worked at fluxions, and three years before the publication of Newton's Principia, Leibniz published, in the Leipzig Acts, his first paper on the differential calculus. He was unwilling to give to the world all his treasures, but chose those parts of his work which were most abstruse and least perspicuous. This epoch-making paper of only six pages bears the title: "Nova methodus pro maximis et minimis, itemque tangentibus, quæ nec fractas nec irrationales quantitates moratur, et singulare pro illis calculi genus." The rules of calculation are briefly stated without proof, and the meaning of dx and dy is not made clear. It has been inferred from this that Leibniz himself had no definite and settled ideas on this

A HISTORY OF MATHEMATICS. 262 subject. Are dy and dx finite or infinitesimal quantities? At first they appear, indeed, to have been taken as finite, when he says: "We now call any line selected at random dx, then we designate the line which is todx as y is to the sub-tangent, by dy, which is the difference of y." Leibniz then ascertains, by his calculus, in what way a ray of light passing through two differently refracting media, can travel easiest from one point to another; and then closes his article by giving his solution, in a few words, of De Beaune's problem. Two years later (1686) Leibniz published in theActa Eruditorum a paper containing the rudiments of the integral calculus. The quantities dx and dy are there treated as infinitely small. He showed that by the use of his notation, the properties of curves could be fully expressed by equations. Thus the equation y = √ 2x− x2 + ∫ dx√ 2x− x2 characterises the cycloid. [38] The great invention of Leibniz, now made public by his articles in the Leipzig Acts, made little impression upon the mass of mathematicians. In Germany no

one comprehended the new calculus except Tschirnhaus, who remained indifferent to it. The author’s statements were too short and succinct to make the calculus generally understood. The first to recognise its importance and to take up the study of it were two foreigners,—the Scotchman John Craig , and the Swiss James Bernoulli . The latter wrote Leibniz a letter in 1687, wishing to be initiated into the mysteries of the new analysis. Leibniz was then travelling abroad, so that

NEWTON TO EULER. 263 this letter remained unanswered till 1690. James Bernoulli succeeded, meanwhile, by close application, in uncovering the secrets of the differential calculus without assistance. He and his brother John proved to be mathematicians of exceptional power. They applied themselves to the new science with a success and to an extent which made Leibniz declare that it was as much theirs as his. Leibniz carried on an extensive correspondence with them, as well as with other mathematicians. In a letter to John Bernoulli he suggests, among other things, that the integral calculus be improved by reducing integrals back to certain fundamental irreducible forms. The integration of logarithmic expressions was then studied. The writings of Leibniz contain many innovations, and anticipations of since prominent methods. Thus he made use of variable parameters, laid the foundation of analysis in situ , introduced the first notion of determinants in his effort to simplify the expression arising in the elimination of the unknown quantities from a set of linear equations. He resorted to the device of breaking up certain fractions into the sum of other fractions for the purpose of easier integration; he explicitly assumed the principle of continuity; he gave the first instance of a “singular solution,” and laid the foundation to the theory of envelopes in two papers, one of which contains for the first time the terms co-ordinate and axes of co-ordinates. He wrote on osculating curves, but his paper contained the error (pointed out by John Bernoulli, but not admitted by him) that an osculating circle will necessarily cut a curve in four consecutive points. Well known is his theorem on the

A HISTORY OF MATHEMATICS. 264 nth differential coefficient of the product of two functions of a variable. Of his many papers on mechanics, some are valuable, while others contain grave errors. Before tracing the further development of the calculus we shall sketch the history of that long and bitter controversy between English and Continental mathematicians on the invention of the calculus. The question was, did Leibniz invent it independently of Newton, or was he a plagiarist? We must begin with the early correspondence between the parties appearing in this dispute. Newton had begun using his notation of fluxions in 1666. [41] In 1669 Barrow sent Collins Newton’s tract, De Analysi per Equationes, etc. The first visit of Leibniz to London extended from the 11th of January until March, 1673. He was in the habit of committing to writing important scientific communications received from others. In 1890 Gerhardt discovered in the royal library at Hanover a sheet of manuscript with notes taken by Leibniz during this journey. [40] They are headed “Observata Philosophica in itinere Anglicano sub initium anni 1673.” The sheet is divided by horizontal lines into sec- tions. The sections given to Chymica, Mechanica, Magnetica, Botanica, Anatomica, Medica, Miscellanea, contain extensive memoranda, while those devoted to mathematics have very few notes. Under Geometrica he says only this: “Tangentes omnium figurarum. Figurarum geometricarum explicatio per

motum puncti in moto lati." We suspect from this that Leibniz had read Barrow's lectures. Newton is referred to only under Optica. Evidently Leibniz did not obtain a knowledge

NEWTON TO EULER. 265 of fluxions during this visit to London, nor is it claimed that he did by his opponents. Various letters of Newton, Collins, and others, up to the beginning of 1676, state that Newton invented a method by which tangents could be drawn without the necessity of freeing their equations from irrational terms. Leibniz announced in 1674 to Oldenburg, then secretary of the Royal Society, that he possessed very general analytical methods, by which he had found theorems of great importance on the quadrature of the circle by means of series. In answer, Oldenburg stated Newton and James Gregory had also discovered methods of quadratures, which extended to the circle. Leibniz desired to have these methods communicated to him; and Newton, at the request of Oldenburg and Collins, wrote to the former the celebrated letters of June 13 and October 24, 1676. The first contained the Binomial Theorem and a variety of other matters relating to infinite series and quadratures; but nothing directly on the method of fluxions. Leibniz in reply speaks in the highest terms of what Newton had done, and requests further explanation. Newton in his second letter just mentioned explains the way in which he found the Binomial Theorem, and also communicates his method of fluxions and fluents in form of an anagram in which all the letters in the sentence communicated were placed in alphabetical order. Thus Newton says that his method of drawing tangents was 6 a cc d æ 13 e ff 7 i 3 l 9 n 4 o 4 q rr 4 s 9 t 12 v x. The sentence was, "Data æquatione quotcunque fluentes

A HISTORY OF MATHEMATICS. 266 quantitates involvente fluxiones invenire, et vice versa." ("Having any given equation involving never so many flowing quantities, to find the fluxions, and vice versa.") Surely this anagram afforded no hint. Leibniz wrote a reply to Collins, in which, without any desire of concealment, he explained the principle, notation, and the use of the differential calculus. The death of Oldenburg brought this correspondence to a close. Nothing material happened till 1684, when Leibniz published his first paper on the differential calculus in the Leipzig Acts, so that while Newton's claim to the priority of invention must be admitted by all, it must also be granted that Leibniz was the first to give the full benefit of the calculus to the world. Thus, while Newton's invention remained a secret, communicated only to a few friends, the calculus of Leibniz was spreading over the Continent. No rivalry or hostility existed, as yet, between the illustrious scientists. Newton expressed a very favourable opinion of Leibniz's inventions, known to him through the above correspondence with Oldenburg, in the following celebrated scholium (Principia, first edition, 1687, Book II., Prop. 7, scholium):— "In letters which went between me and that most excellent geometer, G. G. Leibniz, ten years ago, when I signified that I was in the knowledge of a method of determining maxima and minima, of drawing tangents, and the like, and when I concealed it in transposed letters involving this sentence (Data æquatione, etc., above cited), that most distinguished man wrote back that he had also fallen upon a method of the same kind, and communicated his method, which hardly

NEWTON TO EULER. 267 differed from mine, except in his forms of words and symbols.” As regards this passage, we shall see that Newton was afterwards weak enough, as De Morgan says: “First, to deny the plain and obvious meaning, and secondly, to omit it entirely from the third edition of the Principia.” On the Continent, great progress was made in the calculus by Leibniz and his coadjutors, the brothers James and John Bernoulli, and Marquis de l’Hospital. In 1695 Wallis informed Newton by letter that “he had heard that his notions of fluxions passed in Holland with great applause by the name of ‘Leibniz’s Calculus Differentialis.’ ” Accordingly Wallis stated in the preface to a volume of his works that the calculus differentialis was Newton’s method of fluxions which had been communicated to Leibniz in the Oldenburg letters. A review of Wallis’ works, in the Leipzig Acts for 1696, reminded the reader of Newton’s own admission in the scholium above cited. For fifteen years Leibniz had enjoyed unchallenged the honour of being the inventor of his calculus. But in 1699 Fato de Duillier, a Swiss, who had settled in England, stated in a mathematical paper, presented to the Royal Society, his conviction that Newton was the first inventor; adding that, whether Leibniz, the second inventor, had borrowed anything from the other, he would leave to the judgment of those who had seen the letters and manuscripts of Newton. This was the first distinct insinuation of plagiarism. It would seem that the English mathematicians had for some time been cherishing suspicions unfavourable to Leibniz. A feeling had doubtless long prevailed that Leibniz, during his second

A HISTORY OF MATHEMATICS. 268 visit to London in 1676, had or might have seen among the papers of Collins, Newton’sAnalysis per æquationes, etc., which contained applications of the fluxionary method, but no systematic development or explanation of it. Leibniz certainly did see at least part of this tract. During the week spent in London, he took note of whatever interested him among the letters and papers of Collins. His memoranda discovered by Gerhardt in 1849 in the Hanover library fill two sheets. [40] The one bearing on our question is headed “Excerpta ex tractatu Newtoni Msc. de Analysi per æquationes numero terminorum infinitas.” The notes are very brief, excepting those De Resolutione æquationum affectarum, of which there is an almost complete copy. This part was evidently new to him. If he examined Newton’s entire tract, the other parts did not particularly impress him. From it he seems to have gained nothing pertaining to the infinitesimal calculus. By the previous introduction of his own algorithm he had made greater progress than by what came to his knowledge in London. Nothing mathematical that he had received engaged his thoughts in the immediate future, for on his way back to Holland he composed a lengthy dialogue on mechanical subjects. Duillier’s insinuations lighted up a flame of discord which a wholecenturywashardlysufficienttoextinguish. Leibniz, who had never contested the priority of Newton’s discovery, and who appeared to be quite satisfied with Newton’s admission in his scholium, now appears for the first time in the controversy. Hemadeananimatedreplyinthe Leipzig Acts, andcomplained

NEWTON TO EULER. 269 to the Royal Society of the injustice done him. Here the affair rested for some time. In the Quadrature of Curves, published 1704, for the first time, a formal exposition of the method and notation of fluxions was made public. In 1705 appeared an unfavourable review of this in theLeipzig Acts, stating that Newton

uses and always has used fluxions for the differences of Leibniz. This was considered by Newton's friends an imputation of plagiarism on the part of their chief, but this interpretation was always strenuously resisted by Leibniz. Keill, professor of astronomy at Oxford, undertook with more zeal than judgment the defence of Newton. In a paper inserted in the Philosophical Transactions of 1708, he claimed that Newton was the first inventor of fluxions and "that the same calculus was afterward published by Leibniz, the name and the mode of notation being changed." Leibniz complained to the secretary of the Royal Society of bad treatment and requested the interference of that body to induce Keill to disavow the intention of imputing fraud. Keill was not made to retract his accusation; on the contrary, was authorised by Newton and the Royal Society to explain and defend his statement. This he did in a long letter. Leibniz thereupon complained that the charge was now more open than before, and appealed for justice to the Royal Society and to Newton himself. The Royal Society, thus appealed to as a judge, appointed a committee which collected and reported upon a large mass of documents—mostly letters from and to Newton, Leibniz, Wallis, Collins, etc. This report, called the Commercium Epistolicum, appeared in the year 1712 and

A HISTORY OF MATHEMATICS. 270 again in 1725, with a Recensio prefixed, and additional notes by Keill. The final conclusion in theCommercium Epistolicum was that Newton was the first inventor. But this was not to the point. The question was not whether Newton was the first inventor, but whether Leibniz had stolen the method. The committee had not formally ventured to assert their belief that Leibniz was a plagiarist. Yet there runs throughout the document a desire of proving Leibniz guilty of more than they meant positively to affirm. Leibniz protested only in private letters against the proceeding of the Royal Society, declaring that he would not answer an argument so weak. John Bernoulli, in a letter to Leibniz, which was published later in an anonymous tract, is as decidedly unfair towards Newton as the friends of the latter had been towards Leibniz. Keill replied, and then Newton and Leibniz appear as mutual accusers in several letters addressed to third parties. In a letter to Conti, April 9, 1716, Leibniz again reminded Newton of the admission he had made in the scholium, which he was now desirous of disavowing; Leibniz also states that he always believed Newton, but that, seeing him connive at accusations which he must have known to be false, it was natural that he (Leibniz) should begin to doubt. Newton did not reply to this letter, but circulated some remarks among his friends which he published immediately after hearing of the death of Leibniz, November 14, 1716. This paper of Newton gives the following explanation pertaining to the scholium in question: "He [Leibniz] pretends that in my book of principles I allowed him the invention of the calculus differentialis, independently

NEWTON TO EULER. 271 of my own; and that to attribute this invention to myself is contrary to my knowledge there avowed. But in the paragraph there referred unto I do not find one word to this purpose." In the third edition of the Principia, 1726, Newton omitted the scholium and substituted in its place another, in which the name of Leibniz does not appear. National pride and party feeling long prevented the adoption of impartial opinions in England, but now it is generally admitted by nearly all familiar with the matter, that Leibniz really was an independent inventor. Perhaps

the most telling evidence to show that Leibniz was an independent inventor is found in the study of his mathematical papers (collected and edited by C. I. Gerhardt, in six volumes, Berlin, 1849–1860), which point out a gradual and natural evolution of the rules of the calculus in his own mind. “There was throughout the whole dispute,” says De Morgan, “a confusion between the knowledge of fluxions or differentials and that of a calculus of fluxions or differentials; that is, a digested method with general rules.” This controversy is to be regretted on account of the long and bitter alienation which it produced between English and Continental mathematicians. It stopped almost completely all interchange of ideas on scientific subjects. The English adhered closely to Newton’s methods and, until about 1820, remained, in most cases, ignorant of the brilliant mathematical discoveries that were being made on the Continent. The loss in point of scientific advantage was almost entirely on the side of Britain. The only way in which this dispute may

A HISTORY OF MATHEMATICS. 272 be said, in a small measure, to have furthered the progress of mathematics, is through the challenge problems by which each side attempted to annoy its adversaries. The recurring practice of issuing challenge problems was inaugurated at this time by Leibniz. They were, at first, not intended as defiances, but merely as exercises in the new calculus. Such was the problem of the isochronous curve (to find the curve along which a body falls with uniform velocity), proposed by him to the Cartesians in 1687, and solved by James Bernoulli, himself, and John Bernoulli. James Bernoulli proposed in the LeipzigJournal the question to find the curve (the catenary) formed by a chain of uniform weight suspended freely from its ends. It was resolved by Huygens, Leibniz, and himself. In 1697 John Bernoulli challenged the best mathematicians in Europe to solve the difficult problem, to find the curve (the cycloid) along which a body falls from one point to another in the shortest possible time. Leibniz solved it the day he received it. Newton, de l’Hospital, and the two Bernoullis gave solutions. Newton’s appeared anonymously in the Philosophical Transactions, but John Bernoulli recognised in it his powerful mind, “anquam,” he says, “ex ungue leonem.” The problem of orthogonal trajectories (a system of curves described by a known law being given, to describe a curve which shall cut them all at right angles) had been long proposed in theActa Eruditorum, but failed at first to receive much attention. It was again proposed in 1716 by Leibniz, to feel the pulse of the English mathematicians.

NEWTON TO EULER. 273 This may be considered as the first defiance problem professedly aimed at the English. Newton solved it the same evening on which it was delivered to him, although he was much fatigued by the day’s work at the mint. His solution, as published, was a general plan of an investigation rather than an actual solution, and was, on that account, criticised by Bernoulli as being of no value. Brook Taylor undertook the defence of it, but ended by using very reprehensible language. Bernoulli was not to be outdone in incivility, and made a bitter reply. Not long afterwards Taylor sent an open defiance to Continental mathematicians of a problem on the integration of a fluxion of complicated form which was known to very few geometers in England and supposed to be beyond the power of their adversaries. The selection was injudicious, for Bernoulli had long before explained the method of this and similar integrations. It served only to display the skill and augment the triumph

of the followers of Leibniz. The last and most unskilful challenge was by John Keill. The problem was to find the path of a projectile in a medium which resists proportionally to the square of the velocity. Without first making sure that he himself could solve it, Keill boldly challenged Bernoulli to produce a solution. The latter resolved the question in very short time, not only for a resistance proportional to the square, but to any power of the velocity. Suspecting the weakness of the adversary, he repeatedly offered to send his solution to a confidential person in London, provided Keill would do the same. Keill never made a reply, and Bernoulli abused him and cruelly exulted

A HISTORY OF MATHEMATICS. 274 over him. [26] The explanations of the fundamental principles of the calculus, as given by Newton and Leibniz, lacked clearness and rigour. For that reason it met with opposition from several quarters. In 1694 Bernard Nieuwentyt of Holland denied the existence of differentials of higher orders and objected to the practice of neglecting infinitely small quantities. These objections Leibniz was not able to meet satisfactorily. In his reply he said the value of dy dx in geometry could be expressed as the ratio of finite quantities. In the interpretation of dx and dy Leibniz vacillated. At one time they appear in his writings as finite lines; then they are called infinitely small quantities, and again,quantitates inassignabiles, which spring from quantitates assignabiles by the law of continuity. In this last presentation Leibniz approached nearest to Newton. In England the principles of fluxions were boldly attacked by Bishop Berkeley, the eminent metaphysician, who argued with great acuteness, contending, among other things, that the fundamental idea of supposing a finite ratio to exist between terms absolutely evanescent—“the ghosts of departed quantities,” as he called them—was absurd and unintelligible. The reply made by Jurin failed to remove all the objections. Berkeley was the first to point out what was again shown later by Lazare Carnot, that correct answers were reached by a “compensation of errors.” Berkeley’s attack was not devoid of good results, for it was the immediate cause of the work on fluxions by Maclaurin. In France Michel Rolle rejected the differential calculus and had a controversy with Varignon on

NEWTON TO EULER. 275 the subject. Among the most vigorous promoters of the calculus on the Continent were the Bernoullis. They and Euler made Basel in Switzerland famous as the cradle of great mathematicians. The family of Bernoullis furnished in course of a century eight members who distinguished themselves in mathematics. We subjoin the following genealogical table:— Nicolaus Bernoulli, the Father□ □□ □ Jacob, 1654–1705 Nicolaus Johann, 1667–1748 | | Nicolaus, 1687–1759 Nicolaus, 1695–1726 Daniel, 1700–1782 Johann, 1710–1790 □ □□ □ Daniel Johann , 1744–1807 Jacob, 1758–1789 Most celebrated were the two brothers Jacob (James) and Johann (John), and Daniel, the son of John. James and John were staunch friends of Leibniz and worked hand in hand with him. James Bernoulli (1654–1705) was born in Basel. Becoming interested in the calculus, he mastered it without aid from a teacher. From 1687 until his death he occupied the mathematical chair at the University of Basel. He was the first to give a solution to Leibniz’s problem of the isochronous curve. In his solution, published in the Acta Eruditorum, 1690, we meet for the first time with the wordintegral. Leibniz had called the integral calculus

calculus summatorius, but in 1696 the term calculus integralis was agreed upon between Leibniz and John Bernoulli. James proposed the problem

A HISTORY OF MATHEMATICS. 276 of the catenary, then proved the correctness of Leibniz's construction of this curve, and solved the more complicated problems, supposing the string to be (1) of variable density, (2) extensible, (3) acted upon at each point by a force directed to a fixed centre. Of these problems he published answers without explanations, while his brother John gave in addition their theory. He determined the shape of the "elastic curve" formed by an elastic plate or rod fixed at one end and bent by a weight applied to the other end; of the "lintearia," a flexible rectangular plate with two sides fixed horizontally at the same height, filled with a liquid; of the "volaria," a rectangular sail filled with wind. He studied the loxodromic and logarithmic spirals, in the last of which he took particular delight from its remarkable property of reproducing itself under a variety of conditions. Following the example of Archimedes, he willed that the curve be engraved upon his tomb-stone with the inscription " eadem mutata resurgo ." In 1696 he proposed the famous problem of isoperimetrical figures, and in 1701 published his own solution. He wrote a work on Ars Conjectandi, which is a development of the calculus of probabilities and contains the investigation now called "Bernoulli's theorem" and the so-called "numbers of Bernoulli," which are in fact (though not so considered by him) the coefficients of xn n! in the expansion of (ex− 1)−1. Of his collected works, in three volumes, one was printed in 1713, the other two in 1744. John Bernoulli (1667–1748) was initiated into mathe- matics by his brother. He afterwards visited France, where

NEWTON TO EULER. 277 he met Malebranche, Cassini, De Lahire, Varignon, and de l'Hospital. For ten years he occupied the mathematical chair at Gr¨ oningen and then succeeded his brother at Basel. He was one of the most enthusiastic teachers and most successful original investigators of his time. He was a member of almost every learned society in Europe. His controversies were almost as numerous as his discoveries. He was ardent in his friend- ships, but unfair, mean, and violent toward all who incurred his dislike—even his own brother and son. He had a bitter dispute with James on the isoperimetrical problem. James convicted him of several paralogisms. After his brother's death he attempted to substitute a disguised solution of the former for an incorrect one of his own. John admired the merits of Leibniz and Euler, but was blind to those of Newton. He immensely enriched the integral calculus by his labours. Among his discoveries are the exponential calculus, the line of swiftest descent, and its beautiful relation to the path described by a ray passing through strata of variable density. He treated trigonometry by the analytical method, studied caustic curves and trajectories. Several times he was given prizes by the Academy of Science in Paris. Of his sons, Nicholas and Daniel were appointed pro- fessors of mathematics at the same time in the Academy of St. Petersburg. The former soon died in the prime of life; the latter returned to Basel in 1733, where he assumed the chair of experimental philosophy. His first mathematical publication was the solution of a differential equation proposed by Riccati. He wrote a work on hydrodynamics. His investigations on

A HISTORY OF MATHEMATICS. 278 probability are remarkable for their boldness and originality. He proposed the theory ofmoral expectation, which he thought would give results more in accordance with our ordinary no- tions than the theory of mathematical probability. His “moral expectation” has become classic, but no one ever makes use of it. He applies the theory of probability to insurance; to determine the mortality caused by small-pox at various stages of life; to determine the number of survivors at a given age from a given number of births; to determine how much inoculation lengthens the average duration of life. He showed how the differential calculus could be used in the theory of probability. He and Euler enjoyed the honour of having gained or shared no less than ten prizes from the Academy of Sciences in Paris. Johann Bernoulli(born 1710) succeeded his father in the professorship of mathematics at Basel. He captured three prizes (on the capstan, the propagation of light, and the magnet) from the Academy of Sciences at Paris. Nicolaus Bernoulli (born 1687) held for a time the mathematical chair at Padua which Galileo had once filled. Johann Bernoulli (born 1744) at the age of nineteen was appointed astronomer royal at Berlin, and afterwards director of the mathematical department of the Academy. His brother Jacob took upon himself the duties of the chair of experimental physics at Basel, previously performed by his uncle Jacob, and later was appointed mathematical professor in the Academy at St. Petersburg. Brief mention will now be made of some other mathemati- cians belonging to the period of Newton, Leibniz, and the

NEWTON TO EULER. 279 elder Bernoullis. Guillaume Fran¸ cois Antoine l’Hospital(1661–1704), a pupil of John Bernoulli, has already been mentioned as taking part in the challenges issued by Leibniz and the Bernoullis. He helped powerfully in making the calculus of Leibniz better known to the mass of mathematicians by the publication of a treatise thereon in 1696. This contains for the first time the method of finding the limiting value of a fraction whose two terms tend toward zero at the same time. Another zealous French advocate of the calculus wasPierre Varignon (1654–1722). Joseph Saurin (1659–1737) solved the delicate problem of how to determine the tangents at the multiple points of algebraic curves. Fran¸ cois Nicole (1683–1758) in 1717 issued the first systematic treatise on finite differences, in which he finds the sums of a considerable number of interesting series. He wrote also on roulettes, particularly spherical epicycloids, and their rectification. Also interested in finite differences was Pierre Raymond de Montmort(1678–1719). His chief writings, on the theory of probability, served to stimulate his more distinguished successor, De Moivre. Jean Paul de Gua (1713–1785) gave the demonstration of Descartes’ rule of signs, now given in books. This skilful geometer wrote in 1740 a work on analytical geometry, the object of which was to show that most investigations on curves could be carried on with the analysis of Descartes quite as easily as with the calculus. He shows how to find the tangents, asymptotes, and various singular points of curves of all degrees, and proved by perspective that

A HISTORY OF MATHEMATICS. 280 several of these points can be at infinity. A mathematician who clung to the methods of the ancients was Philippe de Lahire (1640–1718), a pupil of Desargues. His work on conic sections is purely synthetic, but

differs from ancient treatises in deducing the properties of conics from those of the circle in the same manner as did Desargues and Pascal. His innovations stand in close relation with modern synthetic geometry. He wrote on roulettes, on graphical methods, epicycloids, conchoids, and on magic squares. Michel Rolle (1652–1719) is the author of a theorem named after him. Of Italian mathematicians, Riccati and Fagnano must not remain unmentioned. Jacopo Francesco, Count Riccati (1676–1754) is best known in connection with his problem, called Riccati's equation, published in the Acta Eruditorum in 1724. He succeeded in integrating this differential equation for some special cases. A geometrician of remarkable power was Giulio Carlo, Count de Fagnano (1682–1766). He discovered the following formula, π = 2 i log 1− i 1 + i, in which he anticipated Euler in the use of imaginary exponents and logarithms. His studies on the rectification of the ellipse and hyperbola are the starting-points of the theory of elliptic functions. He showed, for instance, that two arcs of an ellipse can be found in an indefinite number of ways, whose difference is expressible by a right line. In Germany the only noted contemporary of Leibniz is Ehrenfried Walter Tschirnhausen(1651–1708), who dis- covered the caustic of reflection, experimented on metallic reflectors and large burning-glasses, and gave us a method of

NEWTON TO EULER. 281 transforming equations named after him. Believing that the most simple methods (like those of the ancients) are the most correct, he concluded that in the researches relating to the properties of curves the calculus might as well be dispensed with. After the death of Leibniz there was in Germany not a single mathematician of note. Christian Wolf (1679–1754), professor at Halle, was ambitious to figure as successor of Leibniz, but he "forced the ingenious ideas of Leibniz into a pedantic scholasticism, and had the unenviable reputation of having presented the elements of the arithmetic, algebra, and analysis developed since the time of the Renaissance in the form of Euclid,—of course only in outward form, for into the spirit of them he was quite unable to penetrate." [16] The contemporaries and immediate successors of Newton in Great Britain were men of no mean merit. We have reference to Cotes, Taylor, Maclaurin, and De Moivre. We are told that at the death ofRogerCotes (1682–1716), Newton exclaimed, "If Cotes had lived, we might have known something." It was at the request of Dr. Bentley that Cotes undertook the publication of the second edition of Newton's Principia. His mathematical papers were published after his death by Robert Smith, his successor in the Plumbian professorship at Trinity College. The title of the work, Harmonia Mensurarum, was suggested by the following theorem contained in it: If on each radius vector, through a fixed point O, there be taken a point R, such that the reciprocal of OR be the arithmetic mean of the reciprocals of OR1, OR2, . . . ORn, then the locus

A HISTORY OF MATHEMATICS. 282 of R will be a straight line. In this work progress was made in the application of logarithms and the properties of the circle to the calculus of fluents. To Cotes we owe a theorem in trigonometry which depends on the forming of factors of xn− 1. Chief among the admirers of Newton were Taylor and Maclaurin. The quarrel between English and Continental mathematicians caused them to work quite independently of their great contemporaries across the Channel. BrookTaylor (1685–1731)wasinterestedinmanybranches of learning, and in the latter

part of his life engaged mainly in religious and philosophic speculations. His principal work, Methodus incrementorum directa et inversa , London, 1715– 1717, added a new branch to mathematics, now called “finite differences.” He made many important applications of it, particularly to the study of the form of movement of vibrating strings, first reduced to mechanical principles by him. This work contains also “Taylor’s theorem,” the importance of which was not recognised by analysts for over fifty years, until Lagrange pointed out its power. His proof of it does not consider the question of convergency, and is quite worth- less. The first rigorous proof was given a century later by Cauchy. Taylor’s work contains the first correct explanation of astronomical refraction. He wrote also a work on linear perspective, a treatise which, like his other writings, suffers for want of fulness and clearness of expression. At the age of twenty-three he gave a remarkable solution of the problem of the centre of oscillation, published in 1714. His claim to priority was unjustly disputed by John Bernoulli.

NEWTON TO EULER. 283 Colin Maclaurin (1698–1746) was elected professor of mathematicsat Aberdeenat theage ofnineteen bycompetitive examination, and in 1725 succeeded James Gregory at the University of Edinburgh. He enjoyed the friendship of Newton, and, inspired by Newton’s discoveries, he published in 1719 his Geometria Organica, containing a new and remarkable mode of generating conics, known by his name. A second tract, De Linearum geometricarum Proprietatibus , 1720, is remarkable for the elegance of its demonstrations. It is based upon two theorems: the first is the theorem of Cotes; the second is Maclaurin’s: If through any point O a line be drawn meeting the curve in n points, and at these points tangents be drawn, and if any other line through O cut the curve in R1, R2, etc., and the system of n tangents in r1, r2, etc., then ∑ 1 OR =∑ 1 Or . This and Cotes’ theorem are generalisations of theorems of Newton. Maclaurin uses these in his treatment of curves of the second and third degree, culminating in the remarkable theorem that if a quadrangle has its vertices and the two points of intersection of its opposite sides upon a curve of the third degree, then the tangents drawn at two opposite vertices cut each other on the curve. He deduced independently Pascal’s theorem on the hexagram. The following is his extension of this theorem (Phil. Trans., 1735): If a polygon move so that each of its sides passes through a fixed point, and if all its summits except one describe curves of the degrees m, n, p, etc., respectively, then the free summit moves on a curve of the degree2mnp··· , which reduces to mnp··· when the fixed points all lie on

A HISTORY OF MATHEMATICS. 284 a straight line. Maclaurin wrote on pedal curves. He is the author of an Algebra. The object of his treatise on Fluxions was to found the doctrine of fluxions on geometric demonstrations after the manner of the ancients, and thus, by rigorous exposition, answer such attacks as Berkeley’s that the doctrine rested on false reasoning. The Fluxions contained for the first time the correct way of distinguishing between maxima and minima, and explained their use in the theory of multiple points. “Maclaurin’s theorem” was previously given by James Stirling, and is but a particular case of “Taylor’s theorem.” Appended to the treatise on Fluxions is the solution of a number of beautiful geometric, mechanical, and astronomical problems, in which he employs ancient methods with such consummate

skill as to induce Clairaut to abandon analytic methods and to attack the problem of the figure of the earth by pure geometry. His solutions commanded the liveliest admiration of Lagrange. Maclaurin investigated the attraction of the ellipsoid of revolution, and showed that a homogeneous liquid mass revolving uniformly around an axis under the action of gravity must assume the form of an ellipsoid of revolution. Newton had given this theorem without proof. Notwithstanding the genius of Maclaurin, his influence on the progress of mathematics in Great Britain was unfortunate; for, by his example, he induced his countrymen to neglect analysis and to be indifferent to the wonderful progress in the higher analysis made on the Continent. It remains for us to speak ofAbraham de Moivre(1667– 1754), who was of French descent, but was compelled to leave

EULER, LAGRANGE, AND LAPLACE. 285 France at the age of eighteen, on the Revocation of the Edict of Nantes. He settled in London, where he gave lessons in mathematics. He lived to the advanced age of eighty-seven and sank into a state of almost total lethargy. His subsistence was latterly dependent on the solution of questions on games of chance and problems on probabilities, which he was in the habit of giving at a tavern in St. Martin’s Lane. Shortly before his death he declared that it was necessary for him to sleep ten or twenty minutes longer every day. The day after he had reached the total of over twenty-three hours, he slept exactly twenty-four hours and then passed away in his sleep. De Moivre enjoyed the friendship of Newton and Halley. His power as a mathematician lay in analytic rather than geometric investigation. He revolutionised higher trigonometry by the discovery of the theorem known by his name and by extending the theorems on the multiplication and division of sectors from the circle to the hyperbola. His work on the theory of probability surpasses anything done by any other mathematician except Laplace. His principal contributions are his investigations respecting the Duration of Play, his Theory of Recurring Series, and his extension of the value of Bernoulli’s theorem by the aid of Stirling’s theorem. [42] His chief works are the Doctrine of Chances , 1716, the Miscellanea Analytica, 1730, and his papers in the Philosophical Transactions.

A HISTORY OF MATHEMATICS. 286 EULER, LAGRANGE, AND LAPLACE. During the epoch of ninety years from 1730 to 1820 the French and Swiss cultivated mathematics with most brilliant success. No previous period had shown such an array of illustrious names. At this time Switzerland had her Euler; France, her Lagrange, Laplace, Legendre, and Monge. The mediocrity of French mathematics which marked the time of Louis XIV. was now followed by one of the very brightest periods of all history. England and Germany, on the other hand, which during the unproductive period in France had their Newton and Leibniz, could now boast of no great mathematician. France now waved the mathematical sceptre. Mathematical studies among the English and German people had sunk to the lowest ebb. Among them the direction of original research was ill-chosen. The former adhered with excessive partiality to ancient geometrical methods; the latter produced the combinatorial school, which brought forth nothing of value. The labours of Euler, Lagrange, and Laplace lay in higher analysis, and this they developed to a wonderful degree. By them analysis came to be completely severed from geometry. During the preceding

period the effort of mathematicians not only in England, but, to some extent, even on the continent, had been directed toward the solution of problems clothed in geometric garb, and the results of calculation were usually reduced to geometric form. A change now took place. Euler brought about an emancipation of the

EULER, LAGRANGE, AND LAPLACE. 287 analytical calculus from geometry and established it as an independent science. Lagrange and Laplace scrupulously adhered to this separation. Building on the broad foundation laid for higher analysis and mechanics by Newton and Leibniz, Euler, with matchless fertility of mind, erected an elaborate structure. There are few great ideas pursued by succeeding analysts which were not suggested by Euler, or of which he did not share the honour of invention. With, perhaps, less exuberance of invention, but with more comprehensive genius and profounder reasoning, Lagrange developed the infinitesimal calculus and put analytical mechanics into the form in which we now know it. Laplace applied the calculus and mechanics to the elaboration of the theory of universal gravitation, and thus, largely extending and supplementing the labours of Newton, gave a full analytical discussion of the solar system. He also wrote an epoch-marking work on Probability. Among the analytical branches created during this period are the calculus of Variations by Euler and Lagrange, Spherical Harmonics by Laplace and Legendre, and Elliptic Integrals by Legendre. Comparing the growth of analysis at this time with the growth during the time of Gauss, Cauchy, and recent mathematicians, we observe an important difference. During the former period we witness mainly a development with refer- ence to form. Placing almost implicit confidence in results of calculation, mathematicians did not always pause to discover rigorous proofs, and were thus led to general propositions, some of which have since been found to be true in only special

A HISTORY OF MATHEMATICS. 288 cases. The Combinatorial School in Germany carried this tendency to the greatest extreme; they worshipped formalism and paid no attention to the actual contents of formulæ. But in recent times there has been added to the dexterity in the formal treatment of problems, a much-needed rigour of demonstration. A good example of this increased rigour is seen in the present use of infinite series as compared to that of Euler, and of Lagrange in his earlier works. The ostracism of geometry, brought about by the master- minds of this period, could not last permanently. Indeed, a new geometric school sprang into existence in France before the close of this period. Lagrange would not permit a single diagram to appear in his M´ ecanique analytique, but thirteen years before his death, Monge published his epoch-making G´ eometrie descriptive. Leonhard Euler (1707–1783) was born in Basel. His father, a minister, gave him his first instruction in mathematics and then sent him to the University of Basel, where he became a favourite pupil of John Bernoulli. In his nineteenth year he composed a dissertation on the masting of ships, which received the second prize from the French Academy of Sciences. When John Bernoulli’s two sons, Daniel and Nicolaus, went to Russia, they induced Catharine I., in 1727, to invite their friend Euler to St. Petersburg, where Daniel, in 1733, was assigned to the chair of mathematics. In 1735 the solving of an astronomical problem, proposed by the Academy, for which several eminent mathematicians had demanded some months’ time, was achieved in three days by Euler with aid of improved

EULER, LAGRANGE, AND LAPLACE. 289 methods of his own. But the effort threw him into a fever and deprived him of the use of his right eye. With still superior methods this same problem was solved later by the illustrious Gauss in one hour! [47] The despotism of Anne I. caused the gentle Euler to shrink from public affairs and to devote all his time to science. After his call to Berlin by Frederick the Great in 1747, the queen of Prussia, who received him kindly, wondered how so distinguished a scholar should be so timid and reticent. Euler na¨ ıvely replied, “Madam, it is because I come from a country where, when one speaks, one is hanged.” In 1766 he with difficulty obtained permission to depart from Berlin to accept a call by Catharine II. to St. Petersburg. Soon after his return to Russia he became blind, but this did not stop his wonderful literary productiveness, which continued for seventeen years, until the day of his death. [45] He dictated to his servant his Anleitung zur Algebra , 1770, which, though purely elementary, is meritorious as one of the earliest attempts to put the fundamental processes on a sound basis. Euler wrote an immense number of works, chief of which are thefollowing: Introductio in analysin infinitorum, 1748, awork that caused a revolution in analytical mathematics, a subject which had hitherto never been presented in so general and systematic manner; Institutiones calculi differentialis, 1755, and Institutiones calculi integralis, 1768–1770, which were the most complete and accurate works on the calculus of that time, and contained not only a full summary of everything then knownonthissubject, butalsotheBetaandGammaFunctions

A HISTORY OF MATHEMATICS. 290 and other original investigations; Methodus inveniendi lineas curvas maximi minimive proprietate gaudentes, 1744, which, displaying an amount of mathematical genius seldom rivalled, contained his researches on the calculus of variations (a subject afterwards improved by Lagrange), to the invention of which Euler was led by the study of isoperimetrical curves, the brachistochrone in a resisting medium, and the theory of geodesics (subjects which had previously engaged the attention of the elder Bernoullis and others); the Theoria motuum planetarum et cometarum , 1744, Theoria motus lunæ, 1753, Theoria motuum lunæ, 1772, are his chief works on astronomy; Ses lettres ` a une princesse d’Allemagne sur quelques sujets de Physique et de Philosophie , 1770, was a work which enjoyed great popularity. We proceed to mention the principal innovations and inventions of Euler. He treated trigonometry as a branch of analysis, introduced (simultaneously with Thomas Simpson in England) the now current abbreviations for trigonometric functions, and simplified formulæ by the simple expedient of designating the angles of a triangle byA, B, C, and the opposite sides by a, b, c, respectively. He pointed out the relation between trigonometric and exponential functions. In a paper of 1737 we first meet the symbol π to denote 3.14159 . . .. [21] Euler laid down the rules for the transformation of co-ordinates in space, gave a methodic analytic treatment of plane curves and of surfaces of the second order. He was the first to discuss the equation of the second degree in three variables, and to classify the surfaces represented by it. By criteria analogous

EULER, LAGRANGE, AND LAPLACE. 291 to those used in the classification of conics he obtained five species. He devised a method of solving biquadratic equations by assumingx =√p+√q+√r, with the hope that it would lead him to a general solution

of algebraic equations. The method of elimination by solving a series of linear equations (invented independently by B´ ezout) and the method of elimination by symmetric functions, are due to him. [20] Far reaching are Euler's researches on logarithms. Leibniz and John Bernoulli once argued the question whether a negative number has a logarithm. Bernoulli claimed that since (−a)2 = (+ a)2, we have log(−a)2 = log(+ a)2 and 2 log(−a) = 2 log(+ a), and finally log(−a) = log(+ a). Euler proved that a has really an infinite number of logarithms, all of which are imaginary when a is negative, and all except one when a is positive. He then explained how log(−a)2 might equal log(+a)2, and yetlog(−a) not equal log(+a). The subject of infinite series received new life from him. To his researches on series we owe the creation of the theory of definite integrals by the development of the so-calledEulerian integrals. He warns his readers occasionally against the use of divergent series, but is nevertheless very careless himself. The rigid treatment to which infinite series are subjected now was then undreamed of. No clear notions existed as to what constitutes a convergent series. Neither Leibniz nor Jacob and John Bernoulli had entertained any serious doubt of the correctness of the expression 1 2 = 1− 1 + 1− 1 +⋯ . Guido Grandi went so far as to conclude from this that 1 2 = 0+0+0+ ⋯ . In the treatment of series Leibniz advanced

A HISTORY OF MATHEMATICS. 292 a metaphysical method of proof which held sway over the minds of the elder Bernoullis, and even of Euler. [46] The tendency of that reasoning was to justify results which seem to us now highly absurd. The looseness of treatment can best be seen from examples. The very paper in which Euler cautions against divergent series contains the proof that ⋯ 1 n2 + 1 n + 1 + n + n2 +⋯ = 0 as follows: n + n2 +⋯ = n 1− n , 1 + 1 n + 1 n2 +⋯ = n n− 1; these added give zero. Euler has no hesitation to write 1− 3 + 5− 7 +⋯ = 0, and no one objected to such results excepting Nicolaus Bernoulli, the nephew of John and Jacob. Strange to say, Euler finally succeeded in converting Nicolaus Bernoulli to his own erroneous views. At the present time it is difficult to believe that Euler should have confidently written sin φ− 2 sin 2φ + 3 sin 3φ− 4 sin 4φ +⋯ = 0, but such examples afford striking illustrations of the want of scientific basis of certain parts of analysis at that time. Euler's proof of the binomial formula for negative and fractional exponents, which has been reproduced in elementary text-books of even recent years, is faulty. A remarkable development, due to Euler, is what he named the hypergeometric series, the summation of which he observed to be dependent upon the integration of a linear differential equation of the second order, but it remained for Gauss to point out that for special values of its letters, this series represented nearly all functions then known. Euler developed the calculus of finite differences in the first

EULER, LAGRANGE, AND LAPLACE. 293 chapters of his Institutiones calculi differentialis , and then deduced the differential calculus from it. He established a theorem on homogeneous functions, known by his name, and contributed largely to the theory of differential equations, a subject which had received the attention of Newton, Leibniz, and the Bernoullis, but was still undeveloped. Clairaut, Fontaine, and Euler about the same time observed criteria of integrability, but Euler in addition showed how to employ them to determine integrating factors. The principles on

which the criteria rested involved some degree of obscurity. The celebrated addition-theorem for elliptic integrals was first established by Euler. He invented a new algorithm for continued fractions, which he employed in the solution of the indeterminate equation ax + by = c. We now know that substantially the same solution of this equation was given 1000 years earlier, by the Hindoos. By giving the factors of the number 22n + 1 when n = 5, he pointed out that this expression did not always represent primes, as was supposed by Fermat. He first supplied the proof to "Fermat's theorem," and to a second theorem of Fermat, which states that every prime of the form 4n + 1 is expressible as the sum of two squares in one and only one way. A third theorem of Fermat, that xn + yn = zn, has no integral solution for values of n greater than 2, was proved by Euler to be correct whenn = 3. Euler discovered four theorems which taken together make out the great law of quadratic reciprocity, a law independently discovered by Legendre. [48] Euler enunciated and proved a well-known theorem, giving the relation between the number

A HISTORY OF MATHEMATICS. 294 of vertices, faces, and edges of certain polyhedra, which, however, appears to have been known to Descartes. The powers of Euler were directed also towards the fascinating subject of the theory of probability, in which he solved some difficult problems. Of no little importance are Euler's labours in analytical mechanics. Says Whewell: "The person who did most to give to analysis the generality and symmetry which are now its pride, was also the person who made mechanics analytical; I mean Euler." [11] He worked out the theory of the rotation of a body around a fixed point, established the general equations of motion of a free body, and the general equation of hydrodynamics. He solved an immense number and variety of mechanical problems, which arose in his mind on all occasions. Thus, on reading Virgil's lines, "The anchor drops, the rushing keel is staid," he could not help inquiring what would be the ship's motion in such a case. About the same time as Daniel Bernoulli he published the Principle of the Conservation of Areas and defended the principle of "least action," advanced by Maupertius. He wrote also on tides and on sound. Astronomy owes to Euler the method of the variation of arbitrary constants. By it he attacked the problem of perturbations, explaining, in case of two planets, the secular variations of eccentricities, nodes, etc. He was one of the first to take up with success the theory of the moon's motion by giving approximate solutions to the "problem of three bodies." He laid a sound basis for the calculation of tables of the moon. These researches on the moon's motion, which

EULER, LAGRANGE, AND LAPLACE. 295 captured two prizes, were carried on while he was blind, with the assistance of his sons and two of his pupils. Most of his memoirs are contained in the transactions of the Academy of Sciences at St. Petersburg, and in those of the Academy at Berlin. From 1728 to 1783 a large portion of the Petropolitan transactions were filled by his writings. He had engaged to furnish the Petersburg Academy with memoirs in sufficient number to enrich its acts for twenty years—a promise more than fulfilled, for down to 1818 the volumes usually contained one or more papers of his. It has been said that an edition of Euler's complete works would fill 16, 000 quarto pages. His mode of working was, first to concentrate his powers upon a special problem, then to solve separately all problems growing out of the first. No one excelled him in dexterity of accommodating methods

to special problems. It is easy to see that mathematicians could not long continue in Euler’s habit of writing and publishing. The material would soon grow to such enormous proportions as to be unmanageable. We are not surprised to see almost the opposite in Lagrange, his great successor. The great Frenchman delighted in the general and abstract, rather than, like Euler, in the special and concrete. His writings are condensed and give in a nutshell what Euler narrates at great length. Jean-le-Rond D’Alembert (1717–1783) was exposed, when an infant, by his mother in a market by the church of St. Jean-le-Rond, near the Nˆ otre-Dame in Paris, from which he derived his Christian name. He was brought up by the wife of a

A HISTORY OF MATHEMATICS. 296 poor glazier. It is said that when he began to show signs of great talent, his mother sent for him, but received the reply, “You are only my step-mother; the glazier’s wife is my mother.” His father provided him with a yearly income. D’Alembert entered upon the study of law, but such was his love for mathematics, that law was soon abandoned. At the age of twenty-four his reputation as a mathematician secured for him admission to the Academy of Sciences. In 1743 appeared his Trait´ e de dynamique, founded upon the important general principle bearing his name: The impressed forces are equivalent to the effective forces. D’Alembert’s principle seems to have been recognised before him by Fontaine, and in some measure by John Bernoulli and Newton. D’Alembert gave it a clear mathematical form and made numerous applications of it. It enabled the laws of motion and the reasonings depending on them to be represented in the most general form, in analytical language. D’Alembert applied it in 1744 in a treatise on the equilibrium and motion of fluids, in 1746 to a treatise on the general causes of winds, which obtained a prize from the Berlin Academy. In both these treatises, as also in one of 1747, discussing the famous problem of vibrating chords, he was led to partial differential equations. He was a leader among the pioneers in the study of such equations. To the equation ∂2y ∂t2 = a2 ∂2y ∂x2 , arising in the problem of vibrating chords, he gave as the general solution, y = f(x + at) + φ(x− at), and showed that there is only one arbitrary function, if y be

EULER, LAGRANGE, AND LAPLACE. 297 supposed to vanish for x = 0 and x = l. Daniel Bernoulli, starting with a particular integral given by Brook Taylor, showed that this differential equation is satisfied by the trigonometric series y = α sin πx l · cos πt l + β sin 2πx l · cos 2πt l +··· , and claimed this expression to be the most general solution. Euler denied its generality, on the ground that, if true, the doubtful conclusion would follow that the above series represents any arbitrary function of a variable. These doubts were dispelled by Fourier. Lagrange proceeded to find the sum of the above series, but D’Alembert rightly objected to his process, on the ground that it involved divergent series. [46] A most beautiful result reached by D’Alembert, with aid of his principle, was the complete solution of the problem of the precession of the equinoxes, which had baffled the talents of the best minds. He sent to the French Academy in 1747, on the same day with Clairaut, a solution of the problem of three bodies. This had become a question of universal interest to mathematicians, in which each vied to outdo all others. The problem of two bodies, requiring the determination of their motion when they attract each other with forces inversely proportional to the square of the distance between them, had been

completely solved by Newton. The “problem of three bodies” asks for the motion of three bodies attracting each other according to the law of gravitation. Thus far, the complete solution of this has transcended the power of analysis. The general differential equations of motion were

A HISTORY OF MATHEMATICS. 298 stated by Laplace, but the difficulty arises in their integration. The “solutions” hitherto given are merely convenient methods of approximation in special cases when one body is the sun, disturbing the motion of the moon around the earth, or where a planet moves under the influence of the sun and another planet. In the discussion of the meaning of negative quantities, of the fundamental processes of the calculus, and of the theory of probability, D’Alembert paid some attention to the philosophy of mathematics. His criticisms were not always happy. In 1754 he was made permanent secretary of the French Academy. During the last years of his life he was mainly occupied with the great French encyclopædia, which was begun by Diderot and himself. D’Alembert declined, in 1762, an invitation of Catharine II. to undertake the education of her son. Frederick the Great pressed him to go to Berlin. He made a visit, but declined a permanent residence there. Alexis Claude Clairaut (1713–1765) was a youthful prodigy. He read l’Hospital’s works on the infinitesimal calculus and on conic sections at the age of ten. In 1731 was published his Recherches sur les courbes ` a double courbure, which he had ready for the press when he was sixteen. It was a work of remarkable elegance and secured his admission to the Academy of Sciences when still under legal age. In 1731 he gave a proof of the theorem enunciated by Newton, that every cubic is a projection of one of five divergent parabolas. Clairaut formed the acquaintance of Maupertius, whom he accompanied on an expedition to Lapland to measure the

EULER, LAGRANGE, AND LAPLACE. 299 length of a degree of the meridian. At that time the shape of the earth was a subject of serious disagreement. Newton and Huygens had concluded from theory that the earth was flattened at the poles. About 1713 Dominico Cassini measured an arc extending from Dunkirk to Perpignan and arrived at the startling result that the earth is elongated at the poles. To decide between the conflicting opinions, measurements were renewed. Maupertius earned by his work in Lapland the title of “earth flattener” by disproving the Cassinian tenet that the earth was elongated at the poles, and showing that Newton was right. On his return, in 1743, Clairaut published a work, Th´ eorie de la figure de la Terre, which was based on the results of Maclaurin on homogeneous ellipsoids. It contains a remarkable theorem, named after Clairaut, that the sum of the fractions expressing the ellipticity and the increase of gravity at the pole is equal to2 1 2 times the fraction expressing the centrifugal force at the equator, the unit of force being represented by the force of gravity at the equator. This theorem is independent of any hypothesis with respect to the law of densities of the successive strata of the earth. It embodies most of Clairaut’s researches. Todhunter says that “in the figure of the earth no other person has accomplished so much as Clairaut, and the subject remains at present substantially as he left it, though the form is different. The splendid analysis which Laplace supplied, adorned but did not really alter the theory which started from the creative hands of Clairaut.” In 1752 he gained a prize of the St. Petersburg Academy for

A HISTORY OF MATHEMATICS. 300 his paper on Th´ eorie de la Lune, in which for the first time modern analysis is applied to lunar motion. This contained the explanation of the motion of the lunar apsides. This motion, left unexplained by Newton, seemed to him at first inexplicable by Newton's law, and he was on the point of advancing a new hypothesis regarding gravitation, when, taking the precaution to carry his calculation to a higher degree of approximation, he reached results agreeing with observation. The motion of the moon was studied about the same time by Euler and D'Alembert. Clairaut predicted that "Halley's Comet," then expected to return, would arrive at its nearest point to the sun on April 13, 1759, a date which turned out to be one month too late. He was the first to detect singular solutions in differential equations of the first order but of higher degree than the first. In their scientific labours there was between Clairaut and D'Alembert great rivalry, often far from friendly. The growing ambition of Clairaut to shine in society, where he was a great favourite, hindered his scientific work in the latter part of his life. Johann Heinrich Lambert (1728–1777), born at M¨ uhl- hausen in Alsace, was the son of a poor tailor. While working at his father's trade, he acquired through his own unaided efforts a knowledge of elementary mathematics. At the age of thirty he became tutor in a Swiss family and secured leisure to continue his studies. In his travels with his pupils through Europe he became acquainted with the leading mathematicians. In 1764 he settled in Berlin,

EULER, LAGRANGE, AND LAPLACE. 301 where he became member of the Academy, and enjoyed the society of Euler and Lagrange. He received a small pension, and later became editor of the Berlin Ephemeris. His many-sided scholarship reminds one of Leibniz. In his Cosmological Letters he made some remarkable prophecies regarding the stellar system. In mathematics he made several discoveries which were extended and overshadowed by his great contemporaries. His first research on pure mathematics developed in an infinite series the root x of the equation $xm + px = q$. Since each equation of the form $axr + bxs = d$ can be reduced to $xm + px = q$ in two ways, one or the other of the two resulting series was always found to be convergent, and to give a value of x. Lambert's results stimulated Euler, who extended the method to an equation of four terms, and particularly Lagrange, who found that a function of a root of $a- x + \varphi(x) = 0$ can be expressed by the series bearing his name. In 1761 Lambert communicated to the Berlin Academy a memoir, in which he proves that π is irrational. This proof is given in Note IV. of Legendre'sG´ eometrie, where it is extended to π2. To the genius of Lambert we owe the introduction into trigonometry of hyperbolic functions, which he designated by sinh x, cosh x, etc. His Freye Perspective, 1759 and 1773, contains researches on descriptive geometry, and entitle him to the honour of being the forerunner of Monge. In his effort to simplify the calculation of cometary orbits, he was led geometrically to some remarkable theorems on conics, for instance this: "If in two ellipses having a common major axis we take two such arcs that their chords

A HISTORY OF MATHEMATICS. 302 are equal, and that also the sums of the radii vectores, drawn respectively from the foci to the extremities of these arcs, are equal to each other, then the sectors formed in each ellipse by the arc and the two radii vectores are to each other as the square roots of the parameters of the ellipses." [13]

John Landen(1719–1790) was an English mathematician whose writings served as the starting-point of investigations by Euler, Lagrange, and Legendre. Landen's capital discovery, contained in a memoir of 1755, was that every arc of the hyperbola is immediately rectified by means of two arcs of an ellipse. In his "residual analysis" he attempted to obviate the metaphysical difficulties of fluxions by adopting a purely algebraic method. Lagrange's Calcul des Fonctions is based upon this idea. Landen showed how the algebraic expression for the roots of a cubic equation could be derived by application of the differential and integral calculus. Most of the time of this suggestive writer was spent in the pursuits of active life. ´Etienne B´ ezout(1730–1783) was a French writer of popular mathematical school-books. In his Th´ eorie g´ en´ erale des ´Equations Alg´ ebriques, 1779, he gave the method of elimination by linear equations (invented also by Euler). This method was first published by him in a memoir of 1764, in which he uses determinants, without, however, entering upon their theory. A beautiful theorem as to the degree of the resultant goes by his name. Louis Arbogaste (1759–1803) of Alsace was professor of mathematics at Strasburg. His chief work, the Calcul des D´ erivations, 1800, gives the method known by his name,

EULER, LAGRANGE, AND LAPLACE. 303 by which the successive coefficients of a development are derived from one another when the expression is complicated. De Morgan has pointed out that the true nature of derivation is differentiation accompanied by integration. In this book for the first time are the symbols of operation separated from those of quantity. The notation Dxy for dy/dx is due to him. Maria Gaetana Agnesi (1718–1799) of Milan, distin- guished as a linguist, mathematician, and philosopher, filled the mathematical chair at the University of Bologna during her father's sickness. In 1748 she published her Instituzioni Analitiche, which was translated into English in 1801. The "witch of Agnesi" or "versiera" is a plane curve containing a straight line, x = 0, and a cubic (y c)2 + 1 = c x. Joseph Louis Lagrange(1736–1813), one of the greatest mathematicians of all times, was born at Turin and died at Paris. He was of French extraction. His father, who had charge of the Sardinian military chest, was once wealthy, but lost all he had in speculation. Lagrange considered this loss his good fortune, for otherwise he might not have made mathematics the pursuit of his life. While at the college in Turin his genius did not at once take its true bent. Cicero and Virgil at first attracted him more than Archimedes and Newton. He soon came to admire the geometry of the ancients, but the perusal of a tract of Halley roused his enthusiasm for the analytical method, in the development of which he was destined to reap undying glory. He now applied himself to mathematics, and in his seventeenth year he became professor of mathematics in the royal military academy at

A HISTORY OF MATHEMATICS. 304 Turin. Without assistance or guidance he entered upon a course of study which in two years placed him on a level with the greatest of his contemporaries. With aid of his pupils he established a society which subsequently developed into the Turin Academy. In the first five volumes of its transactions appear most of his earlier papers. At the age of nineteen he communicated to Euler a general method of dealing with "isoperimetrical problems," known now as the Calculus of Variations. This commanded Euler's lively admiration,

and he courteously withheld for a time from publication some researches of his own on this subject, so that the youthful Lagrange might complete his investigations and claim the invention. Lagrange did quite as much as Euler towards the creation of the Calculus of Variations. As it came from Euler it lacked an analytic foundation, and this Lagrange supplied. He separated the principles of this calculus from geometric considerations by which his predecessor had derived them. Euler had assumed as fixed the limits of the integral, i.e. the extremities of the curve to be determined, but Lagrange removed this restriction and allowed all co-ordinates of the curve to vary at the same time. Euler introduced in 1766 the name “calculus of variations,” and did much to improve this science along the lines marked out by Lagrange. Another subject engaging the attention of Lagrange at Turin was the propagation of sound. In his papers on this subject in the Miscellanea Taurinensia, the young mathematician appears as the critic of Newton, and the arbiter between Euler and D’Alembert. By considering only the particles which are

EULER, LAGRANGE, AND LAPLACE. 305 in a straight line, he reduced the problem to the same partial differential equation that represents the motions of vibrating strings. The general integral of this was found by D’Alembert to contain two arbitrary functions, and the question now came to be discussed whether an arbitrary function may be discontinuous. D’Alembert maintained the negative against Euler, Daniel Bernoulli, and finally Lagrange,—arguing that in order to determine the position of a point of the chord at a time t, the initial position of the chord must be continuous. Lagrange settled the question in the affirmative. By constant application during nine years, Lagrange, at the age of twenty-six, stood at the summit of European fame. But his intense studies had seriously weakened a constitution never robust, and though his physicians induced him to take rest and exercise, his nervous system never fully recovered its tone, and he was thenceforth subject to fits of melancholy. In 1764 the French Academy proposed as the subject of a prize the theory of the libration of the moon. It demanded an explanation, on the principle of universal gravitation, why the moon always turns, with but slight variations, the same face to the earth. Lagrange secured the prize. This success encouraged the Academy to propose as a prize the theory of the four satellites of Jupiter,—a problem of six bodies, more difficult than the one of three bodies previously solved by Clairaut, D’Alembert, and Euler. Lagrange overcame the difficulties, but the shortness of time did not permit him to exhaust the subject. Twenty-four years afterwards it was completed by Laplace. Later astronomical investigations of

A HISTORY OF MATHEMATICS. 306 Lagrange are on cometary perturbations (1778 and 1783), on Kepler’s problem, and on a new method of solving the problem of three bodies. Being anxious to make the personal acquaintance of leading mathematicians, Lagrange visited Paris, where he enjoyed the stimulating delight of conversing with Clairaut, D’Alembert, Condorcet, the Abb´ e Marie, and others. He had planned a visit to London, but he fell dangerously ill after a dinner in Paris, and was compelled to return to Turin. In 1766 Euler left Berlin for St. Petersburg, and he pointed out Lagrange as the only man capable of filling the place. D’Alembert recommended him at the same time. Frederick the Great thereupon sent a message to Turin, expressing the wish of “the greatest king of Europe” to have “the greatest

mathematician” at his court. Lagrange went to Berlin, and staid there twenty years. Finding all his colleagues married, and being assured by their wives that the marital state alone is happy, he married. The union was not a happy one. His wife soon died. Frederick the Great held him in high esteem, and frequently conversed with him on the advantages of perfect regularity of life. This led Lagrange to cultivate regular habits. He worked no longer each day than experience taught him he could without breaking down. His papers were carefully thought out before he began writing, and when he wrote he did so without a single correction. During the twenty years in Berlin he crowded the trans- actions of the Berlin Academy with memoirs, and wrote also the epoch-making work called theM´ ecanique Analytique.

EULER, LAGRANGE, AND LAPLACE. 307 He enriched algebra by researches on the solution of equa- tions. There are two methods of solving directly algebraic equations,—that of substitution and that of combination. The formermethodwasdevelopedbyFerrari, Vieta, Tchirnhausen, Euler, B´ ezout, and Lagrange; the latter by Vandermonde and Lagrange. [20] In the method of substitution the original forms are so transformed that the determination of the roots is made to depend upon simpler functions (resolvents). In the method of combination auxiliary quantities are substituted for certain simple combinations (“types”) of the unknown roots of the equation, and auxiliary equations (resolvents) are obtained for these quantities with aid of the coefficients of the given equation. Lagrange traced all known algebraic solutions of equations to the uniform principle consisting in the formation and solution of equations of lower degree whose roots are linear functions of the required roots, and of the roots of unity. He showed that the quintic cannot be reduced in this way, its resolvent being of the sixth degree. His researches on the theory of equations were continued after he left Berlin. In the R´ esolution des ´ equations num´ eriques (1798) he gave a method of approximating to the real roots of numerical equations by continued fractions. Among other things, it contains also a proof that every equation must have a root,—a theorem which appears before this to have been considered self-evident. Other proofs of this were given by Argand, Gauss, and Cauchy. In a note to the above work Lagrange uses Fermat’s theorem and certain suggestions of Gauss in effecting a complete algebraic solution of any

A HISTORY OF MATHEMATICS. 308 binomial equation. While in Berlin Lagrange published several papers on the theory of numbers. In 1769 he gave a solution in integers of indeterminate equations of the second degree, which resembles the Hindoo cyclic method; he was the first to prove, in 1771, “Wilson’s theorem,” enunciated by an Englishman, John Wilson, and first published by Waring in his Meditationes Algebraicæ; he investigated in 1775 under what conditions ±2 and±5 (−1 and±3 having been discussed by Euler) are quadratic residues, or non-residues of odd prime numbers, q; he proved in 1770 M´ eziriac’s theorem that every integer is equal to the sum of four, or a less number, of squares. He proved Fermat’s theorem on xn + yn = zn, for the case n = 4, also Fermat’s theorem that, if a2 + b2 = c2, then ab is not a square. In his memoir on Pyramids, 1773, Lagrange made consider- able use of determinants of the third order, and demonstrated that the square of a determinant is itself a determinant. He never, however, dealt explicitly and directly with determi- nants; he simply obtained accidentally identities which are now

recognised as relations between determinants. Lagrange wrote much on differential equations. Though the subject of contemplation by the greatest mathemati- cians (Euler, D'Alembert, Clairaut, Lagrange, Laplace), yet more than other branches of mathematics did they resist the systematic application of fixed methods and principles. Lagrange established criteria for singular solutions (Calcul des Fonctions, Lessons 14–17), which are, however, erroneous.

EULER, LAGRANGE, AND LAPLACE. 309 He was the first to point out the geometrical significance of such solutions. He generalised Euler's researches on total differential equations of two variables, and of the ninth order; he gave a solution of partial differential equations of the first order (Berlin Memoirs, 1772 and 1774), and spoke of their singular solutions, extending their solution in Memoirs of 1779 and 1785 to equations of any number of variables. The discussion on partial differential equations of the second order, carried on by D'Alembert, Euler, and Lagrange, has already been referred to in our account of D'Alembert. While in Berlin, Lagrange wrote the " M´ ecanique Analy- tique," the greatest of his works (Paris, 1788). From the principle of virtual velocities he deduced, with aid of the cal- culus of variations, the whole system of mechanics so elegantly and harmoniously that it may fitly be called, in Sir William Rowan Hamilton's words, "a kind of scientific poem." It is a most consummate example of analytic generality. Geo- metrical figures are nowhere allowed. "On ne trouvera point de figures dans cet ouvrage" (Preface). The two divisions of mechanics—statics and dynamics—are in the first four sections of each carried out analogously, and each is prefaced by a historic sketch of principles. Lagrange formulated the principle of least action. In their original form, the equations of motion involve the co-ordinates x, y, z, of the different particles m or dm of the system. But x, y, z, are in general not independent, and Lagrange introduced in place of them any variables ξ, ψ, φ, whatever, determining the position of the point at the time. These may be taken to be independent.

A HISTORY OF MATHEMATICS. 310 The equations of motion may now assume the form d dt dT dξ′− dT dξ + Ξ = 0; or whenΞ, Ψ, Φ, . . .are the partial differential coefficients with respect to ξ, ψ, φ, . . . of one and the same functionV , then the form d dt dT dξ′− dT dξ + dV dξ = 0. The latter is par excellence the Lagrangian form of the equations of motion. With Lagrange originated the remark that mechanics may be regarded as a geometry of four dimensions. To him falls the honour of the introduction of the potential into dynamics. [49] Lagrange was anxious to have his M´ ecanique Analytiquepublished in Paris. The work was ready for print in 1786, but not till 1788 could he find a publisher, and then only with the condition that after a few years he would purchase all the unsold copies. The work was edited by Legendre. After the death of Frederick the Great, men of science were no longer respected in Germany, and Lagrange accepted an invitation of Louis XVI. to migrate to Paris. The French queen treated him with regard, and lodging was procured for him in the Louvre. But he was seized with a long attack of melancholy which destroyed his taste for mathematics. For two years his printed copy of the M´ ecanique, fresh from the press,—the work of a quarter of a century,—lay unopened on his desk. Through Lavoisier he became interested in chemistry, which he found "as easy as algebra." The disastrous crisis of the

EULER, LAGRANGE, AND LAPLACE. 311 French Revolution aroused him again to activity. About this time the young and accomplished daughter of the astronomer Lemonnier took compassion on the sad, lonely Lagrange, and insisted upon marrying him. Her devotion to him constituted the one tie to life which at the approach of death he found it hard to break. He was made one of the commissioners to establish weights and measures having units founded on nature. Lagrange strongly favoured the decimal subdivision, the general idea of which was obtained from a work of Thomas Williams, London, 1788. Such was the moderation of Lagrange's character, and such the universal respect for him, that he was retained as president of the commission on weights and measures even after it had been purified by the Jacobins by striking out the names of Lavoisier, Laplace, and others. Lagrange took alarm at the fate of Lavoisier, and planned to return to Berlin, but at the establishment of the ´Ecole Normale in 1795 in Paris, he was induced to accept a professorship. Scarcely had he time to elucidate the foundations of arithmetic and algebra to young pupils, when the school was closed. His additions to the algebra of Euler were prepared at this time. In 1797 the ´Ecole Polytechnique was founded, with Lagrange as one of the professors. The earliest triumph of this institution was the restoration of Lagrange to analysis. His mathematical activity burst out anew. He brought forth the Th´ eorie des fonctions analytiques (1797), Le¸ cons sur le calcul des fonctions, a treatise on the same lines as the preceding (1801), and the R´ esolution des ´ equations num´ eriques(1798). In 1810

A HISTORY OF MATHEMATICS. 312 he began a thorough revision of hisM´ ecanique analytique, but he died before its completion. The Th´ eorie des fonctions, the germ of which is found in a memoir of his of 1772, aimed to place the principles of the calculus upon a sound foundation by relieving the mind of the difficult conception of a limit or infinitesimal. John Landen's residual calculus, professing a similar object, was unknown to him. Lagrange attempted to prove Taylor's theorem (the power of which he was the first to point out) by simple algebra, and then to develop the entire calculus from that theorem. The principles of the calculus were in his day involved in philosophic difficulties of a serious nature. The infinitesimals of Leibniz had no satisfactory metaphysical basis. In the differential calculus of Euler they were treated as absolute zeros. In Newton's limiting ratio, the magnitudes of which it is the ratio cannot be found, for at the moment when they should be caught and equated, there is neither arc nor chord. The chord and arc were not taken by Newton as equal before vanishing, nor after vanishing, but when they vanish. "That method," said Lagrange, "has the great inconvenience of considering quantities in the state in which they cease, so to speak, to be quantities; for though we can always well conceive the ratios of two quantities, as long as they remain finite, that ratio offers to the mind no clear and precise idea, as soon as its terms become both nothing at the same time." D'Alembert's method of limits was much the same as the method of prime and ultimate ratios. D'Alembert taught that a variable actually reached its limit. When Lagrange

EULER, LAGRANGE, AND LAPLACE. 313 endeavoured to free the calculus of its metaphysical difficulties, by resorting to common algebra, he avoided the whirlpool of Charybdis only to suffer wreck against the rocks of Scylla. The algebra of his day, as

handed down to him by Euler, was founded on a false view of infinity. No correct theory of infinite series had then been established. Lagrange proposed to define the differential coefficient of f(x) with respect to x as the coefficient of h in the expansion of f(x + h) by Taylor's theorem, and thus to avoid all reference to limits. But he used infinite series without ascertaining that they were convergent, and his proof that f(x + h) can always be expanded in a series of ascending powers of h, labours under serious defects. Though Lagrange's method of developing the calculus was at first greatly applauded, its defects were fatal, and to-day his "method of derivatives," as it was called, has been generally abandoned. He introduced a notation of his own, but it was inconvenient, and was abandoned by him in the second edition of his M´ ecanique, in which he used infinitesimals. The primary object of the Th´ eorie des fonctionswas not attained, but its secondary results were far-reaching. It was a purely abstract mode of regarding functions, apart from geometrical or mechanical considerations. In the further development of higher analysis a function became the leading idea, and Lagrange's work may be regarded as the starting-point of the theory of functions as developed by Cauchy, Riemann, Weierstrass, and others. In the treatment of infinite series Lagrange displayed in his earlier writings that laxity common to all mathematicians of

A HISTORY OF MATHEMATICS. 314 his time, excepting Nicolaus Bernoulli II. and D'Alembert. But his later articles mark the beginning of a period of greater rigour. Thus, in the Calcul de fonctions he gives his theorem on the limits of Taylor's theorem. Lagrange's mathematical researches extended to subjects which have not been mentioned here—such as probabilities, finite differences, ascending continued fractions, elliptic integrals. Everywhere his wonderful powers of generalisation and abstraction are made manifest. In that respect he stood without a peer, but his great contemporary, Laplace, surpassed him in practical sagacity. Lagrange was content to leave the application of his general results to others, and some of the most important researches of Laplace (particularly those on the velocity of sound and on the secular acceleration of the moon) are implicitly contained in Lagrange's works. Lagrange was an extremely modest man, eager to avoid controversy, and even timid in conversation. He spoke in tones of doubt, and his first words generally were, "Je ne sais pas." He would never allow his portrait to be taken, and the only ones that were secured were sketched without his knowledge by persons attending the meetings of the Institute. Pierre Simon Laplace (1749–1827) was born at Beau- mont-en-Auge in Normandy. Very little is known of his early life. When at the height of his fame he was loath to speak of his boyhood, spent in poverty. His father was a small farmer. Some rich neighbours who recognised the boy's talent assisted him in securing an education. As an extern he attended the military school in Beaumont, where at an early age he became

EULER, LAGRANGE, AND LAPLACE. 315 teacher of mathematics. At eighteen he went to Paris, armed with letters of recommendation to D'Alembert, who was then at the height of his fame. The letters remained unnoticed, but young Laplace, undaunted, wrote the great geometer a letter on the principles of mechanics, which brought the following enthusiastic response: "You needed no introduction; you have recommended yourself; my support is your due." D'Alembert secured him a position

at the ´Ecole Militaire of Paris as professor of mathematics. His future was now assured, and he entered upon those profound researches which brought him the title of “the Newton of France.” With wonderful mastery of analysis, Laplace attacked the pending problems in the application of the law of gravitation to celestial motions. During the succeeding fifteen years appeared most of his original contributions to astronomy. His career was one of almost uninterrupted prosperity. In 1784 he succeeded B´ ezout as examiner to the royal artillery, and the following year he became member of the Academy of Sciences. He was made president of the Bureau of Longitude; he aided in the introduction of the decimal system, and taught, with Lagrange, mathematics in the ´Ecole Normale. When, during the Revolution, there arose a cry for the reform of everything, even of the calendar, Laplace suggested the adoption of an era beginning with the year 1250, when, according to his calculation, the major axis of the earth’s orbit had been perpendicular to the equinoctial line. The year was to begin with the vernal equinox, and the zero meridian was to be located east of Paris by 185.30 degrees of the centesimal

A HISTORY OF MATHEMATICS. 316 division of the quadrant, for by this meridian the beginning of his proposed era fell at midnight. But the revolutionists rejected this scheme, and made the start of the new era coincide with the beginning of the glorious French Republic. [50] Laplace was justly admired throughout Europe as a most sagacious and profound scientist, but, unhappily for his reputation, he strove not only after greatness in science, but also after political honours. The political career of this eminent scientist was stained by servility and suppleness. After the 18th of Brumaire, the day when Napoleon was made emperor, Laplace’s ardour for republican principles suddenly gave way to a great devotion to the emperor. Napoleon rewarded this devotion by giving him the post of minister of the interior, but dismissed him after six months for incapacity. Said Napoleon, ”Laplace ne saisissait aucune question sous son v´ eritable point de vue; il cherchait des subtilit´ es partout, n’avait que des id´ ees problematiques, et portait enfin l’esprit des infiniment petits jusque dans l’administration.” Desirous to retain his allegiance, Napoleon elevated him to the Senate and bestowed various other honours upon him. Nevertheless, he cheerfully gave his voice in 1814 to the dethronement of his patron and hastened to tender his services to the Bourbons, thereby earning the title of marquis. This pettiness of his character is seen in his writings. The first edition of the Syst` eme du monde was dedicated to the Council of Five Hundred. To the third volume of the M´ ecanique C´ elesteis prefixed a note that of all the truths contained in the book, that most precious to the author was the declaration he thus

EULER, LAGRANGE, AND LAPLACE. 317 made of gratitude and devotion to the peace-maker of Europe. After this outburst of affection, we are surprised to find in the editions of the Th´ eorie analytique des probabilit´ es, which appeared after the Restoration, that the original dedication to the emperor is suppressed. Though supple and servile in politics, it must be said that in religion and science Laplace never misrepresented or concealed his own convictions however distasteful they might be to others. In mathematics and astronomy his genius shines with a lustre excelled by few. Three great works did he give to the scientific world,—the M´ ecanique C´ eleste, the Exposition du syst` eme du monde, and the Th´ eorie analytique des

probabilit´ es. Besides these he contributed important memoirs to the French Academy. We first pass in brief review his astronomical researches. In 1773 he brought out a paper in which he proved that the mean motions or mean distances of planets are invariable or merely subject to small periodic changes. This was the first and most important step in establishing the stability of the solar system. [51] To Newton and also to Euler it had seemed doubtful whether forces so numerous, so variable in position, so different in intensity, as those in the solar system, could be capable of maintaining permanently a condition of equilibrium. Newton was of the opinion that a powerful hand must intervene from time to time to repair the derangements occasioned by the mutual action of the different bodies. This paper was the beginning of a series of profound researches by Lagrange and Laplace on the limits of variation of the

A HISTORY OF MATHEMATICS. 318 various elements of planetary orbits, in which the two great mathematicians alternately surpassed and supplemented each other. Laplace's first paper really grew out of researches on the theory of Jupiter and Saturn. The behaviour of these planets had been studied by Euler and Lagrange without receiving satisfactory explanation. Observation revealed the existence of a steady acceleration of the mean motions of our moon and of Jupiter and an equally strange diminution of the mean motion of Saturn. It looked as though Saturn might eventually leave the planetary system, while Jupiter would fall into the sun, and the moon upon the earth. Laplace finally succeeded in showing, in a paper of 1784–1786, that these variations (called the "great inequality") belonged to the class of ordinary periodic perturbations, depending upon the law of attraction. The cause of so influential a perturbation was found in the commensurability of the mean motion of the two planets. In the study of the Jovian system, Laplace was enabled to determine the masses of the moons. He also discovered certain very remarkable, simple relations between the movements of those bodies, known as "Laws of Laplace." His theory of these bodies was completed in papers of 1788 and 1789. These, as well as the other papers here mentioned, were published in the M´ emoires pr´ esent´ es par divers savans. The year 1787 was made memorable by Laplace's announcement that the lunar acceleration depended upon the secular changes in the eccentricity of the earth's orbit. This removed all doubt then existing as to the stability of the solar system. The universal

EULER, LAGRANGE, AND LAPLACE. 319 validity of the law of gravitation to explain all motion in the solar system was established. That system, as then known, was at last found to be a complete machine. In 1796 Laplace published his Exposition du syst` eme du monde, a non-mathematical popular treatise on astronomy, ending with a sketch of the history of the science. In this work he enunciates for the first time his celebrated nebular hypothesis. A similar theory had been previously proposed by Kant in 1755, and by Swedenborg; but Laplace does not appear to have been aware of this. Laplace conceived the idea of writing a work which should contain a complete analytical solution of the mechanical prob- lem presented by the solar system, without deriving from observation any but indispensable data. The result was the M´ ecanique C´ eleste, which is a systematic presentation em- bracing all the discoveries of Newton, Clairaut, D'Alembert, Euler, Lagrange, and of Laplace

himself, on celestial mechan- ics. The first and second volumes of this work were published in 1799; the third appeared in 1802, the fourth in 1805. Of the fifth volume, Books XI. and XII. were published in 1823; Books XIII., XIV., XV. in 1824, and Book XVI. in 1825. The first two volumes contain the general theory of the motions and figure of celestial bodies. The third and fourth volumes give special theories of celestial motions,—treating particularly of motions of comets, of our moon, and of other satellites. The fifth volume opens with a brief history of celestial mechanics, and then gives in appendices the results of the author's later researches. The M´ ecanique C´ elestewas such a master-piece,

A HISTORY OF MATHEMATICS. 320 and so complete, that Laplace's successors have been able to add comparatively little. The general part of the work was translated into German by Joh. Karl Burkhardt, and appeared in Berlin, 1800–1802. Nathaniel Bowditch brought out an edition in English, with an extensive commentary, in Boston, 1829–1839. The M´ ecanique C´ elesteis not easy reading. The difficulties lie, as a rule, not so much in the subject itself as in the want of verbal explanation. A complicated chain of reasoning receives often no explanation whatever. Biot, who assisted Laplace in revising the work for the press, tells that he once asked Laplace some explanation of a passage in the book which had been written not long before, and that Laplace spent an hour endeavouring to recover the reasoning which had been carelessly suppressed with the remark, "Il est facile de voir." Notwithstanding the important researches in the work, which are due to Laplace himself, it naturally contains a great deal that is drawn from his predecessors. It is, in fact, the organised result of a century of patient toil. But Laplace frequently neglects to properly acknowledge the source from which he draws, and lets the reader infer that theorems and formulæ due to a predecessor are really his own. We are told that when Laplace presented Napoleon with a copy of the M´ ecanique C´ eleste, the latter made the remark, "M. Laplace, they tell me you have written this large book on the system of the universe, and have never even mentioned its Creator." Laplace is said to have replied bluntly, "Je n'avais pas besoin de cette hypoth` ese-la." This assertion, taken literally, is impious, but may it not have been intended

EULER, LAGRANGE, AND LAPLACE. 321 to convey a meaning somewhat different from its literal one? Newton was not able to explain by his law of gravitation all questions arising in the mechanics of the heavens. Thus, being unable to show that the solar system was stable, and suspecting in fact that it was unstable, Newton expressed the opinion that the special intervention, from time to time, of a powerful hand was necessary to preserve order. Now Laplace was able to prove by the law of gravitation that the solar system is stable, and in that sense may be said to have felt no necessity for reference to the Almighty. We now proceed to researches which belong more properly to pure mathematics. Of these the most conspicuous are on the theory of probability. Laplace has done more towards advancing this subject than any one other investigator. He published a series of papers, the main results of which were collected in his Th´ eorie analytique des probabilit´ es, 1812. The third edition (1820) consists of an introduction and two books. The introduction was published separately under the title, Essai philosophique sur les probabilit´ es, and is an admirable and masterly exposition without the aid of analytical formulæ of the

principles and applications of the science. The first book contains the theory of generating functions, which are applied, in the second book, to the theory of probability. Laplace gives in his work on probability his method of approximation to the values of definite integrals. The solution of linear differential equations was reduced by him to definite integrals. One of the most important parts of the work is the application of probability to the method of least squares, which is shown to

A HISTORY OF MATHEMATICS. 322 give the most probable as well as the most convenient results. The first printed statement of the principle of least squares was made in 1806 by Legendre, without demonstration. Gauss had used it still earlier, but did not publish it until 1809. The first deduction of the law of probability of error that appeared in print was given in 1808 by Robert Adrain in theAnalyst, a journal published by himself in Philadelphia. [2] Proofs of this law have since been given by Gauss, Ivory, Herschel, Hagen, and others; but all proofs contain some point of difficulty. Laplace's proof is perhaps the most satisfactory. Laplace's work on probability is very difficult reading, particularly the part on the method of least squares. The analytical processes are by no means clearly established or free from error. "No one was more sure of giving the result of analytical processes correctly, and no one ever took so little care to point out the various small considerations on which correctness depends" (De Morgan). Of Laplace's papers on the attraction of ellipsoids, the most important is the one published in 1785, and to a great extent reprinted in the third volume of theM´ ecanique C´ eleste. It gives an exhaustive treatment of the general problem of attraction of any ellipsoid upon a particle situated outside or upon its surface. Spherical harmonics, or the so-called "Laplace's coefficients," constitute a powerful analytic engine in the theory of attraction, in electricity, and magnetism. The theory of spherical harmonics for two dimensions had been previously given by Legendre. Laplace failed to make due acknowledgment of this, and there existed, in consequence,

EULER, LAGRANGE, AND LAPLACE. 323 between the two great men, "a feeling more than coldness." The potential function, V , is much used by Laplace, and is shown by him to satisfy the partial differential equation ∂2V ∂x2 + ∂2V ∂y 2 + ∂2V ∂z 2 = 0. This is known as Laplace's equation, and was first given by him in the more complicated form which it assumes in polar co-ordinates. The notion of potential was, however, not introduced into analysis by Laplace. The honour of that achievement belongs to Lagrange. [49] Among the minor discoveries of Laplace are his method of solving equations of the second, third, and fourth degrees, his memoir on singular solutions of differential equations, his researches in finite differences and in determinants, the establishment of the expansion theorem in determinants which had been previously given by Vandermonde for a special case, the determination of the complete integral of the linear differential equation of the second order. In the M´ ecanique C´ elestehe made a generalisation of Lagrange's theorem on the development of functions in series known as Laplace's theorem. Laplace's investigations in physics were quite extensive. We mention here his correction of Newton's formula on the velocity of sound in gases by taking into account the changes of elasticity due to the heat of compression and cold of rarefaction; his researches on the theory of tides; his mathematical theory of capillarity; his

explanation of astronomical refraction; his formulæ for measuring heights by the barometer.

A HISTORY OF MATHEMATICS. 324 Laplace’s writings stand out in bold contrast to those of Lagrange in their lack of elegance and symmetry. Laplace looked upon mathematics as the tool for the solution of physical problems. The true result being once reached, he spent little time in explaining the various steps of his analysis, or in polishing his work. The last years of his life were spent mostly at Arcueil in peaceful retirement on a country-place, where he pursued his studies with his usual vigour until his death. He was a great admirer of Euler, and would often say, “Lisez Euler, lisez Euler, c’est notre maˆ ıtre ` a tous.” Abnit-Th´ eophile Vandermonde(1735–1796) studied music during his youth in Paris and advocated the theory that all art rested upon one general law, through which any one could become a composer with the aid of mathematics. He was the first to give a connected and logical exposition of the theory of determinants, and may, therefore, almost be regarded as the founder of that theory. He and Lagrange originated the method of combinations in solving equations. [20] Adrien Marie Legendre (1752–1833) was educated at the Coll` ege Mazarin in Paris, where he began the study of mathematics under Abb´ e Marie. His mathematical genius secured for him the position of professor of mathematics at the military school of Paris. While there he prepared an essay on the curve described by projectiles thrown into resisting media (ballistic curve), which captured a prize offered by the Royal Academy of Berlin. In 1780 he resigned his position in order to reserve more time for the study of higher mathematics. He was then made member of several public commissions. In

EULER, LAGRANGE, AND LAPLACE. 325 1795 he was elected professor at the Normal School and later was appointed to some minor government positions. Owing to his timidity and to Laplace’s unfriendliness toward him, but few important public offices commensurate with his ability were tendered to him. As an analyst, second only to Laplace and Lagrange, Legendre enriched mathematics by important contributions, mainly on elliptic integrals, theory of numbers, attraction of ellipsoids, and least squares. The most important of Legendre’s works is his Fonctions elliptiques, issued in two volumes in 1825 and 1826. He took up the subject where Euler, Landen, and Lagrange had left it, and for forty years was the only one to cultivate this new branch of analysis, until at last Jacobi and Abel stepped in with admirable new discoveries. [52] Legendre imparted to the subject that connection and arrangement which belongs to an independent science. Starting with an integral depending upon the square root of a polynomial of the fourth degree in x, he showed that such integrals can be brought back to three canonical forms, designated by F (φ), E(φ), and Π(φ), the radical being expressed in the form $\Delta(\varphi) = \sqrt{1 - k^2 \sin^2 \varphi}$. He also undertook the prodigious task of calculating tables of arcs of the ellipse for different degrees of amplitude and eccentricity, which supply the means of integrating a large number of differentials. An earlier publication which contained part of his researches on elliptic functions was his Calcul int´ egralin three volumes (1811, 1816, 1817), in which he treats also at length of the

A HISTORY OF MATHEMATICS. 326 two classes of definite integrals named by him Eulerian. He tabulated the values oflog Γ(p) for values ofp between 1 and 2. One of the earliest subjects of research was the attraction of spheroids, which suggested to Legendre the function Pn, named after him. His memoir was presented to the Academy of Sciences in 1783. The researches of Maclaurin and Lagrange suppose the point attracted by a spheroid to be at the surface or within the spheroid, but Legendre showed that in order to determine the attraction of a spheroid on any external point it suffices to cause the surface of another spheroid described upon the same foci to pass through that point. Other memoirs on ellipsoids appeared later. The two household gods to which Legendre sacrificed with ever-renewed pleasure in the silence of his closet were the elliptic functions and the theory of numbers. His researches on the latter subject, together with the numerous scattered fragments on the theory of numbers due to his predecessors in this line, were arranged as far as possible into a systematic whole, and published in two large quarto volumes, entitled Th´ eorie des nombres, 1830. Before the publication of this work Legendre had issued at divers times preliminary articles. Its crowning pinnacle is the theorem of quadratic reciprocity, previously indistinctly given by Euler without proof, but for the first time clearly enunciated and partly proved by Legendre. [48] While acting as one of the commissioners to connect Greenwich and Paris geodetically, Legendre calculated all the triangles in France. This furnished the occasion of establishing

EULER, LAGRANGE, AND LAPLACE. 327 formulæ and theorems on geodesics, on the treatment of the spherical triangle as if it were a plane triangle, by applying certain corrections to the angles, and on the method of least squares, published for the first time by him without demonstration in 1806. Legendre wrote an ´El´ ements de G´ eom´ etrie, 1794, which enjoyed great popularity, being generally adopted on the Continent and in the United States as a substitute for Euclid. This great modern rival of Euclid passed through numerous editions; the later ones containing the elements of trigonometry and a proof of the irrationality ofπ and π2. Much attention was given by Legendre to the subject of parallel lines. In the earlier editions of the ´El´ ements, he made direct appeal to the senses for the correctness of the “parallel-axiom.” He then attempted to demonstrate that “axiom,” but his proofs did not satisfy even himself. In Vol. XII. of the Memoirs of the Institute is a paper by Legendre, containing his last attempt at a solution of the problem. Assuming space to be infinite, he proved satisfactorily that it is impossible for the sum of the three angles of a triangle to exceed two right angles; and that if there be any triangle the sum of whose angles is two right angles, then the same must be true of all triangles. But in the next step, to show that this sum cannot be less than two right angles, his demonstration necessarily failed. If it could be granted that the sum of the three angles is always equal to two right angles, then the theory of parallels could be strictly deduced. Joseph Fourier (1768–1830) was born at Auxerre, in

A HISTORY OF MATHEMATICS. 328 central France. He became an orphan in his eighth year. Through the influence of friends he was admitted into the military school in his native place, then conducted by the Benedictines of the Convent of St. Mark. He there prosecuted his studies, particularly mathematics, with surprising success. He wished to enter the artillery, but, being of low birth (the son of a tailor), his

application was answered thus: “Fourier, not being noble, could not enter the artillery, although he were a second Newton.” [53] He was soon appointed to the mathematical chair in the military school. At the age of twenty-one he went to Paris to read before the Academy of Sciences a memoir on the resolution of numerical equations, which was an improvement on Newton’s method of approximation. This investigation of his early youth he never lost sight of. He lectured upon it in the Polytechnic School; he developed it on the banks of the Nile; it constituted a part of a work entitled Analyse des equationes determines (1831), which was in press when death overtook him. This work contained “Fourier’s theorem” on the number of real roots between two chosen limits. Budan had published this result as early as 1807, but there is evidence to show that Fourier had established it before Budan’s publication. These brilliant results were eclipsed by the theorem of Sturm, published in 1835. Fourier took a prominent part at his home in promoting the Revolution. Under the French Revolution the arts and sciences seemed for a time to flourish. The reformation of the weights and measures was planned with grandeur of

EULER, LAGRANGE, AND LAPLACE. 329 conception. The Normal School was created in 1795, of which Fourier became at first pupil, then lecturer. His brilliant success secured him a chair in the Polytechnic School, the duties of which he afterwards quitted, along with Monge and Berthollet, to accompany Napoleon on his campaign to Egypt. Napoleon founded the Institute of Egypt, of which Fourier became secretary. In Egypt he engaged not only in scientific work, but discharged important political functions. After his return to France he held for fourteen years the prefecture of Grenoble. During this period he carried on his elaborate investigations on the propagation of heat in solid bodies, published in 1822 in his work entitled La Theorie Analytique de la Chaleur . This work marks an epoch in the history of mathematical physics. “Fourier’s series” constitutes its gem. By this research a long controversy was brought to a close, and the fact established that any arbitrary function can be represented by a trigonometric series. The first announcement of this great discovery was made by Fourier in 1807, before the French Academy. The trigonometric series $n=\infty\sum n=0$ (an sin nx + bn cos nx) represents the function φ(x) for every value of x, if the coefficients an = $1 \pi \int \pi -\pi$ φ(x) sinnx dx, and bn be equal to a similar integral. The weak point in Fourier’s analysis lies in his failure to prove generally that the trigonometric series actually converges to the value of the function. In 1827 Fourier succeeded Laplace as president of the council of the Polytechnic School. Before proceeding to the origin of modern geometry we

A HISTORY OF MATHEMATICS. 330 shall speak briefly of the introduction of higher analysis into Great Britain. This took place during the first quarter of this century. The British began to deplore the very small progress that science was making in England as compared with its racing progress on the Continent. In 1813 the “Analytical Society” was formed at Cambridge. This was a small club established by George Peacock, John Herschel, Charles Babbage, and a few other Cambridge students, to promote, as it was humorously expressed, the principles of pure “D-ism,” that is, the Leibnizian notation in the calculus against those of “dot-age,” or of the Newtonian notation. This struggle ended in the introduction into Cambridge of the notationdy dx, to the exclusion of the fluxional notation ˙y. This was a great step in

advance, not on account of any great superiority of the Leibnizian over the Newtonian notation, but because the adoption of the former opened up to English students the vast storehouses of continental discoveries. Sir William Thomson, Tait, and some other modern writers find it frequently convenient to use both notations. Herschel, Peacock, and Babbage translated, in 1816, from the French, Lacroix's treatise on the differential and integral calculus, and added in 1820 two volumes of examples. Lacroix's was one of the best and most extensive works on the calculus of that time. Of the three founders of the "Analytical Society," Peacock afterwards did most work in pure mathematics. Babbage became famous for his invention of a calculating engine superior to Pascal's. It was never finished, owing to a misunderstanding with the government, and a consequent failure to secure funds. John

EULER, LAGRANGE, AND LAPLACE. 331 Herschel, the eminent astronomer, displayed his mastery over higher analysis in memoirs communicated to the Royal Society on new applications of mathematical analysis, and in articles contributed to cyclopædias on light, on meteorology, and on the history of mathematics. George Peacock (1791–1858) was educated at Trinity College, Cambridge, became Lowndean professor there, and later, dean of Ely. His chief publications are hisAlgebra, 1830 and 1842, and hisReport on Recent Progress in Analysis, which was the first of several valuable summaries of scientific progress printed in the volumes of the British Association. He was one of the first to study seriously the fundamental principles of algebra, and to fully recognise its purely symbolic character. He advances, though somewhat imperfectly, the "principle of the permanence of equivalent forms." It assumes that the rules applying to the symbols of arithmetical algebra apply also in symbolical algebra. About this time D. F. Gregory wrote a paper "on the real nature of symbolical algebra," which brought out clearly the commutative and distributive laws. These laws had been noticed years before by the inventors of symbolic methods in the calculus. It was Servois who introduced the names commutative and distributive in 1813. Peacock's investigations on the foundation of algebra were considerably advanced by De Morgan and Hankel. JamesIvory (1765–1842)wasaScotchmathematicianwho for twelve years, beginning in 1804, held the mathematical chair in the Royal Military College at Marlow (now at Sandhurst). He was essentially a self-trained mathematician,

A HISTORY OF MATHEMATICS. 332 and almost the only one in Great Britain previous to the organisation of the Analytical Society who was well versed in continental mathematics. Of importance is his memoir (Phil. Trans., 1809) in which the problem of the attraction of a homogeneous ellipsoid upon an external point is reduced to the simpler problem of the attraction of a related ellipsoid upon a corresponding point interior to it. This is known as "Ivory's theorem." He criticised with undue severity Laplace's solution of the method of least squares, and gave three proofs of the principle without recourse to probability; but they are far from being satisfactory. The Origin of Modern Geometry. By the researches of Descartes and the invention of the calculus, the analytical treatment of geometry was brought into great prominence for over a century. Notwithstanding the efforts to revive synthetic methods made by Desargues, Pascal, De Lahire, Newton, and Maclaurin, the analytical method retained almost undisputed supremacy. It was reserved for the

genius of Monge to bring synthetic geometry in the foreground, and to open up new avenues of progress. His G´ eom´ etrie descriptivemarks the beginning of a wonderful development of modern geometry. Of the two leading problems of descriptive geometry, the one—to represent by drawings geometrical magnitudes—was brought to a high degree of perfection before the time of Monge; the other—to solve problems on figures in space by

EULER, LAGRANGE, AND LAPLACE. 333 constructions in a plane—had received considerable attention before his time. His most noteworthy predecessor in descriptive geometry was the Frenchman Fr´ ezier (1682–1773). But it remained for Monge to create descriptive geometry as a distinct branch of science by imparting to it geometric generality and elegance. All problems previously treated in a special and uncertain manner were referred back to a few general principles. He introduced the line of intersection of the horizontal and the vertical plane as the axis of projection. By revolving one plane into the other around this axis or ground-line, many advantages were gained. [54] Gaspard Monge (1746–1818) was born at Beaune. The construction of a plan of his native town brought the boy under the notice of a colonel of engineers, who procured for him an appointment in the college of engineers at M´ ezi` eres. Being of low birth, he could not receive a commission in the army, but he was permitted to enter the annex of the school, where surveying and drawing were taught. Observing that all the operations connected with the construction of plans of fortification were conducted by long arithmetical processes, he substituted a geometrical method, which the commandant at first refused even to look at, so short was the time in which it could be practised; when once examined, it was received with avidity. Monge developed these methods further and thus created his descriptive geometry. Owing to the rivalry between the French military schools of that time, he was not permitted to divulge his new methods to any one outside of this institution. In 1768 he was made professor of

A HISTORY OF MATHEMATICS. 334 mathematics at M´ ezi` eres. In 1780, when conversing with two of his pupils, S. F. Lacroix and Gayvernon in Paris, he was obliged to say, “All that I have here done by calculation, I could have done with the ruler and compass, but I am not allowed to reveal these secrets to you.” But Lacroix set himself to examine what the secret could be, discovered the processes, and published them in 1795. The method was published by Monge himself in the same year, first in the form in which the short-hand writers took down his lessons given at the Normal School, where he had been elected professor, and then again, in revised form, in the Journal des ´ ecoles normales. The next edition occurred in 1798–1799. After an ephemeral existence of only four months the Normal School was closed in 1795. In the same year the Polytechnic School was opened, in the establishing of which Monge took active part. He taught there descriptive geometry until his departure from France to accompany Napoleon on the Egyptian campaign. He was the first president of the Institute of Egypt. Monge was a zealous partisan of Napoleon and was, for that reason, deprived of all his honours by Louis XVIII. This and the destruction of the Polytechnic School preyed heavily upon his mind. He did not long survive this insult. Monge’s numerous papers were by no means confined to descriptive geometry. His analytical discoveries are hardly less remarkable. He

introduced into analytic geometry the methodic use of the equation of a line. He made important contributions to surfaces of the second degree (previously studied by Wren and Euler) and discovered between the theory

EULER, LAGRANGE, AND LAPLACE. 335 of surfaces and the integration of partial differential equations, a hidden relation which threw new light upon both subjects. He gave the differential of curves of curvature, established a general theory of curvature, and applied it to the ellipsoid. He found that the validity of solutions was not impaired when imaginaries are involved among subsidiary quantities. Monge published the following books: Statics, 1786; Applications de l'alg` ebre ` a la g´ eom´ etrie, 1805; Application de l'analyse ` a la g´ eom´ etrie. The last two contain most of his miscellaneous papers. Monge was an inspiring teacher, and he gathered around him a large circle of pupils, among which were Dupin, Servois, Brianchon, Hachette, Biot, and Poncelet. Charles Dupin (1784–1873), for many years professor of mechanics in the Conservatoire des Arts et M´ etiers in Paris, published in 1813 an important work on D´ eveloppements de g´ eom´ etrie, in which is introduced the conception of conjugate tangents of a point of a surface, and of the indicatrix. [53] It contains also the theorem known as "Dupin's theorem." Surfaces of the second degree and descriptive geometry were successfully studied by Jean Nicolas Pierre Hachette (1769– 1834), who became professor of descriptive geometry at the Polytechnic School after the departure of Monge for Rome and Egypt. In 1822 he published his Trait´ e de g´ eom´ etrie descriptive. Descriptive geometry, which arose, as we have seen, in technical schools in France, was transferred to Germany at the foundation of technical schools there. G. Schreiber,

A HISTORY OF MATHEMATICS. 336 professor in Karlsruhe, was the first to spread Monge's geometry in Germany by the publication of a work thereon in 1828–1829. [54] In the United States descriptive geometry was introduced in 1816 at the Military Academy in West Point by Claude Crozet, once a pupil at the Polytechnic School in Paris. Crozet wrote the first English work on the subject. [2] Lazare Nicholas Marguerite Carnot (1753–1823) was born at Nolay in Burgundy, and educated in his native province. He entered the army, but continued his mathematical studies, and wrote in 1784 a work on machines, containing the earliest proof that kinetic energy is lost in collisions of bodies. With the advent of the Revolution he threw himself into politics, and when coalesced Europe, in 1793, launched against France a million soldiers, the gigantic task of organising fourteen armies to meet the enemy was achieved by him. He was banished in 1796 for opposing Napoleon's coup d'´ etat. The refugee went to Geneva, where he issued, in 1797, a work still frequently quoted, entitled, R´ eflexions sur la M´ etaphysique du Calcul Infinit´ esimal. He declared himself as an "irreconcilable enemy of kings." After the Russian campaign he offered to fight for France, though not for the empire. On the restoration he was exiled. He died in Magdeburg. His G´ eom´ etrie de position, 1803, and hisEssay on Transversals, 1806, are important contributions to modern geometry. While Monge revelled mainly in three-dimensional geometry, Carnot confined himself to that of two. By his effort to explain the meaning of the negative sign in geometry he established a "geometry of position," which, however,

EULER, LAGRANGE, AND LAPLACE. 337 is different from the “Geometrie der Lage” of to-day. He invented a class of general theorems on projective properties of figures, which have since been pushed to great extent by Poncelet, Chasles, and others. Jean Victor Poncelet (1788–1867), a native of Metz, took part in the Russian campaign, was abandoned as dead on the bloody field of Krasnoi, and taken prisoner to Saratoff. Deprived there of all books, and reduced to the remembrance of what he had learned at the Lyceum at Metz and the Polytechnic School, where he had studied with predilection the works of Monge, Carnot, and Brianchon, he began to study mathematics from its elements. He entered upon original researches which afterwards made him illustrious. While in prison he did for mathematics what Bunyan did for literature,—produced a much-read work, which has remained of great value down to the present time. He returned to France in 1814, and in 1822 published the work in question, entitled, Trait´ e des Propri´ et´ es projectives des figures. In it he investigated the properties of figures which remain unaltered by projection of the figures. The projection is not effected here by parallel rays of prescribed direction, as with Monge, but by central projection. Thus perspective projection, used before him by Desargues, Pascal, Newton, and Lambert, was elevated by him into a fruitful geometric method. In the same way he elaborated some ideas of De Lahire, Servois, and Gergonne into a regular method—the method of “reciprocal polars.” To him we owe the Law of Duality as a consequence of reciprocal polars. As an independent principle it is due to

A HISTORY OF MATHEMATICS. 338 Gergonne. Poncelet wrote much on applied mechanics. In 1838 the Faculty of Sciences was enlarged by his election to the chair of mechanics. While in France the school of Monge was creating modern geometry, efforts were made in England to revive Greek geometry by Robert Simson (1687–1768) and Matthew Stewart (1717–1785). Stewart was a pupil of Simson and Maclaurin, and succeeded the latter in the chair at Edinburgh. During the eighteenth century he and Maclaurin were the only prominent mathematicians in Great Britain. His genius was ill-directed by the fashion then prevalent in England to ignore higher analysis. In his Four Tracts, Physical and Mathematical , 1761, he applied geometry to the solution of difficult astronomical problems, which on the Continent were approached analytically with greater success. He published, in 1746, General Theorems, and in 1763, his Propositiones geometricæ more veterum demonstratæ . The former work contains sixty-nine theorems, of which only five are accompanied by demonstrations. It gives many interesting new results on the circle and the straight line. Stewart extended some theorems on transversals due to Giovanni Ceva (1648–1737), an Italian, who published in 1678 at Mediolani a work containing the theorem now known by his name.

RECENT TIMES. Never more zealously and successfully has mathematics been cultivated than in this century. Nor has progress, as in previous periods, been confined to one or two countries. While the French and Swiss, who alone during the preceding epoch carried the torch of progress, have continued to develop mathematics with great success, from other countries whole armies of enthusiastic workers have wheeled into the front rank. Germany awoke from her lethargy by bringing forward Gauss, Jacobi, Dirichlet, and hosts of more recent men; Great

Britain produced her De Morgan, Boole, Hamilton, besides champions who are still living; Russia entered the arena with her Lobatchewsky; Norway with Abel; Italy with Cremona; Hungary with her two Bolyais; the United States with Benjamin Peirce. The productiveness of modern writers has been enormous. “It is difficult,” says Professor Cayley, [56] “to give an idea of the vast extent of modern mathematics. This word ‘extent’ is not the right one: I mean extent crowded with beautiful detail,—not an extent of mere uniformity such as an objectless plain, but of a tract of beautiful country seen at first in the distance, but which will bear to be rambled through and studied in every detail of hillside and valley, stream, rock, wood, and flower.” It is pleasant to the mathematician to think that in his, as in no other science, the achievements of 339

A HISTORY OF MATHEMATICS. 340 every age remain possessions forever; new discoveries seldom disprove older tenets; seldom is anything lost or wasted. If it be asked wherein the utility of some modern extensions of mathematics lies, it must be acknowledged that it is at present difficult to see how they are ever to become applicable to questions of common life or physical science. But our inability to do this should not be urged as an argument against the pursuit of such studies. In the first place, we know neither the day nor the hour when these abstract developments will find application in the mechanic arts, in physical science, or in other branches of mathematics. For example, the whole subject of graphical statics, so useful to the practical engineer, was made to rest upon von Staudt’s Geometrie der Lage ; Hamilton’s “principle of varying action” has its use in astronomy; complex quantities, general integrals, and general theorems in integration offer advantages in the study of electricity and magnetism. “The utility of such researches,” says Spottiswoode, [57] “can in no case be discounted, or even imaginedbeforehand. Who, forinstance, wouldhavesupposed that the calculus of forms or the theory of substitutions would have thrown much light upon ordinary equations; or that Abelian functions and hyperelliptic transcendents would have told us anything about the properties of curves; or that the calculus of operations would have helped us in any way towards the figure of the earth?” A second reason in favour of the pursuit of advanced mathematics, even when there is no promise of practical application, is this, that mathematics, like poetry and music, deserves cultivation for its own sake.

SYNTHETIC GEOMETRY. 341 The great characteristic of modern mathematics is its generalising tendency. Nowadays little weight is given to isolated theorems, “except as affording hints of an unsuspected new sphere of thought, like meteorites detached from some undiscovered planetary orb of speculation.” In mathematics, as in all true sciences, no subject is considered in itself alone, but always as related to, or an outgrowth of, other things. The development of the notion of continuity plays a leading part in modern research. In geometry the principle of continuity, the idea of correspondence, and the theory of projection constitute the fundamental modern notions. Continuity asserts itself in a most striking way in relation to the circular points at infinity in a plane. In algebra the modern idea finds expression in the theory of linear transformations and invariants, and in the recognition of the value of homogeneity and symmetry. SYNTHETIC GEOMETRY. The conflict between geometry and analysis which arose near the close of the last century and the

beginning of the present has now come to an end. Neither side has come out victorious. The greatest strength is found to lie, not in the suppression of either, but in the friendly rivalry between the two, and in the stimulating influence of the one upon the other. Lagrange prided himself that in his M´ ecanique Analytique he had succeeded in avoiding all figures; but since his time mechanics has received much help from geometry. Modern synthetic geometry was created by several investi-

A HISTORY OF MATHEMATICS. 342 gators about the same time. It seemed to be the outgrowth of a desire for general methods which should serve as threads of Ariadne to guide the student through the labyrinth of theo- rems, corollaries, porisms, and problems. Synthetic geometry was first cultivated by Monge, Carnot, and Poncelet in France; it then bore rich fruits at the hands of M¨ obius and Steiner in Germany and Switzerland, and was finally developed to still higher perfection by Chasles in France, von Staudt in Germany, and Cremona in Italy. Augustus Ferdinand M¨ obius(1790–1868) was a native of Schulpforta in Prussia. He studied at G¨ ottingen under Gauss, also at Leipzig and Halle. In Leipzig he became, in 1815, privat-docent, the next year extraordinary professor of astronomy, and in 1844 ordinary professor. This position he held till his death. The most important of his researches are on geometry. They appeared in Crelle's Journal, and in his celebrated work entitled Der Barycentrische Calcul, Leipzig, 1827. As the name indicates, this calculus is based upon properties of the centre of gravity. [58] Thus, that the pointS is the centre of gravity of weightsa, b, c, d placed at the points A, B, C, D respectively, is expressed by the equation (a + b + c + d)S = aA + bB + cC + dD. His calculus is the beginning of a quadruple algebra, and contains the germs of Grassmann's marvellous system. In designating segments of lines we find throughout this work for the first time consistency in the distinction of positive and negative by the order of letters AB, BA. Similarly

SYNTHETIC GEOMETRY. 343 for triangles and tetrahedra. The remark that it is always possible to give three points A, B, C such weights α, β, γ that any fourth point M in their plane will become a centre of mass, led M¨ obius to a new system of co-ordinates in which the position of a point was indicated by an equation, and that of a line by co-ordinates. By this algorithm he found by algebra many geometric theorems expressing mainly invariantal properties,—for example, the theorems on the anharmonic relation. M¨ obius wrote also on statics and astronomy. He generalised spherical trigonometry by letting the sides or angles of triangles exceed180◦. Jacob Steiner (1796–1863), "the greatest geometrician since the time of Euclid," was born in Utzendorf in the Canton of Bern. He did not learn to write till he was fourteen. At eighteen he became a pupil of Pestalozzi. Later he studied at Heidelberg and Berlin. When Crelle started, in 1826, the celebrated mathematical journal bearing his name, Steiner and Abel became leading contributors. In 1832 Steiner published his Systematische Entwickelung der Abh¨ angigkeit geometrischer Gestalten von einander, "in which is uncovered the organism by which the most diverse phenomena (Erscheinungen) in the world of space are united to each other." Through the influence of Jacobi and others, the chair of geometry was founded for him at Berlin in 1834. This position he occupied until his death, which occurred after years of bad health. In his Systematische Entwickelungen , for the first time, is the

principle of duality introduced at the outset. This book and von Staudt's lay the foundation

A HISTORY OF MATHEMATICS. 344 on which synthetic geometry in its present form rests. Not only did he fairly complete the theory of curves and surfaces of the second degree, but he made great advances in the theory of those of higher degrees. In his hands synthetic geometry made prodigious progress. New discoveries followed each other so rapidly that he often did not take time to record their demonstrations. In an article in Crelle's Journal on Allgemeine Eigenschaften Algebraischer Curven he gives without proof theorems which were declared by Hesse to be "like Fermat's theorems, riddles to the present and future generations." Analytical proofs of some of them have been given since by others, but Cremona finally proved them all by a synthetic method. Steiner discovered synthetically the two prominent properties of a surface of the third order; viz. that it contains twenty-seven straight lines and a pentahedron which has the double points for its vertices and the lines of the Hessian of the given surface for its edges. [55] The first property was discovered analytically somewhat earlier in England by Cayley and Salmon, and the second by Sylvester. Steiner's work on this subject was the starting-point of important researches by H. Schr¨ oter, F. August, L. Cremona, and R. Sturm. Steiner made investigations by synthetic methods on maxima and minima, and arrived at the solution of problems which at that time altogether surpassed the analytic power of the calculus of variations. He generalised the hexagrammum mysticum and also Malfatti's problem. [59] Malfatti, in 1803, proposed the problem, to cut three cylindrical holes out of a three-sided prism in such a way that the cylinders and

SYNTHETIC GEOMETRY. 345 the prism have the same altitude and that the volume of the cylinders be a maximum. This problem was reduced to another, now generally known as Malfatti's problem: to inscribe three circles in a triangle that each circle will be tangent to two sides of a triangle and to the other two circles. Malfatti gave an analytical solution, but Steiner gave without proof a construction, remarked that there were thirty-two solutions, generalised the problem by replacing the three lines by three circles, and solved the analogous problem for three dimensions. This general problem was solved analytically by C. H. Schellbach (1809–1892) and Cayley, and by Clebsch with the aid of the addition theorem of elliptic functions. [60] Steiner's researches are confined to synthetic geometry. He hated analysis as thoroughly as Lagrange disliked geometry. Steiner's Gesammelte Werke were published in Berlin in 1881 and 1882. MichelChasles (1793–1880) was born at Epernon, entered the Polytechnic School of Paris in 1812, engaged afterwards in business, which he later gave up that he might devote all his time to scientific pursuits. In 1841 he became professor of geodesy and mechanics at the Polytechnic School; later, "Professeur de G´ eom´ etrie sup´ erieure ` a la Facult´ e des Sciences de Paris." He was a voluminous writer on geometrical subjects. In 1837 he published his admirableAper¸ cu historique sur l'origine et le d´ eveloppement des m´ ethodes en g´ eom´ etrie, containing a history of geometry and, as an appendix, a treatise "sur deux principes g´ en´ eraux de la Science." The Aper¸ cu historique is still a standard historical work; the appendix

A HISTORY OF MATHEMATICS. 346 contains the general theory of Homography (Collineation) and of duality (Reciprocity). The name duality is due to Joseph Diaz Gergonne (1771–1859). Chasles introduced the term anharmonic ratio , corresponding to the German Doppelverh¨ altnissand to Clifford's cross-ratio. Chasles and Steiner elaborated independently the modern synthetic or projective geometry. Numerous original memoirs of Chasles were published later in the Journal de l' ´Ecole Polytechnique. He gave a reduction of cubics, different from Newton's in this, that the five curves from which all others can be projected are symmetrical with respect to a centre. In 1864 he began the publication, in theComptes rendus, of articles in which he solves by his "method of characteristics" and the "principle of correspondence" an immense number of problems. He determined, for instance, the number of intersections of two curves in a plane. The method of characteristics contains the basis of enumerative geometry. The application of the principle of correspondence was extended by Cayley, A. Brill, H. G. Zeuthen, H. A. Schwarz, G. H. Halphen (1844–1889), and others. The full value of these principles of Chasles was not brought out until the appearance, in 1879, of the Kalk¨ ul der Abz¨ ahlenden Geometrieby Hermann Schubert of Hamburg. This work contains a masterly discussion of the problem of enumerative geometry, viz. to determine how many geometric figures of given definition satisfy a sufficient number of conditions. Schubert extended his enumerative geometry to n-dimensional space. [55] To Chasles we owe the introduction into projective geom-

SYNTHETIC GEOMETRY. 347 etry of non-projective properties of figures by means of the infinitely distant imaginary sphero-circle. [61] Remarkable is his complete solution, in 1846, by synthetic geometry, of the difficult question of the attraction of an ellipsoid on an external point. This was accomplished analytically by Poisson in 1835. The labours of Chasles and Steiner raised synthetic geometry to an honoured and respected position by the side of analysis. Karl Georg Christian von Staudt(1798–1867) was born in Rothenburg on the Tauber, and, at his death, was professor in Erlangen. His great works are the Geometrie der Lage , N¨ urnberg, 1847, and his Beitr¨ age zur Geometrie der Lage, 1856–1860. The author cut loose from algebraic formulæ and from metrical relations, particularly the anharmonic ratio of Steiner and Chasles, and then created a geometry of position, which is a complete science in itself, independent of all measurements. He shows that projective properties of figures have no dependence whatever on measurements, and can be established without any mention of them. In his theory of what he calls "W¨ urfe," he even gives a geometrical definition of a number in its relation to geometry as determining the position of a point. The Beitr¨ agecontains the first complete and general theory of imaginary points, lines, and planes in projective geometry. Representation of an imaginary point is sought in the combination of an involution with a determinate direction, both on the real line through the point. While purely projective, von Staudt's method is intimately related to the problem of representing by actual points and lines the

A HISTORY OF MATHEMATICS. 348 imaginaries of analytical geometry. This was systematically undertaken by C. F. Maximilien Marie, who worked, however, on entirely different lines. An independent attempt has been made recently (1893) by F.

H. Loud of Colorado College. Von Staudt's geometry of position was for a long time disregarded, mainly, no doubt, because his book is extremely condensed. An impulse to the study of this subject was given by Culmann, who rests his graphical statics upon the work of von Staudt. An interpreter of von Staudt was at last found in Theodor Reye of Strassburg, who wrote aGeometrie der Lage in 1868. Synthetic geometry has been studied with much success by Luigi Cremona, professor in the University of Rome. In his Introduzione ad una teoria geometrica delle curve piane he developed by a uniform method many new results and proved synthetically all important results reached before that time by analysis. His writings have been translated into German by M. Curtze, professor at the gymnasium in Thorn. The theory of the transformation of curves and of the correspondence of points on curves was extended by him to three dimensions. Ruled surfaces, surfaces of the second order, space-curves of the third order, and the general theory of surfaces have received much attention at his hands. Karl Culmann, professor at the Polytechnicum in Z¨ urich, published an epoch-making work on Die graphische Statik , Z¨ urich, 1864, which has rendered graphical statics a great rival of analytical statics. Before Culmann, B. E. Cousinery had turned his attention to the graphical calculus, but he

SYNTHETIC GEOMETRY. 349 made use of perspective, and not of modern geometry. [62] Culmann is the first to undertake to present the graphical calculus as a symmetrical whole, holding the same relation to the new geometry that analytical mechanics does to higher analysis. He makes use of the polar theory of reciprocal figures as expressing the relation between the force and the funicular polygons. He deduces this relation without leaving the plane of the two figures. But if the polygons be regarded as projections of lines in space, these lines may be treated as reciprocal elements of a "Nullsystem." This was done by Clerk Maxwell in 1864, and elaborated further by Cremona. [63] The graphical calculus has been applied by O. Mohr of Dresden to the elastic line for continuous spans. Henry T. Eddy , of the Rose Polytechnic Institute, gives graphical solutions of problems on the maximum stresses in bridges under concentrated loads, with aid of what he calls "reaction polygons." A standard work,La Statique graphique, 1874, was issued by Maurice Levy of Paris. Descriptive geometry (reduced to a science by Monge in France, and elaborated further by his successors, Hachette, Dupin, Olivier, J. de la Gournerie) was soon studied also in other countries. The French directed their attention mainly to the theory of surfaces and their curvature; the Germans and Swiss, through Schreiber, Pohlke, Schlessinger, and particularly Fiedler, interwove projective and descriptive geometry. Bellavitis in Italy worked along the same line. The theory of shades and shadows was first investigated by the French writers just quoted, and in Germany treated most

A HISTORY OF MATHEMATICS. 350 exhaustively by Burmester. [62] During the present century very remarkable generalisations have been made, which reach to the very root of two of the oldest branches of mathematics,—elementary algebra and geometry. In algebra the laws of operation have been extended; in geometry the axioms have been searched to the bottom, and the conclusion has been reached that the space defined by Euclid's axioms is not the only possible non-contradictory space. Euclid proved (I. 27) that "if a straight line falling on two other straight lines make

the alternate angles equal to one another, the two straight lines shall be parallel to one another." Being unable to prove that in every other case the two lines are not parallel, he assumed this to be true in what is generally called the 12th "axiom," by some the 11th "axiom." But this so-called axiom is far from axiomatic. After centuries of desperate but fruitless attempts to prove Euclid's assumption, the bold idea dawned upon the minds of several mathematicians that a geometry might be built up without assuming the parallel-axiom. While Legendre still endeavoured to establish the axiom by rigid proof, Lobatchewsky brought out a publication which assumed the contradictory of that axiom, and which was the first of a series of articles destined to clear up obscurities in the fundamental concepts, and to greatly extend the field of geometry. Nicholaus Ivanovitch Lobatchewsky(1793–1856) was born at Makarief, in Nischni-Nowgorod, Russia, studied at Kasan, and from 1827 to 1846 was professor and rector of the University of Kasan. His views on the foundation of geometry

SYNTHETIC GEOMETRY. 351 were first made public in a discourse before the physical and mathematical faculty at Kasan, and first printed in the Kasan Messenger for 1829, and then in theGelehrte Schriften der Universit¨ at Kasan, 1836–1838, under the title, "New Elements of Geometry, with a complete theory of Parallels." Being in the Russian language, the work remained unknown to foreigners, but even at home it attracted no notice. In 1840 he published a brief statement of his researches in Berlin. Lobatchewsky constructed an "imaginary geometry," as he called it, which has been described by Clifford as "quite simple, merely Euclid without the vicious assumption." A remarkable part of this geometry is this, that through a point an indefinite number of lines can be drawn in a plane, none of which cut a given line in the same plane. A similar system of geometry was deduced independently by the Bolyais in Hungary, who called it "absolute geometry." Wolfgang Bolyai de Bolya (1775–1856) was born in Szekler-Land, Transylvania. After studying at Jena, he went to G¨ ottingen, where he became intimate with Gauss, then nineteen years old. Gauss used to say that Bolyai was the only man who fully understood his views on the metaphysics of mathematics. Bolyai became professor at the Reformed College of Maros-V´ as´ arhely, where for forty-seven years he had for his pupils most of the present professors of Transylvania. The first publications of this remarkable genius were dramas and poetry. Clad in old-time planter's garb, he was truly original in his private life as well as in his mode of thinking. He was extremely modest. No monument, said he, should

A HISTORY OF MATHEMATICS. 352 stand over his grave, only an apple-tree, in memory of the three apples; the two of Eve and Paris, which made hell out of earth, and that of Newton, which elevated the earth again into the circle of heavenly bodies. [64] His son, Johann Bolyai (1802–1860), was educated for the army, and distinguished himself as a profound mathematician, an impassioned violin- player, and an expert fencer. He once accepted the challenge of thirteen officers on condition that after each duel he might play a piece on his violin, and he vanquished them all. The chief mathematical work of Wolfgang Bolyai appeared in two volumes, 1832–1833, entitled Tentamen juventutem studiosam in elementa matheseos puræ. . . introducendi. It is followed by an appendix composed by his son Johann onThe Science Absolute of Space . Its twenty-six pages make the name of Johann Bolyai

immortal. He published nothing else, but he left behind one thousand pages of manuscript which have never been read by a competent mathematician! His father seems to have been the only person in Hungary who really appreciated the merits of his son's work. For thirty- five years this appendix, as also Lobatchewsky's researches, remained in almost entire oblivion. Finally Richard Baltzer of the University of Giessen, in 1867, called attention to the wonderful researches. Johann Bolyai's Science Absolute of Space and Lobatchewsky's Geometrical Researches on the Theory of Parallels (1840) were rendered easily accessible to American readers by translations into English made in 1891 by George Bruce Halsted of the University of Texas. The Russian and Hungarian mathematicians were not

SYNTHETIC GEOMETRY. 353 the only ones to whom pangeometry suggested itself. A copy of the Tentamen reached Gauss, the elder Bolyai's former room-mate at G¨ ottingen, and this Nestor of German mathematicians was surprised to discover in it worked out what he himself had begun long before, only to leave it after him in his papers. As early as 1792 he had started on researches of that character. His letters show that in 1799 he was trying to prove a priori the reality of Euclid's system; but some time within the next thirty years he arrived at the conclusion reached by Lobatchewsky and Bolyai. In 1829 he wrote to Bessel, stating that his "conviction that we cannot found geometry completely a priori has become, if possible, still firmer," and that "if number is merely a product of our mind, space has also areality beyond our mind of which we cannot fully foreordain the laws a priori." The term non-Euclidean geometry is due to Gauss. It has recently been brought to notice that Geronimo Saccheri , a Jesuit father of Milan, in 1733 anticipated Lobatchewsky's doctrine of the parallel angle. Moreover, G. B. Halsted has pointed out that in 1766 Lambert wrote a paper "Zur Theorie der Parallellinien," published in the Leipziger Magazin f¨ ur reine und angewandte Mathematik, 1786, in which: (1) The failure of the parallel-axiom in surface-spherics gives a geometry with angle-sum > 2 right angles; (2) In order to make intuitive a geometry with angle-sum < 2 right angles we need the aid of an "imaginary sphere" (pseudo-sphere); (3) In a space with the angle-sum differing from 2 right angles, there is an absolute measure (Bolyai's natural unit for length).

A HISTORY OF MATHEMATICS. 354 In 1854, nearly twenty years later, Gauss heard from his pupil, Riemann, a marvellous dissertation carrying the discussion one step further by developing the notion of n-ply extended magnitude, and the measure-relations of which a manifoldness of n dimensions is capable, on the assumption that every line may be measured by every other. Riemann applied his ideas to space. He taught us to distinguish between "unboundedness" and "infinite extent." According to him we have in our mind a more general notion of space, i.e. a notion of non-Euclidean space; but we learn by experience that our physical space is, if not exactly, at least to high degree of approximation, Euclidean space. Riemann's profound dissertation was not published until 1867, when it appeared in the G¨ ottingen Abhandlungen. Before this the idea of n-dimensions had suggested itself under various aspects to Lagrange, Pl¨ ucker, and H. Grassmann. About the same time with Riemann's paper, others were published from the pens of Helmholtz and Beltrami. These contributed powerfully to the victory of logic over excessive

empiricism. This period marks the beginning of lively discussions upon this subject. Some writers—Bellavitis, for example—were able to see in non-Euclidean geometry and n-dimensional space nothing but huge caricatures, or diseased outgrowths of mathematics. Helmholtz's article was entitledThatsachen, welche der Geometrie zu Grunde liegen, 1868, and contained many of the ideas of Riemann. Helmholtz popularised the subject in lectures, and in articles for various magazines. Eugenio Beltrami, born at Cremona, Italy, in 1835, and

SYNTHETIC GEOMETRY. 355 now professor at Rome, wrote the classical paper Saggio di interpretazione della geometria non-euclidea (Giorn. di Matem., 6), which is analytical (and, like several other papers, should be mentioned elsewhere were we to adhere to a strict separation between synthesis and analysis). He reached the brilliant and surprising conclusion that the theorems of non- Euclidean geometry find their realisation upon surfaces of constant negative curvature. He studied, also, surfaces of constant positive curvature, and ended with the interesting theorem that the space of constant positive curvature is contained in the space of constant negative curvature. These researches of Beltrami, Helmholtz, and Riemann culminated in the conclusion that on surfaces of constant curvature we may have three geometries,—the non-Euclidean on a surface of constant negative curvature, the spherical on a surface of constant positive curvature, and the Euclidean geometry on a surface of zero curvature. The three geometries do not contradict each other, but are members of a system,—a geometrical trinity. The ideas of hyperspace were brilliantly expounded and popularised in England by Clifford. William Kingdon Clifford(1845–1879) was born at Ex- eter, educated at Trinity College, Cambridge, and from 1871 until his death professor of applied mathematics in University College, London. His premature death left incomplete several brilliant researches which he had entered upon. Among these are his paper On Classification of Loci and his Theory of Graphs. He wrote articles On the Canonical Form and Dissection of a Riemann's Surface, on Biquaternions, and an

A HISTORY OF MATHEMATICS. 356 incomplete work on the Elements of Dynamic. The theory of polars of curves and surfaces was generalised by him and by Reye. His classification of loci, 1878, being a general study of curves, was an introduction to the study of n-dimensional space in a direction mainly projective. This study has been continued since chiefly by G. Veronese of Padua, C. Segre of Turin, E. Bertini, F. Aschieri, P. Del Pezzo of Naples. Beltrami's researches on non-Euclidean geometry were followed, in 1871, by important investigations of Felix Klein, resting upon Cayley's Sixth Memoir on Quantics, 1859. The question whether it is not possible to so express the metrical properties of figures that they will not vary by projection (or linear transformation) had been solved for special projections by Chasles, Poncelet, and E. Laguerre (1834–1886) of Paris, but it remained for Cayley to give a general solution by defining the distance between two points as an arbitrary constant multiplied by the logarithm of the anharmonic ratio in which the line joining the two points is divided by the fundamental quadric. Enlarging upon this notion, Klein showed the independence of projective geometry from the parallel-axiom, and by properly choosing the law of the measurement of distance deduced from projective geometry the spherical, Euclidean, and pseudospherical geometries, named by him

respectively the elliptic, parabolic, and hyperbolic geometries. This suggestive investigation was followed up by numerous writers, particularly by G. Battaglini of Naples, E. d’Ovidio of Turin, R. de Paolis of Pisa, F. Aschieri, A. Cayley, F.LindemannofMunich, E.ScheringofG¨ ottingen, W.Storyof

ANALYTIC GEOMETRY. 357 Clark University, H. Stahl of T¨ ubingen, A. Voss of W¨ urzburg, Homersham Cox, A. Buchheim. [55] The geometry of n dimensions was studied along a line mainly metrical by a host of writers, among whom may be mentioned Simon Newcomb of the Johns Hopkins University, L. Schl¨ afli of Bern, W. I. Stringham of the University of California, W. Killing of M¨ unster, T. Craig of the Johns Hopkins, R. Lipschitz of Bonn. R. S. Heath and Killing investigated the kinematics and mechanics of such a space. Regular solids in n-dimensional space were studied by Stringham, Ellery W. Davis of the University of Nebraska, R. Hoppe of Berlin, and others. Stringham gave pictures of projections upon our space of regular solids in four dimensions, and Schlegel at Hagen constructed models of such projections. These are among the most curious of a series of models published by L. Brill in Darmstadt. It has been pointed out that if a fourth dimension existed, certain motions could take place which we hold to be impossible. Thus Newcomb showed the possibility of turning a closed material shell inside out by simple flexure without either stretching or tearing; Klein pointed out that knots could not be tied; Veronese showed that a body could be removed from a closed room without breaking the walls; C. S. Peirce proved that a body in four-fold space either rotates about two axes at once, or cannot rotate without losing one of its dimensions.

A HISTORY OF MATHEMATICS. 358 ANALYTIC GEOMETRY. In the preceding chapter we endeavoured to give a flash- light view of the rapid advance of synthetic geometry. In connection with hyperspace we also mentioned analytical treatises. Modern synthetic and modern analytical geometry have much in common, and may be grouped together under the common name “projective geometry.” Each has advantages over the other. The continual direct viewing of figures as existing in space adds exceptional charm to the study of the former, but the latter has the advantage in this, that a well- established routine in a certain degree may outrun thought itself, and thereby aid original research. While in Germany Steiner and von Staudt developed synthetic geometry, Pl¨ ucker laid the foundation of modern analytic geometry. Julius Pl¨ ucker(1801–1868) was born at Elberfeld, in Prussia. After studying at Bonn, Berlin, and Heidelberg, he spent a short time in Paris attending lectures of Monge and his pupils. Between 1826 and 1836 he held positions successively at Bonn, Berlin, and Halle. He then became professor of physics at Bonn. Until 1846 his original researches were on geometry. In 1828 and in 1831 he published his Analytisch-Geometrische Entwicklungen in two volumes. Therein he adopted the abbreviated notation (used before him in a more restricted way by Bobillier), and avoided the tedious process of algebraic elimination by a geometric consideration. In the second volume the principle of duality is formulated analytically. With him duality and homogeneity

ANALYTIC GEOMETRY. 359 found expression already in his system of co-ordinates. The homogenous or tri-linear system used by him is much the same as

the co-ordinates of M¨ obius. In the identity of analytical operation and geometric construction Pl¨ ucker looked for the source of his proofs. The System der Analytischen Geometrie, 1835, contains a complete classification of plane curves of the third order, based on the nature of the points at infinity. The Theorie der Algebraischen Curven , 1839, contains, besides an enumeration of curves of the fourth order, the analytic relations between the ordinary singularities of plane curves known as “Pl¨ ucker’s equations,” by which he was able to explain “Poncelet’s paradox.” The discovery of these relations is, says Cayley, “the most important one beyond all comparison in the entire subject of modern geometry.” But in Germany Pl¨ ucker’s researches met with no favour. His method was declared to be unproductive as compared with the synthetic method of Steiner and Poncelet! His relations with Jacobi were not altogether friendly. Steiner once declared that he would stop writing forCrelle’s Journal if Pl¨ ucker continued to contribute to it. [66] The result was that many of Pl¨ ucker’s researches were published in foreign journals, and that his work came to be better known in France and England than in his native country. The charge was also brought against Pl¨ ucker that, though occupying the chair of physics, he was no physicist. This induced him to relinquish mathematics, and for nearly twenty years to devote his energies to physics. Important discoveries on Fresnel’s wave-surface, magnetism, spectrum-analysis were made by him. But towards the close

A HISTORY OF MATHEMATICS. 360 of his life he returned to his first love,—mathematics,—and enriched it with new discoveries. By considering space as made up of lines he created a “new geometry of space.” Regarding a right line as a curve involving four arbitrary parameters, one has the whole system of lines in space. By connecting them by a single relation, he got a “complex” of lines; by connecting them with a twofold relation, he got a “congruency” of lines. His first researches on this subject were laid before the Royal Society in 1865. His further investigations thereon appeared in 1868 in a posthumous work entitled Neue Geometrie des Raumes gegr¨ undet auf die Betrachtung der geraden Linie als Raumelement, edited by Felix Klein. Pl¨ ucker’s analysis lacks the elegance found in Lagrange, Jacobi, Hesse, and Clebsch. For many years he had not kept up with the progress of geometry, so that many investigations in his last work had already received more general treatment on the part of others. The work contained, nevertheless, much that was fresh and original. The theory of complexes of the second degree, left unfinished by Pl¨ ucker, was continued by Felix Klein, who greatly extended and supplemented the ideas of his master. Ludwig Otto Hesse(1811–1874) was born at K¨ onigsberg, and studied at the university of his native place under Bessel, Jacobi, Richelot, and F. Neumann. Having taken the doctor’s degree in 1840, he became docent at K¨ onigsberg, and in 1845 extraordinary professor there. Among his pupils at that time were Dur` ege, Carl Neumann, Clebsch, Kirchhoff. The K¨ onigsberg period was one of great activity for Hesse. Every new discovery increased his zeal for still

ANALYTIC GEOMETRY. 361 greater achievement. His earliest researches were on surfaces of the second order, and were partly synthetic. He solved the problem to construct any tenth point of such a surface when nine points are given. The analogous problem for a conic had been solved by Pascal by means of the hexagram.

A difficult problem confronting mathematicians of this time was that of elimination. Pl¨ ucker had seen that the main advantage of his special method in analytic geometry lay in the avoidance of algebraic elimination. Hesse, however, showed how by determinants to make algebraic elimination easy. In his earlier results he was anticipated by Sylvester, who published his dialytic method of elimination in 1840. These advances in algebra Hesse applied to the analytic study of curves of the third order. By linear substitutions, he reduced a form of the third degree in three variables to one of only four terms, and was led to an important determinant involving the second differential coefficient of a form of the third degree, called the "Hessian." The "Hessian" plays a leading part in the theory of invariants, a subject first studied by Cayley. Hesse showed that his determinant gives for every curve another curve, such that the double points of the first are points on the second, or "Hessian." Similarly for surfaces (Crelle, 1844). Many of the most important theorems on curves of the third order are due to Hesse. He determined the curve of the 14th order, which passes through the 56 points of contact of the 28 bi-tangents of a curve of the fourth order. His great memoir on this subject (Crelle, 1855) was published at the same time as was a paper by Steiner treating of the same subject.

A HISTORY OF MATHEMATICS. 362 Hesse's income at K¨ onigsberg had not kept pace with his growing reputation. Hardly was he able to support himself and family. In 1855 he accepted a more lucrative position at Halle, and in 1856 one at Heidelberg. Here he remained until 1868, when he accepted a position at a technic school in Munich. [67] At Heidelberg he revised and enlarged upon his previous researches, and published in 1861 his Vorlesungen ¨ uber die Analytische Geometrie des Raumes, insbesondere ¨ uber Fl¨ achen 2. Ordnung. More elementary works soon followed. While in Heidelberg he elaborated a principle, his "Uebertragungsprincip." According to this, there corresponds to every point in a plane a pair of points in a line, and the projective geometry of the plane can be carried back to the geometry of points in a line. The researches of Pl¨ ucker and Hesse were continued in England by Cayley, Salmon, and Sylvester. It may be premised here that among the early writers on analytical geometry in England was James Booth (1806–1878), whose chief results are embodied in his Treatise on Some New Geometrical Methods; and James MacCullagh (1809–1846), who was professor of natural philosophy at Dublin, and made some valuable discoveries on the theory of quadrics. The influence of these men on the progress of geometry was insignificant, for the interchange of scientific results between different nations was not so complete at that time as might have been desired. In further illustration of this, we mention that Chasles in France elaborated subjects which had previously been disposed of by Steiner in Germany, and Steiner published researches which

ANALYTIC GEOMETRY. 363 had been given by Cayley, Sylvester, and Salmon nearly five years earlier. Cayley and Salmon in 1849 determined the straight lines in a cubic surface, and studied its principal properties, while Sylvester in 1851 discovered the pentahedron of such a surface. Cayley extended Pl¨ ucker's equations to curves of higher singularities. Cayley's own investigations, and those of M. N¨ other of Erlangen, G. H. Halphen (1844– 1889) of the Polytechnic School in Paris, De La Gournerie of Paris, A. Brill of T¨ ubingen, lead to the conclusion that each higher

singularity of a curve is equivalent to a certain number of simple singularities,—the node, the ordinary cusp, the double tangent, and the inflection. Sylvester studied the “twisted Cartesian,” a curve of the fourth order. Salmon helped powerfully towards the spreading of a knowledge of the new algebraic and geometric methods by the publication of an excellent series of text-books (Conic Sections, Modern Higher Algebra , Higher Plane Curves , Geometry of Three Dimensions), which have been placed within easy reach of German readers by a free translation, with additions, made by Wilhelm Fiedler of the Polytechnicum in Z¨ urich. The next great worker in the field of analytic geometry was Clebsch. Rudolf Friedrich Alfred Clebsch(1833–1872) was born at K¨ onigsberg in Prussia, studied at the university of that place under Hesse, Richelot, F. Neumann. From 1858 to 1863 he held the chair of theoretical mechanics at the Polytechnicum in Carlsruhe. The study of Salmon’s works led him into algebra and geometry. In 1863 he accepted a position at the University of Giessen, where he worked in conjunction with

A HISTORY OF MATHEMATICS. 364 Paul Gordan (now of Erlangen). In 1868 Clebsch went to G¨ ottingen, and remained there until his death. He worked successively at the following subjects: Mathematical physics, the calculus of variations and partial differential equations of the first order, the general theory of curves and surfaces, Abelian functions and their use in geometry, the theory of invariants, and “Fl¨ achenabbildung.” [68] He proved theorems on the pentahedron enunciated by Sylvester and Steiner; he made systematic use of “deficiency” (Geschlecht) as a fundamental principle in the classification of algebraic curves. The notion of deficiency was known before him to Abel and Riemann. At the beginning of his career, Clebsch had shown how elliptic functions could be advantageously applied to Malfatti’s problem. The idea involved therein, viz. the use of higher transcendentals in the study of geometry, led him to his greatest discoveries. Not only did he apply Abelian functions to geometry, but conversely, he drew geometry into the service of Abelian functions. Clebsch made liberal use of determinants. His study of curves and surfaces began with the determination of the points of contact of lines which meet a surface in four consecutive points. Salmon had proved that these points lie on the intersection of the surface with a derived surface of the degree 11n− 24, but his solution was given in inconvenient form. Clebsch’s investigation thereon is a most beautiful piece of analysis. The representation of one surface upon another (Fl¨ achen- abbildung), so that they have a (1, 1) correspondence, was

ANALYTIC GEOMETRY. 365 thoroughly studied for the first time by Clebsch. The rep- resentation of a sphere on a plane is an old problem which drew the attention of Ptolemæus, Gerard Mercator, Lambert, Gauss, Lagrange. Its importance in the construction of maps is obvious. Gauss was the first to represent a surface upon another with a view of more easily arriving at its properties. Pl¨ ucker, Chasles, Cayley, thus represented on a plane the geometry of quadric surfaces; Clebsch and Cremona, that of cubic surfaces. Other surfaces have been studied in the same way by recent writers, particularly M. N¨ other of Erlangen, Armenante, Felix Klein, Kornd¨ orfer, Caporali, H. G. Zeuthen of Copenhagen. A fundamental question which has as yet received only a partial answer is this: What surfaces can be represented by a (1, 1) correspondence upon a given surface? This and the analogous question for

curves was studied by Clebsch. Higher correspondences between surfaces have been investigated by Cayley and N¨ other. The theory of surfaces has been studied also byJoseph Alfred Serret(1819–1885), professor at the Sorbonne in Paris, Jean Gaston Darboux of Paris, John Casey of Dublin (died 1891),W. R. W. Roberts of Dublin, H. Schr¨ oter(1829–1892) of Breslau. Surfaces of the fourth order were investigated by Kummer, and Fresnel's wave-surface, studied by Hamilton, is a particular case of Kummer's quartic surface, with sixteen canonical points and sixteen singular tangent planes. [56] The infinitesimal calculus was first applied to the determi- nation of the measure of curvature of surfaces by Lagrange, Euler, and Meusnier (1754–1793) of Paris. Then followed the

A HISTORY OF MATHEMATICS. 366 researches of Monge and Dupin, but they were eclipsed by the work of Gauss, who disposed of this difficult subject in a way that opened new vistas to geometricians. His treatment is embodied in the Disquisitiones generales circa superfi- cies curvas (1827) and Untersuchungen ¨ uber gegenst¨ ande der h¨ oheren Geod¨ asieof 1843 and 1846. He defined the measure of curvature at a point to be the reciprocal of the product of the two principal radii of curvature at that point. From this flows the theorem ofJohann August Grunert (1797–1872; professor in Greifswald), that the arithmetical mean of the radii of curvature of all normal sections through a point is the radius of a sphere which has the same measure of cur- vature as has the surface at that point. Gauss's deduction of the formula of curvature was simplified through the use of determinants by Heinrich Richard Baltzer (1818–1887) of Giessen. [69] Gauss obtained an interesting theorem that if one surface be developed (abgewickelt) upon another, the measure of curvature remains unaltered at each point. The question whether two surfaces having the same curvature in corresponding points can be unwound, one upon the other, was answered by F. Minding in the affirmative only when the curvature is constant. The case of variable curvature is diffi- cult, and was studied by Minding, J. Liouville (1806–1882) of the Polytechnic School in Paris, Ossian Bonnet of Paris (died 1892). Gauss's measure of curvature, expressed as a function of curvilinear co-ordinates, gave an impetus to the study of differential-invariants, or differential-parameters, which have been investigated by Jacobi, C. Neumann, Sir James Cockle,

ALGEBRA. 367 Halphen, and elaborated into a general theory by Beltrami, S. Lie, and others. Beltrami showed also the connection between the measure of curvature and the geometric axioms. Various researches have been brought under the head of "analysis situs." The subject was first investigated by Leibniz, and was later treated by Gauss, whose theory of knots (Verschlingungen) has been employed recently by J. B. Listing, O.Simony, F.Dingeldey, andothersintheir"topologic studies." Tait was led to the study of knots by Sir William Thomson's theory of vortex atoms. In the hands of Riemann the analysis situs had for its object the determination of what remains unchanged under transformations brought about by a combination of infinitesimal distortions. In continuation of his work, Walter Dyck of Munich wrote on the analysis situs of three-dimensional spaces. Of geometrical text-books not yet mentioned, reference should be made to Alfred Clebsch's Vorlesungen ¨ uber Geometrie, edited by Ferdinand Lindemann, now of Munich; Frost's Solid Geometry ; Dur` ege'sEbene Curven dritter Ordnung. ALGEBRA. The progress of algebra in recent times may be

considered under three principal heads: the study of fundamental laws and the birth of new algebras, the growth of the theory of equations, and the development of what is called modern higher algebra.

A HISTORY OF MATHEMATICS. 368 We have already spoken of George Peacock and D. F. Gregory in connection with the fundamental laws of algebra. Much was done in this line by De Morgan. Augustus De Morgan (1806–1871) was born at Madura (Madras), and educated at Trinity College, Cambridge. His scruples about the doctrines of the established church prevented him from proceeding to the M.A. degree, and from sitting for a fellowship. In 1828 he became professor at the newly established University of London, and taught there until 1867, except for five years, from 1831–1835. De Morgan was a unique, manly character, and pre-eminent as a teacher. The value of his original work lies not so much in increasing our stock of mathematical knowledge as in putting it all upon a thoroughly logical basis. He felt keenly the lack of close reasoning in mathematics as he received it. He said once: “We know that mathematicians care no more for logic than logicians for mathematics. The two eyes of exact science are mathematics and logic: the mathematical sect puts out the logical eye, the logical sect puts out the mathematical eye; each believing that it can see better with one eye than with two.” De Morgan saw with both eyes. He analysed logic mathematically, and studied the logical analysis of the laws, symbols, and operations of mathematics; he wrote a Formal Logic as well as aDouble Algebra, and corresponded both with Sir William Hamilton, the metaphysician, and Sir William Rowan Hamilton, the mathematician. Few contemporaries were as profoundly read in the history of mathematics as was De Morgan. No subject was too insignificant to receive

ALGEBRA. 369 his attention. The authorship of “Cocker’s Arithmetic” and the work of circle-squarers was investigated as minutely as was the history of the invention of the calculus. Numerous articles of his lie scattered in the volumes of the Penny and English Cyclopædias. His Differential Calculus, 1842, is still a standard work, and contains much that is original with the author. For the Encyclopædia Metropolitana he wrote on the calculus of functions (giving principles of symbolic reasoning) and on the theory of probability. Celebrated is his Budget of Paradoxes, 1872. He published memoirs “On the Foundation of Algebra” (Trans. of Cam. Phil. Soc. , 1841, 1842, 1844, and 1847). In Germany symbolical algebra was studied by Martin Ohm, who wrote a System der Mathematik in 1822. The ideas of Peacock and De Morgan recognise the possibility of algebras which differ from ordinary algebra. Such algebras were indeed not slow in forthcoming, but, like non-Euclidean geometry, some of them were slow in finding recognition. This is true of Grassmann’s, Bellavitis’s, and Peirce’s discoveries, but Hamilton’s quaternions met with immediate appreciation in England. These algebras offer a geometrical interpretation of imaginaries. During the times of Descartes, Newton, and Euler, we have seen the negative and the imaginary, $\sqrt{-1}$, accepted as numbers, but the latter was still regarded as an algebraic fiction. The first to give it a geometric picture, analogous to the geometric interpretation of the negative, was H. K¨ uhn, a teacher in Danzig, in a publication of 1750– 1751. He represented a$\sqrt{-1}$ by a line perpendicular to the

A HISTORY OF MATHEMATICS. 370 line a, and equal to a in length, and construed √−1 as the mean proportional between +1 and−1. This same idea was developed further, so as to give a geometric interpretation of a + √ −b, by Jean-Robert Argand (1768–?) of Geneva, in a remarkable Essai (1806). [70] The writings of K¨ uhn and Argand were little noticed, and it remained for Gauss to break down the last opposition to the imaginary. He introducedi as an independent unit co-ordinate to1, and a + ib as a “complex number.” The connection between complex numbers and points on a plane, though artificial, constituted a powerful aid in the further study of symbolic algebra. The mind required a visual representation to aid it. The notion of what we now call vectors was growing upon mathematicians, and the geometric addition of vectors in space was discovered independently by Hamilton, Grassmann, and others, about the same time. William Rowan Hamilton (1805–1865) was born of Scotch parents in Dublin. His early education, carried on at home, was mainly in languages. At the age of thirteen he is said to have been familiar with as many languages as he had lived years. About this time he came across a copy of Newton’s Universal Arithmetic. After reading that, he took up successively analytical geometry, the calculus, Newton’s Principia, Laplace’sM´ ecanique C´ eleste. At the age of eighteen he published a paper correcting a mistake in Laplace’s work. In 1824 he entered Trinity College, Dublin, and in 1827, while he was still an undergraduate, he was appointed to the chair of astronomy. His early papers were on optics. In 1832 he predicted conical refraction, a discovery by aid of mathematics

ALGEBRA. 371 which ranks with the discovery of Neptune by Le Verrier and Adams. Then followed papers on the Principle of Varying Action (1827) and a general method of dynamics (1834–1835). He wrote also on the solution of equations of the fifth degree, the hodograph, fluctuating functions, the numerical solution of differential equations. The capital discovery of Hamilton is his quaternions, in which his study of algebra culminated. In 1835 he published in the Transactions of the Royal Irish Academyhis Theory of Algebraic Couples. He regarded algebra “as being no mere art, nor language, nor primarily a science of quantity, but rather as the science of order of progression.” Time appeared to him as the picture of such a progression. Hence his definition of algebra as “the science of pure time.” It was the subject of years’ meditation for him to determine what he should regard as the product of each pair of a system of perpendicular directed lines. At last, on the 16th of October, 1843, while walking with his wife one evening, along the Royal Canal in Dublin, the discovery of quaternions flashed upon him, and he then engraved with his knife on a stone in Brougham Bridge the fundamental formula i2 = j2 = k2 = ijk =−1. At the general meeting of the Irish Academy, a month later, he made the first communication on quaternions. An account of the discovery was given the following year in the Philosophical Magazine. Hamilton displayed wonderful fertility in their development. His Lectures on Quaternions, delivered in Dublin, were printed in 1852. His Elements of Quaternions appeared in 1866. Quaternions were greatly

A HISTORY OF MATHEMATICS. 372 admired in England from the start, but on the Continent they received less attention. P. G. Tait’sElementary Treatise helped powerfully to spread a knowledge of them in England. Cayley, Clifford, and Tait advanced the subject somewhat by original contributions. But there has been little

progress in recent years, except that made by Sylvester in the solution of quaternion equations, nor has the application of quaternions to physics been as extended as was predicted. The change in notation made in France by Ho¨ uel and by Laisant has been considered in England as a wrong step, but the true cause for the lack of progress is perhaps more deep-seated. There is indeed great doubt as to whether the quaternionic product can claim a necessary and fundamental place in a system of vector analysis. Physicists claim that there is a loss of naturalness in taking the square of a vector to be negative. In order to meet more adequately their wants, J. W. Gibbs of Yale University and A. Macfarlane of the University of Texas, have each suggested an algebra of vectors with a new notation. Each gives a definition of his own for the product of two vectors, but in such a way that the square of a vector is positive. A third system of vector analysis has been used by Oliver Heaviside in his electrical researches. Hermann Grassmann (1809–1877) was born at Stettin, attended a gymnasium at his native place (where his father was teacher of mathematics and physics), and studied theology in Berlin for three years. In 1834 he succeeded Steiner as teacher of mathematics in an industrial school in Berlin, but returned to Stettin in 1836 to assume the duties of teacher

ALGEBRA. 373 of mathematics, the sciences, and of religion in a school there. [71] Up to this time his knowledge of mathematics was pretty much confined to what he had learned from his father, who had written two books on “Raumlehre” and “Gr¨ ossenlehre.” But now he made his acquaintance with the works of Lacroix, Lagrange, and Laplace. He noticed that Laplace’s results could be reached in a shorter way by some new ideas advanced in his father’s books, and he proceeded to elaborate this abridged method, and to apply it in the study of tides. He was thus led to a new geometric analysis. In 1840 he had made considerable progress in its development, but a new book of Schleiermacher drew him again to theology. In 1842 he resumed mathematical research, and becoming thoroughly convinced of the importance of his new analysis, decided to devote himself to it. It now became his ambition to secure a mathematical chair at a university, but in this he never succeeded. In 1844 appeared his great classical work, the Lineale Ausdehnungslehre, which was full of new and strange matter, and so general, abstract, and out of fashion in its mode of exposition, that it could hardly have had less influence on European mathematics during its first twenty years, had it been published in China. Gauss, Grunert, and M¨ obius glanced over it, praised it, but complained of the strange terminology and its “philosophische Allgemeinheit.” Eight years afterwards, Bretschneider of Gotha was said to be the only man who had read it through. An article in Crelle’s Journal, in which Grassmann eclipsed the geometers of that time by constructing, with aid of his method, geometrically

A HISTORY OF MATHEMATICS. 374 any algebraic curve, remained again unnoticed. Need we marvel if Grassmann turned his attention to other subjects,— to Schleiermacher’s philosophy, to politics, to philology? Still, articles by him continued to appear inCrelle’s Journal, and in 1862 came out the second part of hisAusdehnungslehre. It was intended to show better than the first part the broad scope of the Ausdehnungslehre, by considering not only geometric applications, but by treating also of algebraic functions, infinite series, and the differential and integral

calculus. But the second part was no more appreciated than the first. At the age of fifty-three, this wonderful man, with heavy heart, gave up mathematics, and directed his energies to the study of Sanskrit, achieving in philology results which were better appreciated, and which vie in splendour with those in mathematics. Common to the Ausdehnungslehre and to quaternions are geometric addition, the function of two vectors represented in quaternions by Sαβ and V αβ, and the linear vector func- tions. The quaternion is peculiar to Hamilton, while with Grassmann we find in addition to the algebra of vectors a geo- metrical algebra of wide application, and resembling M¨ obius's Barycentrische Calcul, in which the point is the fundamental element. Grassmann developed the idea of the "external product," the "internal product," and the "open product." The last we now call a matrix. His Ausdehnungslehre has very great extension, having no limitation to any particular number of dimensions. Only in recent years has the wonderful richness of his discoveries begun to be appreciated. A second

ALGEBRA. 375 edition of the Ausdehnungslehre of 1844 was printed in 1877. C. S. Peirce gave a representation of Grassmann's system in the logical notation, and E. W. Hyde of the University of Cincinnati wrote the first text-book on Grassmann's calculus in the English language. Discoveries of less value, which in part covered those of Grassmann and Hamilton, were made by Saint-Venant (1797–1886), who described the multiplication of vectors, and the addition of vectors and oriented areas; by Cauchy, whose "clefs alg´ ebriques" were units subject to combinatorial multiplication, and were applied by the author to the theory of elimination in the same way as had been done earlier by Grassmann; by Justus Bellavitis (1803–1880), who published in 1835 and 1837 in the Annali delle Scienze his calculus of æquipollences. Bellavitis, for many years professor at Padua, was a self-taught mathematician of much power, who in his thirty-eighth year laid down a city office in his native place, Bassano, that he might give his time to science. [65] The first impression of Grassmann's ideas is marked in the writings of Hermann Hankel (1839–1873), who published in 1867 his Vorlesungen ¨ uber die Complexen Zahlen. Hankel, then docent in Leipzig, had been in correspondence with Grassmann. The "alternate numbers" of Hankel are subject to his law of combinatorial multiplication. In considering the foundations of algebra Hankel affirms the principle of the permanence of formal laws previously enunciatedincompletely by Peacock. Hankel was a close student of mathematical history, and left behind an unfinished work thereon. Before

A HISTORY OF MATHEMATICS. 376 his death he was professor at T¨ ubingen. HisComplexen Zahlen was at first little read, and we must turn toVictor Schlegel of Hagen as the successful interpreter of Grassmann. Schlegel was at one time a young colleague of Grassmann at the Marienstifts-Gymnasium in Stettin. Encouraged by Clebsch, Schlegel wrote a System der Raumlehre which explained the essential conceptions and operations of the Ausdehnungslehre. Multiple algebra was powerfully advanced by Peirce, whose theory is not geometrical, as are those of Hamilton and Grassmann. Benjamin Peirce (1809–1880) was born at Salem, Mass., and graduated at Harvard College, having as undergraduate carried the study of mathematics far beyond the limits of the college course. [2] When Bowditch was

preparing his translation and commentary of the M´ ecanique C´ eleste, young Peirce helped in reading the proof-sheets. He was made professor at Harvard in 1833, a position which he retained until his death. For some years he was in charge of the Nautical Almanac and superintendent of the United States Coast Survey. He published a series of college text- books on mathematics, an Analytical Mechanics, 1855, and calculated, together with Sears C. Walker of Washington, the orbit of Neptune. Profound are his researches on Linear Associative Algebra. The first of several papers thereon was read at the first meeting of the American Association for the Advancement of Science in 1864. Lithographed copies of a memoir were distributed among friends in 1870, but so small seemed to be the interest taken in this subject that the memoir was not printed until 1881 (Am. Jour. Math. ,

ALGEBRA. 377 Vol. IV., No. 2). Peirce works out the multiplication tables, first of single algebras, then of double algebras, and so on up to sextuple, making in all 162 algebras, which he shows to be possible on the consideration of symbols A, B, etc., which are linear functions of a determinate number of letters or units i, j, k, l, etc., with coefficients which are ordinary analytical magnitudes, real or imaginary,—the letters i, j, etc., being such that every binary combination i2, ij, ji, etc., is equal to a linear function of the letters, but under the restriction of satisfying the associative law. [56] Charles S. Peirce , a son of Benjamin Peirce, and one of the foremost writers on mathematical logic, showed that these algebras were all defective forms of quadrate algebras which he had previously discovered by logical analysis, and for which he had devised a simple notation. Of these quadrate algebras quaternions is a simple example; nonions is another. C. S. Peirce showed that of all linear associative algebras there are only three in which division is unambiguous. These are ordinary single algebra, ordinary double algebra, and quaternions, from which the imaginary scalar is excluded. He showed that his father’s algebras are operational and matricular. Lectures on multiple algebra were delivered by J. J. Sylvester at the Johns Hopkins University, and published in various journals. They treat largely of the algebra of matrices. The theory of matrices was developed as early as 1858 by Cayley in an important memoir which, in the opinion of Sylvester, ushered in the reign of Algebra the Second. Clifford, Sylvester, H. Taber, C. H. Chapman, carried the investigations much further.

A HISTORY OF MATHEMATICS. 378 The originator of matrices is really Hamilton, but his theory, published in his Lectures on Quaternions, is less general than that of Cayley. The latter makes no reference to Hamilton. The theory of determinants [73] was studied by Ho¨ en´ e Wronski in Italy and J. Binet in France; but they were fore-stalled by the great master of this subject, Cauchy. In a paper (Jour. de l’ecole Polyt. , IX., 16) Cauchy developed several general theorems. He introduced the name determinant, a term previously used by Gauss in the functions considered by him. In 1826 Jacobi began using this calculus, and he gave brilliant proof of its power. In 1841 he wrote extended memoirs on determinants inCrelle’s Journal, which rendered the theory easily accessible. In England the study of linear transformations of quantics gave a powerful impulse. Cayley developed skew-determinants and Pfaffians, and introduced the use of determinant brackets, or the familiar pair of upright lines. More recent researches on determinants appertain to special forms.

“Continuants” are due to Sylvester; “alter- nants,” originated by Cauchy, have been developed by Jacobi, N. Trudi, H. N¨ agelbach, and G. Garbieri; “axisymmetric determinants,” first used by Jacobi, have been studied by V. A. Lebesgue, Sylvester, and Hesse; “circulants” are due to E. Catalan of Li` ege, W. Spottiswoode (1825–1883), J. W. L. Glaisher, and R. F. Scott; for “centro-symmetric determi- nants” we are indebted to G. Zehfuss. E. B. Christoffel of Strassburg and G. Frobenius discovered the properties of “Wronskians,” first used by Wronski. V. Nachreiner and S. G¨ unther, both of Munich, pointed out relations between

ALGEBRA. 379 determinants and continued fractions; Scott uses Hankel’s al- ternate numbers in his treatise. Text-books on determinants were written by Spottiswoode (1851), Brioschi (1854), Baltzer (1857), G¨ unther (1875), Dostor (1877), Scott (1880), Muir (1882), Hanus (1886). Modern higher algebra is especially occupied with the theory of linear transformations. Its development is mainly the work of Cayley and Sylvester. Arthur Cayley, born at Richmond, in Surrey, in 1821, was educated at Trinity College, Cambridge. [74] He came out Senior Wrangler in 1842. He then devoted some years to the study and practice of law. On the foundation of the Sadlerian professorship at Cambridge, he accepted the offer of that chair, thus giving up a profession promising wealth for a very modest provision, but which would enable him to give all his time to mathematics. Cayley began his mathematical publications in the Cambridge Mathematical Journal while he was still an undergraduate. Some of his most brilliant discoveries were made during the time of his legal practice. There is hardly any subject in pure mathematics which the genius of Cayley has not enriched, but most important is his creation of a new branch of analysis by his theory of invariants. Germs of the principle of invariants are found in the writings of Lagrange, Gauss, and particularly of Boole, who showed, in 1841, that invariance is a property of discriminants generally, and who applied it to the theory of orthogonal substitution. Cayley set himself the problem to determine a priori what functions of the coefficients of a given equation possess this

A HISTORY OF MATHEMATICS. 380 property of invariance, and found, to begin with, in 1845, that the so-called “hyper-determinants” possessed it. Boole made a number of additional discoveries. Then Sylvester began his papers in the Cambridge and Dublin Mathematical Journal on the Calculus of Forms. After this, discoveries followed in rapid succession. At that time Cayley and Sylvester were both residents of London, and they stimulated each other by frequent oral communications. It has often been difficult to determine how much really belongs to each. James Joseph Sylvester was born in London in 1814, and educated at St. Johns College, Cambridge. He came out Second Wrangler in 1837. His Jewish origin incapacitated him from taking a degree. In 1846 he became a student at the Inner Temple, and was called to the bar in 1850. He became professor of natural philosophy at University College, London; then, successively, professor of mathematics at the University of Virginia, at the Royal Military Academy in Woolwich, at the Johns Hopkins University in Baltimore, and is, since 1883, professor of geometry at Oxford. His first printed paper was on Fresnel’s optic theory, 1837. Then followed his researches on invariants, the theory of equations, theory of partitions, multiple algebra, the theory of numbers, and other subjects mentioned elsewhere. About 1874 he took part in the development of the

geometrical theory of link-work movements, originated by the beautiful discovery of A. Peaucellier, Capitaine du G´ enie ` a Nice (published in Nouvelles Annales , 1864 and 1873), and made the subject of close study by A. B. Kempe. To Sylvester is ascribed

ALGEBRA. 381 the general statement of the theory of contravariants, the discovery of the partial differential equations satisfied by the invariants and covariants of binary quantics, and the subject of mixed concomitants. In the American Journal of Mathematics are memoirs on binary and ternary quantics, elaborated partly with aid of F. Franklin, now professor at the Johns Hopkins University. At Oxford, Sylvester has opened up a new subject, the theory of reciprocants, treating of the functions of a dependent variable y and the functions of its differential coefficients in regard to x, which remain unaltered by the interchange of x and y. This theory is more general than one on differential invariants by Halphen (1878), and has been developed further by J. Hammond of Oxford, McMahon of Woolwich, A. R. Forsyth of Cambridge, and others. Sylvester playfully lays claim to the appellation of the Mathematical Adam, for the many names he has introduced into mathematics. Thus the terms invariant, discriminant, Hessian, Jacobian, are his. The great theory of invariants, developed in England mainly by Cayley and Sylvester, came to be studied earnestly in Germany, France, and Italy. One of the earliest in the field was Siegfried Heinrich Aronhold (1819–1884), who demonstrated the existence of invariants, S and T, of the ternary cubic. Hermite discovered evectants and the theorem of reciprocity named after him. Paul Gordan showed, with the aid of symbolic methods, that the number of distinct forms for a binary quantic is finite. Clebsch proved this to be true for quantics with any number of variables. A very much

A HISTORY OF MATHEMATICS. 382 simpler proof of this was given in 1891, by David Hilbert of K¨ onigsberg. In Italy, F. Brioschi of Milan andFa` a de Bruno (1825–1888) contributed to the theory of invariants, the latter writing a text-book on binary forms, which ranks by the side of Salmon’s treatise and those of Clebsch and Gordan. Among other writers on invariants are E. B. Christoffel, Wilhelm Fiedler, P. A. McMahon, J. W. L. Glaisher of Cambridge, Emory McClintock of New York. McMahon discovered that the theory of semi-invariants is a part of that of symmetric functions. The modern higher algebra has reached out and indissolubly connected itself with several other branches of mathematics—geometry, calculus of variations, mechanics. Clebsch extended the theory of binary forms to ternary, and applied the results to geometry. Clebsch, Klein, Weierstrass, Burckhardt, and Bianchi have used the theory of invariants in hyperelliptic and Abelian functions. In the theory of equations Lagrange, Argand, and Gauss furnished proof to the important theorem that every algebraic equation has a real or a complex root. Abel proved rigorously that the general algebraic equation of the fifth or of higher degrees cannot be solved by radicals (Crelle, I., 1826). A modification of Abel’s proof was given by Wantzel. Before Abel, an Italian physician, Paolo Ruffini (1765–1822), had printed proofs of the insolvability, which were criticised by his countryman Malfatti. Though inconclusive, Ruffini’s papers are remarkable as containing anticipations of Cauchy’s theory of groups. [76] A transcendental solution of the quintic involving elliptic integrals was given by Hermite (Compt.

ALGEBRA. 383 Rend., 1858, 1865, 1866). After Hermite's first publication, Kronecker, in 1858, in a letter to Hermite, gave a second solution in which was obtained a simple resolvent of the sixth degree. Jerrard, in hisMathematical Researches(1832–1835), reduced the quintic to the trinomial form by an extension of the method of Tschirnhausen. This important reduction had been effected as early as 1786 byE. S. Bring, a Swede, and brought out in a publication of the University of Lund. Jerrard, like Tschirnhausen, believed that his method furnished a general algebraic solution of equations of any degree. In 1836 William R. Hamilton made a report on the validity of Jerrard's method, and showed that by his process the quintic could be transformed to any one of the four trinomial forms. Hamilton defined the limits of its applicability to higher equations. Sylvester investigated this question, What is the lowest degree an equation can have in order that it may admit of being deprived of i consecutive terms by aid of equations not higher than ith degree. He carried the investigation as far as i = 8, and was led to a series of numbers which he named "Hamilton's numbers." A transformation of equal importance to Jerrard's is that of Sylvester, who expressed the quintic as the sum of three fifth-powers. The covariants and invariants of higher equations have been studied much in recent years. Abel's proof that higher equations cannot always be solved algebraically led to the inquiry as to what equations of a given degree can be solved by radicals. Such equations are the ones discussed by Gauss in considering the division of the circle.

A HISTORY OF MATHEMATICS. 384 Abel advanced one step further by proving that an irreducible equation can always be solved in radicals, if, of two of its roots, the one can be expressed rationally in terms of the other, provided that the degree of the equation is prime; if it is not prime, then the solution depends upon that of equations of lower degree. Through geometrical considerations, Hesse came upon algebraically solvable equations of the ninth degree, not included in the previous groups. The subject was powerfully advanced in Paris by the youthful Evariste Galois (born, 1811; killed in a duel, 1832), who introduced the notion of a group of substitutions. To him are due also some valuable results in relation to another set of equations, presenting themselves in the theory of elliptic functions, viz. the modular equations. Galois's labours gave birth to the important theory of substitutions, which has been greatly advanced by C. Jordan of Paris, J. A. Serret (1819–1885) of the Sorbonne in Paris, L. Kronecker (1823–1891) of Berlin, Klein of G¨ ottingen, M. N¨ other of Erlangen, C. Hermite of Paris, A. Capelli of Naples, L. Sylow of Friedrichshald, E. Netto of Giessen. Netto's book, the Substitutionstheorie, has been translated into English by F. N. Cole of the University of Michigan, who contributed to the theory. A simple group of 504 substitutions of nine letters, discovered by Cole, has been shown by E. H. Moore of the University of Chicago to belong to a doubly-infinite system of simple groups. The theory of substitutions has important applications in the theory of differential equations. Kronecker published, in 1882, his Grundz¨ uge einer Arithmetischen Theorie der Algebraischen

ANALYSIS. 385 Gr¨ ossen. Since Fourier and Budan, the solution of numerical equa- tions has been advanced by W. G. Horner of Bath, who gave an improved method of approximation (Philosophical Transactions, 1819). Jacques Charles Fran¸ cois Sturm (1803–1855), a native of Geneva, Switzerland, and the suc- cessor of

Poisson in the chair of mechanics at the Sorbonne, published in 1829 his celebrated theorem determining the number and situation of roots of an equation comprised between given limits. Sturm tells us that his theorem stared him in the face in the midst of some mechanical investigations connected with the motion of a compound pendulum. [77] This theorem, and Horner's method, offer together sure and ready means of finding the real roots of a numerical equation. The symmetric functions of the sums of powers of the roots of an equation, studied by Newton and Waring, was con- sidered more recently by Gauss, Cayley, Sylvester, Brioschi. Cayley gives rules for the "weight" and "order" of symmetric functions. ThetheoryofeliminationwasgreatlyadvancedbySylvester, Cayley, Salmon, Jacobi, Hesse, Cauchy, Brioschi, and Gordan. Sylvester gave the dialytic method (Philosophical Magazine, 1840), and in 1852 established a theorem relating to the expression of an eliminant as a determinant. Cayley made a new statement of B´ ezout's method of elimination and established a general theory of elimination (1852).

A HISTORY OF MATHEMATICS. 386 ANALYSIS. Under this head we find it convenient to consider the subjects of the differential and integral calculus, the calculus of variations, infinite series, probability, and differential equations. Prominent in the development of these subjects was Cauchy. Augustin-Louis Cauchy [78] (1789–1857) was born in Paris, and received his early education from his father. Lagrange and Laplace, with whom the father came in frequent contact, foretold the future greatness of the young boy. At the ´Ecole Centrale du Panth´ eon he excelled in ancient classical studies. In 1805 he entered the Polytechnic School, and two years later the ´Ecole des Ponts et Chauss´ ees. Cauchy left for Cherbourg in 1810, in the capacity of engineer. Laplace's M´ ecanique C´ eleste andLagrange's Fonctions Analytiques were among his book companions there. Considerations of health induced him to return to Paris after three years. Yielding to the persuasions of Lagrange and Laplace, he renounced engineering in favour of pure science. We find him next holding a professorship at the Polytechnic School. On the expulsion of Charles X., and the accession to the throne of Louis Philippe in 1830, Cauchy, being exceedingly conscientious, found himself unable to take the oath demanded of him. Being, in consequence, deprived of his positions, he went into voluntary exile. At Fribourg in Switzerland, Cauchy resumed his studies, and in 1831 was induced by the king of Piedmont to accept the chair of mathematical physics,

ANALYSIS. 387 especially created for him at the university of Turin. In 1833 he obeyed the call of his exiled king, Charles X., to undertake the education of a grandson, the Duke of Bordeaux. This gave Cauchy an opportunity to visit various parts of Europe, and to learn how extensively his works were being read. Charles X. bestowed upon him the title of Baron. On his return to Paris in 1838, a chair in the College de France was offered to him, but the oath demanded of him prevented his acceptance. He was nominated member of the Bureau of Longitude, but declared ineligible by the ruling power. During the political events of 1848 the oath was suspended, and Cauchy at last became professor at the Polytechnic School. On the establishment of the second empire, the oath was re-instated, but Cauchy and Arago were exempt from it. Cauchy was a man of great piety, and in two of his publications staunchly defended the Jesuits. Cauchy was a prolific and profound mathematician.

By a prompt publication of his results, and the preparation of stan- dard text-books, he exercised a more immediate and beneficial influence upon the great mass of mathematicians than any con- temporary writer. He was one of the leaders in infusing rigour into analysis. His researches extended over the field of series, of imaginaries, theory of numbers, differential equations, theory of substitutions, theory of functions, determinants, math- ematical astronomy, light, elasticity, etc.,—covering pretty much the whole realm of mathematics, pure and applied. Encouraged by Laplace and Poisson, Cauchy published in 1821 his Cours d'Analyse de l' ´Ecole Royale Polytechnique, a work of great merit. Had it been studied more diligently by

A HISTORY OF MATHEMATICS. 388 writers of text-books in England and the United States, many a lax and loose method of analysis hardly as yet eradicated from elementary text-books would have been discarded over half a century ago. Cauchy was the first to publish a rigorous proof of Taylor's theorem. He greatly improved the exposition of fundamental principles of the differential calculus by his mode of considering limits and his new theory on the continuity of functions. The method of Cauchy and Duhamel was accepted with favour by Ho¨ uel and others. In England special attention to the clear exposition of fundamental principles was given by De Morgan. Recent American treatises on the calculus introduce time as an independent variable, and the allied notions of velocity and acceleration—thus virtually returning to the method of fluxions. Cauchy made some researches on the calculus of variations. This subject is now in its essential principles the same as when it came from the hands of Lagrange. Recent studies pertain to the variation of a double integral when the limits are also variable, and to variations of multiple integrals in general. Memoirs were published by Gauss in 1829, Poisson in 1831, and Ostrogradsky of St. Petersburg in 1834, without, however, determining in a general manner the number and form of the equations which must subsist at the limits in case of a double or triple integral. In 1837 Jacobi published a memoir, showing that the difficult integrations demanded by the discussion of the second variation, by which the existence of a maximum or minimum can be ascertained, are included in the integrations of the first variation, and thus are

ANALYSIS. 389 superfluous. This important theorem, presented with great brevity by Jacobi, was elucidated and extended by V. A. Lebesgue, C. E. Delaunay, Eisenlohr, S. Spitzer, Hesse, and Clebsch. An important memoir by Sarrus on the question of determining the limiting equations which must be combined with the indefinite equations in order to determine completely the maxima and minima of multiple integrals, was awarded a prize by the French Academy in 1845, honourable mention being made of a paper by Delaunay. Sarrus's method was simplified by Cauchy. In 1852 G. Mainardi attempted to exhibit a new method of discriminating maxima and minima, and extended Jacobi's theorem to double integrals. Mainardi and F. Brioschi showed the value of determinants in exhibiting the terms of the second variation. In 1861 Isaac Todhunter (1820–1884) of St. John's College, Cambridge, published his valuable work on theHistory of the Progress of the Calculus of Variations, which contains researches of his own. In 1866 he published a most important research, developing the theory of discontinuous solutions (discussed in particular cases by Legendre), and doing for this subject what Sarrus had done for multiple integrals.

The following are the more important authors of systematic treatises on the calculus of variations, and the dates of publication: Robert Woodhouse, Fellow of Caius College, Cambridge, 1810; Richard Abbatt in London, 1837; John Hewitt Jellett (1817–1888), once Provost of Trinity College, Dublin, 1850; G. W. Strauch in Z¨ urich, 1849; Moigno and Lindel¨ of, 1861; Lewis Buffett Carll of Flushing in New York,

A HISTORY OF MATHEMATICS. 390 1881. The lectures on definite integrals, delivered by Dirichlet in 1858, have been elaborated into a standard work by G. F. Meyer. The subject has been treated most exhaustively by D. Bierens de Haan of Leiden in his Expos´ e de la th´ eorie des int´ egrals d´ efinies, Amsterdam, 1862. The history of infinite series illustrates vividly the salient feature of the new era which analysis entered upon during the first quarter of this century. Newton and Leibniz felt the necessity of inquiring into the convergence of infinite series, but they had no proper criteria, excepting the test advanced by Leibniz for alternating series. By Euler and his contemporaries the formal treatment of series was greatly extended, while the necessity for determining the convergence was generally lost sight of. Euler reached some very pretty results on infinite series, now well known, and also some very absurd results, now quite forgotten. The faults of his time found their culmination in the Combinatorial School in Germany, which has now passed into deserved oblivion. At the beginning of the period now under consideration, the doubtful, or plainly absurd, results obtained from infinite series stimulated profounder inquiries into the validity of operations with them. Their actual contents came to be the primary,form a secondary, consideration. The first important and strictly rigorous investigation of series was made by Gauss in connection with the hypergeometric series. The criterion developed by him settles the question of convergence in every case which it is intended to cover, and thus bears the stamp of

ANALYSIS. 391 generality so characteristic of Gauss’s writings. Owing to the strangeness of treatment and unusual rigour, Gauss’s paper excited little interest among the mathematicians of that time. More fortunate in reaching the public was Cauchy, whose Analyse Alg´ ebriqueof 1821 contains a rigorous treatment of series. All series whose sum does not approach a fixed limit as the number of terms increases indefinitely are called divergent. Like Gauss, he institutes comparisons with geometric series, and finds that series with positive terms are convergent or not, according as the nth root of the nth term, or the ratio of the (n + 1)th term and the nth term, is ultimately less or greater than unity. To reach some of the cases where these expressions become ultimately unity and fail, Cauchy established two other tests. He showed that series with negative terms converge when the absolute values of the terms converge, and then deduces Leibniz’s test for alternating series. The product of two convergent series was not found to be necessarily convergent. Cauchy’s theorem that the product of two absolutely convergent series converges to the product of the sums of the two series was shown half a century later by F. Mertens of Graz to be still true if, of the two convergent series to be multiplied together, only one is absolutely convergent. The most outspoken critic of the old methods in series was Abel. His letter to his friend Holmboe (1826) contains severe criticisms. It is very interesting reading, even to modern students. In his demonstration of the binomial theorem he established the theorem that if two series

and

A HISTORY OF MATHEMATICS. 392 their product series are all convergent, then the product series will converge towards the product of the sums of the two given series. This remarkable result would dispose of the whole problem of multiplication of series if we had a universal practical criterion of convergency for semi-convergent series. Since we do not possess such a criterion, theorems have been recently established by A. Pringsheim of Munich and A. Voss of W¨ urzburg which remove in certain cases the necessity of applying tests of convergency to the product series by the application of tests to easier related expressions. Pringsheim reaches the following interesting conclusions: The product of two semi-convergent series can never converge absolutely, but a semi-convergent series, or even a divergent series, multiplied by an absolutely convergent series, may yield an absolutely convergent product. The researches of Abel and Cauchy caused a considerable stir. We are told that after a scientific meeting in which Cauchy had presented his first researches on series, Laplace hastened home and remained there in seclusion until he had examined the series in his M´ ecanique C´ eleste. Luckily, every one was found to be convergent! We must not conclude, however, that the new ideas at once displaced the old. On the contrary, the new views were generally accepted only after a severe and long struggle. As late as 1844 De Morgan began a paper on “divergent series” in this style: “I believe it will be generally admitted that the heading of this paper describes the only subject yet remaining, of an elementary character, on which a serious schism exists among mathematicians as to

ANALYSIS. 393 the absolute correctness or incorrectness of results.” First in time in the evolution of more delicate criteria of convergence and divergence come the researches of Josef Lud- wig Raabe (Crelle, Vol. IX.); then follow those of De Morgan as given in his calculus. De Morgan established the loga- rithmic criteria which were discovered in part independently by J. Bertrand. The forms of these criteria, as given by Bertrand and by Ossian Bonnet, are more convenient than De Morgan’s. It appears from Abel’s posthumous papers that he had anticipated the above-named writers in establishing logarithmic criteria. It was the opinion of Bonnet that the logarithmic criteria never fail; but Du Bois-Reymond and Pringsheim have each discovered series demonstrably conver- gent in which these criteria fail to determine the convergence. The criteria thus far alluded to have been called by Pringsheim special criteria, because they all depend upon a comparison of the nth term of the series with special functions an, nx, n(log n)x, etc. Among the first to suggest general criteria, and to consider the subject from a still wider point of view, culminating in a regular mathematical theory, was Kummer. He established a theorem yielding a test consisting of two parts, the first part of which was afterwards found to be superfluous. The study of general criteria was continued by U. Dini of Pisa, Paul Du Bois-Reymond, G. Kohn of Minden, and Pringsheim. Du Bois-Reymond divides criteria into two classes: criteria of the first kind and criteria of the second kind, according as the general nth term, or the ratio of the (n + 1)th term and thenth term, is made the basis of research.

A HISTORY OF MATHEMATICS. 394 Kummer’s is a criterion of the second kind. A criterion of the first kind, analogous to this, was invented by Pringsheim. From the general criteria established by Du Bois-Reymond and Pringsheim respectively, all the special criteria can be derived. The theory of Pringsheim is very complete, and offers, in addition to the criteria of the first kind and second kind, entirely new criteria of athird kind, and also generalised criteria of the second kind, which apply, however, only to series with never increasing terms. Those of the third kind rest mainly on the consideration of the limit of the difference either of consecutive terms or of their reciprocals. In the generalised criteria of the second kind he does not consider the ratio of two consecutive terms, but the ratio of any two terms however far apart, and deduces, among others, two criteria previously given by Kohn and Ermakoff respectively. Difficult questions arose in the study of Fourier’s series. [79] Cauchy was the first who felt the necessity of inquiring into its convergence. But his mode of proceeding was found by Dirichlet to be unsatisfactory. Dirichlet made the first thorough researches on this subject (Crelle, Vol. IV.). They culminate in the result that whenever the function does not become infinite, does not have an infinite number of discontinuities, and does not possess an infinite number of maxima and minima, then Fourier’s series converges toward the value of that function at all places, except points of discontinuity, and there it converges toward the mean of the two boundary values. Schl¨ afli of Bern and Du Bois-Reymond expressed doubts as to the correctness of the mean value, which

ANALYSIS. 395 were, however, not well founded. Dirichlet’s conditions are sufficient, but not necessary. Lipschitz, of Bonn, proved that Fourier’s series still represents the function when the number of discontinuities is infinite, and established a condition on which it represents a function having an infinite number of maxima and minima. Dirichlet’s belief that all continuous functions can be represented by Fourier’s series at all points was shared by Riemann and H. Hankel, but was proved to be false by Du Bois-Reymond and H. A. Schwarz. Riemann inquired what properties a function must have, so that there may be a trigonometric series which, whenever it is convergent, converges toward the value of the function. He found necessary and sufficient conditions for this. They do not decide, however, whether such a series actually represents the function or not. Riemann rejected Cauchy’s definition of a definite integral on account of its arbitrariness, gave a new definition, and then inquired when a function has an integral. His researches brought to light the fact that continuous functions need not always have a differential coefficient. But this property, which was shown by Weierstrass to belong to large classes of functions, was not found necessarily to exclude them from being represented by Fourier’s series. Doubts on some of the conclusions about Fourier’s series were thrown by the observation, made by Weierstrass, that the integral of an infinite series can be shown to be equal to the sum of the integrals of the separate terms only when the series converges uniformly within the region in question. The subject of uniform convergence was investigated by Philipp Ludwig

A HISTORY OF MATHEMATICS. 396 Seidel (1848) and G. G. Stokes (1847), and has assumed great importance in Weierstrass’ theory of functions. It became necessary to prove that a trigonometric series representing a continuous function

converges uniformly. This was done by Heinrich Eduard Heine (1821–1881), of Halle. Later researches on Fourier's series were made by G. Cantor and Du Bois-Reymond. As compared with the vast development of other mathe- matical branches, the theory of probability has made very insignificant progress since the time of Laplace. Improvements and simplifications in the mode of exposition have been made by A. De Morgan, G. Boole, A. Meyer (edited by E. Czuber), J. Bertrand. Cournot's and Westergaard's treatment of insur- ance and the theory of life-tables are classical. Applications of the calculus to statistics have been made by L. A. J. Quetelet (1796–1874), director of the observatory at Brussels; by Lexis; Harald Westergaard, of Copenhagen; and D¨ using. Worthy of note is the rejection of inverse probability by the best authorities of our time. This branch of probability had been worked out by Thomas Bayes (died 1761) and by Laplace (Bk. II., Ch. VI. of his Th´ eorie Analytique). By it some logicians have explained induction. For example, if a man, who has never heard of the tides, were to go to the shore of the Atlantic Ocean and witness on m successive days the rise of the sea, then, says Quetelet, he would be entitled to conclude that there was a probability equal to m + 1 m + 2 that the sea would rise next day. Putting m = 0, it is seen that this view rests upon the unwarrantable assumption that the

ANALYSIS. 397 probability of a totally unknown event is 1 2, or that of all theories proposed for investigation one-half are true. W. S. Jevons in his Principles of Science founds induction upon the theory of inverse probability, and F. Y. Edgeworth also accepts it in his Mathematical Psychics. The only noteworthy recent addition to probability is the subject of "local probability," developed by several English and a few American and French mathematicians. The earliest problem on this subject dates back to the time of Buffon, the naturalist, who proposed the problem, solved by himself and Laplace, to determine the probability that a short needle, thrown at random upon a floor ruled with equidistant parallel lines, will fall on one of the lines. Then came Sylvester's four- point problem: to find the probability that four points, taken at random within a given boundary, shall form a re-entrant quadrilateral. Local probability has been studied in England by A. R. Clarke, H. McColl, S. Watson, J. Wolstenholme, but with greatest success byM. W. Crofton of the military school at Woolwich. It was pursued in America by E. B. Seitz; in France by C. Jordan, E. Lemoine, E. Barbier, and others. Through considerations of local probability, Crofton was led to the evaluation of certain definite integrals. Thefirstfullscientifictreatmentofdifferentialequationswas given by Lagrange and Laplace. This remark is especially true of partial differential equations. The latter were investigated in more recent time by Monge, Pfaff, Jacobi, ´Emile Bour (1831–1866) of Paris, A. Weiler, Clebsch, A. N. Korkine of St. Petersburg, G. Boole, A. Meyer, Cauchy, Serret, Sophus

A HISTORY OF MATHEMATICS. 398 Lie, and others. In 1873 their researches, on partial differential equations of the first order, were presented in text-book form by Paul Mansion, of the University of Gand. The keen researches of Johann Friedrich Pfaff (1795–1825) marked a decided advance. He was an intimate friend of young Gauss at G¨ ottingen. Afterwards he was with the astronomer Bode. Later he became professor at Helmst¨ adt, then at Halle. By a peculiar method, Pfaff found the general

integration of partial differential equations of the first order for any number of variables. Starting from the theory of ordinary differential equations of the first order in n variables, he gives first their general integration, and then considers the integration of the partial differential equations as a particular case of the former, assuming, however, as known, the general integration of differential equations of any order between two variables. His researches led Jacobi to introduce the name “Pfaffian problem.” From the connection, observed by Hamilton, between a system of ordinary differential equations (in analytical mechanics) and a partial differential equation, Jacobi drew the conclusion that, of the series of systems whose successive integration Pfaff’s method demanded, all but the first system were entirely superfluous. Clebsch considered Pfaff’s problem from a new point of view, and reduced it to systems of simultaneous linear partial differential equations, which can be established independently of each other without any integration. Jacobi materially advanced the theory of differential equations of the first order. The problem to determine unknown functions in such a way that an integral

ANALYSIS. 399 containing these functions and their differential coefficients, in a prescribed manner, shall reach a maximum or minimum value, demands, in the first place, the vanishing of the first variation of the integral. This condition leads to differential equations, the integration of which determines the functions. To ascertain whether the value is a maximum or a minimum, the second variation must be examined. This leads to new and difficult differential equations, the integration of which, for the simpler cases, was ingeniously deduced by Jacobi from the integration of the differential equations of the first variation. Jacobi’s solution was perfected by Hesse, while Clebsch extended to the general case Jacobi’s results on the second variation. Cauchy gave a method of solving partial differential equations of the first order having any number of variables, which was corrected and extended by Serret, J. Bertrand, O. Bonnet in France, and Imschenetzky in Russia. Fundamental is the proposition of Cauchy that every ordinary differential equation admits in the vicinity of any non-singular point of an integral, which is synectic within a certain circle of convergence, and is developable by Taylor’s theorem. Allied to the point of view indicated by this theorem is that of Riemann, who regards a function of a single variable as defined by the position and nature of its singularities, and who has applied this conception to that linear differential equation of the second order, which is satisfied by the hypergeometric series. This equation was studied also by Gauss and Kummer. Its general theory, when no restriction is imposed upon the value of the variable,

A HISTORY OF MATHEMATICS. 400 has been considered by J. Tannery, of Paris, who employed Fuchs’ method of linear differential equations and found all of Kummer’s twenty-four integrals of this equation. This study has been continued by ´Edouard Goursat of Paris. A standard text-book on Differential Equations, including original matter on integrating factors, singular solutions, and especially on symbolical methods, was prepared in 1859 by George Boole(1815–1864), at one time professor in Queen’s University, Cork, Ireland. He was a native of Lincoln, and a self-educated mathematician of great power. His treatise on Finite Differences (1860) and his Laws of Thought (1854) are works of high merit. The fertility of the

conceptions of Cauchy and Riemann with regard to differential equations is attested by the researches to which they have given rise on the part of Lazarus Fuchs of Berlin (born 1835), Felix Kleinof G¨ ottingen (born 1849), Henri Poincar´ eof Paris (born 1854), and others. The study of linear differential equations entered a new period with the publication of Fuchs' memoirs of 1866 and 1868. Before this, linear equations with constant coefficients were almost the only ones for which general methods of integration were known. While the general theory of these equations has recently been presented in a new light by Hermite, Darboux, and Jordan, Fuchs began the study from the more general standpoint of the linear differential equations whose coefficients are not constant. He directed his attention mainly to those whose integrals are all regular. If the variable be made to describe all possible paths enclosing one or more of the

ANALYSIS. 401 critical points of the equation, we have a certain substitution corresponding to each of the paths; the aggregate of all these substitutions being called a group. The forms of integrals of such equations were examined by Fuchs and by G. Frobenius by independent methods. Logarithms generally appear in the integrals of a group, and Fuchs and Frobenius investigated the conditions under which no logarithms shall appear. Through the study of groups the reducibility or irreducibility of linear differential equations has been examined by Frobenius and Leo K¨ onigsberger. The subject of linear differential equations, not all of whose integrals are regular, has been attacked by G. Frobenius of Berlin, W. Thom´ e of Greifswald (born 1841), and Poincar´ e, but the resulting theory of irregular integrals is as yet in very incomplete form. The theory of invariants associated with linear differential equations has been developed by Halphen and by A. R. Forsyth. The researches above referred to are closely connected with the theory of functions and of groups. Endeavours have thus been made to determine the nature of the function defined by a differential equation from the differential equation itself, and not from any analytical expression of the function, obtained first by solving the differential equation. Instead of studying the properties of the integrals of a differential equation for all the values of the variable, investigators at first contented themselves with the study of the properties in the vicinity of a given point. The nature of the integrals at singular points and at ordinary points is entirely different. Albert Briot (1817–

A HISTORY OF MATHEMATICS. 402 1882) and Jean Claude Bouquet (1819–1885), both of Paris, studied the case when, near a singular point, the differential equations take the form (x− x0) dy dx = ∫ (xy). Fuchs gave the development in series of the integrals for the particular case of linear equations. Poincar´ e did the same for the case when the equations are not linear, as also for partial differential equations of the first order. The developments for ordinary points were given by Cauchy and Madame Kowalevsky. The attempt to express the integrals by developments that are always convergent and not limited to particular points in a plane necessitates the introduction of new transcendents, for the old functions permit the integration of only a small number of differential equations. Poincar´ e tried this plan with linear equations, which were then the best known, having been studied in the vicinity of given points by Fuchs, Thom´ e, Frobenius, Schwarz, Klein, and Halphen. Confining himself to those with rational algebraical coefficients, Poincar´ e was able to

integrate them by the use of functions named by him Fuchsians. [81] He divided these equations into “families.” If the integral of such an equation be subjected to a certain transformation, the result will be the integral of an equation belonging to the same family. The new transcendents have a great analogy to elliptic functions; while the region of the latter may be divided into parallelograms, each representing a group, the former may be divided into curvilinear polygons, so that the knowledge of the function inside of one polygon carries with it the knowledge of it inside the others. Thus Poincar´ e arrives at what he callsFuchsian groups. He found, moreover,

ANALYSIS. 403 that Fuchsian functions can be expressed as the ratio of two transcendents (theta-fuchsians) in the same way that elliptic functions can be. If, instead of linear substitutions with real coefficients, as employed in the above groups, imaginary coefficients be used, then discontinuous groups are obtained, which he called Kleinians. The extension to non-linear equations of the method thus applied to linear equations has been begun by Fuchs and Poincar´ e. We have seen that among the earliest of the several kinds of “groups” are the finite discontinuous groups (groups in the theory of substitution), which since the time of Galois have become the leading concept in the theory of algebraic equations; that since 1876 Felix Klein, H. Poincar´ e, and others have applied the theory of finite and infinite discontinuous groups to the theory of functions and of differential equations. The finite continuous groups were first made the subject of general research in 1873 by Sophus Lie, now of Leipzig, and applied by him to the integration of ordinary linear partial differential equations. Much interest attaches to the determination of those linear differential equations which can be integrated by simpler functions, such as algebraic, elliptic, or Abelian. This has been studied by C. Jordan, P. Appel of Paris (born 1858), and Poincar´ e. The mode of integration above referred to, which makes known the properties of equations from the standpoint of the theory of functions, does not suffice in the application of differential equations to questions of mechanics. If we

A HISTORY OF MATHEMATICS. 404 consider the function as defining a plane curve, then the general form of the curve does not appear from the above mode of investigation. It is, however, often desirable to construct the curves defined by differential equations. Studies having this end in view have been carried on by Briot and Bouquet, and by Poincar´ e. [81] The subject of singular solutions of differential equations has been materially advanced since the time of Boole by G. Darboux and Cayley. The papers prepared by these mathematicians point out a difficulty as yet unsurmounted: whereas a singular solution, from the point of view of the integrated equation, ought to be a phenomenon of universal, or at least of general occurrence, it is, on the other hand, a very special and exceptional phenomenon from the point of view of the differential equation. [89] A geometrical theory of singular solutions resembling the one used by Cayley was previously employed by W. W. Johnson of Annapolis. An advanced Treatise on Linear Differential Equations (1889) was brought out by Thomas Craig of the Johns Hopkins University. He chose the algebraic method of presentation followed by Hermite and Poincar´ e, instead of the geometric method preferred by Klein and Schwarz. A notable work, the Trait´ e d’Analyse, is now being published by ´Emile Picard of Paris, the interest of which is made to centre in the subject of differential equations.

THEORY OF FUNCTIONS. 405 THEORY OF FUNCTIONS. We begin our sketch of the vast progress in the theory of functions by considering the special class called elliptic functions. These were richly developed by Abel and Jacobi. Niels Henrick Abel (1802–1829) was born at Findo¨ e in Norway, and was prepared for the university at the cathedral school in Christiania. He exhibited no interest in mathematics until 1818, when B. Holmboe became lecturer there, and aroused Abel's interest by assigning original problems to the class. Like Jacobi and many other young men who became eminent mathematicians, Abel found the first exercise of his talent in the attempt to solve by algebra the general equation of the fifth degree. In 1821 he entered the University in Christiania. The works of Euler, Lagrange, and Legendre were closely studied by him. The idea of the inversion of elliptic functions dates back to this time. His extraordinary success in mathematical study led to the offer of a stipend by the government, that he might continue his studies in Germany and France. Leaving Norway in 1825, Abel visited the astronomer, Schumacher, in Hamburg, and spent six months in Berlin, where he became intimate with August Leopold Crelle (1780–1855), and met Steiner. Encouraged by Abel and Steiner, Crelle started his journal in 1826. Abel began to put some of his work in shape for print. His proof of the impossibility of solving the general equation of the fifth degree by radicals,—first printed in 1824 in a very concise form, and difficult of apprehension,—was elaborated

A HISTORY OF MATHEMATICS. 406 in greater detail, and published in the first volume. He entered also upon the subject of infinite series (particularly the binomial theorem, of which he gave in Crelle's Journal a rigid general investigation), the study of functions, and of the integral calculus. The obscurities everywhere encountered by him owing to the prevailing loose methods of analysis he endeavoured to clear up. For a short time he left Berlin for Freiberg, where he had fewer interruptions to work, and it was there that he made researches on hyperelliptic and Abelian functions. In July, 1826, Abel left Germany for Paris without having met Gauss! Abel had sent to Gauss his proof of 1824 of the impossibility of solving equations of the fifth degree, to which Gauss never paid any attention. This slight, and a haughtiness of spirit which he associated with Gauss, prevented the genial Abel from going to G¨ ottingen. A similar feeling was entertained by him later against Cauchy. Abel remained ten months in Paris. He met there Dirichlet, Legendre, Cauchy, and others; but was little appreciated. He had already published several important memoirs in Crelle's Journal, but by the French this new periodical was as yet hardly known to exist, and Abel was too modest to speak of his own work. Pecuniary embarrassments induced him to return home after a second short stay in Berlin. At Christiania he for some time gave private lessons, and served as docent. Crelle secured at last an appointment for him at Berlin; but the news of it did not reach Norway until after the death of Abel at Froland. [82] At nearly the same time with Abel, Jacobi published articles

THEORY OF FUNCTIONS. 407 on elliptic functions. Legendre's favourite subject, so long neglected, was at last to be enriched by some extraordinary discoveries. The advantage to be derived by inverting the elliptic integral of the first kind and treating it as a function of its amplitude (now called elliptic function) was recognised by Abel,

and a few months later also by Jacobi. A second fruitful idea, also arrived at independently by both, is the introduction of imaginaries leading to the observation that the new functions simulated at once trigonometric and expo- nential functions. For it was shown that while trigonometric functions had only a real period, and exponential only an imaginary, elliptic functions had both sorts of periods. These two discoveries were the foundations upon which Abel and Jacobi, each in his own way, erected beautiful new structures. Abel developed the curious expressions representing elliptic functions by infinite series or quotients of infinite products. Great as were the achievements of Abel in elliptic functions, they were eclipsed by his researches on what are now called Abelian functions. Abel's theorem on these functions was given by him in several forms, the most general of these being that in his M´ emoire sur une propri´ et´ e g´ en´ erale d'une classe tr` es-´ etendue de fonctions transcendentes(1826). The history of this memoir is interesting. A few months after his arrival in Paris, Abel submitted it to the French Academy. Cauchy and Legendre were appointed to examine it; but said nothing about it until after Abel's death. In a brief statement of the discoveries in question, published by Abel inCrelle's Journal, 1829, reference is made to that memoir. This led Jacobi to

A HISTORY OF MATHEMATICS. 408 inquire of Legendre what had become of it. Legendre says that the manuscript was so badly written as to be illegible, and that Abel was asked to hand in a better copy, which he neglected to do. The memoir remained in Cauchy's hands. It was not published until 1841. By a singular mishap, the manuscript was lost before the proof-sheets were read. In its form, the contents of the memoir belongs to the integral calculus. Abelian integrals depend upon an irrational function y which is connected withx by an algebraic equation $F(x, y) = 0$. Abel's theorem asserts that a sum of such integrals can be expressed by a definite number p of similar integrals, where p depends merely on the properties of the equation $F(x, y) = 0$. It was shown later that p is the deficiency of the curve $F(x, y) = 0$. The addition theorems of elliptic integrals are deducible from Abel's theorem. The hyperelliptic integrals introduced by Abel, and proved by him to possess multiple periodicity, are special cases of Abelian integrals whenever p = or > 3. The reduction of Abelian to elliptic integrals has been studied mainly by Jacobi, Hermite, K¨ onigsberger, Brioschi, Goursat, E. Picard, and O. Bolza of the University of Chicago. Two editions of Abel's works have been published: the first by Holmboe in 1839, and the second by Sylow and Lie in 1881. Abel's theorem was pronounced by Jacobi the greatest discovery of our century on the integral calculus. The aged Legendre, who greatly admired Abel's genius, called it "monumentum aere perennius." During the few years of work allotted to the young Norwegian, he penetrated new fields of

THEORY OF FUNCTIONS. 409 research, the development of which has kept mathematicians busy for over half a century. Some of the discoveries of Abel and Jacobi were anticipated by Gauss. In theDisquisitiones Arithmeticæ he observed that the principles which he used in the division of the circle were applicable to many other functions, besides the circular, and particularly to the transcendents dependent on the integral$\int \frac{dx}{\sqrt{1 - x^4}}$. From this Jacobi [83] concluded that Gauss had thirty years earlier considered the nature and properties of elliptic functions and had discovered their double periodicity. The papers in the collected works of Gauss

confirm this conclusion. Carl Gustav Jacob Jacobi[84] (1804–1851) was born of Jewish parents at Potsdam. Like many other mathematicians he was initiated into mathematics by reading Euler. At the University of Berlin, where he pursued his mathematical studies independently of the lecture courses, he took the degree of Ph.D. in 1825. After giving lectures in Berlin for two years, he was elected extraordinary professor at K¨ onigsberg, and two years later to the ordinary professorship there. After the publication of his Fundamenta Nova he spent some time in travel, meeting Gauss in G¨ ottingen, and Legendre, Fourier, Poisson, in Paris. In 1842 he and his colleague, Bessel, attended the meetings of the British Association, where they made the acquaintance of English mathematicians. His early researches were on Gauss' approximation to the value of definite integrals, partial differential equations, Legendre's coefficients, and cubic residues. He read Legendre's

A HISTORY OF MATHEMATICS. 410 Exercises, which give an account of elliptic integrals. When he returned the book to the library, he was depressed in spirits and said that important books generally excited in him new ideas, but that this time he had not been led to a single original thought. Though slow at first, his ideas flowed all the richer afterwards. Many of his discoveries in elliptic functions were made independently by Abel. Jacobi communicated his first researches to Crelle's Journal. In 1829, at the age of twenty- five, he published hisFundamenta Nova Theoriæ Functionum Ellipticarum, which contains in condensed form the main results in elliptic functions. This work at once secured for him a wide reputation. He then made a closer study of theta-functions and lectured to his pupils on a new theory of elliptic functions based on the theta-functions. He developed a theory of transformation which led him to a multitude of formulæ containing q, a transcendental function of the modulus, defined by the equationq = e−πk′/k. He was also led by it to consider the two new functionsH and Θ, which taken each separately with two different arguments are the four (single) theta-functions designated by theΘ1, Θ2, Θ3, Θ4. [56] In a short but very important memoir of 1832, he shows that for the hyperelliptic integral of any class the direct functions to which Abel's theorem has reference are not functions of a single variable, such as the ellipticsn, cn, dn, but functions of p variables. [56] Thus in the casep = 2, which Jacobi especially considers, it is shown that Abel's theorem has reference to two functions λ(u, v), λ1(u, v), each of two variables, and gives in effect an addition-theorem for the expression of the functions

THEORY OF FUNCTIONS. 411 λ(u + u′, v + v′), λ1(u + u′, v + v′) algebraically in terms of the functions λ(u, v), λ1(u, v), λ(u′, v′), λ1(u′, v′). By the memoirs of Abel and Jacobi it may be considered that the notion of the Abelian function of p variables was established and the addition-theorem for these functions given. Recent studies touching Abelian functions have been made by Weierstrass, E. Picard, Madame Kowalevski, and Poincar´ e. Jacobi's work on differential equations, determinants, dynamics, and the theory of numbers is mentioned elsewhere. In 1842 Jacobi visited Italy for a few months to recuperate his health. At this time the Prussian government gave him a pension, and he moved to Berlin, where the last years of his life were spent. The researches on functions mentioned thus far have been greatly extended. In 1858 Charles Hermite of Paris (born 1822), introduced in place of the variable q of

Jacobi a new variable ω connected with it by the equation q = eiπω, so that ω = ik′/k, and was led to consider the functions φ(ω), ψ(ω), χ(ω). [56] Henry Smith regarded a theta-function with the argument equal to zero, as a function of ω. This he called an omega-function, while the three functions φ(ω), ψ(ω), χ(ω), are his modular functions. Researches on theta-functions with respect to real and imaginary arguments have been made by Meissel of Kiel, J. Thomae of Jena, Alfred Enneper of G¨ ottingen (1830–1885). A general formula for the product of two theta-functions was given in 1854 by H. Schr¨ oter of Breslau (1829–1892). These functions have been studied also by Cauchy, K¨ onigsberger of Heidelberg (born 1837),

A HISTORY OF MATHEMATICS. 412 F. S. Richelot of K¨ onigsberg (1808–1875), Johann Georg Rosenhain of K¨ onigsberg (1816–1887), L. Schl¨ afli of Bern (born 1818). [85] Legendre's method of reducing an elliptic differential to its normal form has called forth many investigations, most important of which are those of Richelot and of Weierstrass of Berlin. The algebraic transformations of elliptic functions involve a relation between the old modulus and the new one which Jacobi expressed by a differential equation of the third order, and also by an algebraic equation, called by him "modular equation." The notion of modular equations was familiar to Abel, but the development of this subject devolved upon later investigators. These equations have become of importance in the theory of algebraic equations, and have been studied by Sohnke, E. Mathieu, L. K¨ onigsberger, E. Betti of Pisa (died 1892), C. Hermite of Paris, Joubert of Angers, Francesco Brioschi of Milan, Schl¨ afli, H. Schr¨ oter, M. Gudermann of Cleve, G¨ utzlaff. Felix Klein of G¨ ottingen has made an extensive study of modular functions, dealing with a type of operations lying between the two extreme types, known as the theory of substitutions and the theory of invariants and covariants. Klein's theory has been presented in book-form by his pupil, Robert Fricke. The bolder features of it were first published in his Ikosaeder, 1884. His researches embrace the theory of modular functions as a specific class of elliptic functions, the statement of a more general problem as based on the doctrine

THEORY OF FUNCTIONS. 413 of groups of operations, and the further development of the subject in connection with a class of Riemann's surfaces. The elliptic functions were expressed by Abel as quotients of doubly infinite products. He did not, however, inquire rigorously into the convergency of the products. In 1845 Cayley studied these products, and found for them a complete theory, based in part upon geometrical interpretation, which he made the basis of the whole theory of elliptic functions. Eisenstein discussed by purely analytical methods the general doubly infinite product, and arrived at results which have been greatly simplified in form by the theory of primary factors, due to Weierstrass. A certain function involving a doubly infinite product has been called by Weierstrass the sigma- function, and is the basis of his beautiful theory of elliptic functions. The first systematic presentation of Weierstrass' theory of elliptic functions was published in 1886 by G. H. Halphen in his Th´ eorie des fonctions elliptiques et des leurs applications. Applications of these functions have been given also by A. G. Greenhill. Generalisations analogous to those of Weierstrass on elliptic functions have been made by Felix Klein on hyperelliptic functions. Standard works on elliptic functions have been published by Briot and Bouquet (1859), by K¨ onigsberger, Cayley, Heinrich Dur` egeof Prague (1821–1893),

and others. Jacobi's work on Abelian and theta-functions was greatly extended by Adolph G¨ opel(1812–1847), professor in a gymnasium near Potsdam, and Johann Georg Rosenhain of K¨ onigsberg (1816–1887). G¨ opel in hisTheoriæ transcen-

A HISTORY OF MATHEMATICS. 414 dentium primi ordinis adumbratio levis (Crelle, 35, 1847) and Rosenhain in several memoirs established each independently, on the analogy of the single theta-functions, the functions of two variables, called double theta-functions, and worked out in connection with them the theory of the Abelian functions of two variables. The theta-relations established by G¨ opel and Rosenhain received for thirty years no further development, notwithstanding the fact that the double theta series came to be of increasing importance in analytical, geometrical, and mechanical problems, and that Hermite and K¨ onigsberger had considered the subject of transformation. Finally, the investigations of C. W. Borchardt of Berlin (1817–1880), treating of the representation of Kummer's surface by G¨ opel's biquadratic relation between four theta-functions of two vari- ables, and researches of H. H. Weber of Marburg, F. Prym of W¨ urzburg, Adolf Krazer, and Martin Krause of Dresden led to broader views. Researches on double theta-functions, made by Cayley, were extended to quadruple theta-functions by Thomas Craig of the Johns Hopkins University. Starting with the integrals of the most general form and considering the inverse functions corresponding to these integrals (the Abelian functions of p variables), Riemann defined the theta-functions of p variables as the sum of a p-tuply infinite series of exponentials, the general term depending on p variables. Riemann shows that the Abelian functions are algebraically connected with theta-functions of the proper arguments, and presents the theory in the broadest form. [56] He rests the theory of the multiple theta-functions

THEORY OF FUNCTIONS. 415 upon the general principles of the theory of functions of a complex variable. Through the researches of A. Brill of T¨ ubingen, M. N¨ other of Erlangen, and Ferdinand Lindemann of Munich, made in connection with Riemann-Roch's theorem and the theory of residuation, there has grown out of the theory of Abelian functions a theory of algebraic functions and point-groups on algebraic curves. Before proceeding to the general theory of functions, we make mention of the "calculus of functions," studied chiefly by C. Babbage, J. F. W. Herschel, and De Morgan, which was not so much a theory of functions as a theory of the solution of functional equations by means of known functions or symbols. The history of the general theory of functions begins with the adoption of new definitions of a function. With the Bernoullis and Leibniz, y was called a function of x, if there existed an equation between these variables which made it possible to calculate y for any given value ofx lying anywhere between$-\infty$ and $+\infty$. The study of Fourier's theory of heat led Dirichlet to a new definition: y is called a function of x, if y possess one or more definite values for each of certain values that x is assumed to take in an interval x0 to x1. In functions thus defined, there need be no analytical connection between y and x, and it becomes necessary to look for possible discontinuities. A great revolution in the ideas of a function was brought about by Cauchy when, in a function as defined by Dirichlet, he gave the variables imaginary values, and when he extended the notion of a definite integral by letting the

A HISTORY OF MATHEMATICS. 416 variable pass from one limit to the other by a succession of imaginary values along arbitrary paths. Cauchy established several fundamental theorems, and gave the first great impulse to the study of the general theory of functions. His researches were continued in France by Puiseux and Liouville. But more profound investigations were made in Germany by Riemann. Georg Friedrich Bernhard Riemann(1826–1866) was born at Breselenz in Hanover. His father wished him to study theology, and he accordingly entered upon philological and theological studies at G¨ ottingen. He attended also some lectures on mathematics. Such was his predilection for this science that he abandoned theology. After studying for a time under Gauss and Stern, he was drawn, in 1847, to Berlin by a galaxy of mathematicians, in which shone Dirichlet, Jacobi, Steiner, and Eisenstein. Returning to G¨ ottingen in 1850, he studied physics under Weber, and obtained the doctorate the following year. The thesis presented on that occasion, Grundlagen f¨ ur eine allgemeine Theorie der Funktionen einer ver¨ anderlichen complexen Gr¨ osse, excited the admiration of Gauss to a very unusual degree, as did also Riemann's trial lecture, Ueber die Hypothesen welche der Geometrie zu Grunde liegen . Riemann's Habilitationsschrift was on the Representation of a Function by means of a Trigonometric Series, in which he advanced materially beyond the position of Dirichlet. Our hearts are drawn to this extraordinarily gifted but shy genius when we read of the timidity and nervousness displayed when he began to lecture at G¨ ottingen, and of his jubilation over the unexpectedly large audience of eight

THEORY OF FUNCTIONS. 417 students at his first lecture on differential equations. Later he lectured on Abelian functions to a class of three only,—Schering, Bjerknes, and Dedekind. Gauss died in 1855, and was succeeded by Dirichlet. On the death of the latter, in 1859, Riemann was made ordinary professor. In 1860 he visited Paris, where he made the acquaintance of French mathematicians. The delicate state of his health induced him to go to Italy three times. He died on his last trip at Selasca, and was buried at Biganzolo. Like all of Riemann's researches, those on functions were profound and far-reaching. He laid the foundation for a general theory of functions of a complex variable. The theory of potential, which up to that time had been used only in mathematical physics, was applied by him in pure mathematics. He accordingly based his theory of functions on the partial differential equation, ∂2u ∂x2 + ∂2u ∂y 2 = Δu = 0, which must hold for the analytical functionw = u+iv of z = x+iy. It had been proved by Dirichlet that (for a plane) there is always one, and only one, function ofx and y, which satisfies Δu = 0, and which, together with its differential quotients of the first two orders, is for all values of x and y within a given area one-valued and continuous, and which has for points on the boundary of the area arbitrarily given values. [86] Riemann called this "Dirichlet's principle," but the same theorem was stated by Green and proved analytically by Sir William Thomson. It follows then that w is uniquely determined for all points within a closed surface, ifu is arbitrarily given for all

A HISTORY OF MATHEMATICS. 418 points on the curve, whilst v is given for one point within the curve. In order to treat the more complicated case wherew has n values for one value ofz, and to observe the conditions about continuity, Riemann invented the celebrated surfaces, known as "Riemann's surfaces," consisting of n

coincident planes or sheets, such that the passage from one sheet to another is made at the branch-points, and that the n sheets form together a multiply-connected surface, which can be dissected by cross-cuts into a singly-connected surface. The n-valued function w becomes thus a one-valued function. Aided by researches of J. L¨ uroth of Freiburg and of Clebsch, W. K. Clifford brought Riemann's surface for algebraic functions to a canonical form, in which only the two last of then leaves are multiply-connected, and then transformed the surface into the surface of a solid withp holes. A. Hurwitz of Z¨ urich discussed the question, how far a Riemann's surface is determinate by the assignment of its number of sheets, its branch-points and branch-lines. [62] Riemann's theory ascertains the criteria which will deter- mine an analytical function by aid of its discontinuities and boundary conditions, and thus defines a function indepen- dently of a mathematical expression. In order to show that two different expressions are identical, it is not necessary to transform one into the other, but it is sufficient to prove the agreement to a far less extent, merely in certain critical points. Riemann's theory, as based on Dirichlet's principle (Thom- son's theorem), is not free from objections. It has become evident that the existence of a derived function is not a con-

THEORY OF FUNCTIONS. 419 sequence of continuity, and that a function may be integrable without being differentiable. It is not known how far the meth- ods of the infinitesimal calculus and the calculus of variations (by which Dirichlet's principle is established) can be applied to an unknown analytical function in its generality. Hence the use of these methods will endow the functions with properties which themselves require proof. Objections of this kind to Riemann's theory have been raised by Kronecker, Weierstrass, and others, and it has become doubtful whether his most im- portant theorems are actually proved. In consequence of this, attempts have been made to graft Riemann's speculations on the more strongly rooted methods of Weierstrass. The latter developed a theory of functions by starting, not with the theory of potential, but with analytical expressions and operations. Both applied their theories to Abelian functions, but there Riemann's work is more general. [86] The theory of functions of one complex variable has been studied since Riemann's time mainly by Karl Weierstrass of Berlin (born 1815), Gustaf Mittag-Leffler of Stockholm (born 1846), and Poincar´ e of Paris. Of the three classes of such functions (viz. functions uniform throughout, functions uniform only in lacunary spaces, and non-uniform functions) Weierstrass showed that those functions of the first class which can be developed according to ascending powers of x into converging series, can be decomposed into a product of an infinite number of primary factors. A primary factor of the species n is the product (1− x a) eP (x), P (x) being an entire polynomial of the nth degree. A function of the species n is

A HISTORY OF MATHEMATICS. 420 one, all the primary factors of which are of species n. This classification gave rise to many interesting problems studied also by Poincar´ e. The first of the three classes of functions of a complex variable embraces, among others, functions having an infinite number of singular points, but no singular lines, and at the same time no isolated singular points. These are Fuchsian functions, existing throughout the whole extent. Poincar´ e first gave an example of such a function. Uniform functions of two variables, unaltered by certain linear

substitutions, called hyperfuchsian functions, have been studied by E. Picard of Paris, and by Poincar´ e. [81] Functions of the second class, uniform only in lacunary spaces, were first pointed out by Weierstrass. The Fuchsian and the Kleinian functions do not generally exist, except in the interior of a circle or of a domain otherwise bounded, and are therefore examples of functions of the second class. Poincar´ e has shown how to generate functions of this class, and has studied them along the lines marked out by Weierstrass. Important is his proof that there is no way of generalising them so as to get rid of the lacunæ. Non-uniform functions are much less developed than the preceding classes, even though their properties in the vicinity of a given point have been diligently studied, and though much light has been thrown on them by the use of Riemann's surfaces. With the view of reducing their study to that of uniform transcendents, Poincar´ e proved that if y is any

THEORY OF FUNCTIONS. 421 analytical non-uniform function of x, one can always find a variable z, such that x and y are uniform functions of z. Weierstrass and Darboux have each given examples of continuous functions having no derivatives. Formerly it had been generally assumed that every function had a derivative. Amp` ere was the first who attempted to prove analytically (1806) the existence of a derivative, but the demonstration is not valid. In treating of discontinuous functions, Darboux established rigorously the necessary and sufficient condition that a continuous or discontinuous function be susceptible of integration. He gave fresh evidence of the care that must be exercised in the use of series by giving an example of a series always convergent and continuous, such that the series formed by the integrals of the terms is always convergent, and yet does not represent the integral of the first series. [87] The general theory of functions of two variables has been investigated to some extent by Weierstrass and Poincar´ e. H. A. Schwarz of Berlin (born 1845), a pupil of Weier- strass, has given the conform representation (Abbildung) of various surfaces on a circle. In transforming by aid of certain substitutions a polygon bounded by circular arcs into another also bounded by circular arcs, he was led to a remarkable differential equation $\psi(u', t) = \psi(u, t)$, where $\psi(u, t)$ is the ex- pression which Cayley calls the "Schwarzian derivative," and which led Sylvester to the theory of reciprocants. Schwarz's developments on minimum surfaces, his work on hyperge- ometric series, his inquiries on the existence of solutions to important partial differential equations under prescribed

A HISTORY OF MATHEMATICS. 422 conditions, have secured a prominent place in mathematical literature. The modern theory of functions of one real variable was first worked out by H. Hankel, Dedekind, G. Cantor, Dini, and Heine, and then carried further, principally, by Weierstrass, Schwarz, Du Bois-Reymond, Thomae, and Darboux. Hankel established the principle of the condensation of singularities; Dedekind and Cantor gave definitions for irrational numbers; definite integrals were studied by Thomae, Du Bois-Reymond, and Darboux along the lines indicated by the definitions of such integrals given by Cauchy, Dirichlet, and Riemann. Dini wrote a text-book on functions of a real variable (1878), which was translated into German, with additions, by J. L¨ uroth and A. Schepp. Important works on the theory of functions are the Cours de M. Hermite , Tannery's Th´ eorie des Fonctions d'une variable seule, A Treatise on the Theory of Functionsby James Harkness and Frank

Morley, and Theory of Functions of a Complex Variable by A. R. Forsyth. THEORY OF NUMBERS. “Mathematics, the queen of the sciences, and arithmetic, the queen of mathematics.” Such was the dictum of Gauss, who was destined to revolutionise the theory of numbers. When asked who was the greatest mathematician in Germany, Laplace answered, Pfaff. When the questioner said he should have thought Gauss was, Laplace replied, “Pfaff is by far the greatest mathematician in Germany; but Gauss is the greatest

THEORY OF NUMBERS. 423 in all Europe.” [83] Gauss is one of the three greatest masters of modern analysis,—Lagrange, Laplace, Gauss. Of these three contemporaries he was the youngest. While the first two belong to the period in mathematical history preceding the one now under consideration, Gauss is the one whose writings may truly be said to mark the beginning of our own epoch. In him that abundant fertility of invention, displayed by mathematicians of the preceding period, is combined with an absolute rigorousness in demonstration which is too often wanting in their writings, and which the ancient Greeks might have envied. Unlike Laplace, Gauss strove in his writings after perfection of form. He rivals Lagrange in elegance, and surpasses this great Frenchman in rigour. Wonderful was his richness of ideas; one thought followed another so quickly that he had hardly time to write down even the most meagre outline. At the age of twenty Gauss had overturned old theories and old methods in all branches of higher mathematics; but little pains did he take to publish his results, and thereby to establish his priority. He was the first to observe rigour in the treatment of infinite series, the first to fully recognise and emphasise the importance, and to make systematic use of determinants and of imaginaries, the first to arrive at the method of least squares, the first to observe the double periodicity of elliptic functions. He invented the heliotrope and, together with Weber, the bifilar magnetometer and the declination instrument. He reconstructed the whole of magnetic science. Carl Friedrich Gauss [47] (1777–1855), the son of a

A HISTORY OF MATHEMATICS. 424 bricklayer, was born at Brunswick. He used to say, jokingly, that he could reckon before he could talk. The marvellous aptitude for calculation of the young boy attracted the attention of Bartels, afterwards professor of mathematics at Dorpat, who brought him under the notice of Charles William, Duke of Brunswick. The duke undertook to educate the boy, and sent him to the Collegium Carolinum. His progress in languages there was quite equal to that in mathematics. In 1795 he went to G¨ ottingen, as yet undecided whether to pursue philology or mathematics. Abraham Gotthelf K¨ astner, then professor of mathematics there, and now chiefly remembered for hisGeschichte der Mathematik(1796), was not an inspiring teacher. At the age of nineteen Gauss discovered a method of inscribing in a circle a regular polygon of seventeen sides, and this success encouraged him to pursue mathematics. He worked quite independently of his teachers, and while a student at G¨ ottingen made several of his greatest discoveries. Higher arithmetic was his favourite study. Among his small circle of intimate friends was Wolfgang Bolyai. After completing his course he returned to Brunswick. In 1798 and 1799 he repaired to the university at Helmst¨ adt to consult the library, and there made the acquaintance of Pfaff, a mathematician of much power. In 1807 the Emperor of Russia offered Gauss a chair in the Academy at

St. Petersburg, but by the advice of the astronomer Olbers, who desired to secure him as director of a proposed new observatory at G¨ ottingen, he declined the offer, and accepted the place at G¨ ottingen. Gauss had a marked objection to a mathematical chair, and preferred the

THEORY OF NUMBERS. 425 post of astronomer, that he might give all his time to science. He spent his life in G¨ ottingen in the midst of continuous work. In 1828 he went to Berlin to attend a meeting of scientists, but after this he never again left G¨ ottingen, except in 1854, when a railroad was opened between G¨ ottingen and Hanover. He had a strong will, and his character showed a curious mixture of self-conscious dignity and child-like simplicity. He was little communicative, and at times morose. A new epoch in the theory of numbers dates from the publication of his Disquisitiones Arithmeticæ, Leipzig, 1801. The beginning of this work dates back as far as 1795. Some of its results had been previously given by Lagrange and Euler, but were reached independently by Gauss, who had gone deeply into the subject before he became acquainted with the writings of his great predecessors. The Disquisitiones Arithmeticæ was already in print when Legendre's Th´ eorie des Nombresappeared. The great law of quadratic reciprocity, given in the fourth section of Gauss' work, a law which involves the whole theory of quadratic residues, was discovered by him by induction before he was eighteen, and was proved by him one year later. Afterwards he learned that Euler had imperfectly enunciated that theorem, and that Legendre had attempted to prove it, but met with apparently insuperable difficulties. In the fifth section Gauss gave a second proof of this "gem" of higher arithmetic. In 1808 followed a third and fourth demonstration; in 1817, a fifth and sixth. No wonder that he felt a personal attachment to this theorem. Proofs were given also by Jacobi, Eisenstein, Liouville, Lebesgue, A. Genocchi,

A HISTORY OF MATHEMATICS. 426 Kummer, M. A. Stern, Chr. Zeller, Kronecker, Bouniakowsky, E. Schering, J. Petersen, Voigt, E. Busche, and Th. Pepin. [48] The solution of the problem of the representation of numbers by binary quadratic forms is one of the great achievements of Gauss. He created a new algorithm by introducing the theory of congruences. The fourth section of the Disquisitiones Arithmeticæ, treating of congruences of the second degree, and the fifth section, treating of quadratic forms, were, until the time of Jacobi, passed over with universal neglect, but they have since been the starting-point of a long series of important researches. The seventh or last section, developing the theory of the division of the circle, was received from the start with deserved enthusiasm, and has since been repeatedly elaborated for students. A standard work on Kreistheilung was published in 1872 by Paul Bachmann, then of Breslau. Gauss had planned an eighth section, which was omitted to lessen the expense of publication. His papers on the theory of numbers were not all included in his great treatise. Some of them were published for the first time after his death in his collected works (1863–1871). He wrote two memoirs on the theory of biquadratic residues (1825 and 1831), the second of which contains a theorem of biquadratic reciprocity. Gauss was led to astronomy by the discovery of the planet Ceres at Palermo in 1801. His determination of the elements of its orbit with sufficient accuracy to enable Olbers to re-discover it, made the name of Gauss generally known. In 1809 he published the Theoria motus corporum coelestium,

which contains a discussion of the problems

THEORY OF NUMBERS. 427 arising in the determination of the movements of planets and comets from observations made on them under any circumstances. In it are found four formulæ in spherical trigonometry, now usually called “Gauss’ Analogies,” but which were published somewhat earlier by Karl Brandon Mollweide of Leipzig (1774–1825), and earlier still by Jean Baptiste Joseph Delambre (1749–1822). [44] Many years of hard work were spent in the astronomical and magnetic observatory. He founded the German Magnetic Union, with the object of securing continuous observations at fixed times. He took part in geodetic observations, and in 1843 and 1846 wrote two memoirs,Ueber Gegenst¨ ande der h¨ oheren Geodesie. He wrote on the attraction of homogeneous ellipsoids, 1813. In a memoir on capillary attraction, 1833, he solves a problem in the calculus of variations involving the variation of a certain double integral, the limits of integration being also variable; it is the earliest example of the solution of such a problem. He discussed the problem of rays of light passing through a system of lenses. Among Gauss’ pupils were Christian Heinrich Schumacher, Christian Gerling, Friedrich Nicolai, August Ferdinand M¨ obius, Georg Wilhelm Struve, Johann Frantz Encke. Gauss’ researches on the theory of numbers were the starting-point for a school of writers, among the earliest of whom was Jacobi. The latter contributed to Crelle’s Journal an article on cubic residues, giving theorems without proofs. After the publication of Gauss’ paper on biquadratic residues, giving the law of biquadratic reciprocity, and his treatment

A HISTORY OF MATHEMATICS. 428 of complex numbers, Jacobi found a similar law for cubic residues. By the theory of elliptical functions, he was led to beautiful theorems on the representation of numbers by 2, 4, 6, and 8 squares. Next come the researches of Dirichlet, the expounder of Gauss, and a contributor of rich results of his own. Peter Gustav Lejeune Dirichlet [88] (1805–1859) was born in D¨ uren, attended the gymnasium in Bonn, and then the Jesuit gymnasium in Cologne. In 1822 he was attracted to Paris by the names of Laplace, Legendre, Fourier, Poisson, Cauchy. The facilities for a mathematical education there were far better than in Germany, where Gauss was the only great figure. He read in Paris Gauss’ Disquisitiones Arithmeticæ, a work which he never ceased to admire and study. Much in it was simplified by Dirichlet, and thereby placed within easier reach of mathematicians. His first memoir on the impossibility of certain indeterminate equations of the fifth degree was presented to the French Academy in 1825. He showed that Fermat’s equation, xn + yn = zn, cannot exist when n = 5. Some parts of the analysis are, however, Legendre’s. Euler and Lagrange had proved this when n is 3 and 4, and Lam´ e proved it when n = 7. Dirichlet’s acquaintance with Fourier led him to investigate Fourier’s series. He became docent in Breslau in 1827. In 1828 he accepted a position in Berlin, and finally succeeded Gauss at G¨ ottingen in 1855. The general principles on which depends the average number of classes of binary quadratic forms of positive and negative determinant (a subject first investigated

THEORY OF NUMBERS. 429 by Gauss) were given by Dirichlet in a memoir, Ueber die Bestimmung der mittleren Werthe in der Zahlentheorie , 1849. More recently F. Mertens of Graz has determined the asymptotic values of several numerical

functions. Dirichlet gave some attention to prime numbers. Gauss and Legendre had given expressions denoting approximately the asymptotic value of the number of primes inferior to a given limit, but it remained for Riemann in his memoir, Ueber die Anzahl der Primzahlen unter einer gegebenen Gr¨ osse, 1859, to give an investigation of the asymptotic frequency of primes which is rigorous. Approaching the problem from a different direction, Patnutij Tchebycheff , formerly professor in the University of St. Petersburg (born 1821), established, in a celebrated memoir, Sur les Nombres Premiers , 1850, the existence of limits within which the sum of the logarithms of the primesP, inferior to a given number x, must be comprised. [89] This paper depends on very elementary considerations, and, in that respect, contrasts strongly with Riemann's, which involves abstruse theorems of the integral calculus. Poincar´ e's papers, Sylvester's contraction of Tchebycheff's limits, with reference to the distribution of primes, and researches of J. Hadamard (awarded the Grand prix of 1892), are among the latest researches in this line. The enumeration of prime numbers has been undertaken at different times by various mathematicians. In 1877 the British Association began the preparation of factor-tables, under the direction of J. W. L. Glaisher. The printing, by the Association, of tables for the sixth million marked the completion of tables, to the preparation of which

A HISTORY OF MATHEMATICS. 430 Germany, France, and England contributed, and which enable us to resolve into prime factors every composite number less than 9, 000, 000. Miscellaneous contributions to the theory of numbers were made by Cauchy. He showed, for instance, how to find all the infinite solutions of a homogeneous indeterminate equation of the second degree in three variables when one solution is given. He established the theorem that if two congruences, which have the same modulus, admit of a common solution, the modulus is a divisor of their resultant.Joseph Liouville (1809–1882), professor at the Coll` ege de France, investigated mainly questions on the theory of quadratic forms of two, and of a greater number of variables. Profound researches were instituted byFerdinand Gotthold Eisenstein(1823– 1852), of Berlin. Ternary quadratic forms had been studied somewhat by Gauss, but the extension from two to three indeterminates was the work of Eisenstein who, in his memoir, Neue Theoreme der h¨ oheren Arithmetik, defined the ordinal and generic characters of ternary quadratic forms of uneven determinant; and, in case of definite forms, assigned the weight of any order or genus. But he did not publish demonstrations of his results. In inspecting the theory of binary cubic forms, he was led to the discovery of the first covariant ever considered in analysis. He showed that the series of theorems, relating to the presentation of numbers by sums of squares, ceases when the number of squares surpasses eight. Many of the proofs omitted by Eisenstein were supplied by Henry Smith, who was one of the few Englishmen who devoted themselves to the

THEORY OF NUMBERS. 431 study of higher arithmetic. Henry John Stephen Smith [90] (1826–1883) was born in London, and educated at Rugby and at Balliol College, Oxford. Before 1847 he travelled much in Europe for his health, and at one time attended lectures of Arago in Paris, but after that year he was never absent from Oxford for a single term. In 1861 he was elected Savilian professor of geometry. His first paper on the theory of numbers appeared in 1855. The results of ten years' study

of everything published on the theory of numbers are contained in his Reports which appeared in the British Association volumes from 1859 to 1865. These reports are a model of clear and precise exposition and perfection of form. They contain much original matter, but the chief results of his own discoveries were printed in the Philosophical Transactions for 1861 and 1867. They treat of linear indeterminate equations and congruences, and of the orders and genera of ternary quadratic forms. He established the principles on which the extension to the general case of n indeterminates of quadratic forms depends. He contributed also two memoirs to the Proceedings of the Royal Society of 1864 and 1868, in the second of which he remarks that the theorems of Jacobi, Eisenstein, and Liouville, relating to the representation of numbers by4, 6, 8 squares, and other simple quadratic forms are deducible by a uniform method from the principles indicated in his paper. Theorems relating to the case of 5 squares were given by Eisenstein, but Smith completed the enunciation of them, and added the corresponding theorems for 7 squares. The solution of the cases of 2, 4, 6 squares

A HISTORY OF MATHEMATICS. 432 may be obtained by elliptic functions, but when the number of squares is odd, it involves processes peculiar to the theory of numbers. This class of theorems is limited to 8 squares, and Smith completed the group. In ignorance of Smith's investigations, the French Academy offered a prize for the demonstration and completion of Eisenstein's theorems for 5 squares. This Smith had accomplished fifteen years earlier. He sent in a dissertation in 1882, and next year, a month after his death, the prize was awarded to him, another prize being also awarded to H. Minkowsky of Bonn. The theory of numbers led Smith to the study of elliptic functions. He wrote also on modern geometry. His successor at Oxford was J. J. Sylvester. Ernst Eduard Kummer (1810–1893), professor in the University of Berlin, is closely identified with the theory of numbers. Dirichlet's work on complex numbers of the form a + ib, introduced by Gauss, was extended by him, by Eisenstein, and Dedekind. Instead of the equation x4− 1 = 0, the roots of which yield Gauss' units, Eisenstein used the equation x3−1 = 0 and complex numbersa+bρ (ρ being a cube root of unity), the theory of which resembles that of Gauss' numbers. Kummer passed to the general case xn− 1 = 0 and got complex numbers of the formα = a1A1+a2A2+a3A3+⋯ , where ai are whole real numbers, and Ai roots of the above equation. [59] Euclid's theory of the greatest common divisor is not applicable to such complex numbers, and their prime factors cannot be defined in the same way as prime factors of common integers are defined. In the effort to overcome

THEORY OF NUMBERS. 433 this difficulty, Kummer was led to introduce the conception of "ideal numbers." These ideal numbers have been applied by G. Zolotareff of St. Petersburg to the solution of a problem of the integral calculus, left unfinished by Abel (Liouville's Journal , Second Series, 1864, Vol. IX.). Julius Wilhelm Richard Dedekindof Braunschweig (born 1831) has given in the second edition of Dirichlet'sVorlesungen ¨ uber Zahlentheorie a new theory of complex numbers, in which he to some extent deviates from the course of Kummer, and avoids the use of ideal numbers. Dedekind has taken the roots of any irreducible equation with integral coefficients as the units for his complex numbers. Attracted by Kummer's investigations, his pupil, Leopold Kronecker (1823–1891) made researches which he

applied to algebraic equations. On the other hand, efforts have been made to utilise in the theory of numbers the results of the modern higher algebra. Following up researches of Hermite, Paul Bachmann of M¨ unster investigated the arithmetical formula which gives the automorphics of a ternary quadratic form. [89] The problem of the equivalence of two positive or definite ternary quadratic forms was solved by L. Seeber; and that of the arithmetical automorphics of such forms, by Eisenstein. The more difficult problem of the equivalence for indefinite ternary forms has been investigated by Edward Selling of W¨ urzburg. On quadratic forms of four or more indeterminates little has yet been done. Hermite showed that the number of non-equivalent classes of quadratic forms having integral coefficients and a given discriminant is finite, while Zolotareff

A HISTORY OF MATHEMATICS. 434 and A. N. Korkine, both of St. Petersburg, investigated the minima of positive quadratic forms. In connection with binary quadratic forms, Smith established the theorem that if the joint invariant of two properly primitive forms vanishes, the determinant of either of them is represented primitively by the duplicate of the other. The interchange of theorems between arithmetic and alge- bra is displayed in the recent researches of J. W. L. Glaisher of Trinity College (born 1848) and Sylvester. Sylvester gave a Constructive Theory of Partitions, which received additions from his pupils, F. Franklin and G. S. Ely. The conception of “number” has been much extended in our time. With the Greeks it included only the ordinary positive whole numbers; Diophantus added rational fractions to the domain of numbers. Later negative numbers and imaginaries came gradually to be recognised. Descartes fully grasped the notion of the negative; Gauss, that of the imaginary. With Euclid, a ratio, whether rational or irrational, was not a number. The recognition of ratios and irrationals as numbers took place in the sixteenth century, and found expression with Newton. By the ratio method, the continuity of the real number system has been based on the continuity of space, but in recent time three theories of irrationals have been advanced by Weierstrass, J. W. R. Dedekind, G. Cantor, and Heine, which prove the continuity of numbers without borrowing it from space. They are based on the definition of numbers by regular sequences, the use of series and limits, and some new mathematical conceptions.

APPLIED MATHEMATICS. 435 APPLIED MATHEMATICS. Notwithstanding the beautiful developments of celestial mechanics reached by Laplace at the close of the eighteenth century, there was made a discovery on the first day of the present century which presented a problem seemingly beyond the power of that analysis. We refer to the discovery of Ceres by Piazzi in Italy, which became known in Germany just after the philosopher Hegel had published a dissertation proving a priori that such a discovery could not be made. From the positions of the planet observed by Piazzi its orbit could not be satisfactorily calculated by the old methods, and it remained for the genius of Gauss to devise a method of calculating elliptic orbits which was free from the assumption of a small eccentricity and inclination. Gauss’ method was developed further in his Theoria Motus . The new planet was re-discovered with aid of Gauss’ data by Olbers, an astronomer who promoted science not only by his own astronomical studies, but also by discerning and directing towards astronomical pursuits the genius of Bessel. Friedrich Wilhelm Bessel[91]

(1784–1846) was a native of Minden in Westphalia. Fondness for figures, and a distaste for Latin grammar led him to the choice of a mercantile career. In his fifteenth year he became an apprenticed clerk in Bremen, and for nearly seven years he devoted his days to mastering the details of his business, and part of his nights to study. Hoping some day to become a supercargo on trading expeditions, he became interested in observations at

A HISTORY OF MATHEMATICS. 436 sea. With a sextant constructed by him and an ordinary clock he determined the latitude of Bremen. His success in this inspired him for astronomical study. One work after another was mastered by him, unaided, during the hours snatched from sleep. From old observations he calculated the orbit of Halley's comet. Bessel introduced himself to Olbers, and submitted to him the calculation, which Olbers immediately sent for publication. Encouraged by Olbers, Bessel turned his back to the prospect of affluence, chose poverty and the stars, and became assistant in J. H. Schr¨ oter's observatory at Lilienthal. Four years later he was chosen to superintend the construction of the new observatory at K¨ onigsberg. [92] In the absence of an adequate mathematical teaching force, Bessel was obliged to lecture on mathematics to prepare students for astronomy. He was relieved of this work in 1825 by the arrival of Jacobi. We shall not recount the labours by which Bessel earned the title of founder of modern practical astronomy and geodesy. As an observer he towered far above Gauss, but as a mathematician he reverently bowed before the genius of his great contemporary. Of Bessel's papers, the one of greatest mathematical interest is an " Untersuchung des Theils der planetarischen St¨ orungen, welcher aus der Bewegung der Sonne ensteht" (1824), in which he introduces a class of transcendental functions, Jn(x), much used in applied mathematics, and known as "Bessel's functions." He gave their principal properties, and constructed tables for their evaluation. Recently it has been observed that Bessel's functions appear much earlier in mathematical literature. [98]

APPLIED MATHEMATICS. 437 Such functions of the zero order occur in papers of Daniel Bernoulli (1732) and Euler on vibration of heavy strings suspended from one end. All of Bessel's functions of the first kind and of integral orders occur in a paper by Euler (1764) on the vibration of a stretched elastic membrane. In 1878 Lord Rayleigh proved that Bessel's functions are merely particular cases of Laplace's functions. J. W. L. Glaisher illustrates by Bessel's functions his assertion that mathematical branches growing out of physical inquiries as a rule "lack the easy flow or homogeneity of form which is characteristic of a mathematical theory properly so called." These functions have been studied by C. Th. Anger of Danzig, O. Schl¨ omilch of Dresden, R. Lipschitz of Bonn (born 1832), Carl Neumann of Leipzig (born 1832), Eugen Lommel of Leipzig, I. Todhunter of St. John's College, Cambridge. Prominent among the successors of Laplace are the follow- ing: Sim´ eon Denis Poisson(1781–1840), who wrote in 1808 a classic M´ emoire sur les in´ egalit´ es s´ eculaires des moyens mouvements des plan` etes. Giovanni Antonio Amadeo Plana (1781–1864) of Turin, a nephew of Lagrange, who published in 1811 a Memoria sulla teoria dell' attrazione degli sferoidi ellitici , and contributed to the theory of the moon. Peter Andreas Hansen (1795–1874) of Gotha, at one time a clockmaker in Tondern, then Schumacher's assistant at Al- tona, and finally director of the

observatory at Gotha, wrote on various astronomical subjects, but mainly on the lunar theory, which he elaborated in his work Fundamenta nova investigationes orbitæ veræ quam Luna perlustrat(1838), and

A HISTORY OF MATHEMATICS. 438 in subsequent investigations embracing extensive lunar ta- bles. George Biddel Airy(1801–1892), royal astronomer at Greenwich, published in 1826 his Mathematical Tracts on the Lunar and Planetary Theories . These researches have since been greatly extended by him. August Ferdinand M¨ obius (1790–1868) of Leipzig wrote, in 1842,Elemente der Mechanik des Himmels. Urbain Jean Joseph Le Verrier(1811–1877) of Paris wrote the Recherches Astronomiques, constituting in part a new elaboration of celestial mechanics, and is fa- mous for his theoretical discovery of Neptune. John Couch Adams (1819–1892) of Cambridge divided with Le Verrier the honour of the mathematical discovery of Neptune, and pointed out in 1853 that Laplace's explanation of the secular acceleration of the moon's mean motion accounted for only half the observed acceleration. Charles Eug` ene Delaunay (born 1816, and drowned off Cherbourg in 1872), professor of mechanics at the Sorbonne in Paris, explained most of the remaining acceleration of the moon, unaccounted for by Laplace's theory as corrected by Adams, by tracing the effect of tidal friction, a theory previously suggested independently by Kant, Robert Mayer, and William Ferrel of Kentucky. George Howard Darwin of Cambridge (born 1845) made some very remarkable investigations in 1879 on tidal friction, which trace with great certainty the history of the moon from its origin. He has since studied also the effects of tidal friction upon other bodies in the solar system. Criticisms on some parts of his researches have been made by James Nolan of Victoria. Simon Newcomb (born 1835), superintendent

APPLIED MATHEMATICS. 439 of the Nautical Almanac at Washington, and professor of mathematics at the Johns Hopkins University, investigated the errors in Hansen's tables of the moon. For the last twelve years the main work of the U. S. Nautical Almanac office has been to collect and discuss data for new tables of the planets which will supplant the tables of Le Verrier.G. W. Hill of that office has contributed an elegant paper on certain possible abbreviations in the computation of the long-period of the moon's motion due to the direct action of the planets, and has made the most elaborate determination yet undertaken of the inequalities of the moon's motion due to the figure of the earth. He has also computed certain lunar inequalities due to the action of Jupiter. The mathematical discussion of Saturn's rings was taken up first by Laplace, who demonstrated that a homogeneous solid ring could not be in equilibrium, and in 1851 by B. Peirce, who proved their non-solidity by showing that even an irregular solid ring could not be in equilibrium about Saturn. The mechanism of these rings was investigated by James Clerk Maxwell in an essay to which the Adams prize was awarded. He concluded that they consisted of an aggregate of unconnected particles. The problem of three bodies has been treated in various ways since the time of Lagrange, but no decided advance towards a more complete algebraic solution has been made, and the problem stands substantially where it was left by him. He had made a reduction in the differential equations to the seventh order. This was elegantly accomplished in a different

A HISTORY OF MATHEMATICS. 440 way by Jacobi in 1843. R. Radau (Comptes Rendus, LXVII., 1868, p. 841) and All´ egret(Journal de Math´ ematiques, 1875, p. 277) showed that the reduction can be performed on the equations in their original form. Noteworthy transformations and discussions of the problem have been given by J. L. F. Bertrand, by ´Emile Bour (1831–1866) of the Polytechnic School in Paris, by Mathieu, Hesse, J. A. Serret. H. Bruns of Leipzig has shown that no advance in the problem of three or of n bodies may be expected by algebraic integrals, and that we must look to the modern theory of functions for a complete solution (Acta Math., XI., p. 43). [93] Among valuable text-books on mathematical astronomy rank the following works: Manual of Spherical and Practical Astronomy by Chauvenet (1863), Practical and Spherical Astronomy by Robert Main of Cambridge, Theoretical As- tronomy by James C. Watson of Ann Arbor (1868), Trait´ e ´ el´ ementaire de M´ ecanique C´ elesteof H. Resal of the Poly- technic School in Paris, Cours d'Astronomie de l' ´Ecole Poly- technique by Faye, Trait´ e de M´ ecanique C´ elesteby Tisserand, Lehrbuch der Bahnbestimmung by T. Oppolzer, Mathematis- che Theorien der Planetenbewegungby O. Dziobek, translated into English by M. W. Harrington and W. J. Hussey. During the present century we have come to recognise the advantages frequently arising from a geometrical treatment of mechanical problems. To Poinsot, Chasles, and M¨ obius we owe the most important developments made in geometrical mechanics. Louis Poinsot (1777–1859), a graduate of the Polytechnic School in Paris, and for many years member of

APPLIED MATHEMATICS. 441 the superior council of public instruction, published in 1804 his ´El´ ements de Statique. This work is remarkable not only as being the earliest introduction to synthetic mechanics, but also as containing for the first time the idea of couples, which was applied by Poinsot in a publication of 1834 to the theory of rotation. A clear conception of the nature of rotary motion was conveyed by Poinsot's elegant geometrical representation by means of an ellipsoid rolling on a certain fixed plane. This construction was extended by Sylvester so as to measure the rate of rotation of the ellipsoid on the plane. A particular class of dynamical problems has recently been treated geometrically bySir Robert Stawell Ball, formerly astronomer royal of Ireland, now Lowndean Professor of Astronomy and Geometry at Cambridge. His method is given in a work entitled Theory of Screws , Dublin, 1876, and in subsequent articles. Modern geometry is here drawn upon, as was done also by Clifford in the related subject of Biquaternions. Arthur Buchheim of Manchester (1859–1888), showed that Grassmann's Ausdehnungslehre supplies all the necessary materials for a simple calculus of screws in elliptic space. Horace Lamb applied the theory of screws to the question of the steady motion of any solid in a fluid. Advances in theoretical mechanics, bearing on the integra- tion and the alteration in form of dynamical equations, were made since Lagrange by Poisson, William Rowan Hamilton, Jacobi, Madame Kowalevski, and others. Lagrange had estab- lished the "Lagrangian form" of the equations of motion. He had given a theory of the variation of the arbitrary constants

A HISTORY OF MATHEMATICS. 442 which, however, turned out to be less fruitful in results than a theory advanced by Poisson. [99] Poisson's theory of the vari- ation of the arbitrary constants and the method of integration thereby afforded marked the

first onward step since Lagrange. Then came the researches of Sir William Rowan Hamilton. His discovery that the integration of the dynamic differential equations is connected with the integration of a certain partial differential equation of the first order and second degree, grew out of an attempt to deduce, by the undulatory theory, results in geometrical optics previously based on the conceptions of the emission theory. The Philosophical Transactions of 1833 and 1834 contain Hamilton's papers, in which appear the first applications to mechanics of the principle of varying action and the characteristic function, established by him some years previously. The object which Hamilton proposed to himself is indicated by the title of his first paper, viz. the discovery of a function by means of which all integral equations can be actually represented. The new form obtained by him for the equation of motion is a result of no less importance than that which was the professed object of the memoir. Hamilton's method of integration was freed by Jacobi of an unnecessary complication, and was then applied by him to the determina- tion of a geodetic line on the general ellipsoid. With aid of elliptic co-ordinates Jacobi integrated the partial differential equation and expressed the equation of the geodetic in form of a relation between two Abelian integrals. Jacobi applied to differential equations of dynamics the theory of the ulti- mate multiplier. The differential equations of dynamics are

APPLIED MATHEMATICS. 443 only one of the classes of differential equations considered by Jacobi. Dynamic investigations along the lines of Lagrange, Hamilton, and Jacobi were made by Liouville, A. Desboves, Serret, J. C. F. Sturm, Ostrogradsky, J. Bertrand, Donkin, Brioschi, leading up to the development of the theory of a system of canonical integrals. An important addition to the theory of the motion of a solid body about a fixed point was made by Madame Sophie de Kowalevski[96] (1853–1891), who discovered a new case in which the differential equations of motion can be integrated. By the use of theta-functions of two independent variables she furnished a remarkable example of how the modern theory of functions may become useful in mechanical problems. She was a native of Moscow, studied under Weierstrass, obtained the doctor's degree at G¨ ottingen, and from 1884 until her death was professor of higher mathematics at the University of Stockholm. The research above mentioned received the Bordin prize of the French Academy in 1888, which was doubled on account of the exceptional merit of the paper. There are in vogue three forms for the expression of the kinetic energy of a dynamical system: the Lagrangian, the Hamiltonian, and a modified form of Lagrange's equations in which certain velocities are omitted. The kinetic energy is expressed in the first form as a homogeneous quadratic function of the velocities, which are the time-variations of the co-ordinates of the system; in the second form, as a homogeneous quadratic function of the momenta of the system; the third form, elaborated recently by Edward

A HISTORY OF MATHEMATICS. 444 John Routh of Cambridge, in connection with his theory of "ignoration of co-ordinates," and by A. B. Basset, is of importance in hydrodynamical problems relating to the motion of perforated solids in a liquid, and in other branches of physics. In recent time great practical importance has come to be attached to the principle of mechanical similitude. By it one can determine from the performance of a model the action of the machine constructed on a larger scale. The principle was first enunciated by Newton (Principia, Bk. II., Sec. VIII., Prop. 32),

and was derived by Bertrand from the principle of virtual velocities. A corollary to it, applied in ship-building, goes by the name of William Froude's law, but was enunciated also by Reech. The present problems of dynamics differ materially from those of the last century. The explanation of the orbital and axial motions of the heavenly bodies by the law of universal gravitation was the great problem solved by Clairaut, Euler, D'Alembert, Lagrange, and Laplace. It did not involve the consideration of frictional resistances. In the present time the aid of dynamics has been invoked by the physical sciences. The problems there arising are often complicated by the presence of friction. Unlike astronomical problems of a century ago, they refer to phenomena of matter and motion that are usually concealed from direct observation. The great pioneer in such problems is Lord Kelvin. While yet an undergraduate at Cambridge, during holidays spent at the seaside, he entered upon researches of this kind by working

APPLIED MATHEMATICS. 445 out the theory of spinning tops, which previously had been only partially explained by Jellet in hisTreatise on the Theory of Friction (1872), and by Archibald Smith. Among standard works on mechanics are Jacobi's Vor- lesungen ¨ uber Dynamik, edited by Clebsch, 1866;Kirchhoff's Vorlesungen ¨ uber mathematische Physik, 1876; Benjamin Peirce's Analytic Mechanics, 1855; Somoff's Theoretische Mechanik, 1879; Tait and Steele's Dynamics of a Particle , 1856; Minchin's Treatise on Statics; Routh's Dynamics of a System of Rigid Bodies ; Sturm's Cours de M´ ecanique de l' ´Ecole Polytechnique. The equations which constitute the foundation of the theory of fluid motion were fully laid down at the time of Lagrange, but the solutions actually worked out were few and mainly of the irrotational type. A powerful method of attacking problems in fluid motion is that of images, introduced in 1843 by George Gabriel Stokes of Pembroke College, Cambridge. It received little attention until Sir William Thomson's discovery of electrical images, whereupon the theory was extended by Stokes, Hicks, and Lewis. In 1849, Thomson gave the maximum and minimum theorem peculiar to hydrodynamics, which was afterwards extended to dynamical problems in general. A new epoch in the progress of hydrodynamics was created, in 1856, by Helmholtz, who worked out remarkable properties of rotational motion in a homogeneous, incompressible fluid, devoid of viscosity. He showed that the vortex filaments in such a medium may possess any number of knottings and

A HISTORY OF MATHEMATICS. 446 twistings, but are either endless or the ends are in the free surface of the medium; they are indivisible. These results suggested to Sir William Thomson the possibility of founding on them a new form of the atomic theory, according to which every atom is a vortex ring in a non-frictional ether, and as such must be absolutely permanent in substance and duration. The vortex-atom theory is discussed by J. J. Thomson of Cambridge (born 1856) in his classical treatise on the Motion of Vortex Rings , to which the Adams Prize was awarded in 1882. Papers on vortex motion have been published also by Horace Lamb, Thomas Craig, Henry A. Rowland, and Charles Chree. The subject of jets was investigated by Helmholtz, Kirch- hoff, Plateau, and Rayleigh; the motion of fluids in a fluid by Stokes, Sir W. Thomson, K¨ opcke, Greenhill, and Lamb; the theory of viscous fluids by Navier, Poisson, Saint-Venant, Stokes, O. E. Meyer, Stefano, Maxwell, Lipschitz,

Craig, Helmholtz, and A. B. Basset. Viscous fluids present great difficulties, because the equations of motion have not the same degree of certainty as in perfect fluids, on account of a deficient theory of friction, and of the difficulty of connecting oblique pressures on a small area with the differentials of the velocities. Waves in liquids have been a favourite subject with English mathematicians. The early inquiries of Poisson and Cauchy were directed to the investigation of waves produced by disturbing causes acting arbitrarily on a small portion of the fluid. The velocity of the long wave was given approximately

APPLIED MATHEMATICS. 447 by Lagrange in 1786 in case of a channel of rectangular cross-section, by Green in 1839 for a channel of triangular section, and by P. Kelland for a channel of any uniform section. Sir George B. Airy, in his treatise on Tides and Waves, discarded mere approximations, and gave the exact equation on which the theory of the long wave in a channel of uniform rectangular section depends. But he gave no general solutions. J. McCowan of University College at Dundee discusses this topic more fully, and arrives at exact and complete solutions for certain cases. The most important application of the theory of the long wave is to the explanation of tidal phenomena in rivers and estuaries. The mathematical treatment of solitary waves was first taken up by S. Earnshaw in 1845, then by Stokes; but the first sound approximate theory was given by J. Boussinesq in 1871, who obtained an equation for their form, and a value for the velocity in agreement with experiment. Other methods of approximation were given by Rayleigh and J. McCowan. In connection with deep-water waves, Osborne Reynolds gave in 1877 the dynamical explanation for the fact that a group of such waves advances with only half the rapidity of the individual waves. The solution of the problem of the general motion of an ellipsoid in a fluid is due to the successive labours of Green (1833), Clebsch (1856), and Bjerknes (1873). The free motion of a solid in a liquid has been investigated by W. Thomson, Kirchhoff, and Horace Lamb. By these labours, the motion of a single solid in a fluid has come to be pretty well understood,

A HISTORY OF MATHEMATICS. 448 but the case of two solids in a fluid is not developed so fully. The problem has been attacked by W. M. Hicks. The determination of the period of oscillation of a rotating liquid spheroid has important bearings on the question of the origin of the moon. G. H. Darwin’s investigations thereon, viewed in the light of Riemann’s and Poincar´ e’s researches, seem to disprove Laplace’s hypothesis that the moon separated from the earth as a ring, because the angular velocity was too great for stability; Darwin finds no instability. The explanation of the contracted vein has been a point of much controversy, but has been put in a much better light by the application of the principle of momentum, originated by Froude and Rayleigh. Rayleigh considered also the reflection of waves, not at the surface of separation of two uniform media, where the transition is abrupt, but at the confines of two media between which the transition is gradual. The first serious study of the circulation of winds on the earth’s surface was instituted at the beginning of the second quarter of this century byH. W. Dov´ e, William C. Redfield, and James P. Espy, followed by researches ofW. Reid, Piddington, and Elias Loomis. But the deepest insight into the wonderful correlations that exist among the varied motions of the atmosphere was obtained by William Ferrel (1817–1891). He was born in Fulton

County, Pa., and brought up on a farm. Though in unfavourable surroundings, a burning thirst for knowledge spurred the boy to the mastery of one branch after another. He attended Marshall College, Pa., and graduated in 1844 from Bethany College. While teaching school he became

APPLIED MATHEMATICS. 449 interested in meteorology and in the subject of tides. In 1856 he wrote an article on “the winds and currents of the ocean.” The following year he became connected with the Nautical Almanac. A mathematical paper followed in 1858 on “the motion of fluids and solids relative to the earth’s surface.” The subject was extended afterwards so as to embrace the mathematical theory of cyclones, tornadoes, water-spouts, etc. In 1885 appeared his Recent Advances in Meteorology. In the opinion of a leading European meteorologist (Julius Hann of Vienna), Ferrel has “contributed more to the advance of the physics of the atmosphere than any other living physicist or meteorologist.” Ferrel teaches that the air flows in great spirals toward the poles, both in the upper strata of the atmosphere and on the earth’s surface beyond the 30th degree of latitude; while the return current blows at nearly right angles to the above spirals, in the middle strata as well as on the earth’s surface, in a zone comprised between the parallels 30◦ N. and 30◦ S. The idea of three superposed currents blowing spirals was first advanced by James Thomson, but was published in very meagre abstract. Ferrel’s views have given a strong impulse to theoretical re- search in America, Austria, and Germany. Several objections raised against his argument have been abandoned, or have been answered by W. M. Davis of Harvard. The mathematical analysis of F. Waldo of Washington, and of others, has further confirmed the accuracy of the theory. The transport of Krakatoa dust and observations made on clouds point toward

A HISTORY OF MATHEMATICS. 450 the existence of an upper east current on the equator, and Pernter has mathematically deduced from Ferrel’s theory the existence of such a current. Another theory of the general circulation of the atmosphere was propounded by Werner Siemens of Berlin, in which an attempt is made to apply thermodynamics to a¨ erial currents. Important new points of view have been introduced recently by Helmholtz, who concludes that when two air currents blow one above the other in different directions, a system of air waves must arise in the same way as waves are formed on the sea. He and A. Oberbeck showed that when the waves on the sea attain lengths of from 16 to 33 feet, the air waves must attain lengths of from10 to 20 miles, and proportional depths. Superposed strata would thus mix more thoroughly, and their energy would be partly dissipated. From hydrodynamical equations of rotation Helmholtz established the reason why the observed velocity from equatorial regions is much less in a latitude of, say, 20◦ or 30◦, than it would be were the movements unchecked. About 1860 acoustics began to be studied with renewed zeal. The mathematical theory of pipes and vibrating strings had been elaborated in the eighteenth century by Daniel Bernoulli, D’Alembert, Euler, and Lagrange. In the first part of the present century Laplace corrected Newton’s theory on the velocity of sound in gases, Poisson gave a mathematical discussion of torsional vibrations; Poisson, Sophie Germain, and Wheatstone studied Chladni’s figures; Thomas Young and the brothers Weber developed the wave-theory of sound.

APPLIED MATHEMATICS. 451 Sir J. F. W. Herschel wrote on the mathematical theory of sound for the Encyclopædia Metropolitana, 1845. Epoch- making were Helmholtz's experimental and mathematical researches. In his hands and Rayleigh's, Fourier's series received due attention. Helmholtz gave the mathematical theory of beats, difference tones, and summation tones.Lord Rayleigh (John William Strutt) of Cambridge (born 1842) made extensive mathematical researches in acoustics as a part of the theory of vibration in general. Particular mention may be made of his discussion of the disturbance produced by a spherical obstacle on the waves of sound, and of phenomena, such as sensitive flames, connected with the instability of jets of fluid. In 1877 and 1878 he published in two volumes a treatise on The Theory of Sound . Other mathematical researches on this subject have been made in England by Donkin and Stokes. The theory of elasticity [42] belongs to this century. Before 1800 no attempt had been made to form general equations for the motion or equilibrium of an elastic solid. Particular problems had been solved by special hypotheses. Thus, James Bernoulli considered elastic laminæ; Daniel Bernoulli and Euler investigated vibrating rods; Lagrange and Euler, the equilibrium of springs and columns. The earliest in- vestigations of this century, by Thomas Young ("Young's modulus of elasticity") in England, J. Binet in France, and G. A. A. Plana in Italy, were chiefly occupied in extend- ing and correcting the earlier labours. Between 1830 and 1840 the broad outline of the modern theory of elasticity

A HISTORY OF MATHEMATICS. 452 was established. This was accomplished almost exclusively by French writers,—Louis-Marie-Henri Navier (1785–1836), Poisson, Cauchy, Mademoiselle Sophie Germain (1776–1831), F´ elix Savart (1791–1841). Sim´ eon Denis Poisson[94] (1781–1840) was born at Pithiviers. The boy was put out to a nurse, and he used to tell that when his father (a common soldier) came to see him one day, the nurse had gone out and left him suspended by a thin cord to a nail in the wall in order to protect him from perishing under the teeth of the carnivorous and unclean animals that roamed on the floor. Poisson used to add that his gymnastic efforts when thus suspended caused him to swing back and forth, and thus to gain an early familiarity with the pendulum, the study of which occupied him much in his maturer life. His father destined him for the medical profession, but so repugnant was this to him that he was permitted to enter the Polytechnic School at the age of seventeen. His talents excited the interest of Lagrange and Laplace. At eighteen he wrote a memoir on finite differences which was printed on the recommendation of Legendre. He soon became a lecturer at the school, and continued through life to hold various government scientific posts and professorships. He prepared some 400 publications, mainly on applied mathematics. His Trait´ e de M´ ecanique, 2 vols., 1811 and 1833, was long a standard work. He wrote on the mathematical theory of heat, capillary action, probability of judgment, the mathematical theory of electricity and magnetism, physical astronomy, the attraction of ellipsoids, definite integrals, series, and the

APPLIED MATHEMATICS. 453 theory of elasticity. He was considered one of the leading analysts of his time. His work on elasticity is hardly excelled by that of Cauchy, and second only to that of Saint-Venant. There is hardly a problem in elasticity to which he has not contributed, while many of his inquiries were new. The

equilibrium and motion of a circular plate was first successfully treated by him. Instead of the definite integrals of earlier writers, he used preferably finite summations. Poisson's contour conditions for elastic plates were objected to by Gustav Kirchhoff of Berlin, who established new conditions. But Thomson and Tait in their Treatise on Natural Philosophy have explained the discrepancy between Poisson's and Kirchhoff's boundary conditions, and established a reconciliation between them. Important contributions to the theory of elasticity were made by Cauchy. To him we owe the origin of the theory of stress, and the transition from the consideration of the force upon a molecule exerted by its neighbours to the consideration of the stress upon a small plane at a point. He anticipated Green and Stokes in giving the equations of isotropic elasticity with two constants. The theory of elasticity was presented by Gabrio Piola of Italy according to the principles of Lagrange's M´ ecanique Analytique, but the superiority of this method over that of Poisson and Cauchy is far from evident. The influence of temperature on stress was first investigated experimentally by Wilhelm Weber of G¨ ottingen, and afterwards mathematically by Duhamel, who, assuming Poisson's theory of elasticity, examined the alterations of form which the formulæ undergo

A HISTORY OF MATHEMATICS. 454 when we allow for changes of temperature. Weber was also the first to experiment on elastic after-strain. Other important experiments were made by different scientists, which disclosed a wider range of phenomena, and demanded a more comprehensive theory. Set was investigated by Gerstner (1756–1832) and Eaton Hodgkinson, while the latter physicist in England and Vicat (1786–1861) in France experimented extensively on absolute strength. Vicat boldly attacked the mathematical theories of flexure because they failed to consider shear and the time-element. As a result, a truer theory of flexure was soon propounded by Saint-Venant. Poncelet advanced the theories of resilience and cohesion. Gabriel Lam´ e[94] (1795–1870) was born at Tours, and graduated at the Polytechnic School. He was called to Russia with Clapeyron and others to superintend the construction of bridges and roads. On his return, in 1832, he was elected professor of physics at the Polytechnic School. Subsequently he held various engineering posts and professorships in Paris. As engineer he took an active part in the construction of the first railroads in France. Lam´ e devoted his fine mathematical talents mainly to mathematical physics. In four works:Le¸ cons sur les fonctions inverses des transcendantes et les surfaces isothermes; Sur les coordonn´ ees curvilignes et leurs diverses applications; Sur la th´ eorie analytique de la chaleur; Sur la th´ eorie math´ ematique de l'´ elasticit´ e des corps solides(1852), and in various memoirs he displays fine analytical powers; but a certain want of physical touch sometimes reduces the value of his contributions to elasticity and other physical subjects.

APPLIED MATHEMATICS. 455 In considering the temperature in the interior of an ellipsoid under certain conditions, he employed functions analogous to Laplace's functions, and known by the name of "Lam´ e's functions." A problem in elasticity called by Lam´ e's name, viz. to investigate the conditions for equilibrium of a spherical elastic envelope subject to a given distribution of load on the bounding spherical surfaces, and the determination of the resulting shifts is the only completely

general problem on elasticity which can be said to be completely solved. He deserves much credit for his derivation and transformation of the general elastic equations, and for his application of them to double refraction. Rectangular and triangular membranes were shown by him to be connected with questions in the theory of numbers. The field of photo-elasticity was entered upon by Lam´ e, F. E. Neumann, Clerk Maxwell. Stokes, Wertheim, R. Clausius, Jellett, threw new light upon the subject of “rari-constancy” and “multi-constancy,” which has long divided elasticians into two opposing factions. The uni- constant isotropy of Navier and Poisson had been questioned by Cauchy, and was now severely criticised by Green and Stokes. Barr´ e de Saint-Venant(1797–1886), ing´ enieur des ponts et chauss´ ees, made it his life-work to render the theory of elasticity of practical value. The charge brought by practical engineers, like Vicat, against the theorists led Saint-Venant to place the theory in its true place as a guide to the practical man. Numerous errors committed by his predecessors were removed. He corrected the theory of flexure by the consideration of

A HISTORY OF MATHEMATICS. 456 slide, the theory of elastic rods of double curvature by the introduction of the third moment, and the theory of torsion by the discovery of the distortion of the primitively plane section. His results on torsion abound in beautiful graphic illustrations. In case of a rod, upon the side surfaces of which no forces act, he showed that the problems of flexure and torsion can be solved, if the end-forces are distributed over the end-surfaces by a definite law. Clebsch, in his Lehrbuch der Elasticit¨ at, 1862, showed that this problem is reversible to the case of side-forces without end-forces. Clebsch [68] extended the research to very thin rods and to very thin plates. Saint- Venant considered problems arising in the scientific design of built-up artillery, and his solution of them differs considerably from Lam´ e’s solution, which was popularised by Rankine, and much used by gun-designers. In Saint-Venant’s translation into French of Clebsch’sElasticit¨ at, he develops extensively a double-suffix notation for strain and stresses. Though often advantageous, this notation is cumbrous, and has not been generally adopted. Karl Pearson , professor in University College, London, has recently examined mathematically the permissible limits of the application of the ordinary theory of flexure of a beam. The mathematical theory of elasticity is still in an unsettled condition. Not only are scientists still divided into two schools of “rari-constancy” and “multi-constancy,” but difference of opinion exists on other vital questions. Among the numerous modern writers on elasticity may be mentioned´Emile Mathieu (1835–1891), professor at Besan¸ con, Maurice Levy of Paris,

APPLIED MATHEMATICS. 457 Charles Chree, superintendent of the Kew Observatory, A. B. Basset, Sir William Thomson (Lord Kelvin) of Glasgow, J. Boussinesq of Paris, and others. Sir William Thomson applied the laws of elasticity of solids to the investigation of the earth’s elasticity, which is an important element in the theory of ocean-tides. If the earth is a solid, then its elasticity co-operates with gravity in opposing deformation due to the attraction of the sun and moon. Laplace had shown how the earth would behave if it resisted deformation only by gravity. Lam´ e had investigated how a solid sphere would change if its elasticity only came

into play. Sir William Thomson combined the two results, and compared them with the actual deformation. Thomson, and afterwards G. H. Darwin, computed that the resistance of the earth to tidal deformation is nearly as great as though it were of steel. This conclusion has been confirmed recently by Simon Newcomb, from the study of the observed periodic changes in latitude. For an ideally rigid earth the period would be 360 days, but if as rigid as steel, it would be 441, the observed period being 430 days. Among text-books on elasticity may be mentioned the works of Lam´ e, Clebsch, Winkler, Beer, Mathieu, W. J. Ibbetson, and F. Neumann, edited by O. E. Meyer. Riemann's opinion that ascience of physics only exists since the invention of differential equations finds corroboration even in this brief and fragmentary outline of the progress of mathematical physics. The undulatory theory of light, first advancedbyHuygens, owesmuchtothepowerofmathematics:

A HISTORY OF MATHEMATICS. 458 by mathematical analysis its assumptions were worked out to their last consequences. Thomas Young [95] (1773–1829) was the first to explain the principle of interference, both of light and sound, and the first to bring forward the idea of transverse vibrations in light waves. Young's explanations, not being verified by him by extensive numerical calculations, attracted little notice, and it was not untilAugustin Fresnel (1788–1827) applied mathematical analysis to a much greater extent than Young had done, that the undulatory theory began to carry conviction. Some of Fresnel's mathematical assumptions were not satisfactory; hence Laplace, Poisson, and others belonging to the strictly mathematical school, at first disdained to consider the theory. By their opposition Fresnel was spurred to greater exertion. Arago was the first great convert made by Fresnel. When polarisation and double refraction were explained by Young and Fresnel, then Laplace was at last won over. Poisson drew from Fresnel's formulæ the seemingly paradoxical deduction that a small circular disc, illuminated by a luminous point, must cast a shadow with a bright spot in the centre. But this was found to be in accordance with fact. The theory was taken up by another great mathematician, Hamilton, who from his formulæ predicted conical refraction, verified experimentally by Lloyd. These predictions do not prove, however, that Fresnel's formulæ are correct, for these prophecies might have been made by other forms of the wave-theory. The theory was placed on a sounder dynamical basis by the writings of Cauchy, Biot, Green, C. Neumann, Kirchhoff,

APPLIED MATHEMATICS. 459 McCullagh, Stokes, Saint-Venant, Sarrau, Lorenz, and Sir William Thomson. In the wave-theory, as taught by Green and others, the luminiferous ether was an incompressible elastic solid, for the reason that fluids could not propagate transverse vibrations. But, according to Green, such an elastic solid would transmit a longitudinal disturbance with infinite velocity. Stokes remarked, however, that the ether might act like a fluid in case of finite disturbances, and like an elastic solid in case of the infinitesimal disturbances in light propagation. Fresnel postulated the density of ether to be different in different media, but the elasticity the same, while C. Neumann and McCullagh assume the density uniform and the elasticity different in all substances. On the latter assumption the direction of vibration lies in the plane of polarisation, and not perpendicular to it, as in the theory of Fresnel. While the above writers endeavoured to explain all optical properties of a

medium on the supposition that they arise entirely from difference in rigidity or density of the ether in the medium, there is another school advancing theories in which the mutual action between the molecules of the body and the ether is considered the main cause of refrac- tion and dispersion. [100] The chief workers in this field are J. Boussinesq, W. Sellmeyer, Helmholtz, E. Lommel, E. Ket- teler, W. Voigt, and Sir William Thomson in his lectures delivered at the Johns Hopkins University in 1884. Neither this nor the first-named school succeeded in explaining all the phenomena. A third school was founded by Maxwell.

A HISTORY OF MATHEMATICS. 460 He proposed the electro-magnetic theory, which has received extensive development recently. It will be mentioned again later. According to Maxwell's theory, the direction of vi- bration does not lie exclusively in the plane of polarisation, nor in a plane perpendicular to it, but something occurs in both planes—a magnetic vibration in one, and an electric in the other. Fitzgerald and Trouton in Dublin verified this conclusion of Maxwell by experiments on electro-magnetic waves. Of recent mathematical and experimental contributions to optics, mention must be made of H. A. Rowland's theory of concave gratings, and of A. A. Michelson's work on interference, and his application of interference methods to astronomical measurements. In electricity the mathematical theory and the measure- ments of Henry Cavendish(1731–1810), and in magnetism the measurements of Charles Augustin Coulomb (1736– 1806), became the foundations for a system of measurement. For electro-magnetism the same thing was done by Andr` e Marie Amp` ere(1775–1836). The first complete method of measurement was the system of absolute measurements of terrestrial magnetism introduced by Gauss and Wilhelm Weber (1804–1891) and afterwards extended by Wilhelm Weber and F. Kohlrausch to electro-magnetism and electro- statics. In 1861 the British Association and the Royal Society appointed a special commission with Sir William Thomson at the head, to consider the unit of electrical resistance. The commission recommended a unit in principle like W. Weber's,

APPLIED MATHEMATICS. 461 but greater than Weber's by a factor of107. [101] The discus- sions and labours on this subject continued for twenty years, until in 1881 a general agreement was reached at an electrical congress in Paris. A function of fundamental importance in the mathematical theories of electricity and magnetism is the "potential." It was first used by Lagrange in the determination of gravitational attractions in 1773. Soon after, Laplace gave the celebrated differential equation, $\partial 2V \partial x2 + \partial 2V \partial y\ 2 + \partial 2V \partial z\ 2 = 0$, which was extended by Poisson by writing $-4\pi k$ in place of zero in the right-hand member of the equation, so that it applies not only to a point external to the attracting mass, but to any point whatever. The first to apply the potential function to other than gravitation problems was George Green (1793–1841). He introduced it into the mathematical theory of electricity and magnetism. Green was a self-educated man who started out as a baker, and at his death was fellow of Caius College, Cambridge. In 1828 he published by subscription at Nottingham a paper entitled Essay on the application of mathematical analysis to the theory of electricity and magnetism. It escaped the notice even of English mathematicians until 1846, when Sir William Thomson had it reprinted in Crelle's Journal, vols. xliv. and xlv. It contained what is now known as "Green's theorem" for the treatment of

potential. Meanwhile all of Green’s general theorems had been re-discovered by Sir William Thomson,

A HISTORY OF MATHEMATICS. 462 Chasles, Sturm, and Gauss. The term potential function is due to Green. Hamilton used the word force-function, while Gauss, who about 1840 secured the general adoption of the function, called it simply potential. Large contributions to electricity and magnetism have been made byWilliam Thomson. He was born in 1824 at Belfast, Ireland, but is of Scotch descent. He and his brother James studied in Glasgow. From there he entered Cambridge, and was graduated as Second Wrangler in 1845. William Thomson, Sylvester, Maxwell, Clifford, and J. J. Thomson are a group of great men who were Second Wranglers at Cambridge. At the age of twenty-two W. Thomson was elected professor of natural philosophy in the University of Glasgow, a position which he has held ever since. For his brilliant mathematical and physical achievements he was knighted, and in 1892 was made Lord Kelvin. His researches on the theory of potential are epoch-making. What is called “Dirichlet’s principle” was discovered by him in 1848, somewhat earlier than by Dirichlet. We owe to Sir William Thomson new synthetical methods of great elegance, viz. the theory of electric images and the method of electric inversion founded thereon. By them he determined the distribution of electricity on a bowl, a problem previously considered insolvable. The distribution of static electricity on conductors had been studied before this mainly by Poisson and Plana. In 1845 F. E. Neumann of K¨ onigsberg developed from the experimental laws of Lenz the mathematical theory of magneto-electric induction. In 1855 W. Thomson predicted by mathematical analysis that the

APPLIED MATHEMATICS. 463 discharge of a Leyden jar through a linear conductor would in certain cases consist of a series of decaying oscillations. This was first established experimentally by Joseph Henry of Washington. William Thomson worked out the electro-static induction in submarine cables. The subject of the screening effect against induction, due to sheets of different metals, was worked out mathematically by Horace Lamb and also by Charles Niven. W. Weber’s chief researches were on electro-dynamics. Helmholtz in 1851 gave the mathematical theory of the course of induced currents in various cases. Gustav Robert Kirchhoff [97] (1824–1887) investigated the distribution of a current over a flat conductor, and also the strength of current in each branch of a network of linear conductors. The entire subject of electro-magnetism was revolutionised by James Clerk Maxwell (1831–1879). He was born near Edinburgh, entered the University of Edinburgh, and became a pupil of Kelland and Forbes. In 1850 he went to Trinity College, Cambridge, and came out Second Wrangler, E. Routh being Senior Wrangler. Maxwell then became lecturer at Cambridge, in 1856 professor at Aberdeen, and in 1860 professor at King’s College, London. In 1865 he retired to private life until 1871, when he became professor of physics at Cambridge. Maxwell not only translated into mathematical language the experimental results of Faraday, but established the electro-magnetic theory of light, since verified experimentally by Hertz. His first researches thereon were published in 1864. In 1871 appeared his great Treatise

A HISTORY OF MATHEMATICS. 464 on Electricity and Magnetism . He constructed the electro- magnetic theory from general equations, which are established upon purely dynamical principles, and which determine the state of the electric field. It is a mathematical discussion of the stresses and strains in a dielectric medium subjected to electro- magnetic forces. The electro-magnetic theory has received developments from Lord Rayleigh, J. J. Thomson, H. A. Rowland, R. T. Glazebrook, H. Helmholtz, L. Boltzmann, O. Heaviside, J. H. Poynting, and others. Hermann von Helmholtz turned his attention to this part of the subject in 1871. He was born in 1821 at Potsdam, studied at the University of Berlin, and published in 1847 his pamphlet Ueber die Erhaltung der Kraft. He became teacher of anatomy in the Academy of Art in Berlin. He was elected professor of physiology at K¨ onigsberg in 1849, at Bonn in 1855, at Heidelberg in 1858. It was at Heidelberg that he produced his work on Tonempfindung. In 1871 he accepted the chair of physics at the University of Berlin. From this time on he has been engaged chiefly on inquiries in electricity and hydrodynamics. Helmholtz aimed to determine in what direction experiments should be made to decide between the theories of W. Weber, F. E. Neumann, Riemann, and Clausius, who had attempted to explain electro-dynamic phenomena by the assumption of forces acting at a distance between two portions of the hypothetical electrical fluid,— the intensity being dependent not only on the distance, but also on the velocity and acceleration,—and the theory of Faraday and Maxwell, which discarded action at a distance

APPLIED MATHEMATICS. 465 and assumed stresses and strains in the dielectric. His experiments favoured the British theory. He wrote on abnormal dispersion, and created analogies between electro- dynamics and hydrodynamics. Lord Rayleigh compared electro-magnetic problems with their mechanical analogues, gave a dynamical theory of diffraction, and applied Laplace’s coefficients to the theory of radiation. Rowland made some emendations on Stokes’ paper on diffraction and considered the propagation of an arbitrary electro-magnetic disturbance and spherical waves of light. Electro-magnetic induction has been investigated mathematically by Oliver Heaviside, and he showed that in a cable it is an actual benefit. Heaviside and Poynting have reached remarkable mathematical results in their interpretation and development of Maxwell’s theory. Most of Heaviside’s papers have been published since 1882; they cover a wide field. One part of the theory of capillary attraction, left defective by Laplace, namely, the action of a solid upon a liquid, and the mutual action between two liquids, was made dynamically perfect by Gauss. He stated the rule for angles of contact between liquids and solids. A similar rule for liquids was established by Ernst Franz Neumann. Chief among recent workers on the mathematical theory of capillarity are Lord Rayleigh and E. Mathieu. The great principle of the conservation of energy was estab- lished by Robert Mayer (1814–1878), a physician in Heil- bronn, and again independently by Colding of Copenhagen, Joule, and Helmholtz. James Prescott Joule (1818–1889)

A HISTORY OF MATHEMATICS. 466 determined experimentally the mechanical equivalent of heat. Helmholtz in 1847 applied the conceptions of the transfor- mation and conservation of energy to the various branches of physics, and thereby linked together many well-known phenomena. These labours led to the abandonment of the

corpuscular theory of heat. The mathematical treatment of thermic problems was demanded by practical considerations. Thermodynamics grew out of the attempt to determine math- ematically how much work can be gotten out of a steam engine. Sadi-Carnot, an adherent of the corpuscular theory, gave the first impulse to this. The principle known by his name was published in 1824. Though the importance of his work was emphasised by B. P. E. Clapeyron, it did not meet with general recognition until it was brought forward by William Thomson. The latter pointed out the necessity of modifying Carnot's reasoning so as to bring it into accord with the new theory of heat. William Thomson showed in 1848 that Carnot's principle led to the conception of an absolute scale of temperature. In 1849 he published "an account of Carnot's theory of the motive power of heat, with numerical results deduced from Regnault's experiments." In February, 1850, Rudolph Clausius (1822–1888), then in Z¨ urich (afterwards professor in Bonn), communicated to the Berlin Academy a paper on the same subject which contains the Protean second law of thermodynamics. In the same month William John M. Rankine (1820–1872), professor of engineering and mechanics at Glasgow, read before the Royal Society of Edinburgh a paper in which he declares the nature of heat

APPLIED MATHEMATICS. 467 to consist in the rotational motion of molecules, and arrives at some of the results reached previously by Clausius. He does not mention the second law of thermodynamics, but in a subsequent paper he declares that it could be derived from equations contained in his first paper. His proof of the second law is not free from objections. In March, 1851, appeared a paper of William Thomson which contained a perfectly rigorous proof of the second law. He obtained it before he had seen the researches of Clausius. The statement of this law, as given by Clausius, has been much criticised, particularly by Rankine, Theodor Wand, P. G. Tait, and Tolver Preston. Repeated efforts to deduce it from general mechanical princi- ples have remained fruitless. The science of thermodynamics was developed with great success by Thomson, Clausius, and Rankine. As early as 1852 Thomson discovered the law of the dissipation of energy, deduced at a later period also by Clausius. The latter designated the non-transformable energy by the name entropy, and then stated that the entropy of the universe tends toward a maximum. For entropy Rankine used the term thermodynamic function . Thermodynamic investigations have been carried on also by G. Ad. Hirn of Colmar, and Helmholtz (monocyclic and polycyclic systems). Valuable graphic methods for the study of thermodynamic relations were devised in 1873–1878 byJ. Willard Gibbs of Yale College. Gibbs first gives an account of the advantages of using various pairs of the five fundamental thermodynamic quantities for graphical representation, then discusses the entropy-temperature and entropy-volume diagrams, and the

A HISTORY OF MATHEMATICS. 468 volume-energy-entropy surface (described in Maxwell's The- ory of Heat). Gibbs formulated the energy-entropy criterion of equilibrium and stability, and expressed it in a form applicable to complicated problems of dissociation. Important works on thermodynamics have been prepared by Clausius in 1875, by R. R¨ uhlmann in 1875, and by Poincar´ e in 1892. In the study of the law of dissipation of energy and the principle of least action, mathematics and metaphysics met on common ground. The doctrine of least action was first

propounded by Maupertius in 1744. Two years later he proclaimed it to be a universal law of nature, and the first scientific proof of the existence of God. It was weakly supported by him, violently attacked by K¨ onig of Leipzig, and keenly defended by Euler. Lagrange's conception of the principle of least action became the mother of analytic mechanics, but his statement of it was inaccurate, as has been remarked by Josef Bertrand in the third edition of the M´ ecanique Analytique. The form of the principle of least action, as it now exists, was given by Hamilton, and was extended to electro-dynamics by F. E. Neumann, Clausius, Maxwell, and Helmholtz. To subordinate the principle to all reversible processes, Helmholtz introduced into it the conception of the "kinetic potential." In this form the principle has universal validity. An offshoot of the mechanical theory of heat is the modern kinetic theory of gases, developed mathematically byClausius, Maxwell, Ludwig Boltzmann of Munich, and others. The first suggestions of a kinetic theory of matter go back as far as the

APPLIED MATHEMATICS. 469 time of the Greeks. The earliest work to be mentioned here is that of Daniel Bernoulli, 1738. He attributed to gas-molecules great velocity, explained the pressure of a gas by molecular bombardment, and deduced Boyle's law as a consequence of his assumptions. Over a century later his ideas were taken up by Joule (in 1846), A. K. Kr¨ onig (in 1856), and Clausius (in 1857). Joule dropped his speculations on this subject when he began his experimental work on heat. Kr¨ onig explained by the kinetic theory the fact determined experimentally by Joule that the internal energy of a gas is not altered by expansion when no external work is done. Clausius took an important step in supposing that molecules may have rotary motion, and that atoms in a molecule may move relatively to each other. He assumed that the force acting between molecules is a function of their distances, that temperature depends solely upon the kinetic energy of molecular motions, and that the number of molecules which at any moment are so near to each other that they perceptibly influence each other is comparatively so small that it may be neglected. He calculated the average velocities of molecules, and explained evaporation. Objections to his theory, raised by Buy's-Ballot and by Jochmann, were satisfactorily answered by Clausius and Maxwell, except in one case where an additional hypothesis had to be made. Maxwell proposed to himself the problem to determine the average number of molecules, the velocities of which lie between given limits. His expression therefor constitutes the important law of distribution of velocities named after him. By this law the distribution of molecules according to their velocities

A HISTORY OF MATHEMATICS. 470 is determined by the same formula (given in the theory of probability) as the distribution of empirical observations according to the magnitude of their errors. The average molecular velocity as deduced by Maxwell differs from that of Clausius by a constant factor. Maxwell's first deduction of this average from his law of distribution was not rigorous. A sound derivation was given by O. E. Meyer in 1866. Maxwell predicted that so long as Boyle's law is true, the coefficient of viscosity and the coefficient of thermal conductivity remain independent of the pressure. His deduction that the coefficient of viscosity should be proportional to the square root of the absolute temperature appeared to be at variance with results obtained from pendulum experiments. This induced him to alter the very

foundation of his kinetic theory of gases by assuming between the molecules a repelling force varying inversely as the fifth power of their distances. The founders of the kinetic theory had assumed the molecules of a gas to be hard elastic spheres; but Maxwell, in his second presentation of the theory in 1866, went on the assumption that the molecules behave like centres of forces. He demonstrated anew the law of distribution of velocities; but the proof had a flaw in argument, pointed out by Boltzmann, and recognised by Maxwell, who adopted a somewhat different form of the distributive function in a paper of 1879, intended to explain mathematically the effects observed in Crookes' radiometer. Boltzmann gave a rigorous general proof of Maxwell's law of the distribution of velocities. None of the fundamental assumptions in the kinetic theory

APPLIED MATHEMATICS. 471 of gases leads by the laws of probability to results in very close agreement with observation. Boltzmann tried to estab- lish kinetic theories of gases by assuming the forces between molecules to act according to different laws from those pre- viously assumed. Clausius, Maxwell, and their predecessors took the mutual action of molecules in collision as repulsive, but Boltzmann assumed that they may be attractive. Exper- iments of Joule and Lord Kelvin seem to support the latter assumption. Among the latest researches on the kinetic theory is Lord Kelvin's disproof of a general theorem of Maxwell and Boltzmann, asserting that the average kinetic energy of two given portions of a system must be in the ratio of the number of degrees of freedom of those portions.

ADDENDA. Page 15. The new Akhmim papyrus, written in Greek, is probably the copy of an older papyrus, antedating Heron's works, and is the oldest extant text-book on practical Greek arithmetic. It contains, besides arithmetical examples, a table for finding "unit-fractions," identical in scope with that of Ahmes, and, like Ahmes's, without a clue as to its mode of construction. See Biblioth. Math., 1893, p. 79–89. The papyrus is edited by J. Baillet (M´ emoires publi´ es par les membres de la mission arch´ eologique fran¸ caise au Caire, T. IX., 1 r fascicule, Paris, 1892, p. 1–88). Page 45. Chasles's or Simson's definition of a Porism is preferable to Proclus's, given in the text. See Gow, p. 217–221. Page 132. Nasir Eddin for the first time elaborated trigonometry independently of astronomy and to such great perfection that, had his work been known, Europeans of the 15th century might have spared their labours. See Biblioth. Math., 1893, p. 6. Page 134. This law of sines was probably known before Gabir ben Aflah to Tabit ben Korra and others. SeeBiblioth. Math., 1893, p. 7. Page 145. Athelard was probably not the first to translate Euclid's Elements from the Arabic. See M. Cantor's Vorlesungen, Vol. II., p. 91, 92. Page 279. G. Enestr¨ om argues that Taylor and not Nicole is the real inventor of finite differences. See Biblioth. Math., 1893, p. 91. Page 290. An earlier publication in which 3 .14159 . . . is designated by π, is W. Jones's Synopsis palmariorum matheseos , London, 1706, p. 243, 263 et seq. See Biblioth. Math., 1894, p. 106. Page 391 . Before Gauss a theorem on convergence, usually attributed to Cauchy, was given by Maclaurin (Fluxions,§ 350). A rule of convergence was deduced also by Stirling. See Bull. N. Y. Math. Soc., Vol. III., p. 186. Page 418. The surface of a solid with p holes was considered before Clifford by Tonelli, and was probably used by Riemann himself. See Math. Annalen, Vol. 45, p. 142. Page 421. As early as 1835, Lobachevsky

showed in a memoir the necessity of distinguishing between continuity and differentiability. See G. B. Halsted’s transl. of A. Vasiliev’s Address on Lobachevsky, p. 23. 472

ADDENDA. 473 Recent deaths. Johann Rudolf Wolf, Dec. 6, 1893; Heinrich Hertz, Jan. 1, 1894; Eug` ene Catalan, Feb. 14, 1894; Hermann von Helmholtz, Sept. 8, 1894; Arthur Cayley, Jan. 26, 1895.

INDEX Abacists, 146 Abacus, 8, 13, 73, 91, 94, 138, 141, 146, 150 Abbatt, 389 Abel, 405 ref. to, 170, 325, 339, 364, 382, 391, 393, 408, 412, 433 Abel’s theorem, 410 Abelian functions, 340, 364, 382, 403, 405, 407, 411, 414–417, 419 Abelian integrals, 408, 442 Absolute geometry, 351 Absolutely convergent series, 390, 392, 394 Abul Gud, 128 ref. to, 131 Abul Hasan, 133 Abul Wefa, 127 ref. to, 129, 130 Achilles and tortoise, paradox of, 31 Acoustics, 304, 314, 323, 450 Action varying, 340, 371, 442 Action, least, 294, 426, 468 Adams, 438 ref. to, 249 Addition theorem of elliptic integrals, 293, 408, 462 Adrain, 322 Æquipollences, 375 Agnesi, 303 Agrimensores, 92 Ahmes, 11–16 ref. to, 19, 20, 61, 86, 151 Airy, 438 ref. to, 447 Al Battani, 126 ref. to, 127, 145 Albertus Magnus, 155 Albiruni, 128 ref. to, 118, 120 Alcuin, 137 Alembert, D’, See D’Alembert Alexandrian School (first), 39–62 (second), 62–71 Alfonso’s tables, 147 Algebra, See Notation Arabic, 124, 128, 133 Beginnings in Egypt, 16 Diophantus, 86–89 early Greek, 84 Hindoo, 107–110 Lagrange, 310 Middle Ages, 155, 157 origin of terms, 124, 133 Peacock, 331 recent, 367–385 Renaissance, 162, 165–174, 177 seventeenth century, 192, 218, 224 Algebraic functions, 403 integrals, 439 Algorithm Middle Ages, 146, 149 origin of term, 123 Al Haitam, 133 ref. to, 130 Al Hayyami, 129 ref. to, 131 474

INDEX 475 Al Hazin, 130 Al Hogendi, 128 Al Karhi, 128, 131 Al Kaschi, 132 Al Kuhi, 128 ref. to, 130 All´ egret, 440 Allman, ix, 42 Al Madshriti, 133 Almagest, 65–67 ref. to, 121, 126, 147, 156, 158, 162 Al Mahani, 130 Alphonso’s tables, 147 Al Sagani, 128 Alternate numbers, 375 Amp` ere, 460 ref. to, 421 Amyclas, 38 Analysis (in synthetic geometry), 35, 45 Descartes’, 216 modern, 386–390 Analysis situs, 262, 367 Analytic geometry, 215–219, 222, 224, 279, 334, 358–366 Analytical Society (in Cambridge), 330 Anaxagoras, 21 ref. to, 32 Anaximander, 20 Anaximenes, 21 Angeli, 216 Anger, 437 Anharmonic ratio, 207, 342, 346, 356 Anthology, Palatine, 84, 138 Antiphon, 30 ref. to, 30 Apices of Boethius, 94 ref. to, 73, 119, 138, 146, 150 Apollonian Problem, 57, 179, 218 Apollonius, 51–58 ref. to, 41, 42, 45, 62, 70, 76, 90, 121, 125, 133, 163, 178 Appel, 403 Applied mathematics, See Astronomy, Mechanics, 435–470 Arabic manuscripts, 144–148 Arabic numerals and notation, 3, 84, 100, 118, 129, 147–149, 184 Arabs, 116–135 Arago, xii, 387, 458 Arbogaste, 302 Archimedes, 46–52 ref. to, 2, 40, 42, 44, 51, 53, 56, 57, 62, 71, 75, 85, 90, 104, 121, 125, 163, 167, 196, 201, 212 Archytas, 25 ref. to, 33, 35, 37, 49 Areas, conservation of, 294 Arenarius, 76 Argand, 370 ref. to, 307 Aristæus, 38 ref. to, 53 Aristotle, 39 ref. to, 9, 18, 31, 49, 71, 78, 144 Arithmetic, See Numbers, Notation Arabic, 122 Euclid, 44, 81 Greek, 72–88 Hindoo, 103–106 Middle Ages, 137, 142, 146, 150, 154, 156 Platonists, 33

INDEX 476 Pythagoreans, 23, 77–81 Renaissance, 174, 175, 184–186 Arithmetical machine, 255, 330 Arithmetical triangle, 228 Armemante, 364 Arneth,, x Aronhold,

381 Aryabhatta, 100 ref. to, 102, 106, 114 Aschieri, 356 Assumption, tentative, See Regula falsa, 87, 107 Astrology, 180 Astronomy, See Mechanics Arabic, 115, 117, 121, 134 Babylonian, 9 Egyptian, 10 Greek, 20, 27, 36, 45, 58, 64 Hindoo, 99 Middle Ages, 147 more recent researches, 294, 298, 304, 315–319, 426, 434–440 Newton, 246–251 Athelard of Bath, 145 ref. to, 156 Athenæus, 37 Atomic theory, 446 Attalus, 53 Attraction, See Gravitation, Ellipsoid, 322 August, 344 Ausdehnungslehre, 373, 374, 441 Axioms (of geometry), 34, 42, 43, 327, 350, 367 Babbage, 330, 415 Babylonians, 5–9 ref. to, 21, 59 Bachet de M´ eziriac,See M´ eziriac Bachmann, 433 ref. to, 426 Bacon, R., 156 Baker, Th, 131 Ball, Sir R. S., 441 Ball, W. W. R., xi, 253 Ballistic curve, 324 Baltzer, R., 366 ref. to, 352, 379 Barbier, 397 Barrow, 230 ref. to, 201, 234, 235, 257, 264 Basset, 444, 446 Battaglini, 356 Bauer, xiii Baumgart, xii Baune, De, See De Baune Bayes, 396 Beaumont, xii Bede, the Venerable, 137 Beer, 457 Beha Eddin, 132 Bellavitis, 375 ref. to, 349, 354, 369 Beltrami, 354, 355 ref. to, 367 Ben Junus, 133 Berkeley, 274 Bernelinus, 141 Bernoulli's theorem, 276 Bernoulli, Daniel, 276 ref. to, 297, 305, 450, 469 Bernoulli, James (born 1654), 275, 276 ref. to, 212, 262, 267, 291 Bernoulli, James (born 1758), 278, 415, 451 Bernoulli, John (born 1667), 276 ref. to, 262, 267, 270, 272, 275, 282, 291, 415

INDEX 477 Bernoulli, John (born 1710), 278 Bernoulli, John (born 1744), 278 Bernoulli, Nicolaus (born 1687), 278, 291, 314 Bernoulli, Nicolaus (born 1695), 276 Bernoullis, genealogical table of, 275 Bertini, 356 Bertrand, 393, 396, 399, 440, 443, 444, 468 Bessel, 435–437 ref. to, 353, 360, 409 Bessel's functions, 436 Bessy, 210 Beta function, 289 Betti, 412 Beyer, 186 B´ ezout, 302 ref. to, 290, 307 B´ ezout's method of elimination, 302, 385 Bhaskara, 100 ref. to, 106–110, 112, 177 Bianchi, 382 Billingsley, 160 Binet, 378, 451 Binomial formula, 226, 228, 234, 292, 405 Biot, 320, 335, 459 Biquadratic equation, 130, 169, 172 Biquadratic residues, 426 Biquaternions, 441 Bjerknes, C. A., xiv, 417, 447 Bobillier, 358 Bˆ ocher, xv Bode, 398 Boethius, 94 ref. to, 73, 83, 119, 137, 140, 156 Bois-Reymond, P. du, xiv, 393–396, 422 Boltzmann, 464, 470 Bolyai, Johann, 351 ref. to, 338 Bolyai, Wolfgang, 351 ref. to, 338, 424 Bolza, 408 Bombelli, 169 ref. to, 177 Bonnet, O., 366 ref. to, 393, 399 Boole, 400 ref. to, 339, 379, 396, 398, 404 Booth, 362 Borchardt, 414 Bouniakowsky, 426 Bouquet, 402 ref. to, 404, 413 Bour, 397, 440 Boussinesq, 447, 457, 459 Bowditch, 320, 376 Boyle's law, 469 Brachistochrone (line of swiftest descent), 272, 276 Bradwardine, 156 ref. to, 163 Brahe, Tycho, 127, 161, 195 Brahmagupta, 100 ref. to, 106, 110, 113, 117 Bredon, 157 Bretschneider, ix, 113, 373 Brianchion, 206, 335, 336 Briggs, 190 Brill, A., 346, 363, 415 Brill, L., 357 Bring, 383 Brioschi, 382 ref. to, 379, 385, 389, 408, 412,

INDEX 478 443 Briot, 401 ref. to, 404, 413 Brouncker, 229 Bruno, Fa` a de, 382 Bruns, 440 Bryson of Heraclea, 30 Buchheim, 441 ref. to, 357 Buckley, 185 Budan, 328 Buddha, 103 Buffon, 397 Bungus, 180 B¨ urgi, 186 ref. to, 192 Burkhardt, H., xiii, 382 Burkhardt, J. K., 320 Burmester, 350 Busche, 426 Buteo, 179 Buy's-Ballot, 469 Byrgius, See B¨ urgi Cæsar, Julius, 93 Calculating machines, 255, 330 Calculation, origin of word, 91 Calculus, See Differential Calculus of operations, 340 of variations, 287, 289, 304, 309, 344, 382, 388–389, 415, 427 Calendar, 9, 93, 164, 179, 315 Callisthenes, 9 Canon paschalis, 91 Cantor, G., 396, 422, 434 Cantor, M., ix, x, 129 Capelli, 384 Capillarity, 323, 426, 452, 465 Caporali, 365 Cardan, 167 ref. to, 172,

176, 181, 185 Carll, 390 Carnot, Lazare, 335, 336 ref. to, 64, 274, 342 Carnot, Sadi, 466 Casey, 365 Cassini, D, 299 Cassiodorius, 95, 137 Casting out the 9's, 105, 123 Catalan, E., 378 Cataldi, 185 Catenary, 222, 272, 276 Cattle-problem, 85 Cauchy, 386–388 ref. to, 282, 287, 307, 375, 378, 382, 385, 391, 394, 395, 398, 399, 402, 406–408, 411, 415, 422, 430, 446, 452, 453, 455, 459 Caustics, 276, 280 Cavalieri, 197 ref. to, 194, 225, 257 Cavendish, 460 Cayley, xii, xv, 379 ref. to, 339, 344, 346, 356, 359, 362, 365, 372, 377, 385, 404, 413, 414 Centre of gravity, 205, 222 of oscillation, 222, 282 Centres of osculation, 56 Centrifugal force, 212, 223, 248 Ceulen, van, See Ludolph Ceva, 338 Chapman, 378 Characteristics, method of, 346 Chasles, x, 345–347 ref. to, 45, 54, 56, 60, 200, 337,

INDEX 479 342, 356, 362, 365, 440 Chauvenet, 440 Chess, 107 Cheyne, 239 Chinese, 21 Chladni's figures, 450 Chree, 446, 457 Christoffel, 378, 382 Circle, 21, 27–32, 35, 47, 59, 178, 226 degrees of, 8, 315 division of, 384, 426 Circle-squarers, 2, 21, 220, 369 Cissoid, 58, 222 Clairaut, 298–300 ref. to, 284, 293, 297, 305 Clapeyron, 466 Clarke, 397 Clausius, 466 ref. to, 455, 464, 467–470 Clavius, 180 ref. to, 179 Clebsch, 363, 364 ref. to, xiii, 345, 360, 367, 376, 381, 382, 389, 397, 398, 418, 445, 447, 456–457 Clifford, 355, 356 ref. to, 346, 372, 377, 418, 441, 462 Co-ordinates, 215, 343, 359, 366, 442 first use of term, 263 Cockle, 367 Colburn, Z, 210 Colding, 465 Cole, 384 Colebrooke, 101 Colla, 166, 168 Collins, 235, 260, 264, 265, 268, 269 Colson, 237 Combinatorial School, 288, 390 Commandinus, 177 Commercium epistolicum, 239, 270 Complex of lines, 360 Complex quantities, See Imaginaries, 340, 370 Computus, 137, 138 Comte, x Concentric spheres of Eudoxus, 37 Conchoid, 58 Condensation of singularities, 422 Conform representation of surfaces, 420 Congruencies, theory of, 425 Congruency of lines, 359 Conic sections, See Geometry Arabs, 117, 130 Greek, 36, 38, 45, 46, 52–57, 63 Kepler, 195 more recent researches, 204–206, 224 Renaissance, 178 Conon, 46 ref. to, 48 Conservation of vis viva, 223 of areas, 294 of energy, 463, 465 Continued fractions, 185, 229, 293, 314 Continuity, 197, 224, 263, 341, 388, 419, 434 Contracted vein, 448 Contravariants, 381 Convergence of series, 390–396 Copernican System, 161 Copernicus, 65, 161 Correspondence, principle of, 341,

INDEX 480 346 Cosine, 191 Coss, term for algebra, 176 Cotangent, 163, 191 Cotes, 281 ref. to, 283 Coulomb, 460 Cournot, 396 Cousinery, 348 Covariants, 381, 412, 430 Cox, 357 Craig, J., 262 Craig, T., 357, 404, 414, 446 Cramer, 252 Crelle, 405 ref. to, 406 Crelle's Journal, 344 Cremona, 348 ref. to, 339, 342–344, 349, 365 Cridhara, 100 Criteria of convergence, 389–395 Crofton, 397 Crozet, 336 Ctesibius, 59 Cube numbers, 83, 128, 209 Cube, duplication of, See Duplication of the cube Cubic curves, 252, 298, 345 Cubic equations, See Algebra, 129, 130, 164–168, 172, 176, 177 Cubic residues, 427 Culmann, 348, 349 Curtze, M, 348 Curvature, measure of, 365 Curve of swiftest descent, 272, 277 Curves, See Cubic curves, Rectification, Geometry, Conic sections osculating, 263 quadrature of, 48, 57, 205, 221, 224, 235, 256 theory of, 263, 279, 281, 283, 340, 373 Cusanus, 178 Cyclic method, 111, 112 Cycloid, 199, 201, 205, 217, 221, 222, 262, 272, 279 Cyzicenus, 38 Czuber, 396 D'Alembert, 295–298 ref. to, 295, 300, 304, 308, 312–315, 450 D'Alembert's principle, 295 Damascius, 71 ref. to, 44, 121 Darboux, xiv, 365, 400, 404, 421, 422 Darwin, 438 ref. to, 448, 457 Data (Euclid's), 44 Davis, E. W., 357 Davis, W. M., 449

De Baune, 219 ref. to, 215, 259, 261 Decimal fractions, 184–187 Decimal point, 187 Dedekind, 433 ref. to, 417, 422, 434 Dee, 160 Deficiency of curves, 363 Definite integrals, 196, 390, 395, 397, 409, 422 Deinostratus. seeDinostratus, 36 De Lahire, 332, 337 Delambre, 427 Delaunay, 438 ref. to, 389

INDEX 481 Delian problem, See Duplication of the cube Del Pezzo, 356 Democritus, 32 ref. to, 17 De Moivre, 279, 281, 284 De Morgan, 368 ref. to, xi, 1, 2, 81, 111, 187, 239, 267, 271, 303, 322, 331, 339, 388, 392, 396, 415 De Paolis, 356 Derivatives, method of, 313 Desargues, 206 ref. to, 203, 214, 280, 332, 337 Desboves, 443 Descartes, 213–220 ref. to, 4, 55, 69, 131, 194, 201, 202, 220, 222, 224, 252, 256, 259, 279, 369 rule of signs, 217, 224 Descriptive geometry, 333–336, 349 Determinants, 263, 308, 323, 364, 378, 389, 423 Devanagari-numerals, 119 Dialytic method of elimination, 385 Differences, finite, See Finite differences Differential calculus, See Bernoullis, Euler, Lagrange, Laplace, etc, 233, 257–264, 274–281 alleged invention by Pascal, 202 controversy between Newton and Leibniz, 264–270 philosophy of, 274, 298, 301, 312, 336, 388 Differential equations, 277, 293, 308, 323, 365, 371, 373, 388, 397–404 Differential invariants, 381 Dingeldey, 367 Dini, 393 ref. to, 422 Dinostratus, 36 ref. to, 28 Diocles, 58 Diodorus, 11, 46, 67 Diogenes Laertius, 19, 37 Dionysodorus, 62 Diophantus, 85–89 ref. to, 63, 70, 99, 107, 110, 111, 121, 123, 124, 127, 128, 208, 434 Directrix, 56, 70 Dirichlet, 428–430 ref. to, xiv, 209, 339, 390, 394, 395, 406, 415, 416, 418, 422, 433, 462 Dissipation of energy, 467 Divergent parabolas, 253, 298 Divergent series, 297, 392 Division of the circle, 8, 316, 383, 426 Diwani-numerals, 118 Donkin, 443 Dositheus, 46 Dostor, 379 Dov´ e, 448 D'Ovidio, 356 Dronke, xiii Duality, 337, 346, 358 Duhamel, 388, 453 D¨ uhring, E., xi Duillier, 267 Duodecimals, 144, 147 Dupin, 335, 336 ref. to, 349, 366

INDEX 482 Duplication of the cube, 26–29, 35, 36, 51, 57, 178 Dur` ege, 413 ref. to, 360, 367 D¨ urer, A., 181 D¨ using, 396 Dyck, See Groups, 367 Dynamics, 371, 440–445 Dziobek, xiv, 440 Earnshaw, 447 Earth figure of, 299, 340 rigidity of, 457 size of, 249, 250 Eddy, 349 Edfu, 13, 61 Edgeworth, 397 Egyptians, 10–18, 21 Eisenlohr, 389 Eisenstein, 430 ref. to, 413, 416, 426, 431, 433 Elastic curve, 276 Elasticity, 323, 451–457 Electricity, 460–464 Electro-magnetic theory of light, 460 Elements (Euclid's), See Euclid, 41–45, 70, 121, 132, 145, 147, 148, 154, 156, 157, 159 Elimination, See Equations, 291, 358, 361, 385 Elizabeth, Princess, 219 Ellipsoid (attraction of), 250, 322, 325, 332, 347, 426, 441, 442 motion of, 447 Elliptic co-ordinates, 442 Elliptic functions, 280, 324, 325, 345, 383, 402, 403, 405–413, 423, 427, 432 Elliptic geometry, See Non-Euclidean geometry Elliptic integrals, 287, 293, 383, 406, 408 Ely, 434 Encke, 427 Energy, conservation of, 463, 465 Enestr¨ om, xi Enneper, 411 ref. to, xiv Entropy, 467 Enumerative geometry, 346 Epicycles, 59 Epping, ix, 9 Equations, See Cubic equations, Algebra, Theory of numbers numerical, 170, 307, 328 solution of, 16, 173, 177, 217, 291, 302, 307, 323, 406 theory of, 86, 192, 220, 224, 251, 279, 281, 291, 382–386 Eratosthenes, 51 ref. to, 28, 40, 46, 82 Errors, theory of, See Least squares Espy, 448 Ether, luminiferous, 459 Euclid, 40–45, 81 ref. to, 18, 24, 25, 29, 34, 35, 37, 38, 49, 53, 57, 61, 66, 67, 70, 83, 84, 90, 94, 112, 121, 125, 132, 145, 147, 158, 160, 167, 188, 327, 353 Euclidean space, See Non-Euclidean geometry Eudemian Summary, 18, 23, 34, 37, 38, 40 Eudemus, 18, 25, 52, 53, 80 Eudoxus, 36, 37 ref. to, 17, 32, 35, 36, 40, 42, 58

INDEX 483 Euler, 288–295 ref. to, 89, 111, 209, 278, 280, 286, 290, 300, 301, 304, 307, 308, 311, 312, 317, 324–326, 334, 366, 369, 390, 425, 428, 437, 450, 451, 468 Eulerian integrals, 325 Eutocius, 71 ref. to, 52, 62, 75 Evolutes, 56, 223 Exhaustion, method of, 30, 32, 38, 41, 48, 196 Exponents, 155, 176, 186, 188, 218, 234, 280 Factor-tables, 429 Fagnano, 280 Fahri des Al Karhi, 128 Falsa positio, 107, 170 Faraday, 464 Favaro, xiii Faye, 440 Fermat, 201, 208–212 ref. to, 200, 201, 206, 230, 293, 307, 308, 428 Fermat's theorem, 209, 293 Ferrari, 169 ref. to, 168, 307 Ferrel, 448 ref. to, 438 Ferro, Scipio, 165 Fibonacci, See Leonardo of Pisa Fiedler, 349, 363, 382 Figure of the earth, 299, 340 Finæus, 185 Fine, xiii Finger-reckoning, 72, 137 Finite differences, 279, 282, 293, 314, 323, 400 Fink, xii Fitzgerald, 460 Fl¨ achenabbildung, 364 Flamsteed, 254 Flexure, theory of, 454 Floridas, 165, 167 Fluents, 238, 239 Fluxional controversy, 263–270 Fluxions, 232, 235–247, 388 Focus, 56, 70, 197 Fontaine, 293, 296 Forbes, 463 Force-function, See Potential, 462 Forsyth, xii, 381, 401, 422 Four-point problem, 397 Fourier, 328–330 ref. to, 203, 297, 409, 415, 428 Fourier's series, 329, 394, 395, 428, 451 Fourier's theorem, 328 Fractions, See Arithmetic Babylonian, 7 continued, 184, 229, 293, 314 decimal, 184, 185 duodecimal, 144, 147 Egyptian, 15 Greek, 29, 74, 75 Hindoo, 109 Middle Ages, 139, 144 Roman, 90 sexagesimal, 7, 65, 75, 77, 147 Franklin, 381, 434 Frantz, xiv Fresnel, 458 Fresnel's wave-surface, 243, 365 Fr´ ezier, 333 Fricke, 412 Friction, theory of, 446 Frobenius, 378, 401, 402 Frost, 367 Froude, 444, 448

INDEX 484 Fuchs, 400 ref. to, 401, 402 Fuchsian functions, 403, 419 Fuchsian groups, 402 Functions, See Elliptic functions, Abelian functions, Hyperelliptic functions, Theta functions, Beta function, Gamma function, Omega function, Sigma function, Bessel's function, Potential arbitrary, 305, 329 definition of, 415 theory of, 311, 313, 402, 415–422 Funicular polygons, 348 Gabir ben Aflah, 133 ref. to, 147 Galileo, 212 ref. to, 50, 161, 187, 195, 197, 199, 218 Galois, 384 Gamma function, 289 Garbieri, 378 Gases Kinetic theory of, 468–471 Gauss, 423–429 ref. to, 89, 183, 287, 289, 292, 307, 322, 339, 342, 351, 353, 354, 365–367, 370, 373, 378, 379, 383, 385, 388, 390, 399, 406, 409, 416, 422, 435, 465 Gauss' Analogies, 427 Geber, See Gabir ben Aflah Geber's theorem, 134 Gellibrand, 191 Geminus, 62 ref. to, 52, 58, 66 Genocchi, 426 Geodesics, 290, 442 Geodesy, 426 Geometry, See Curves, Surfaces, Curvature, Quadrature, Rectification, Circle analytic, 216–220, 222, 224, 334, 358–367 Arabic, 120, 124, 127, 130, 132 Babylonian, 9 descriptive, 333–336, 349 Egyptian, 10–14 Greek, 18–71, 79 Hindoo, 112, 113 Middle Ages, 140, 144, 147, 148, 151, 152 modern synthetic, 279, 332–338, 341–358 Renaissance, 159, 177, 179, 184, 194 Roman, 92 Gerard of Cremona, 146 Gerbert, 140–144 Gergonne, 346 ref. to, 207, 337 Gerhardt, xi, 264, 268, 271 Gerling, 427 Germain, Sophie, 452 ref. to, 450 German Magnetic Union, 427 Gerstner, 454 Gibbs, 467 ref. to, xii, 372 Giovanni Campano, 148 Girard, 193 ref. to, 147, 187 Glaisher, 434 ref. to, 378, 382, 429, 437 Glazebrook, 464 ref. to, xv Gobar numerals, 94, 119

INDEX 485 Godfrey, 253 Golden section, 37 G¨ opel, 413 Gordan, 364, 381, 385 Gournerie, 349, 362 Goursat, 400 ref. to, 408 Gow, ix, 40 Graham, xii Grammateus, 175 Grandi, 291 Graphical statics, 340, 348 Grassmann, 372–375 ref. to, 342, 354, 369, 370, 441 Gravitation, theory of, 248, 300, 315, 321 Greeks, 17–88 Green, 461 ref. to, 417, 447, 453, 455, 459, 461 Greenhill, 413, 446 Gregorian Calendar, 178

Gregory, David F, 250, 331, 368 Gregory, James, 265, 283 Gromatici, 92 Groups, theory of, 382–384, 401–403 Grunert, 366 ref. to, 373 Gua, de, 279 Gubar-numerals, 95, 119 Gudermann, 412 Guldin, 194 ref. to, 68, 199 Guldinus, See Guldin Gunter, E., 191 G¨ unther, S., ix–xi, 379 G¨ utzlaff, 412 Haan, 390 Haas, xiii Hachette, 335, 349 Hadamard, 429 Hadley, 254 Hagen, 322 Halifax, 156 ref. to, 157 Halley, 52, 248, 249, 303 Halley's Comet, 300, 436 Halphen, 363 ref. to, 346, 367, 381, 401, 402, 413 Halsted, x, 352 Hamilton's numbers, 383 Hamilton, W., 214, 368 Hamilton, W. R., 370, 371 ref. to, 309, 339, 340, 365, 368, 369, 374, 378, 383, 398, 441, 442, 458, 468 Hammond, J, 381 Hankel, 375 ref. to, ix, x, 32, 108, 111, 331, 379, 395, 422 Hann, 449 Hansen, 437 Hanus, 379 Hardy, 203 Harkness, 422 Harmonics, 64 Haroun-al-Raschid, 121 Harrington, 440 Harriot, 193 ref. to, 171, 177, 188, 218, 224 Hathaway, xii Heat, theory of, 466–469 Heath, 357 Heaviside, 372, 464, 465 Hebrews, 21 Hegel, 435 Heine, 396

INDEX 486 ref. to, 422, 434 Helen of geometers, 217 Helicon, 37 Heliotrope, 423 Helmholtz, 464 ref. to, 354, 355, 445, 450, 459, 463, 467, 468 Henrici, xiv Henry, 463 Heraclides, 52 Hermite, 411 ref. to, xiv, 383, 384, 400, 404, 408, 414, 422, 433 Hermotimus, 38 Herodianic signs, 73 Heron the Elder, 59 ref. to, 58, 62, 75, 93, 113, 121, 152, 163 Herschel, J. F. W., 451 ref. to, x, 322, 330, 415 Hesse, 360–362 ref. to, 344, 360, 363, 378, 384, 385, 389, 399, 440 Hessian, 344, 361, 381 Heuraet, 221 Hexagrammum mysticum, 206, 344 Hicks, 445, 448 Hilbert, 382 Hill, 439 Hindoos, 97–115 ref. to, 3 Hipparchus, 59 ref. to, 62, 65 Hippasus, 24 Hippias of Elis, 28 Hippocrates of Chios, 28, 31, 35 Hippopede, 58 Hirn, 467 History of mathematics, its value, 1–4 Hodgkinson, 454 H¨ older, O.,See Groups Holmboe, 391, 405, 408 Homogeneity, 341, 359 Homological figures, 206 Honein ben Ishak, 121 Hooke, 248 Hoppe, 357 Horner, 171, 385 Hospital, l', 279 Ho¨ uel, 372 Hovarezmi, 122 ref. to, 124, 127, 132, 145, 147 Hudde, 220 ref. to, 235 Hurwitz, 418 Hussey, 440 Huygens, 221–223 ref. to, 206, 212, 219, 248, 249, 255, 272, 299, 458 Hyde, 375 Hydrodynamics, See Mechanics, 277, 296, 443–448 Hydrostatics, See Mechanics, 50, 296 Hypatia, 70 ref. to, 42 Hyperbolic geometry, See Non-Euclidean geometry Hyperelliptic functions, 340, 382, 406, 413, 419 Hyperelliptic integrals, 410 Hypergeometric series, 390, 421 Hyperspace, 354, 355 Hypsicles, 59 ref. to, 7, 44, 82, 121 Iamblichus, 84 ref. to, 11, 24, 79

INDEX 487 Ibbetson, 457 Ideal numbers, 433 Ideler, 37 Iehuda ben Mose Cohen, 147 Ignoration of co-ordinates, 443 Images, theory of, 445 Imaginary geometry, 351 Imaginary points, lines, etc, 347 Imaginary quantities, 169, 193, 280, 334, 407, 423, 434 Imschenetzky, 399 Incommensurables, See Irrationals, 42, 44, 81 Indeterminate analysis, See Theory of numbers, 110, 116, 129 Indeterminate coefficients, 217 Indeterminate equations, See Theory of numbers, 110, 116, 128 Indian mathematics, See Hindoos Indian numerals, See Arabic numerals Indices, See Exponents Indivisibles, 197–200, 205, 224 Induction, 396 Infinite products, 407, 413 Infinite series, 228, 236, 241, 256, 288, 291, 296, 301, 313, 329, 390–395, 406, 407, 420, 423 Infinitesimal calculus, See Differential calculus Infinitesimals, 156, 196, 241, 242, 245 Infinity, 31, 156, 196, 207, 224, 313, 341, 354, 359 symbol for, 224 Insurance, 278, 396 Integral calculus, 199, 259, 405, 408, 429, 433 origin of term, 275 Interpolation, 226 Invariant, 341, 361, 378, 382, 400, 412 Inverse probability, 396 Inverse tangents

(problem of), 197, 219, 255, 258, 259 Involution of points, 70, 205 Ionic School, 19–21 Irrationals, See Incommensurables, 24, 29, 79, 108, 123, 422, 434 Irregular integrals, 401 Ishak ben Honein, 121 Isidorus of Seville, 137 ref. to, 71 Isochronous curve, 272 Isoperimetrical figures, See Calculus of variations, 58, 275, 289, 303 Ivory, 331 ref. to, 322 Ivory's theorem, 332 Jacobi, 409–411 ref. to, 325, 339, 343, 359, 360, 367, 378, 385, 388, 397, 405, 407, 408, 411, 416, 426, 428, 431, 436, 440–442, 445 Jellet, 389 ref. to, 445, 455 Jerrard, 383 Jets, 446, 451 Jevons, 397 Joachim, See Rhæticus Jochmann, 469 John of Seville, 146, 185 Johnson, 404 Jordan, 384 ref. to, 397, 400, 403

INDEX 488 Jordanus Nemorarius, 156 Joubert, 412 Joule, 466 ref. to, 469, 471 Julian calendar, 93 Jurin, 274 Kaestner, 423 ref. to, 252 Kant, 319, 438 Kauffmann, See Mercator, N. Keill, 269, 270, 273 Kelland, 447, 463 Kelvin, Lord, See Thomson, W., 461–462 ref. to, 329, 366, 417, 444–446, 452, 457–459, 461, 465, 467, 471 Kempe, 381 Kepler, 195–197 ref. to, 161, 181, 183, 187, 194, 198, 202, 234, 248, 306 Kepler's laws, 195, 247 Kerbedz, xiv Ketteler, 459 Killing, 357 Kinckhuysen, 237 Kinetic theory of gases, 468–471 Kirchhoff, 463 ref. to, 360, 445–447, 453, 459, 463 Klein, 400 ref. to, 356, 357, 360, 365, 382, 384, 402–404, 412 Kleinian functions, 420 Kleinian groups, 402 Kohlrausch, 460 Kohn, 393 K¨ onig, 468 K¨ onigsberger, 411 ref. to, 401, 408, 413, 414 K¨ opcke, 446 Korkine, 434 ref. to, 397 Kornd¨ orfer, 365 Kowalevsky, 443 ref. to, 402, 410, 440 Kr¨ onig, 469 Krause, 414 Krazer, 414 Kronecker, 384 ref. to, 383, 384, 419, 426 K¨ uhn, H., 369 Kuhn, J., 254 Kummer, 432 ref. to, xiv, 209, 365, 393, 394, 399, 414, 426 Lacroix, 330, 334, 373 Laertius, 11 Lagrange, 303–314 ref. to, 4, 89, 203, 209, 213, 284, 286–288, 295, 297, 301, 318, 323, 325, 326, 341, 345, 354, 360, 365, 379, 423, 425, 428, 441, 447, 450, 451, 468 Laguerre, 356 Lahire, de, 280 Laisant, 372 La Lou` ere, 205 Lamb, 441, 446, 447, 463 Lambert, 300–301 ref. to, 3, 337, 353, 365 Lam´ e, 454 ref. to, 428, 454, 457 Lam´ e's functions, 455 Landen, 302 ref. to, 312, 325 Laplace, 314–324 ref. to, 203, 250, 279, 285, 298, 306, 325, 332, 373, 392, 396,

INDEX 489 422, 423, 435, 437, 438, 448, 450, 458, 461, 465 Laplace's coefficients, 322 Latitude, periodic changes in, 457 Latus rectum, 55 Laws of Laplace, 318 Laws of motion, 212, 218, 247 Least action, 294, 309, 468 Least squares, 322, 327, 332, 423 Lebesgue, 378, 389, 426 Legendre, 324–327 ref. to, 287, 293, 301, 310, 322, 350, 407–409, 412, 425, 428 Legendre's function, 325 Leibniz, 254–274 ref. to, 4, 183, 204, 233, 242, 243, 245, 275, 280, 291–293, 312, 367, 390, 415 Lemoine, 397 Lemonnier, 311 Leodamas, 38 Leon, 38 Leonardo of Pisa, 148 ref. to, 154, 159 Leslie, x Le Verrier, 438 ref. to, 439 Levy, 349, 457 Lewis, 445 Lexis, 396 Leyden jar, 463 L'Hospital, 279 ref. to, 267, 272 Lie, 403 ref. to, 398, 408 Light, theory of, 253, 455 Limits, method of, 246, 312 Lindel¨ of, 389 Lindemann, 367 ref. to, 3, 356, 415 Linear associative algebra, 376 Lintearia, 276 Liouville, 430 ref. to, 366, 416, 426, 431, 443 Lipschitz, 357 ref. to, 395, 437, 446 Listing, 367 Lloyd, 458 Lobatchewsky, 350 ref. to, 339, 352 Local probability, 397 Logarithmic criteria of convergence, 393 Logarithmic series, 230 Logarithms, 184, 187–192, 196, 230, 282, 291 Logic, 43, 368, 377, 399 Lommel, 437, 459 Long wave, 446 Loomis, 448 Lorenz, 459 Loria, xii Loud, 348 Lucas de Burgo, See Pacioli Ludolph, 179 Ludolph's number, 179 Lune, squaring of, 29 L¨ uroth, 418 ref. to, 422 MacCullagh, 362 ref. to, 458 Macfarlane, 372

Machine, arithmetical, 255, 330 Maclaurin, 283 ref. to, 274, 284, 326, 332, 338 Macmahon, 381 Magic squares, 107, 157, 280 Magister matheseos, 158 Main, 440

INDEX 490 Mainardi, 389 Malfatti, 344, 382 Malfatti's problem, 344, 364 Mansion, 398 Marie, Abb´ e, 324 Marie, C. F. M., 348 Marie, M., x, 60, 200 Mathieu, 456 ref. to, 412, 440, 457, 465 Matrices, 373, 377 Matthiessen, x Maudith, 156 ref. to, 163 Maupertius, 294, 298, 468 Maurolycus, 177 ref. to, 181 Maxima and minima, 56, 202, 217, 219, 242, 284, 389, 395, 398 Maxwell, 463 ref. to, 349, 439, 446, 455, 460, 462, 465, 468–470 Mayer, 465 ref. to, 438 McClintock, 382 McColl, 397 McCowan, 447 McCullagh, 362, 459 McMahon, 382 Mechanics, See Dynamics, Hydrodynamics, Hydrostatics, Graphic statics, Laws of motion, Astronomy, D'Alembert's principle Bernoullis, 275, 276 Descartes, Wallis, Wren, Huygens, Newton, 218, 222, 223, 246–251 Euler, 294 Greek, 26, 39, 49 Lagrange, 309 Laplace, 319 Leibniz, 264 more recent work, 338, 382, 404, 439–445, 468 Stevin and Galileo, 184, 211 Taylor, 282 Meissel, 411 Menæchmus, 36 ref. to, 35, 38, 53, 130 Menelaus, 64 ref. to, 66, 182 Mercator, G., 365 Mercator, N., 229 ref. to, 255 M´ er´ e, 211 Mersenne, 209, 223 Mertens, 391, 429 Meteorology, 449–450 Method of characteristics, 346 Method of exhaustion, 32 ref. to, 37, 41, 48, 196 Metius, 179 Meunier, 366 Meyer, A., 396, 398 Meyer, G. F., 390 Meyer, O. E., 446, 457, 470 M´ eziriac, 208 ref. to, 308 Michelson, 460 Middle Ages, 135–158 Midorge, 202 Minchin, 445 Minding, 366 Minkowsky, 432 Mittag-Leffler, 419 M¨ obius, 342 ref. to, 342, 373, 374, 427, 438, 440

INDEX 491 Modern Europe, 160 et seq. Modular equations, 384, 412 Modular functions, 412 Mohammed ben Musa Hovarezmi, 122 ref. to, 124, 127, 132, 145, 147 Mohr, 349 Moigno, 389 Moivre, de, 279, 281, 284 Mollweide, 427 Moments in fluxionary calculus, 239 Monge, 332–336 ref. to, 288, 301, 329, 342, 349, 366, 397 Montmort, de, 279 Montucla, xi, 200 Moon, See Astronomy Moore, 384 Moors, 133, 134, 144 Moral expectation, 278 Morley, 422 Moschopulus, 157 Motion, laws of, 212, 218, 247 Mouton, 255 Muir, xiii, 379 M¨ uller, J.,See Regiomontanus M¨ uller, x Multi-constancy, 455, 456 Multiplication of series, 390, 392 Musa ben Sakir, 125 Musical proportion, 8 Mydorge, 206 Nachreiner, 378 N¨ agelbach, 378 Napier's rule of circular parts, 192 Napier, J., 188, 189 ref. to, 181, 187, 190, 191 Napier, M., x Nasir Eddin, 132 Nautical almanac, United States, 439 Navier, 452 ref. to, 446, 455 Nebular hypothesis, 318 Negative quantities, See Algebra, 108, 176, 218, 298, 433 Negative roots, See Algebra, 108, 130, 169, 172, 176, 193 Neil, 221 ref. to, 230 Neocleides, 38 Neptune, discovery of, 438 Nesselmann, 88 Netto, 384 Neumann, C., 437 ref. to, 360, 367, 459 Neumann, F. E., 464 ref. to, 360, 363, 455, 457, 462, 468 Newcomb, 439 ref. to, 357, 457 Newton, 233–254 ref. to, 4, 58, 69, 171, 201, 216, 222, 223, 227, 232, 277, 283, 284, 293, 296, 298, 300, 304, 312, 328, 332, 337, 346, 352, 369, 385, 390, 434, 444, 450 Newton's discovery of binomial theorem, 227, 228 Newton's discovery of universal gravitation, 248 Newton's parallelogram, 252 Newton's Principia, 222, 242, 246–250, 266, 271, 281 Newton, controversy with Leibniz, 264–271 Nicolai, 427 Nicole, 279

INDEX 492 Nicolo of Brescia, See Tartaglia Nicomachus, 83 ref. to, 67, 94 Nicomedes, 58 Nieuwentyt, 274 Nines, casting out the, 123 Niven, 463 Nolan, 438 Non-Euclidean geometry, 43, 349–357 Nonius, 178 ref. to, 179 Notation, See

Exponents, Algebra Arabic notation, 3, 84, 101, 118, 129, 148–150, 184 Babylonian numbers, 5–7 decimal fractions, 186 differential calculus, 238, 257, 258, 303, 313, 330 Egyptian numbers, 13 Greek numbers, 73 in algebra, 16, 86, 107, 155, 172, 174, 175, 186, 193 Roman, 90 trigonometry, 289 N¨ other, 363, 365, 384, 415 Numbers amicable, 78, 125, 133 defective, 78 definitions of numbers, 434 excessive, 78 heteromecic, 78 perfect, 78 theory of numbers, 63, 87, 109, 125, 138, 152, 207–211, 293, 308, 326, 422–434 triangular, 209 Numbers of Bernoulli, 276 Numerals, See Apices Arabic, 100, 118, 119, 129 Babylonian, 4–7 Egyptian, 14 Greek, 73 Oberbeck, 450 Œnopides, 21 ref. to, 17 Ohm, M, 369 Ohrtmann, x Olbers, 424, 435 Oldenburg, 265 Olivier, 349 Omega-function, 411 Operations, calculus of, 340 Oppolzer, 440 Optics, 45 Oresme, 156 ref. to, 186 Orontius, 178 Oscillation, centre of, 223, 282 Ostrogradsky, 388, 443 Otho, 164 Oughtred, 194 ref. to, 171, 187, 234 Ovals of Descartes, 217 π: values for Arabic, 125 Archimedean, 47 Babylonian and Hebrew, 8 Brouncker's, 229 Egyptian, 12 Fagnano's, 280 Hindoo, 114 Leibniz's, 255 Ludolph's, 179 proved to be irrational, 301, 327 proved to be transcendental, 1 selection of letter π, 290

INDEX 493 Wallis', 226 Pacioli, 157 ref. to, 155, 165, 176, 180, 184, 227 Padmanabha, 100 Palatine anthology, 84, 138 Pappus, 67–70 ref. to, 40, 45, 52, 57, 58, 63, 75, 76, 178, 207, 216 Parabola, See Geometry, 48, 80, 230 semi-cubical, 221 Parabolic geometry, See Non-Euclidean geometry Parallelogram of forces, 213 Parallels, 43, 327, 349, 351, 352, 356 Parameter, 55 Partial differential equations, 241, 296, 335, 397 et seq., 442 Partition of numbers, 433 Pascal, 203–206 ref. to, 207, 211, 228, 256, 280, 330, 332, 337, 361 Pascal's theorem, 207 Peacock, 330 ref. to, X, 150, 154, 187, 330, 368 Pearson, 456 Peaucellier, 380 Peirce, B., 376 ref. to, 339, 369, 439, 445 Peirce, C. S., 376 ref. to, 43, 357, 375 Peletarius, 193 Pell, 171, 175, 210, 255 Pell's problem, 112, 210 Pemberton, 234 Pendulum, 222 Pepin, 426 Perier, Madame, xi Periodicity of functions, 406, 408 Pernter, J. M., 450 Perseus, 58 Perspective, See Geometry, 206 Perturbations, 318 Petersen, 426 Pfaff, 397, 398 ref. to, 422 Pfaffian problem, 398 Pherecydes, 22 Philippus, 38 Philolaus, 25 ref. to, 32, 78 Philonides, 53 Physics, mathematical, See Applied mathematics Piazzi, 435 Picard, E., 404, 408, 420 Picard, J., 249, 250 Piddington, 448 Piola, 453 Pitiscus, 164 Pl¨ ucker, 358–360 ref. to, 354, 359, 365 Plana, 437, 451, 462 Planudes, M., 157 Plateau, 446 Plato, 33–36 ref. to, 4, 10, 17, 25, 36, 37, 39, 40, 72, 78 Plato of Tivoli, 126, 145 Plato Tiburtinus, See Plato of Tivoli Platonic figures, 44 Platonic School, 33–39 Playfair, x, 181 Plectoidal surface, 69 Plus and minus, signs for, 173 Pohlke, 349

INDEX 494 Poincar´ e, 400 ref. to, xiv, 402–404, 411, 419, 429, 448, 468 Poinsot, 440 ref. to, 440 Poisson, 452 ref. to, 203, 347, 385, 388, 409, 437, 441, 446, 450, 452, 455, 458, 461, 462 Poncelet, 337, 338 ref. to, 207, 335, 342, 356, 359, 454 Poncelet's paradox, 359 Porisms, 45 Porphyrius, 63 Potential, 323, 417, 461 Poynting, 464, 465 Preston, 467 Primary factors, Weierstrass' theory of, 412, 419 Prime and ultimate ratios, 230, 246, 312 Prime numbers, 43, 51, 81, 209, 429 Princess Elizabeth, 219 Principia (Newton's), 222, 242, 246–250, 266, 271, 281 Pringsheim, 392–394 Probability, 184, 211, 223, 275, 278, 279, 285, 294, 314, 321, 332, 396, 397 Problem of Pappus, 69 Problem of three bodies, 294, 297, 439 Proclus, 71 ref. to, 18, 21, 38, 40, 43, 44, 58, 62, 67 Progressions, first appearance of arithmetical and geometrical,

8 Projective geometry, 358 Proportion, 19, 25, 26, 29, 37, 42, 44, 77, 79 Propositiones ad acuendos iuvenes, 138 Prym, 414 Ptolemæus, See Ptolemy Ptolemaic System, 64 Ptolemy, 65–67 ref. to, 7, 9, 62, 63, 113, 120, 122, 125, 126, 134, 160, 364 Puiseux, 416 Pulveriser, 110 Purbach, 156 ref. to, 162 Pythagoras, 21–25, 77–80 ref. to, 4, 17, 20, 26, 32, 42, 73, 95, 112, 156 Pythagorean School, 21–25 Quadratic equations, See Algebra, Equations, 88, 107, 124, 128, 130 Quadratic reciprocity, 293, 326, 425 Quadratrix, 28, 36, 68, 69 Quadrature of curves, 48, 56, 205, 220, 224, 256, 258 Quadrature of the circle, See Circle; also see Circle-squarers, π Quaternions, 371 ref. to, 369 Quercu, a, 179 Quetelet, 396 ref. to, x Raabe, 393 Radau, 440 Radiometer, 470 Rahn, 175

INDEX 495 Ramus, 178 Rankine, 466 ref. to, 467 Rari-constancy, 455 Ratios, 434 Rayleigh, Lord, 451 ref. to, 437, 447, 448, 464, 465 Reaction polygons, 349 Reciprocal polars, 337 Reciprocants, 381, 421 Recorde, 175 ref. to, 183 Rectification of curves, See Curves, 196, 206, 221, 230 Redfield, 448 Reductio ad absurdum, 32 Reech, 444 Regiomontanus, 162, 163 ref. to, 161, 173, 177, 178, 181, 184, 185 Regula aurea, See Falsa positio Regula duorum falsorum, 123 Regula falsa, 123 Regular solids, 23, 36, 38, 44, 59, 127, 195 Reid, 448 Reiff, xii Renaissance, 161–181 Resal, 440 Reye, 348 ref. to, 356 Reynolds, 447 Rhæticus, 164 ref. to, 161, 164 Rheticus, See Rhæticus Rhind papyrus, 10–16 Riccati, 280 ref. to, 277 Richard of Wallingford, 156 Richelot, 412 ref. to, 360, 363 Riemann, 416–419 ref. to, 354, 355, 364, 367, 395, 399, 413–415, 422, 429, 448, 457, 464 Riemann’s surfaces, 418 ref. to, 415 Roberts, 365 Roberval, 199 ref. to, 199, 217, 223 Rolle, 280 ref. to, 274 Roman mathematics in Occident, 136–143 Romans, 89–96 Romanus, 179 ref. to, 165, 172, 179 R¨ omer, 232 Rosenberger, xv Rosenhain, 413 ref. to, 412 Roulette, 199 Routh, 444 ref. to, 445, 463 Rowland, 446, 460, 464, 465 Rudolff, 175 Ruffini, 382 R¨ uhlmann, 468 Rule of signs, 218, 224 Rule of three, 107, 123 Saccheri, 353 Sachse, xiii Sacro Bosco, See Halifax Saint-Venant, 455 ref. to, 375, 446, 454, 459 Salmon, xiii, 343, 362–364, 385 Sand-counter, 76, 104 Sarrau, 459 Sarrus, 389

INDEX 496 Saturn’s rings, 223, 439 Saurin, 279 Savart, 452 Scaliger, 179 Schellbach, 345 Schepp, 422 Schering, 357 ref. to, 417, 426 Schiaparelli, 37 Schl¨ afli, 357 ref. to, 394, 412 Schlegel, 376 ref. to, XII, 357 Schlessinger, 349 Schl¨ omilch, 437 Schmidt, xiii Schooten, van, 220 ref. to, 221, 234 Schreiber, 336, 349 Schr¨ oter, H., 365 ref. to, 344, 411 Schr¨ oter, J. H., 436 Schubert, 346 Schumacher, 427 ref. to, 405 Schuster, xiv Schwarz, 421 ref. to, 346, 395, 402, 404, 422 Schwarzian derivative, 421 Scott, 378 Screws, theory of, 441 Secants, 164 Sectio aurea, 37 Section, the golden, 37 Seeber, 433 Segre, 356 Seidel, 396 Seitz, 397 Selling, 433 Sellmeyer, 459 Semi-convergent series, 392 Semi-cubical parabola, 221 Semi-invariants, 382 Serenus, 63 Series, See Infinite series, Trigonometric series, Divergent series, Absolutely convergent series, Semi-convergent series, Fourier’s series, Uniformly convergent series, 129, 285 Serret, 365 ref. to, 398, 399, 440, 443 Servois, 331, 335, 337 Sexagesimal system, 7, 65, 74, 77, 146 Sextant, 253 Sextus Julius Africanus, 67 Siemens, 450 Sigma-function, 413 Signs, rule of, 218, 224 Similitude (mechanical), 444 Simony, 367 Simplicius, 71 Simpson, 290 Simson, 338 ref. to, 42, 45 Sine, 114, 117, 126, 134, 144, 163 origin of term, 126 Singular solutions, 262, 308, 323 Sluze, 220 ref. to, 258, 260 Smith, A., 445 Smith, H., 430,

431 ref. to, xiv, 411, 434 Smith, R., 281 Sohnke, 412 Solid of least resistance, 250 Solitary wave, 446 Somoff, 445

INDEX 497 Sophist School, 26–32 Sosigenes, 93 Sound, velocity of, See Acoustics, 314, 323 Speidell, 192 Spherical Harmonics, 287 Spherical trigonometry, 64, 133, 325, 343 Spheroid (liquid), 448 Spirals, 48, 69, 276 Spitzer, 389 Spottiswoode, 378 ref. to, xii, 340 Square root, 75, 108, 185 Squaring the circle, See Quadrature of the circle Stahl, 357 Star-polygons, 24, 156, 181 Statics, See Mechanics, 50, 212 Statistics, 396 Staudt, von, See Von Staudt Steele, 445 Stefano, 446 Steiner, 343, 344 ref. to, 342, 346, 347, 359, 362, 364, 372, 405, 416 Stereometry, 36, 37, 44, 195 Stern, 416, 426 Stevin, 186 ref. to, 155, 188, 212 Stevinus, See Stevin Stewart, 338 Stifel, 175 ref. to, 173, 175, 180, 188 Stirling, 284 Stokes, 445 ref. to, 396, 445, 447, 451, 453, 455, 459, 465 Story, 357 Strassmaier, ix Strauch, 389 Stringham, 357 Strings, vibrating, 281, 296, 305 Strutt, J. W., See Rayleigh, 451 Struve, 427 Sturm's theorem, 384 Sturm, J. C. F., 385 ref. to, 207, 328, 443, 445 Sturm, R., 344 St. Vincent, Gregory, 221, 229 Substitutions, theory of, 340, 384 Surfaces, theory of, 290, 334, 344, 348, 361, 365 Suter, x Swedenborg, 319 Sylow, 384 ref. to, 408 Sylvester, 380 ref. to, xiii, 344, 361–363, 372, 377, 378, 382, 385, 397, 421, 429, 432, 441, 462 Sylvester II. (Gerbert), 138–143 Sylvester ref. to, 252 Symmetric functions, 291, 382, 385 Synthesis, 35 Synthetic geometry, 341–358 Taber, 377 Tabit ben Korra, 125 ref. to, 121 Tait, 330, 372, 445, 453, 467 Tangents direct problem of, 230, 259 in geometry, 71, 201, 216 in trigonometry, 127, 163, 164 inverse problem of, 197, 220, 256, 258, 259 Tannery, 400 ref. to, 422

INDEX 498 Tartaglia, 166–168 ref. to, 176, 177 Tautochronous curve, 222 Taylor's theorem, 282, 312, 314, 388, 399 Taylor, B., 281 ref. to, 273, 297 Tchebycheff, 429 Tchirnhausen, 280 ref. to, 260, 262, 307, 382 Tentative assumption, See Regula falsa, 86, 107 Thales, 19, 20 ref. to, 17, 22, 24 Theætetus, 38 ref. to, 40, 42, 81 Theodorus, 80 ref. to, 33 Theodosius, 62 ref. to, 125, 145, 147 Theon of Alexandria, 70 ref. to, 42, 59, 63, 75, 94 Theon of Smyrna, 63, 67, 83 Theory of equations, See Equations Theory of functions, See Functions, 311, 313, 401–403, 405–422 Theory of numbers, 63, 87, 109, 125, 138, 151, 207–211, 293, 308, 326, 422–433 Theory of substitutions, 384, 412 Thermodynamics, 450, 464–468 Theta-fuchsians, 403 Theta-functions, 410, 411, 413, 443 Theudius, 38 Thomae, 411, 422 Thom´ e, 401 ref. to, 402 Thomson's theorem, 418 Thomson, J., 449 Thomson, J. J., 445 ref. to, 462, 464 Thomson, Sir William, See Kelvin (Lord), 461, 462 ref. to, 330, 367, 417, 445, 447, 453, 457, 459–461, 466, 467, 470 Three bodies, problem of, 294, 297, 439 Thymaridas, 84 Tides, 323, 447 Timæus of Locri, 33 Tisserand, 440 Todhunter, 389 ref. to, xi, 437 Tonstall, 184 Torricelli, 199 Trajectories, 272, 277 Triangulum characteristicum, 256 Trigonometric series, See Fourier's series, 329, 395, 416 Trigonometry, 59, 65, 114–115, 126, 127, 133, 156, 162, 163, 179, 185, 186, 191, 277, 282, 285, 290, 301 spherical, 66, 133, 325, 343 Trisection of angles, 26, 35, 57, 177 Trochoid, 199 Trouton, 460 Trudi, 378 Tucker, xiv Twisted Cartesian, 363 Tycho Brahe, 127, 161, 195 Ubaldo, 213 Ultimate multiplier, theory of, 442

INDEX 499 Ulug Beg, 132 Undulatory theory of light, 223, 395, 442, 457–459 Universities of Cologne, Leipzig, Oxford, Paris, and Prague, 157 Valson, xiii Van

Ceulen, See Ludolph Vandermonde, 323 ref. to, 307, 323 Van Schooten, 220 ref. to, 221, 234 Variation of arbitrary consonants, 440 Varignon, 279 ref. to, 275 Varying action, principle of, 340, 371, 442 Venturi, 60 Veronese, 356 ref. to, 357 Versed sine, 114 Vibrating rods, 451 Vibrating strings, 281, 296, 305 Vicat, 454 ref. to, 455 Victorius, 91 Vieta, 170 ref. to, 57, 165, 176, 178, 179, 194, 228, 234, 252, 307 Vincent, Gregory St., 221, 229 Virtual velocities, 39, 309 Viviani, 200 Vlacq, 191 Voigt, xiv, 426, 459 Volaria, 276 Von Helmholtz, See Helmholtz Von Staudt, 347, 348 ref. to, 340, 342, 344 Vortex motion, 446 Vortex rings, 445 Voss, 357 ref. to, 392 Waldo, 449 Walker, 376 Wallis, 223–226 ref. to, 113, 187, 205, 208, 218, 219, 229, 234, 267 Waltershausen, xii Wand, 467 Wantzel, 382 Warring, 307, 384 Watson, J. C., 440 Watson, S., 397 Wave theory,See Undulatory theory Waves, 446–450 Weber, H. H., 414 Weber, W. E., 460 ref. to, 416, 423, 453, 463, 464 Weierstrass, 419 ref. to, 382, 395, 411–413, 419, 421, 422, 434 Weigel, 255 Weiler, 397 Werner, 177 Wertheim, 455 Westergaard, 396 Wheatstone, 450 Whewell, ix, 49, 294 Whiston, 251 Whitney, 101 Widmann, 175 Wiener, xii Williams, 311 Wilson, 308 Wilson’s theorem, 308 Winds, 448–450

INDEX 500 Winkler, 457 Witch of Agnesi, 302 Wittstein, xii Woepcke, 96, 119 Wolf, C., 281 ref. to, 194 Wolf, R., xii Wolstenholme, 397 Woodhouse, 389 Wren, 206 ref. to, 219, 230, 248, 334 Wronski, 378 Xenocrates, 33 Xylander, 177 Young, 458 ref. to, 450, 451 Zag, 147 Zahn, xiii Zehfuss, 378 Zeller, 426 Zeno, 31 Zenodorus, 58 Zero (symbol for), 7, 101 origin of term, 149 Zeuthen, 365 ref. to, ix, 346 Zeuxippus, 46 Zolotareff, 433 ref. to, 434

THE MACMILLAN COMPANY’S PUBLICATIONS ON Mathematics and Physics. ALGEBRA. ALDIS: A Text-book of Algebra. By W. S. Aldis. 12mo. $1.90. BALL (W. W. R.): Elementary Algebra. 16mo. $1.25. CHRYSTAL: Algebra. An Elementary Text-book for the Higher Classes of Secondary Schools and Colleges. By G. Chrystal. Part I. New Edition. 8vo. $3.75. Part II. 8vo. $4.00. The Set. Two Volumes. $7.50. DALTON: Rules and Examples in Algebra. By the Rev.T. Dal- ton, M.A. Part I. New Edition. 18mo. 50 cents. Part II. 18mo. 60 cents. KEY TO PART I. $1.90. DUPUIS: Principles of Elementary Algebra. By Nathan F. Dupuis, M.A., F.R.S., Professor of Mathematics in Queen’s College, Kingston, Canada. 12mo. $1.10. ELSEE: Algebra. By the Rev. C. Elsee, M.A. Seventh Edition. $1.00. HALL and KNIGHT: Works by H. S. Hall , M.A., and S. R. KNIGHT, B.A. Elementary Algebra for Schools. 16mo. 90 cents. With Answers. 16mo. $1.10. Key. 12mo. $2.25. Algebraical Exercises and Examination Papers. 16mo. 60 cents. Higher Algebra for Schools. 12mo. $1.90. Key to above. 12mo. $2.60. Algebra for Beginners. 16mo. 60 cents. HAYWARD: The Algebra of Co-Planar Vectors and Trigono- metry. $2.00. 501

MACMILLAN & CO’S PUBLICATIONS ON 502 HEATH: Diaphantos of Alexandria: A Study in the History of Greek Algebra. By T. L. Heath, M.A., Fellow of Trinity College, Cambridge. 8vo. $2.25. HENSLEY: The Scholar’s Algebra. An Introductory Work on Algebra. By Lewis Hensley. 12mo. $1.10. JONES and CHEYNE: Algebraical Exercises. Progressively arranged. By the Rev. C. A. Jones, M.A., and C. H. Cheyne, M.A., F.R.A.S. New Edition. 18mo. 60 cents. Key to the Exercises. By W. Failes, M.A. 12mo. $1.90. KLEIN: On Riemann’s Theory of Algebraic Functions and their

Integrals. A Supplement to the Usual Treatises by Felix Klein . Translated from the German with the author’s permission, by Frances Hardcastle, Girton College, Cambridge. 8vo. $1.40. MACMICHAEL and SMITH: Progressive Course of Examples in Algebra. By the Rev. W. F. Macmichael and R. Prowde Smith, M.A. With Answers. $1.10. SMITH. Works byCharles Smith, M.A., Master of Sidney Sussex College, Cambridge. Elementary Algebra. 16mo. $1.10. American Edition of the Elementary Algebra. Revised by Prof. Irving Stringham, University of California.Nearly ready. Solutions to above. 12mo. $2.25. A Treatise on Algebra. 12mo. $1.90. Solutions to above. 12mo. $2.60. SMITH: Algebra Examples. By R. Prowde Smith. $1.10. TODHUNTER. Works by I. Todhunter, M.A., F.R.S., D.Sc. Algebra for Beginners. With Numerous Examples. New Edition. 18mo. 75 cents. Key. $1.75. Algebra. For the Use of Colleges and Schools. New Edition. 12mo. $1.80. Key. $2.60. ARITHMETIC. ALDIS: The Great Giant Arithmos. A Most Elementary Arithmetic for Children. By Mary Steadman Aldis. With Illustrations. 16mo. 75 cents.

MATHEMATICS AND PHYSICS. 503 ARMY PRELIMINARY EXAMINATION (Specimens of Pa- pers set at the), □□□□–□□. With Answers to the Math- ematical Questions. Subjects: Arithmetic, Algebra, Euclid, Geometrical Drawing, Geography, French, English, Dictation. 12mo. 90 cents. BRADSHAW: A Course of Easy, Arithmetical Examples for Beginners. By J. G. Bradshaw, B.A. 16mo. 60 cents. BROOK-SMITH: Arithmetic in Theory and Practice. By J. Brook-Smith, M.A., LL.B. New Edition, Revised. 12mo. $1.25. Key. 12mo. $2.60. Arithmetic for Beginners. By J. and E. J. Brook-Smith . 12mo. 50 cents. DALTON: Rules and Examples in Arithmetic. By the Rev. T. Dalton, M.A. New Edition. With Answers. 18mo. 60 cents. GOYEN: Higher Arithmetic and Elementary Mensuration. By P. Goyen, Inspector of Schools, Dunedin, New Zealand. 12mo. $1.30. HALL and KNIGHT: Arithmetical Exercises and Examina- tion Papers. With an Appendix containing Questions on Logarithms and Mensuration. By H. S. Hall , M.A., and S. R. Knight, B.A. 16mo. 60 cents. HENSLEY: Figures Made Easy. By Lewis Hensley . Stiff covers. 12mo. 13 cents. Answers to Examples. 12mo. 25 cents. The Scholar’s Arithmetic. With Answers to the Examples. 12mo. $1.10. JACKSON (S.): Commercial Arithmetic. Globe 8vo. $1.10. LOCK. Works by Rev. J. B. Lock, M.A., Senior Fellow, Assistant Tutor and Lecturer in Gonville and Caius College, Cambridge. Arithmetic for Beginners. A School Class-book of Commercial Arithmetic. 16mo. 60 cents. Key. $2.25. LOCK and SCOTT: Arithmetic for Schools. Adapted for Use in American Schools. By Rev. J. B. Lock and Professor Charlotte A. Scott, Bryn Mawr College. 70 cents. PEDLEY: Exercises in Arithmetic for the Use of Schools. Containing more than 7000 Original Examples. By S. Pedley. 12mo. $1.25.

MACMILLAN & CO’S PUBLICATIONS ON 504 PENDLEBURY: Arithmetic. With 8000 Examples. ByCharles Pendlebury, M.A., F.R.A.S. With Answers. $1.10. SMITH: Arithmetic for Schools. By C. Smith, M.A. $1.00. Revised American Edition. (In Preparation.) SMITH. Works by Barnard Smith, M.A. Arithmetic for the Use of Schools. New Edition. $1.10. Key. New Edition. $2.25. School Class-Book of Arithmetic. $1.00. Key. $1.50. Shilling Book of Arithmetic. With Answers. 18mo. 40 cents. Key. 18mo. $1.10. Examination Papers in Arithmetic. With Answers. 60 cents. Key. 18mo. $1.25. The Metric System of Arithmetic. 18mo. Paper. 8 cents. THOMAS:

Enunciations in Arithmetic, Algebra, Euclid, and Trigonometry, with a Few Examples and Notes. By P. A. Thomas, B.A., Assistant Master in Sedbergh School. Globe 8vo. 60 cents. ASTRONOMY. AIRY: Works by Sir G. B. Airy, K.C.B., formerly Astronomer-Royal. Popular Astronomy. With Illustrations. New Edition. 18mo. $1.10. Gravitation. An Elementary Explanation of the Principal Perturbations in the Solar System. New Edition. 12mo. $1.90. CHAMBERS: A Handbook of Descriptive and Practical Astronomy. By George F. Chambers, F.R.A.S. Vol. I. The Sun, Planets, and Comets. 8vo. $5.25. Vol. II. Instruments and Practical Astronomy. 8vo. $5.25. Vol. III. The Starry Heavens. (Completing the work.) 8vo. $3.50. Pictorial Astronomy. By G. F. Chambers . With Numerous Illustrations. $1.25. CHEYNE: An Elementary Treatise on the Planetary Theory. By the late C. H. H. Cheyne, M.A. Third Edition. Edited by Rev. A. Freeman, M.A. 12mo. $1.75.

MATHEMATICS AND PHYSICS. 505 CLARK and SADLER: The Star Guide. A List of the most remarkable Celestial Objects visible with small Telescopes, with their Positions for every Tenth Day in the Year, and other Astronomical Information. By Latimer Clark, F.R.A.S., and Herbert Sadler, F.R.A.S. Royal 8vo. $2.50. CLERKE: A Popular History of Astronomy during the Nine- teenth century. By Agnes M. Clerke . Third Edition, Revised and Enlarged. 8vo. $4.00. CROSSLEY, GLEDHILL, and WILSON: A Handbook of Dou- ble Stars. For the Use of Amateurs. By Edward Crossley, Joseph Gledhill, and James M. Wilson , F.R.A.S. With Illustrations. 8vo. $6.00. Corrections to the Handbook of Double Stars. 8vo. 30 cents. DURHAM: Astronomy—Sun, Moon, Stars, etc. By William Durham, F.R.S.E. 12mo. 50 cents. FORBES: The Transit of Venus. By George Forbes, B.A. With Illustrations. 12mo. $1.00. GODFRAY: Works by Hugh Godfray, M.A. An Elementary Treatise on Lunar Theory.With a Brief Sketch of the Problem up to the Time of Newton. Second Edition, Revised. 12mo. $1.60. A Treatise on Astronomy, for the Use of Colleges and Schools. 8vo. $3.25. LOCKYER: Works by J. Norman Lockyer, F.R.S., etc. The Chemistry of the Sun. With Illustrations. 8vo. $4.50. Contributions to Solar Physics. With Numerous Illustrations. 8vo. $7.50. Elementary Lessons in Astronomy. With Spectra of the Sun, Stars, and Nebulæ, and Numerous Illustrations. New Edition, Revised throughout. 16mo. $1.25. Questions on Lockyer’s Elementary Lessons in Astronomy for Schools. By J. Forbes-Robertson. 18mo. 40 cents. The Meteoritic Hypothesis. A Statement of the Spectroscopic Inquiry into the Origin of Cosmical Systems. With Numerous Illustrations. 8vo. $5.25. The Evolution of the Heavens and the Earth. With Illustrations. (In the Press.) The Dawn of Astronomy. With Illustrations. 8vo. $5.00.

MACMILLAN & CO’S PUBLICATIONS ON 506 LOCKYER and SEABROKE: Star-gazing Past and Present. By J. N. Lockyer and G. W. Seabroke. With Illustrations. 8vo. $6.00. MAIN: An Introduction to Plane Astronomy. By P. T. Main, M.A. Sixth Edition, Revised. $1.00. SMYTH: A Cycle of Celestial Objects. Observed, Reduced, and Discussed. By Admiral W. H. Smyth. Revised and Enlarged by G. F. Chambers, F.R.A.S. 8vo. $3.00. BOOK-KEEPING. FOSTER: Double Entry Elucidated. By B. W. Foster . Fourteenth Edition. Fcap. 4to. 90 cents. HAMILTONand BALL: Book-keeping. By R. G. C. Hamilton and J. Ball. 16mo. 50 cents. MEDHURST: Examination Papers on Book-keeping. Com- piled and Arranged by J. T. Medhurst , A.K.C. Second Edition. 75 cents. THORNTON: Works by J. Thornton. First Lessons in Book-keeping. 8vo. 70

cents. Key. Oblong 4to. Primer of Book-keeping. 18mo. 35 cents. Key. GEOMETRY. ALDIS (W. S.): Solid Geometry. 12mo. $1.50. BESANT (Dr. W. H.): Conic Sections treated Geometrically. 16mo. $1.10. Solutions to the Examples. 16mo. $1.00. COCKSHOTT and WALTERS: A Treatise on Geometrical Conics, in accordance with the Syllabus of the Association for the Improvement of Geometrical Teaching. By A. Cockshott and Rev. F. B. Walters, M.A. 12mo. $1.25. CONSTABLE: Geometrical Exercises for Beginners. By Samuel Constable. 12mo. 90 cents. CUTHBERTSON: Euclidian Geometry. By Francis Cuth- bertson, M.A. 16mo. $1.10.

MATHEMATICS AND PHYSICS. 507 CREMONA: Elements of Projective Geometry. By Luigi Cremona, LL.D. Translated byCharles Leudesdorf, M.A. 8vo. $3.25. DAY: Properties of Conic Sections Proved Geometrically. Part I. The Ellipse. With Problems. By the Rev. H. G. Day, M.A. 12mo. 90 cents. DEIGHTON: Euclid. The Elements. Books I.–IV. and part of Book XI. Newly translated from the Original Text, with numerous Riders and Chapters on Radical Axis, Poles and Polars, Centres of Similitude, Transversals, Harmonic Proportion, Plane Loci, etc. By Horace Deighton, M.A. Revised Edition, with symbols and abbreviations. $1.10. DODGSON: Works by Charles L. Dodgson, M.A. Euclid. Books I. and II. 12mo. 60 cents. Euclid and his Modern Rivals. Second Edition. 12mo. $1.60. Curiosa Mathematica. Part I. A New Theory of Parallels. 12mo. 75 cents. DREW: A Geometrical Treatise on Conic Sections. By W. H. Drew, M.A. Sixth Edition. 12mo. $1.25. DUPUIS; Elementary Synthetic Geometry of the Point, Line, and Circle in the Plane. By N. F. Dupuis, M.A. 16mo. $1.10. Elements of Synthetic Solid Geometry. 12mo. $1.60. DYER: Exercises in Analytical Geometry. By J. M. Dyer , M.A. With Illustrations. 12mo. $1.25. EAGLES: Constructive Geometry of Plane Curves. With Examples. By T. H. Eagles, M.A. 12mo. $3.25. EDGAR and PRITCHARD: Note-book on Practical Solid or Descriptive Geometry. Containing Problems with Help for Solution. By J. H. Edgar, M.A., and G. S. Pritchard. Fourth Edition, Enlarged. By Arthur G. Meeze. 16mo. $1.10. FERRERS: A Treatise on Trilinear Co-ordinates, the Method of Reciprocal Polars, and the Theory of Projections. By the Rev. N. M. Ferrers. Fourth Edition. 12mo. $1.75. FROST: Works by Percival Frost, D.Sc., F.R.S. An Elementary Treatise on Curve Tracing. 8vo. $3.00. Solid Geometry. A New Edition, Revised and Enlarged, of the Treatise by Frost and Wolstenholme. Third Edition. 8vo. $6.00.

MACMILLAN & CO’S PUBLICATIONS ON 508 Hints for the Solution of Problems in the Third Edition of Solid Geometry. 8vo. $3.00. GRAHAM: The Geometry of Position. By Robert H. Graham, C.E., author of “Graphic and Analytical Statics.” Illustrated. 12mo. $1.75. HALL and STEVENS: A Text-book of Euclid’s Elements. Including Alternative Proofs, together with additional Theorems and Exercises, Classified and Arranged. By H. S. Hall, M.A., and F. H. Stevens, M.A. Books I.–VI. Globe 8vo. $1.10. Also sold separately: Book I. 30 cents. Books I. and II. 50 cents. Books I.–IV. 75 cents. Books III.–VI. 75 cents. Book XI. 30 cents. Books V.–VI.–XI. 70 cents. Key to Examples in Books I.–IV., $1.75; Books I.–VI. and XI., $2.25. HAYWARD: The Elements of Solid Geometry. By R. Bald- win Hayward, M.A., F.R.S. 12mo. 75 cents. HEATH (R. S.): Treatise on Geometrical Optics. 8vo. $3.50. An Elementary Treatise on Geometrical Optics. $1.25. JOHNSTON (W. J.) : An Elementary Treatise on Analytic Geometry. $2.60. KITCHENER: Geometrical Note-book. Containing Easy

Problems in Geometrical Drawing, preparatory to the Study of Geometry. By F. E. Kitchener , M.A. New Edition. 4to. 55 cents. LACHLAN (R.): An Elementary Treatise on Modern Pure Geometry. 8vo. $2.25. LOCK: Euclid for Beginners. Being an Introduction to the Existing Text-books. By the Rev. J. B. Lock , M.A. Senior Fellow, Assistant Tutor and Lecturer, Gonville and Caius College, Cambridge, 16mo. (In the Press.) First Book of Euclid's Elements. 16mo. 60 cents. McCLELLAND: The Geometry of the Circle. By W. J. McClelland, M.A., Trinity College, Dublin; Head Master of Santry School. Crown 8vo. Illustrated. $2.60. McDOWELL: Exercises on Euclid and in Modern Geometry, containing Applications of the Principles and Processes of

MATHEMATICS AND PHYSICS. 509 Modern Pure Geometry. By J. McDowell, M.A., F.R.A.S. Third Edition, Revised. $1.50. MILLAR: Elements of Descriptive Geometry. By J. B. Millar, B.E. 12mo. $1.50. MILNE and DAVIS: Geometrical Conics. By J. J. Milne and R. F. Davis. Part I. The Parabola. 12mo. 60 cents. MUKHOPADHAY: Geometrical Conic Sections. By Asu- tosh Mukhopadhay , M.A., Fellow of the University of Calcutta. Globe 8vo. $1.10. NIXON: Euclid. Revised. Containing the Essentials of the Elements of Plane Geometry as given by Euclid. With Additional Propositions and Exercises. Edited by R. C. J. Nixon, M.A. $1.50. Books I. to IV. 12mo. 75 cents. Books V. and VI. 12mo. 75 cents. Geometry in Space. Containing parts of Euclid's Eleventh and Twelfth Books and some Properties of Polyhedra and Solids of Revolution, with Exercises. Edited by R. C. J. Nixon , M.A. 12mo. 90 cents. PLANT: Practical Plane Geometry. By E. C. Plant. (In the Press.) PUCKLE: An Elementary Treatise on Conic Sections and Algebraic Geometry. With Examples and Hints for their Solution. By G. Hale Puckle, M.A. Fifth Edition, Revised and Enlarged. 12mo. $1.90. RICHARDSON: The Progressive Euclid. Books. I.–II. With Notes, Exercises, and Deductions. By A. T. Richardson , M.A. 60 cents. RICHARDSON and RAMSAY: Modern Plane Geometry. By G. Richardson and A. S. Ramsay. $1.00. RUSSELL (J. W.): An Elementary Treatise on Pure Geometry. With Numerous Examples. $2.60. SMITH: Works by Charles Smith, M.A., Fellow and Tutor of Sidney Sussex College, Cambridge. An Elementary Treatise on Conic Sections. Seventh Edition. 12mo. $1.60. Solutions to Conic Sections. 12mo. $2.60. An Elementary Treatise on Solid Geometry.12mo. $2.50.

MACMILLAN & CO'S PUBLICATIONS ON 510 SMITH: Introductory Modern Geometry of the Point, Ray, and Circle. By William B. Smith , Ph.D., Professor of Mathematics in Missouri State University, Columbia, Mo. Part I., 75 cents. Complete Edition, $1.10. SYLLABUS OF PLANE GEOMETRY: (Corresponding to Eu- clid, Books I.–IV.) Revised and brought into correspondence with the text-book prepared by the Association for the Improvement of Geometrical Teaching. New Edition. 16mo. 30 cents. SYLLABUS OF MODERN PLANE GEOMETRY: Paper. 12mo. 30 cents. TAYLOR: Elements of Geometry. 18mo. Books I. and II., 50 cents. Books III. and IV., 50 cents. Books I. to IV., 90 cents. Books V. and VI., 40 cents. Edited by H. M. Taylor, M.A. Solutions to the Exercises in Euclid. Books I.–IV. By W. W. Taylor, M.A. $1.75. TODHUNTER: The Elements of Euclid. By Isaac Tod- hunter, F.R.S. 18mo. 90 cents. Key. 12mo. $1.75. Plane Co-ordinate Geometry, as Applied to the Straight Line and the Conic Sections. 12mo. $1.80. Key. By C. W. Bourne, M.A. 12mo. $2.60. Examples

in Analytical Geometry of Three Dimensions. New Edition, Revised. 12mo. $1.00. VYVYAN: Analytical Geometry for Schools. By T. G. Vyvyan. $1.10. Analytical Geometry for Beginners. Part I. 60 cents. WEEKS: Exercises in Euclid. Graduated and Systematised. By William Weeks, Lecturer on Geometry, St. Luke's Training College, Exeter. 18mo. 60 cents. WILLIS: An Elementary Treatise on Geometrical Conic Sections. By H. G. Willis, M.A. $1.25. WILSON: Works by Rev. J. M. Wilson, M.A. Late Head Master of Clifton College. Elementary Geometry. Books I.–V. Containing the Subjects of Euclid's First Six Books. Following the Syllabus of the Geometrical Association. New Edition. 16mo. $1.10. Solid Geometry and Conic Sections. With Appendices on Transversals and Harmonic Division. 16mo. 90 cents.

MATHEMATICS AND PHYSICS. 511 MENSURATION. MOORE: An Elementary Treatise on Mensuration. By B. T. Moore, M.A. With Numerous Examples. 90 cents. STEVENS: Elementary Mensuration. With Exercises on the Mensuration of Plane and Solid Figures. By F. H. Stevens , M.A. (In the Press.) TODHUNTER: Mensuration for Beginners. By Isaac Tod- hunter, F.R.S. 18mo. 75 cents. Key. By Rev. Fr. L. McCarthy. 12mo. $1.90. TRIGONOMETRY. BEASLEY: An Elementary Treatise on Plane Trigonometry. With Examples. By R. D. Beasley , M.A. Ninth Edition, Revised and Enlarged. 12mo. 90 cents. BOTTOMLEY: Four-Figure Mathematical Tables. Comprising Logarithmic and Trigonometrical Tables, and Tables of Squares, Square Roots, and Reciprocals. By J. T. Bottomley, M.A., F.R.G.S., F.C.S. 8vo. 70 cents. DYER and WHITCOMBE: Trigonometry, The Elements of. By J. M. Dyer, M.A., and the Rev. H. R. Whitcombe, M.A. $1.25. HALL and KNIGHT: Elementary Trigonometry. By H. S. Hall, M.A., and S. R. Knight , B.A., M.B., Ch.B. 16mo. $1.10. HAYWARD: The Algebra of Co-Planar Vectors and Trigono- metry. By R. B. Hayward, M.A., F.R.S. $2.00. HOBSON: Treatise on Plane Trigonometry. By E. W. Hobson. 8vo. $3.00. An Elementary Treatise on Plane Trigonometry for the Use of Schools. By E. W. Hobson , M.A., and C. M. Jessop , M.A. $1.25. JONES: Logarithmic Tables. Royal 8vo. $1.00. JOHNSON: Treatise on Trigonometry. By W. E. Johnson , M.A. Formerly Scholar of King's College, Cambridge. 12mo. $2.25. LEVETT and DAVISON: The Elements of Trigonometry. By Rawdon Levettand A. F. Davison, Masters at King Edward's School, Birmingham. Crown 8vo. $1.60.

MACMILLAN & CO'S PUBLICATIONS ON 512 LOCK: Works by J. B. Lock, M.A., Assistant Tutor and Lecturer in Gonville and Caius College, Cambridge. Trigonometry for Beginners. As far as the Solution of Triangles. 16mo. 75 cents. Key. $1.75. Elementary Trigonometry. Sixth Edition. (In this edition the chapter on Logarithms has been carefully revised.) 16mo. $1.10. Key. $2.25. Higher Trigonometry. Fifth Edition. 16mo. $1.00. Two Partsin one volume. $1.90. The Trigonometry of One Angle. 12mo. Cloth, 65 cents. A Treatise on Elementary and Higher Trigonometry. 16mo. $1.90. LONEY: Plane Trigonometry. By S. L. Loney , M.A. An Elementary Course, excluding the Use of Imaginary Quantities. Fcap. 8vo. Part I. $1.40. Complete. $1.90. McCLELLAND and PRESTON: A Treatise on Spherical Trigonometry. With Applications to Spherical Geometry, and Numerous Examples. By William J. McClelland, M.A., and Thomas Preston, B.A. 12mo. Part I. $1.10. Part II. $1.25. Two Partsin one volume, $2.25. NIXON: Elementary Plane Trigonometry. By R. C. J. Nixon. $1.90. PALMER: Practical Logarithms and Trigonometry, Text-Book of.

By J. H. Palmer. 16mo. $1.10. SNOWBALL: The Elements of Plane and Spherical Trigono- metry. By J. C. Snowball , M.A. Fourteenth Edition. 12mo. $1.90. TODHUNTER: Works by Isaac Todhunter, F.R.S. Trigonometry for Beginners. New Edition. 18mo. 60 cents. Key. $2.25. Plane Trigonometry. 12mo. $1.30. Key. $2.60. A Treatise on Spherical Trigonometry.For the Use of Colleges and Schools. 12mo. $1.10. TODHUNTER and HOGG: Plane Trigonometry. By Isaac Todhunter. New Edition. Revised by R. W. Hogg, M.A., Fellow of St. John's College, Cambridge. 12mo. $1.10.

MATHEMATICS AND PHYSICS. 513 VYVYAN: Introduction to Plane Trigonometry. By the Rev. T. G. Vyvyan , M.A. Third Edition, Revised and Corrected. 90 cents. WARD (G. H.): Trigonometry Examination Papers. 60 cents. WOLSTENHOLME: Examples for Practice in the Use of Seven-Figure Logarithms. By Joseph Wolstenholme, D.Sc. 8vo. $1.25. PROBLEMS AND QUESTIONS IN MATHEMATICS. ARMY PRELIMINARY EXAMINATION (Specimens of Pa- pers set at the), □□□□–□□. With Answers to the Math- ematical Questions. Subjects: Arithmetic, Algebra, Euclid, Geometrical Drawing, Geography, French, English, Dictation. 12mo. 90 cents. BALL: Mathematical Recreations and Problems. By W. W. Rouse Ball. $2.25. CAMBRIDGE Senate-House Problems and Riders, with So- lutions. □□□□. Problems and Riders. Edited by A. G. Greenhill , M.A. 12mo. $2.25. □□□□. Solutions by the Mathematical Moderators and Examiners. Edited by J. W. L. Glaisher, M.A. 8vo. $3.00. CHRISTIE: A Collection of Elementary Test-Questions in Pure and Mixed Mathematics. With Answers and Appendices on Synthetic Division, and on the Solution of Numerical Equations by Horner's Method. By James R. Christie , F.R.S. 12mo. $2.25. DEAKIN: Rider Papers on Euclid. Books I. and II. Graduated and Arranged in Order of Difficulty, with an Introduction on Teaching Euclid. By Robert Deakin, M.A., Balliol College, Oxford, Head Master of King Edward's School, Stourbridge. 18mo. Cloth. 35 cents. DYER: Mathematical Examples. A Collection of Examples in Arithmetic Pure and Mixed, Algebra, Trigonometry, Mensura- tion, Theory of Equations, Analytical Geometry, Statics and Dynamics. With Answers, etc. By J. M. Dyer , M.A., and R. Prowde Smith, M.A. $1.50.

MACMILLAN & CO'S PUBLICATIONS ON 514 FILIPOWSKI (H. E.): A Table of Anti-Logarithms. Natural Numbers answering to all Logarithms from .00001 to .99999, etc. Third Edition. $3.50. LAMB: Hydrodynamics. A Treatise on the Mathematical Theory of Fluid Motion. By Horace Lamb, M.A. 8vo. $3.00. MATTHEWS: Manual of Logarithms. By G. F. Matthews, M.A. 8vo. $1.60. MEDHURST (J. T.): Examination Papers on Book-keeping. Compiled and Arranged by J. T. Medhurst, A.K.C. Second Edition. 75 cents. MILNE: Works by the Rev. John T. Milne. Weekly Problem Papers. With Notes intended for the Use of Students preparing for Mathematical Scholarships. 18mo. $1.00. Solutions to the "Weekly Problem Papers." 12mo. $2.75. A Companion to the "Weekly Problem Papers." Intended for the Use of Students preparing for Mathematical Scholarships, and for the Junior Members of the University who are reading for Mathematical Honors. 12mo. $2.60. RICHARDSON (A. T.): Progressive Mathematical Exercises for Home Work. In Two Parts. By A. T. Richardson , M.A., Senior Mathematical Master at the Isle of Wight College, formerly Scholar of Hertford College, Oxford. Intended for use

in the lower forms of schools together with the ordinary text-books, so that the learner’s progress can from time to time be tested. 60 cents. SANDHURST MATHEMATICAL PAPERS for Admission into the Royal Military College, for the Years□□□□–□□□□. Edited byE. J. Brook-Smith, B.A., LL.M., St. John’s College, Cambridge. 12mo. $1.00. SMITH: Mathematical Examples. By R. Prowde Smith . $1.50. WARD: Trigonometry, Examination Papers in. By G. H. Ward, M.A. 60 cents. Key (for Tutors only). $1.25. WRIGLEY: Collection of Examples and Problems in Arithmetic, Algebra, Geometry, Logarithms, Trigonometry, Conic Sections, Mechanics, etc., with Answers and Occasional Hints. By the Rev. A. Wrigley. Tenth Edition, Twentieth Thousand. 8vo. $2.00.

MATHEMATICS AND PHYSICS. 515 A Key or Companion to the above. Second Edition. $2.60. WOLSTENHOLME: Works of Joseph Wolstenholme, D.Sc. Mathematical Problems on Subjects Included in the First and Second Division of the Schedule of Subjects for the Cambridge Mathematical Tripos Examination. New Edition, Enlarged. 8vo. $4.50. Seven-Figure Logarithms. Examples for Practice in the Use of. For Colleges and Schools. 8vo. $1.25. WOOLWICH MATHEMATICAL PAPERS for Admission into the Royal Military Academy, for the Years □□□□– □□□□. Edited byE. J. Brook-Smith, B.A., LL.M., St. John’s College, Cambridge; Instructor of Mathematics at Royal Military Academy, Woolwich. 12mo. $1.75. HIGHER PURE MATHEMATICS. AIRY: Works by Sir G. B. Airy, K.C.B., formerly Astronomer- Royal. Elementary Treatise on Partial Differential Equations.With Diagrams. Second Edition. 12mo. $1.50. On the Algebraical and Numerical Theory of Errors of Observations and the Combination of Observations. Second Edition, Revised. 12mo. $1.75. BESANT: Notes on Roulettes and Glissettes. By W. H. Besant, D.Sc., F.R.S. Second Edition, Enlarged. $1.25. BOOLE: A Treatise on the Calculus of Finite Differences. By the late George Boole. Edited by J. F. Moulton. Third Edition. 12mo. $2.60. CAYLEY: Elementary Treatise on Elliptic Functions. By Arthur Cayley, D.Sc., F.R.S. (New Edition preparing.) EDWARDS: Differential Calculus. With Applications and Numerous Examples. An Elementary Treatise. By Joseph Edwards, M.A. 12mo. $2.75. Differential Calculus for Beginners. 16mo. $1.10. FERRERS: An Elementary Treatise on Spherical Harmonics and Subjects Connected with Them. By Rev. N. M. Ferrers, D.D., F.R.S. 12mo. $1.90. (Out of print.) A Treatise on Trilinear Co-ordinates.New Edition. $1.75. FORSYTH: Works by Andrew Russell Forsyth, M.A.

MACMILLAN & CO’S PUBLICATIONS ON 516 A Treatise on Differential Equations. 8vo. $3.75. Theory of Differential Equations. Part I. Exact Equations and Pfaff’s Problems. 8vo. $3.75. A Treatise on the Theory of Functionsof a Complex Variable. Royal 8vo. $8.50. FLEISCHER: A System of Volumetric Analysis. With Illustrations. $2.00. FROST: An Elementary Treatise on Curve Tracing. By Percival Frost, M.A. 8vo. $3.00. GREENHILL: Differential and Integral Calculus. With Applications. By Alfred George Greenhill, M.A. 12mo. $2.00. Application of Elliptic Functions. 8vo. $3.00. HEMMING: An Elementary Treatise on the Differential and Integral Calculus. By G. W. Hemming, M.A. 8vo. $2.50. HUNTER (H. St. J.): Decimal Approximations. 18mo. 40 cents. A Key to Dr. Todhunter’s Differential Calculus. $2.60. KELLAND and TAIT: Introduction to Quaternions. With Examples. By P. Kelland , M.A., and P. G. Tait , M.A. Second Edition. 12mo. $2.00. KEMPE: How to

Draw a Straight Line. A Lecture on Linkages. By A. B. Kempe, B.A. With Illustrations. 12mo. 50 cents. KNOX: Differential Calculus for Beginners. With Examples. By Alexander Knox, B.A. 16mo. 90 cents. LOVE: Treatise on the Mathematical Theory of Elasticity. 8vo. Vol. I. $3.00. Vol. II. $3.00. MESSENGER OF MATHEMATICS: Edited by J. W. L. Glaisher. Published Monthly. 35 cents each number. MORLEY and HARKNESS: The Theory of Functions. By Frank Morley, M.A., Professor of Mathematics, Haverford College, Pa., and James Harkness , M.A., Professor of Mathematics, Bryn Mawr College, Pa. 8vo. $5.00. MUIR: Works byThomas Muir, Mathematical Master in the High School, Glasgow. A Treatise on the Theory of Determinants. With Examples. 12mo. New Edition. $2.25.

MATHEMATICS AND PHYSICS. 517 The Theory of Determinants in the Historical Order of its Development. Part I. Determinants in General. Leibnitz (□□□□) to Cayley (□□□□). 8vo. $2.50. PRICE: Treatise on Infinitesimal Calculus. By Bartholomew Price, M.A., F.R.S., Professor of Natural Philosophy, Oxford. Vol. I. Differential Calculus. Second Edition. 8vo. $3.75. Vol. II. Integral Calculus, Calculus of Variations, and Differential Equations. 8vo. (Reprinting.) Vol. III. Statics, including Attractions; Dynamics of a Material Particle. 8vo. $4.00. Vol. IV. Dynamics of Material Systems. Together with a Chapter on Theoretical Dynamics, by W. F. Donkin , M.A. 8vo. $4.50. SCOTT: A Treatise on the Theory of Determinants and their Applications in Analysis and Geometry. By Robert Scott, M.A. 8vo. $3.50. SMALLEY: Facts and Formulæ in Pure Mathematics and Natural Philosophy. Containing Facts, Formulæ, Symbols, and Definitions. By the late G. R. Smalley, B.A., F.R.A.S. New Edition byJ. M'Dowell, M.A., F.R.A.S. 16mo. 70 cents. SMITH: Mathematical Papers of the late Rev. J. S. Smith, Savilian Professor of Geometry in the University of Oxford, With Portrait and Memoir. 2 vols. 4to. (In Preparation.) TAIT: An Elementary Treatise on Quaternions. By P. G. Tait, M.A., Professor of Natural Philosophy in the University of Edinburgh. Third Edition, Much Enlarged. 8vo. $5.50. TODHUNTER: Works by Isaac Todhunter, F.R.S. An Elementary Treatise on the Theory of Equations. 12mo. $1.80. A Treatise on the Differential Calculus. 12mo. $2.60. Key. $2.60. A Treatise on the Integral Calculus and its Applications. 12mo. $2.60. Key. $2.60. An Elementary Treatise on Laplace's Lam´ e's and Bessel's Functions. 12mo. $2.60. (Out of print.) WATSON and BURBURY: A Treatise on the Application of Generalized Co-ordinates to the Kinetics of a Material System. By H. W. Watsonand S. H. Burbury. 8vo. $1.50.

MACMILLAN & CO'S PUBLICATIONS ON 518 WELD: A Short Course in the Theory of Determinants. By Laenas Gifford Weld, B.S., M.A. $1.90. WHITWORTH: Trilinear Co-ordinates, and other methods of Modern Analytical Geometry of Two Dimensions. An Elementary Treatise. By W. Allen Whitworth, M.A. 8vo. $4.00. MECHANICS. ALDIS: Rigid Dynamics, An Introductory Treatise on. By W. Steadman Aldis, M.A. $1.00. ALEXANDER and THOMPSON: Elementary Applied Me- chanics. Part II. Transverse Stress. $2.75. BALL: Experimental Mechanics. A Course of Lectures delivered to the Royal College of Science for Ireland. By Sir R. S. Ball, LL.D., F.R.S. Second Edition. With Illustrations. 12mo. $1.50. BASSET: A Treatise on Hydrodynamics. 2 vols. 8vo. $9.00. An Elementary Treatise on Hydrodynamics and Sound.8vo. $3.00. A Treatise on Physical Optics. 8vo. $6.00. BAYNES: Lessons on Thermodynamics. By R. E. Baynes , M.A. 12mo. $1.90. BESANT: A Treatise on

Hydromechanics. Fifth Edition, Revised. Part I. Hydrostatics. 12mo. $1.25. A Treatise on Dynamics. $1.75. Elementary Hydrostatics. 16mo. $1.00. Solutions to the Examples. (In the Press.) CLIFFORD: Works by W. Kingdon Clifford, F.R.S. Elements of Dynamic. An Introduction to the Study of Motion and Rest in Solid and Fluid Bodies. Part I. Books I.–III. 12mo. $1.90. Part II. Book IV. and Appendix. 12mo. $1.75. COTTERILL: Applied Mechanics. An Elementary General Introduction to the Theory of Structures and Machines. By James H. Cotterill, F.R.S. 8vo. $5.00. COTTERILL and SLADE: Elementary Manual of Applied Mechanics. By Prof. J. H. Cotterill, F.R.S., and J. H. Slade. 12mo. $1.25.

MATHEMATICS AND PHYSICS. 519 CREMONA (Luigi): Graphical Statics. Two Treatises on the Graphical Calculus and Reciprocal Figures in Graphical Calculus. Authorized English Translation by T. Hudson Beare. 8vo. $2.25. GARNETT: Elementary Dynamics, A Treatise on. For the Use of Colleges and Schools. By William Garnett, M.A., D.C.L. Fifth Edition, Revised. $1.50. GOODWIN: Elementary Statics. By H. Goodwin , D.D., Bishop of Carlisle. Second Edition. 75 cents. GREAVES. Works by John Greaves, M.A. A Treatise on Elementary Statics. Second Edition, Revised. 12mo. $1.90. Statics for Beginners. 16mo. 90 cents. Treatise on Elementary Hydrostatics. 12mo. $1.10. GREENHILL: Hydrostatics. By A. G. Greenhill . 12mo. $1.90. GUILLEMIN (A.): The Applications of Physical Forces. Translated and Edited by J. Norman Lockyer , F.R.S. With Colored Plates and Illustrations. Royal 8vo. $6.50. HICKS: Elementary Dynamics of Particles and Solids. By W. M. Hicks. 12mo. $1.60. HOROBIN: Elementary Mechanics. Stage I. By J. C. Horobin, B.A. With Numerous Illustrations. 12mo. Cloth. 50 cents. Stages II. and III. (In Preparation.) Theoretical Mechanics. Division I. (In the Press.) HOSKINS: The Elements of Graphic Statics. A Text-book for Students of Engineering. By L. M. Hoskins, C.E., M.S. 8vo. $2.25. JELLETT: A Treatise on the Theory of Friction. By John H. Jellett, B.D., late Provost of Trinity College, Dublin. 8vo. $2.25. JESSOP: The Elements of Applied Mathematics, including Kinetics, Statics, and Hydrostatics. By C. M. Jessop. $1.25. KENNEDY: The Mechanics of Machinery. By Alexander B. W. Kennedy, F.R.S. With Illustrations. 12mo. $3.50. LAMB: Hydrodynamics. A Treatise on the Mathematical Theory of Fluid Motion. By H. Lamb. 8vo. $3.00. LOCK: Works by the Rev. J. B. Lock, M.A. Dynamics for Beginners. 16mo. $1.00.

MACMILLAN & CO’S PUBLICATIONS ON 520 Elementary Statics. 16mo. $1.10. Key. 12mo. $2.25. Mechanics for Beginners. Part I. 90 cents. Mechanics of Solids. Elementary Hydrostatics. (In Preparation.) Mechanics of Solids. 16mo. (In the Press.) Mechanics of Fluids. 16mo. (In the Press.) LONEY: A Treatise on Elementary Dynamics. New and Enlarged Edition. By S. L. Loney, M.A. 12mo. $1.90. Solutions of the Examples contained in the Above. 12mo. $1.90. The Elements of Statics and Dynamics. Part I. Elements of Statics. $1.25. Part II. Elements of Dynamics. $1.00. Complete in one volume. 12mo. $1.90. Key. 12mo. $1.90. Mechanics and Hydrostatics for Beginners. 16mo. $1.25. MACGREGOR: An Elementary Treatise on Kinematics and Dynamics. By James Gordon Macgregor, M.A., D.Sc., Munro Professor of Physics, Dalhousie College, Halifax. 12mo. $2.60. MINCHIN: Works by G. M. Minchin, M.A. A Treatise on Statics. Third Edition, Corrected and Enlarged. Vol. I. Equilibrium of Coplanar Forces. 8vo. $2.25. Vol. II. Statics. 8vo. $4.00. Uniplanar Kinematics of

Solids and Fluids. 12mo. $1.90. Hydrostatics and Elementary Hydrokinetics. $2.60. PARKINSON (R. M.): Structural Mechanics. $1.10. PARKINSON: A Treatise on Elementary Mechanics. For the use of the Junior Classes at the University and the Higher Classes in Schools. With a collection of Examples by S. Parkinson, F.R.S. Sixth Edition. 12mo. $2.25. PIRIE: Lessons on Rigid Dynamics. By the Rev. G. Pirie , M.A. 12mo. $1.50. RAWLINSON: Elementary Statics. By G. Rawlinson, M.A. Edited by E. Sturges. 8vo. $1.10. ROUTH: Works by E. J. Routh, LL.D., F.R.S. A Treatise on the Dynamics of a System of Rigid Bodies. With Examples. New Edition, Revised and Enlarged. 8vo. In Two Parts. Part I. Elementary. Fifth Edition, Revised and Enlarged. $3.75.

MATHEMATICS AND PHYSICS. 521 Part II. Advanced. $3.75. Stability of a Given State of Motion, Particularly Steady Motion. 8vo. $2.25. A Treatise on Analytical Statics. With Numerous Examples. Vol. I. 8vo. $3.75. SANDERSON: Hydrostatics for Beginners. By F. W. Sanderson, M.A. 16mo. $1.10. SELBY: Elementary Mechanics of Solids and Fluids. $1.90. SYLLABUS OF ELEMENTARY DYNAMICS. Part I. Linear Dynamics. With an Appendix on the Meanings of the Symbols in Physical Equations. Prepared by the Association for the Improvement of Geometrical Teaching. 4to. 30 cents. TAIT and STEELE: A Treatise on Dynamics of a Particle. By Professor Tait, M.A., and W. J. Steele . Sixth Edition, Revised. 12mo. $3.00. TAYLOR: Resistance of Ships, and Screw Propulsion. By D. W. Taylor. $3.75. TODHUNTER. Works by Isaac Todhunter, F.R.S. Mechanics for Beginners. With Numerous Examples. New Edition. 18mo. $1.10. Key. $1.75. A Treatise on Analytical Statics. Fifth Edition. Edited by Professor J. D. Everett, F.R.S. 12mo. $2.60. WALTON: Mechanics, A Collection of Problems in Elementary. By W. Walton, M.A. Second Edition. $1.50. Problems in Theoretical Mechanics. Third Edition, Revised. With the addition of many fresh Problems. By W. Walton, M.A. 8vo. $4.00. WEISBACH and HERRMANN: The Mechanics of Hoisting Machinery, including Accumulators, Excavators, and Pile-Drivers. A Text-Book for Technical Schools, and a Guide for Practical Engineers. By Dr. Julius Weisbach and Professor Gustav Herrmann. Authorized Translation from the Second German Edition. By Karl P. Dahlstrom, M.E., Instructor of Mechanical Engineering in the Lehigh University. With 177 Illustrations. $3.75. ZIWET: An Elementary Treatise on Theoretical Mechanics. In Three Parts: Kinematics, Statics, and Dynamics. By Alexander Ziwet, University of Michigan. Part I. $2.25. Part II. $2.25. Part III. (In Preparation.)

MACMILLAN & CO’S PUBLICATIONS ON 522 PHYSICS. AIRY. Works by Sir G. B. Airy, K.C.B., formerly Astronomer- Royal. On Sound and Atmospheric Vibrations. With the Mathemat- ical Elements of Music. Designed for the Use of Students in the University. Second Edition, Revised and Enlarged. 12mo. $2.50. Gravitation. An Elementary Explanation of the Principal Perturbations in the Solar System. New Edition. 12mo. $1.90. ALDIS: Geometrical Optics. An Elementary Treatise. By W. Steadman Aldis , M.A. Third Edition, Revised. 12mo. $1.00. CLAUSIUS: Mechanical Theory of Heat. By R. Clausius. Translated by Walter R. Browne, M.A. 12mo. $2.60. DANIELL: A Text-Book of the Principles of Physics. By Alfred Daniell, D.Sc. Illustrated. New Edition, Revised and Enlarged. 8vo. $3.50. DAUBENY’S Introduction to the Atomic Theory. 16mo. $1.50. DONKIN (W. F.): Acoustics. Second Edition.

12mo. $1.90. EVERETT: Units and Physical Constants. By J. D. Everett, F.R.S., Professor of Natural Philosophy, Queen's College, Belfast. New Edition. 16mo. $1.25. FERRERS: Spherical Harmonics and Subjects Connected with them. By Rev. N. M. Ferrers, D.D., F.R.S. 12mo. $1.90. FISHER: Physics of the Earth's Crust. By Osmond Fisher. Second Edition, Enlarged. 8vo. $3.50. FOURIER: The Analytical Theory of Heat. By Joseph Fourier. Translated with Notes, by A. Freeman, M.A. 8vo. $4.50. GALLATLY: Physics, Examples in Elementary. Comprising Statics, Dynamics, Hydrostatics, Heat, Light, Chemistry, and Electricity. With Examination Papers. By W. Gallatly, M.A. $1.00. GARNETT: Heat, An Elementary Treatise on. By W. Gar- nett, M.A., D.C.L. Fifth Edition, Revised and Enlarged. $1.10. GLAZEBROOK: Heat. By R. T. Glazebrook, M.A., F.R.S. $1.00. Light. Cambridge Natural Science Manuals. $1.00.

MATHEMATICS AND PHYSICS. 523 HEATH: Treatise on Geometrical Optics. By R. S. Heath. 8vo. $3.50. An Elementary Treatise on Geometrical Optics. By R. S. Heath. 12mo. $1.25. HOGG'S (Jabez) Elements of Experimental and Natural Philosophy. With Index and upwards of 400 Woodcuts. $1.50. IBBETSON: The Mathematical Theory of Perfectly Elastic Solids. With a Short Account of Viscous Fluids. By W. J. Ibbetson, late Senior Scholar of Clare College, Cambridge. 8vo. $5.00. JELLETT (John H. B. D.): A Treatise on the Theory of Friction. 8vo. $2.25. JONES: Examples in Physics. By D. E. Jones, B.Sc. 16mo. 90 cents. Sound, Light, and Heat. An Elementary Text-book. By D. E. Jones, B.Sc., author of "Examples in Physics," etc. With Illustrations. 16mo. 70 cents. Lessons in Heat and Light. 16mo. $1.00. LOEWY. Works by B. Loewy, F.R.A.S. Experimental Physics. Questions and Examples in Physics, Sound, Light, Heat, Electricity, and Magnetism. 16mo. 50 cents. A Graduated Course of Natural Science, Experimental and Theoretical, for Schools and Colleges. Part I. First Year's Course for Elementary Schools and the Junior Classes of Technical Schools and Colleges. 16mo. 60 cents. Part II. 60 cents. LOVE: Treatise on the Mathematical Theory of Elasticity. 8vo. Vol. I. $3.00. Vol. II. $3.00. LUPTON: Numerical Tables and Constants in Elementary Science. By Sydney Lupton. 16mo. 70 cents. MACFARLANE: Physical Arithmetic. By Alexander Mac- farlane, Professor of Physics, University of Texas. 12mo. $1.90. MAXWELL: The Scientific Papers of James Clerk Maxwell, M.A., LL.D. B.Sc., etc., etc. Edited by W. D. Niven, M.A., F.R.S. With Steel Portraits and Page Plates. 2 vols. 4to. $25.00. McAULAY(A.): Utility of Quaternions in Physics. 8vo. $1.60.

MACMILLAN & CO'S PUBLICATIONS ON 524 MOLLOY: Gleanings in Science. A Series of Popular Lectures on Scientific Subjects. By the Rev. Gerard Molloy, D.D., D.Sc. 8vo. $2.25. NEWTON'S Principia. Edited by Professor Sir W. Thomson and Professor Blackburn. (Latin Text.) 4to. $12.00. This volume does not contain an English Translation. First Book. Sections I., II., III. With Notes and Problems. By P. Frost, M.A. Third Edition. 8vo. $3.00. The First Three Sections of Newton's Principia , with an Appendix; and the Ninth and Eleventh Sections. By J. H. Evans, M.A. The Fifth Edition, edited by P. T. Main. $1.00. PARKER: A Treatise on Thermodynamics. By T. Parker, M.A., Fellow of St. John's College, Cambridge. $2.25. PARKINSON: A Treatise on Optics. By S. Parkinson, D.D., F.R.S. Fourth Edition, Revised and Enlarged. 12mo. $2.50. PEARSON: A History of the Theory of Elasticity. By Isaac Todhunter. Edited by Professor Karl Pearson . Vol. I. Galilei to Saint-Venant,

□□□□–□□□□. 8vo. $6.00. Vol. II. Saint-Venant to Lord Kelvin (Sir William Thomson). In Two Parts. $7.50. PERRY: An Elementary Treatise on Steam. By John Perry. With Woodcuts, Numerical Examples, and Exercises. 18mo. $1.10. PRESTON: The Theory of Light. By Thomas Preston. With Illustrations. 8vo. $3.25. The Theory of Heat. By the same author. 8vo. $5.50. RAYLEIGH: The Theory of Sound. By Lord Rayleigh, M.A., F.R.S. 8vo. New edition in two volumes. (In the Press.) Vol. I. $3.25. (Out of Print.) Vol. II. $3.25. Vol. III. (In the Press.) SAINT-VENANT (Barri de): The Elastic Researches of. Edited for the Syndics of the Cambridge University Press by Karl Pearson, M.A. 8vo. $2.75. SHANN: An Elementary Treatise on Heat in Relation to Steam and the Steam-Engine. By G. Shann, M.A. With Illustrations. 12mo. $1.10. SHAW: Practical Work at the Cavendish Laboratory. Edited by W. N. Shaw.

MATHEMATICS AND PHYSICS. 525 Heat. 8vo. 90 cents. SPOTTISWOODE: Polarization of Light. By W. Spottis- woode, LL.D. Illustrated. 12mo. $1.25. STEWART. Works by Balfour Stewart, F.R.S. Lessons in Elementary Physics. With Illustrations and colored Diagrams. 16mo. $1.10. Questions on the Same for Schools. By T. H. Core. 40 cents. A Treatise on Heat. With Numerous Woodcuts and Diagrams. Fourth Edition. 16mo. $1.90. STEWARTand GEE: Lessons on Elementary Practical Physics. By Balfour Stewart, M.A., LL.D., F.R.S., and W. W. Haldane Gee. Vol. I. General Physical Processes. 12mo. $1.50. Vol. II. Electricity and Magnetism. $2.25. Vol. III. Optics, Heat, and Sound. (In the Press.) Practical Physics for Schools and the Junior Students of Colleges. Vol. I. Electricity and Magnetism. 16mo. 60 cents. Vol. II. Optics, Heat, and Sound. (In the Press.) STOKES. Works by George Gabriel Stokes, F.R.S. On Light. Burnett Lectures. On the Nature of Light. On Light as a Means of Investigation. On the Beneficial Effects of Light. 12mo. $2.00. Mathematical and Physical Papers. 8vo. Vol. I. $3.75. Vol. II. $3.75. Vol. III. (In the Press.) STONE: Elementary Lessons on Sound. By W. H. Stone , M.B. With Illustrations. 16mo. 90 cents. TAIT. Works by P. G. Tait, M.A., Sec. R.S.E. Lectures on Some Recent Advances in Physical Science. With Illustrations. Third Edition, Revised and Enlarged, with the Lecture on Force Delivered before the British Association. 12mo. $2.50. Heat. With Numerous Illustrations. 12mo. $2.00. Light. An Elementary Treatise. With Illustrations. 12mo. $2.00. Properties of Matter. Second Edition, Enlarged. 12mo. $2.25. TAYLOR: Sound and Music. An Elementary Treatise on the Physical Constitution of Musical Sounds and Harmony. By Sedley Taylor, M.A. Illustrated. Second Edition. 12mo. $2.50.

MACMILLAN & CO’S PUBLICATIONS ON 526 THOMSON. Works of J. J. Thomson, Professor of Experimental Physics in the University of Cambridge. A Treatise on the Motion of Vortex Rings. An Essay. With Diagrams. 8vo. $1.75. Application of Dynamics to Physics and Chemistry. 12mo. $1.90. THOMSON. Works of Sir W. Thomson , F.R.S. Professor of Natural Philosophy in the University of Glasgow. Mathematical and Physical Papers. Vol. I. 8vo. $5.00. Vol. II. 8vo. $4.50. Vol. III. 8vo. $5.50. Popular Lectures and Addresses on Various Subjects in Physical Science. Vol. I. Constitution of Matter. 12mo. $2.00. Vol. II. Geology and General Physics. 12mo. $2.00. Vol. III. Navigational Affairs. With Illustrations. 12mo. $2.00. On Elasticity. 4to. $1.25. On Heat. 4to. $1.25. TODHUNTER: A History of the Theory of Elasticity. By Isaac Todhunter. Edited by Professor Karl Pearson. Vol. I. Galilei to Saint-Venant,

□□□□–□□□□. 8vo. $6.00. Vol. II. Saint-Venant to Lord Kelvin (Sir William Thomson). In Two Parts. $7.50. TURNER: A Collection of Examples on Heat and Electricity. By H. H. Turner, B.A. 12mo. 75 cents. WALKER: The Theory and Use of a Physical Balance. By James Walker, M.A. With Illustrations in Collotype and Photolithography. 8vo. 90 cents. WATSON and BURBURY. Works by H. W. Watson, D.Sc., and S. H. Burbury, M.A. A Treatise on the Application of Generalized Co-ordinates to the Kinetics of a Material System. 8vo. $1.50. WOOD: Light. By Sir H. Truman Wood. 16mo. 60 cents. WOOLCOMBE: Practical Work in Heat. For Use in Schools and Colleges. By W. G. Woolcombe , M.A., B.Sc., Senior Science Master in King Edward's High School, Birmingham. Crown 8vo. pp. 61. $1.00. WRIGHT: Light. A Course of Experimental Optics, Chiefly with the Lantern. By Lewis Wright. With nearly 200 Illustrations. 12mo. $2.50.

MATHEMATICS AND PHYSICS. 527 ELECTRICITY AND MAGNETISM. ALLSOP (F. C.): Practical Electric Light Fitting. 200 Illustra- tions. $1.50. BENNETT: The Telephoning of Great Cities. Paper. 35 cents. BLAKESLEY: Alternating Currents of Electricity. Third Edition, Enlarged. (In the Press.) BONNEY (G. E.): Induction Coils. $1.00. Electrical Experiments. A Manual of Instructive Amusement. With 144 Illustrations. 12mo. 75 cents. BOTTONE (S. R.): Electricity and Magnetism. With 103 Illustrations. 16mo. 90 cents. How to Manage the Dynamo. A Handbook for Ship Engineers, Electric Light Engineers, etc. 16mo. 60 cents. A Guide to Electric Lighting. By S. R. Bottone , author of "Electric Bells, and All About them," "Electromotors: How Made, and How Used," etc. With Many Illustrations. 75 cents. CAVENDISH: The Electrical Researches of the Honourable Henry Cavendish, F.R.S. Written between □□□□ and □□□□. Edited from the original manuscripts of the late J. Clerk Maxwell, F.R.S. 8vo. $5.00. CUMMING: An Introduction to the Theory of Electricity. By Linnæus Cumming, M.A. With Illustrations. 12mo. $2.25. DAY: Electric Light Arithmetic. By R. E. Day, M.A. 18mo. 40 cents. EMTAGE: An Introduction to the Mathematical Theory of Electricity and Magnetism. By W. T. A. Emtage, M.A., of Pembroke College, Oxford. 12mo. $1.90. GRAY: The Theory and Practice of Absolute Measurements in Electricity and Magnetism. By Andrew Gray, M.A., F.R.S.E. In two volumes. Vol. I., 12mo. $3.25. Vol. II. (in two parts). $6.25. Absolute Measurements in Electricity and Magnetism for Beginners. By Andrew Gray, M.A., F.R.S.E. Students' Edition, Abridged from the larger work. 16mo. $1.25. GUILLEMIN: Electricity and Magnetism. A Popular Treatise. By Am´ed´ee Guillemin , author of "The Forces of Nature," "The Applications of Physical Forces," etc. Translated and edited, with Additions and Notes, by Professor Silvanus P.

MACMILLAN & CO'S PUBLICATIONS ON 528 Thompson, author of "Elementary Lessons in Electricity and Magnetism," etc. With 600 Illustrations. Super royal 8vo. $8.00. HAWKINS and WALLIS: The Dynamo. Its Theory, Design, and Manufacture. By C. C. Hawkins and F. Wallis. With 190 Illustrations. $3.00. HEAVISIDE(Oliver): Electrical Papers. For Advanced Students in Electricity. 2 vols. 8vo. $10.00. HERTZ (H.): Researches in the Propagation of Electrical Forces. Authorized Translation by D. E. Jones , B.Sc. Illustrated. 8vo. $3.00. JACKSON (D. C.): A Text-Book on Electro-Magnetism and the Construction of Dynamos. By Dugald C. Jackson , B.S., C.E., Professor of Electrical Engineering, University of Wisconsin. 12mo. $2.25.

LODGE (Oliver J.): Modern Views of Electricity. By Oliver J. Lodge, LL.D., D.Sc., F.R.S. Illustrated. $2.00. Lightning Conductors and Lightning Guards.With Numerous Illustrations. $4.00. MAXWELL: An Elementary Treatise on Electricity. By James Clerk Maxwell, M.A. Edited byWilliam Garnett, M.A. Second Edition. 8vo. $1.90. A Treatise on Electricity and Magnetism. By James Clerk Maxwell, M.A. 2 vols. 8vo. Second Edition. $8.00. Supplementary volume, by J. J. Thomson, M.A. $4.50. MAYCOCK: A First Book of Electricity and Magnetism. For the Use of Elementary Science and Art and Engineering Students and General Readers. By W. Perren Maycock , M.I.E.E. With 84 Illustrations. Crown 8vo. 60 cents. Electric Lighting and Power Distribution. Illustrated. Com- plete in three parts. 16mo. Paper. 75 cents each. MURDOCK: Notes on Electricity and Magnetism. Designed as a companion to Silvanus P. Thompson's "Elementary Lessons in Electricity and Magnetism." By J. B. Murdock , Lieut. U.S.N. 18mo. New Edition. 60 cents. POOLE: The Practical Telephone Handbook. By Joseph Poole. With 227 Illustrations. Small crown 8vo. $1.00. PREECE and STUBBS: A Manual of Telephony. By Wm. Henry Preece and Arthur J. Stubbs. $4.50.

MATHEMATICS AND PHYSICS. 529 RUSSELL: Electric Light Cables and the Distribution of Electricity. By Stuart A. Russell, A.M., I.C.E. With over 100 Illustrations. 12mo. $2.25. STEWARTand GEE: Practical Physics for Schools and Junior Students of Colleges. By Balfour Stewart, M.A., LL.D., F.R.S., and W. W. Haldane Gee, B.Sc. Vol. I. Electricity and Magnetism. 16mo. 60 cents. Lessons in Elementary Practical Physics. By Balfour Stewart, M.A., LL.D., F.R.S., and W. W. Haldane Gee, B.Sc. Vol. II. Electricity and Magnetism. 12mo. $2.25. THOMSON: Notes on Recent Researches in Electricity and Magnetism. Intended as a Sequel to Professor Clerk Maxwell's "Treatise on Electricity and Magnetism." By J. J. Thomson. 8vo. $4.50. THOMSON: Reprints of Papers of Electrostatics and Mag- netism. By SirWilliam Thomson, D.C.L., LL.D., F.R.S., F.R.S.E. 8vo. $5.00. THOMPSON: Elementary Lessons in Electricity and Mag- netism. By Silvanus P. Thompson, D.Sc., B.A., F.R.A.S. New Edition. With Illustrations. 16mo. $1.25. Notes to the same, by J. B. Murdock. 60 cents. WALKER: How to Light a Colliery by Electricity. 4to. Limp. 75 cents. Town Lighting by Electricity.(In the Press.) WATSON and BURBURY: The Mathematical Theory of Electricity and Magnetism. By H. W. Watson, D.Sc., F.R.S., and S. H. Burbury, M.A. Vol. I. Electrostatics. 8vo. $2.75. II. Magnetism and Electrodynamics. 8vo. $2.60. HISTORICAL. BALL: Works by Walter W. Rouse Ball. A Short Account of the History of Mathematics. 12mo. $2.60. History of the Study of Mathematics at Cambridge. 12mo. $1.90.

MACMILLAN & CO'S PUBLICATIONS ON 530 BARROW: Mathematical Works. Edited by W. Whewell . $2.25. CAYLEY (Arthur): The Collected Mathematical Papers of. To be completed in ten volumes. Vols. I.–VI. published. $5.50 each. GOW: A Short History of Greek Mathematics. By J. Gow. 8vo. $3.00. HEATH: Diophantos of Alexandria. A Study in the History of Greek Algebra. By T. L. Heath, B.A. 8vo. $2.25. KLEIN: Lectures on Mathematics. The Evanston Colloquium. Reported by Alexander Ziwet. $1.50. SMITH (Henry J. S.): Mathematical Papers. With Portrait and Memoir. 2 vols. (In the Press.) WOOLWICH: Mathematical Papers. Edited by E. J. Brook- smith, B.A. $1.75. A SHORT ACCOUNT OF THE HISTORY OF MATHEMATICS. By WALTER

W. ROUSE BALL, Fellow and Tutor of Trinity College, Cambridge. Second Edition, Revised. $3.25. “While technical and exact enough to be of value to the specialist in mathematics as a handy book of reference, it is so clearly and familiarly written, that it is the best work on this subject for the general reader that we know of. . . . From this history, or historical sketch, the intelligent reader can gain a very complete view of the progress of mathematical science from its beginnings until its contemporary differentiation into numerous specialties,—each of them important and difficult enough to detain for a lifetime a brilliant mind,—all of which are fruitful in their applications to the various phases of modern science and modern industry.”—Science. A HISTORY OF THE STUDY OF MATHEMATICS AT CAMBRIDGE.

MATHEMATICS AND PHYSICS. 531 By WALTER W. ROUSE BALL, 12mo. $1.90. A SHORT HISTORY OF GREEK MATHEMATICS. By JAMES GOW, 8vo. $3.00. “. . . Evidently the production of a scholar, and the result of years of laborious research. Mr. Gow divides his history into three parts. The first treats of the decimal scale and Egyptian arithmetic; the second and third parts are concerned with Greek arithmetic and geometry. . . . The largest part of Mr. Gow’s history, and that which will probably be the most interesting to the general mathematical reader, is justly devoted to geometry; for it is in this department of mathematics that the acuteness of the Greek mind is most conspicuously seen, and that the continuity of mathematical discovery can be more fully traced. . . . The interesting character of the notes is quite a feature of the book, which is in this respect distinguished from almost all histories of mathematics. . . . It must be to all students of mathematics a most welcome and instructive volume.”—J. S. Mackay, in The Academy. DIOPHANTOS OF ALEXANDRIA: A STUDY IN THE HISTORY OF GREEK ALGEBRA. By T. S. HEATH, B.A., Scholar of Trinity College, Cambridge. 8vo. $2.00. THE MACMILLAN COMPANY, 66 FIFTH AVENUE, NEW YORK.

LICENSE 532 End of Project Gutenberg’s A History of Mathematics, by Florian Cajori